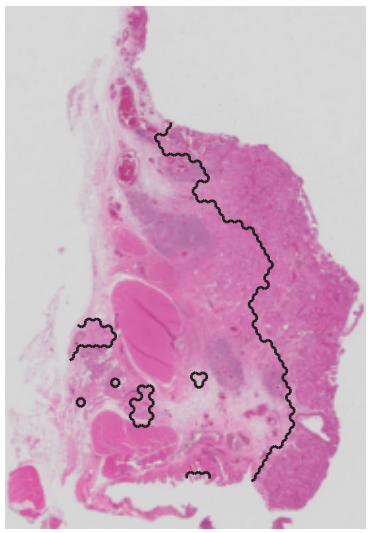
NPJ_sample1 | Tissue context

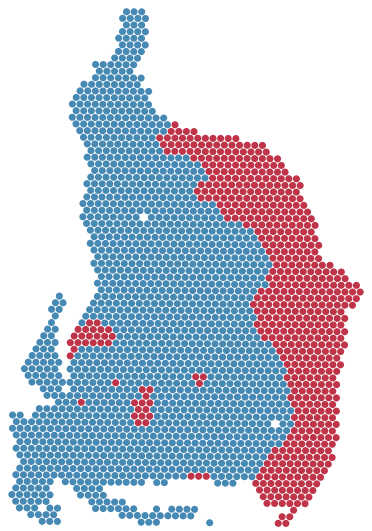
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



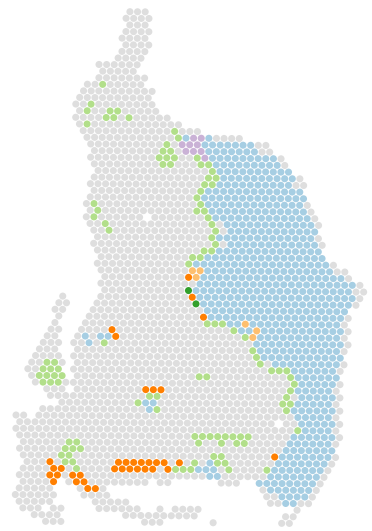
Black line: observed candidate outer edge

Final candidate location



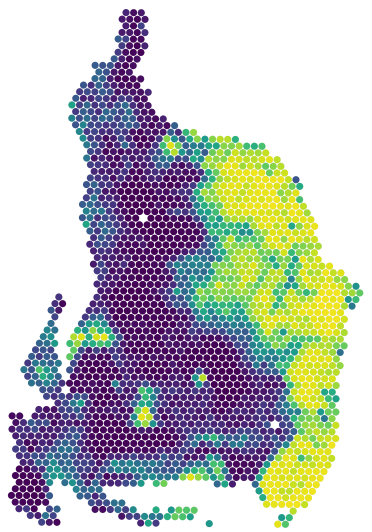
inside outside

Accepted R12 (original 0–8)



0 3 4 5 8 2

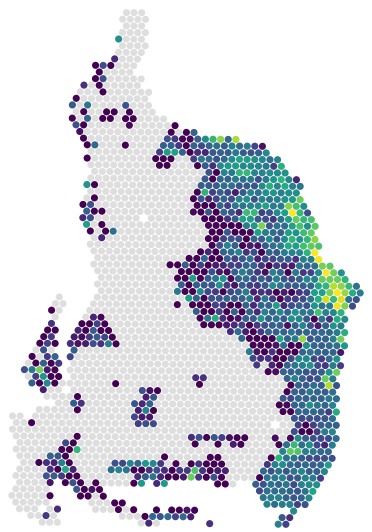
Epithelial support



0 1

Fraction of all 30

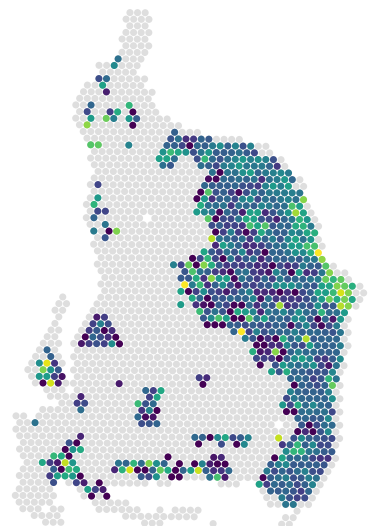
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



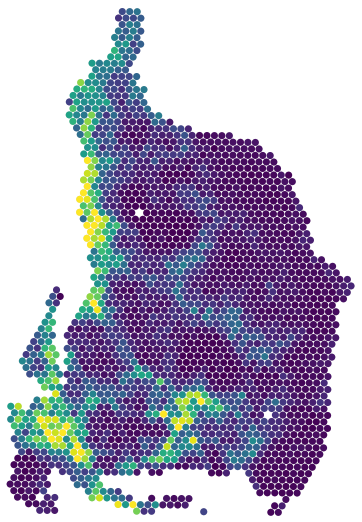
0 0.14

Matched-model AUCell

NPJ_sample1 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

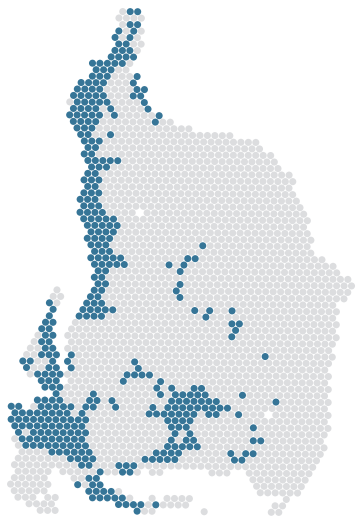
CAF composition



0 0.418

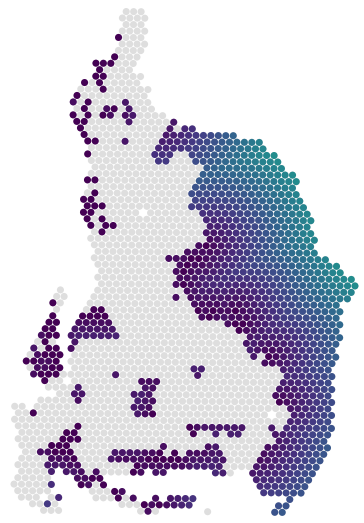
Fraction of all 30

CAF P80 targets



False True

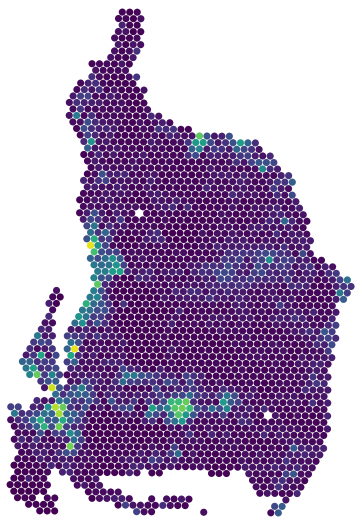
CAF distance



0 3.1e+03

Nearest spot center (μm)

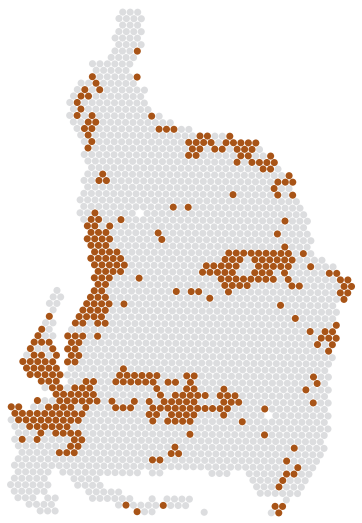
TAM composition



0 0.291

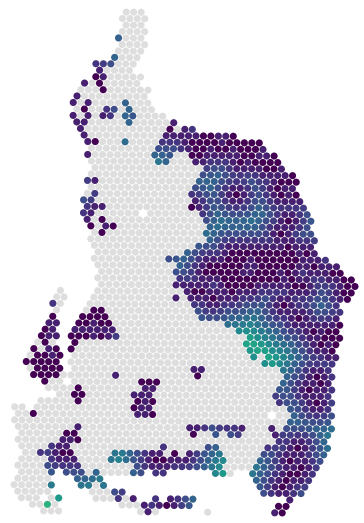
Fraction of all 30

TAM P80 targets



False True

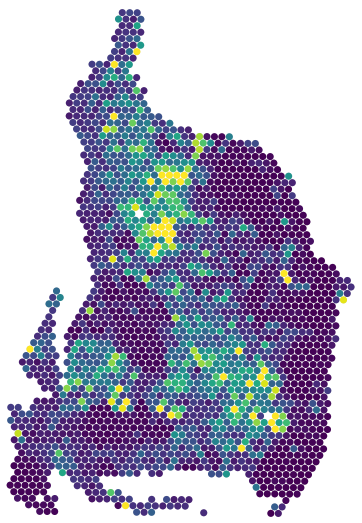
TAM distance



0 1.13e+03

Nearest spot center (μm)

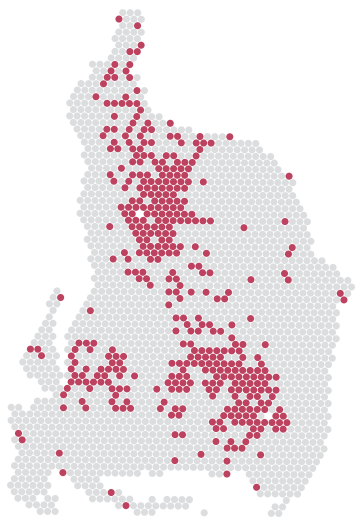
Treg composition



0 0.102

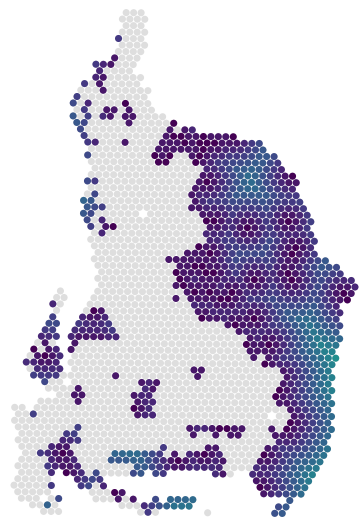
Fraction of all 30

Treg P80 targets



False True

Treg distance

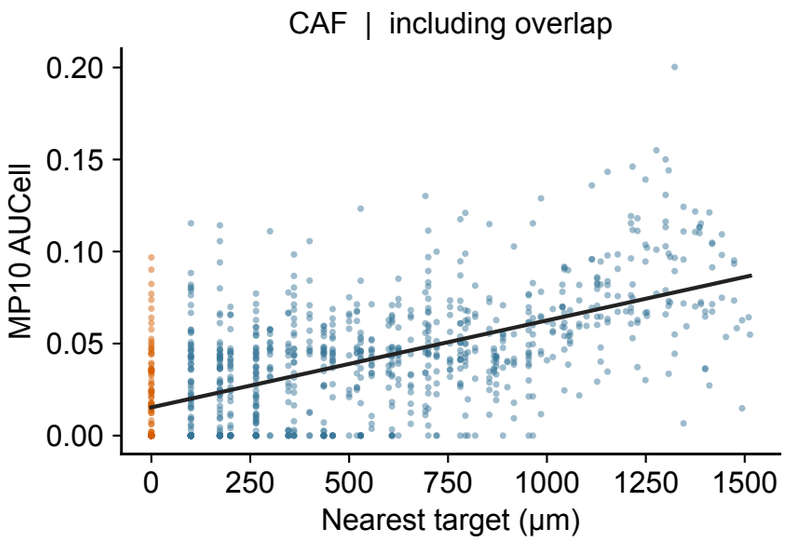


0 1.61e+03

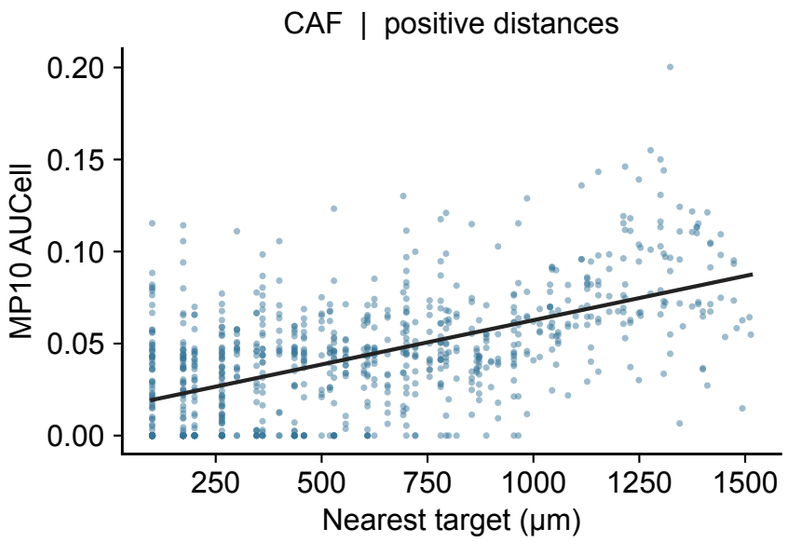
Nearest spot center (μm)

NPJ_sample1 | MP10 versus nearest target

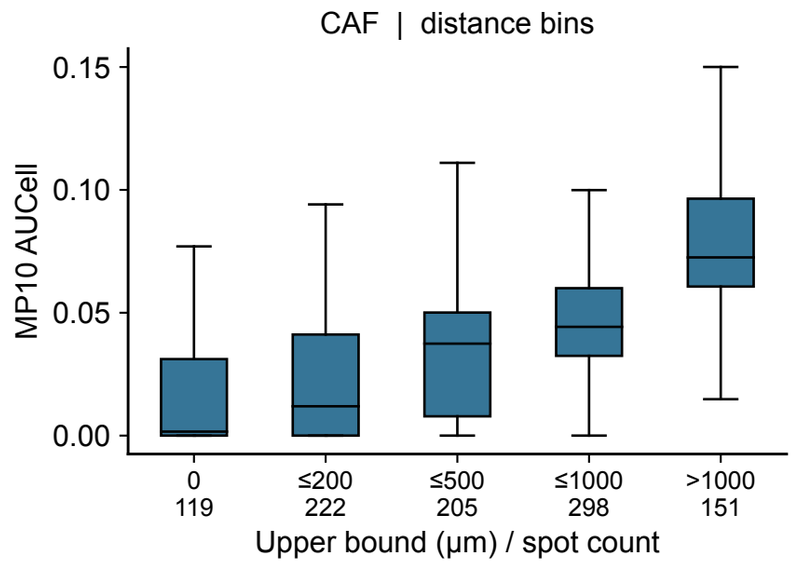
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



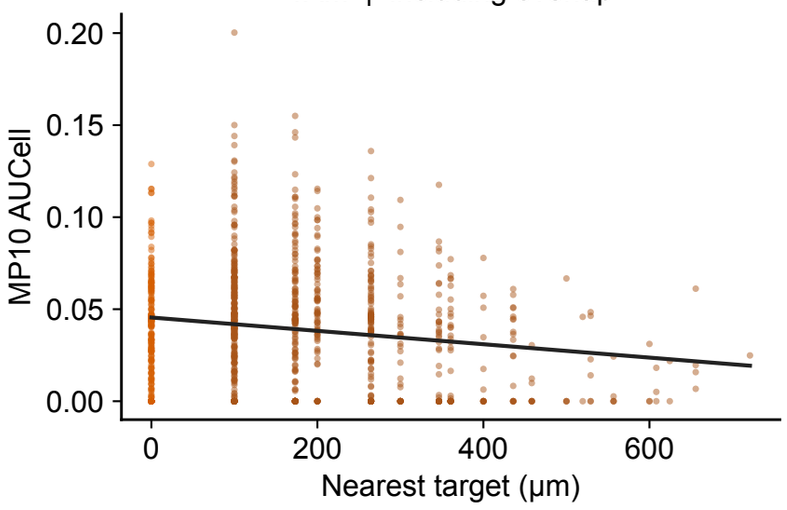
r +0.60 ρ +0.58
n 995/996 queries • 416 targets



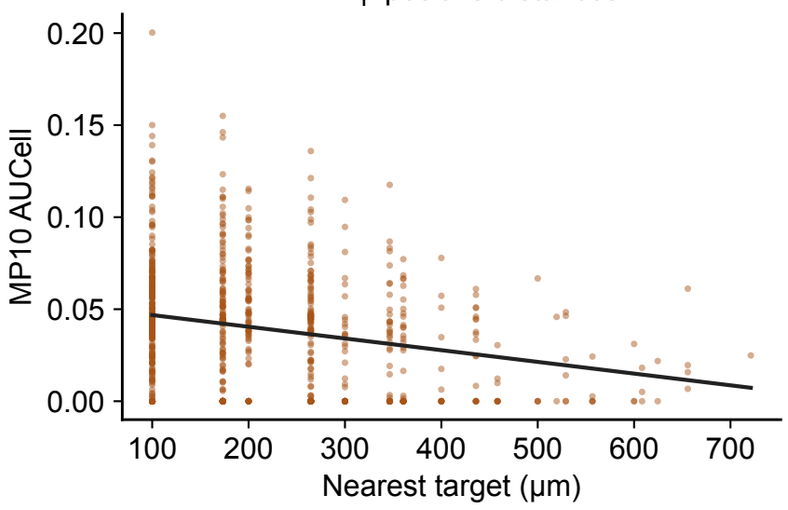
r +0.58 ρ +0.56
n 876/996 queries • 416 targets



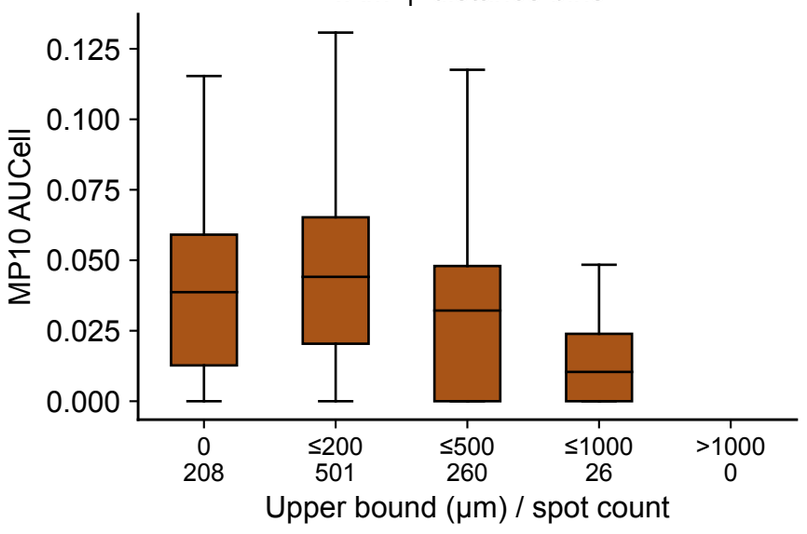
Orange = zero distance
Line = within-section linear fit



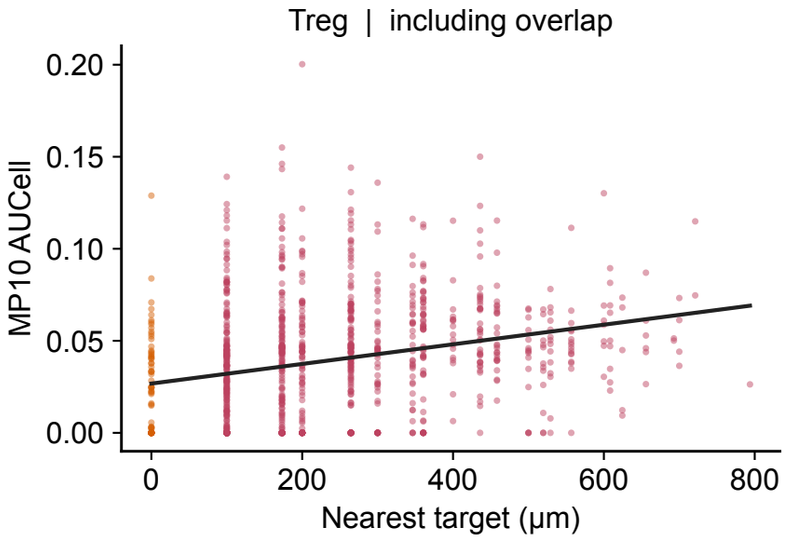
r -0.16 ρ -0.11
n 995/996 queries • 416 targets



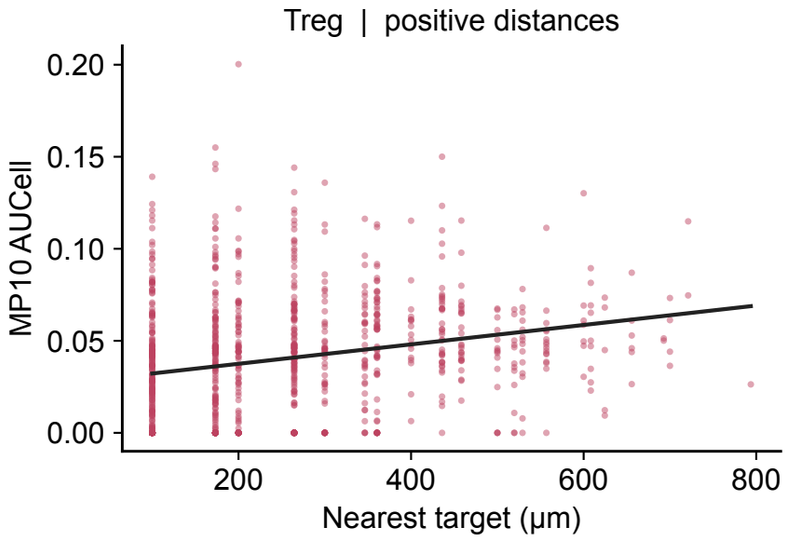
r -0.23 ρ -0.18
n 787/996 queries • 416 targets



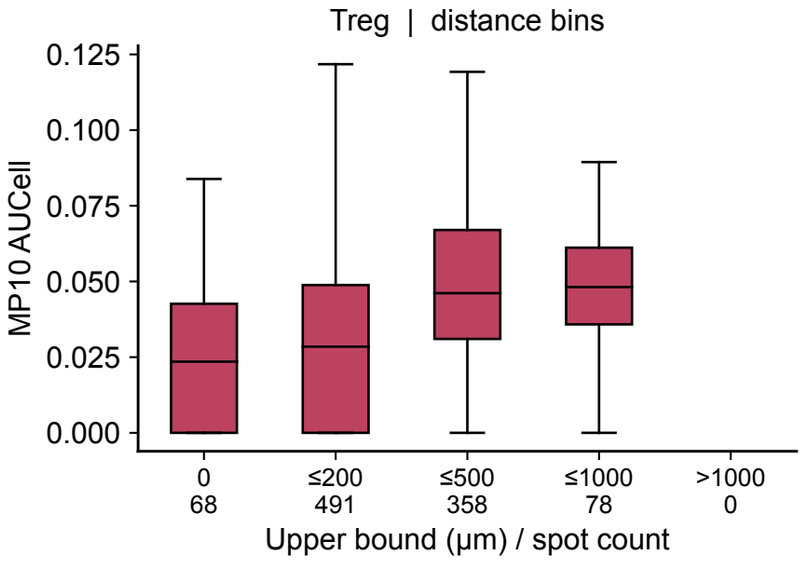
Orange = zero distance
Line = within-section linear fit



r +0.26 ρ +0.32
n 995/996 queries • 416 targets



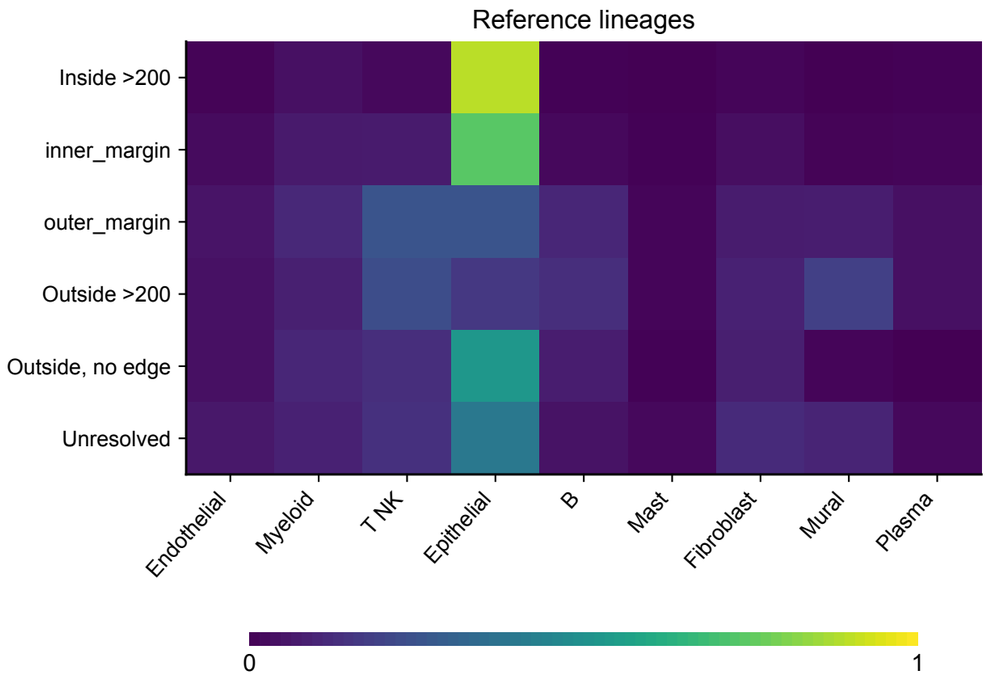
r +0.24 ρ +0.31
n 927/996 queries • 416 targets

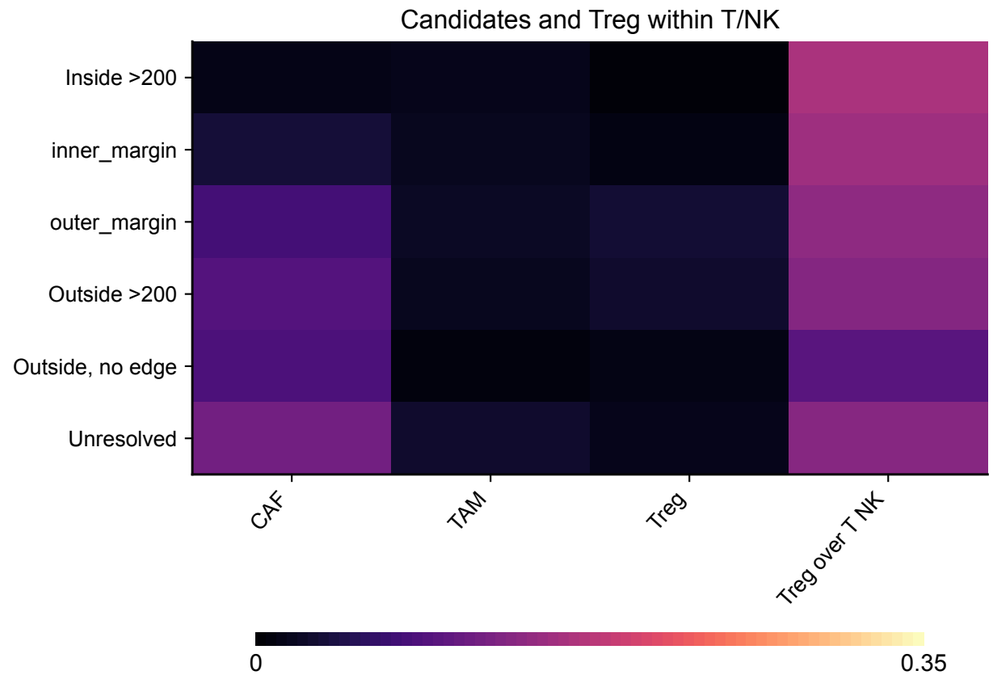


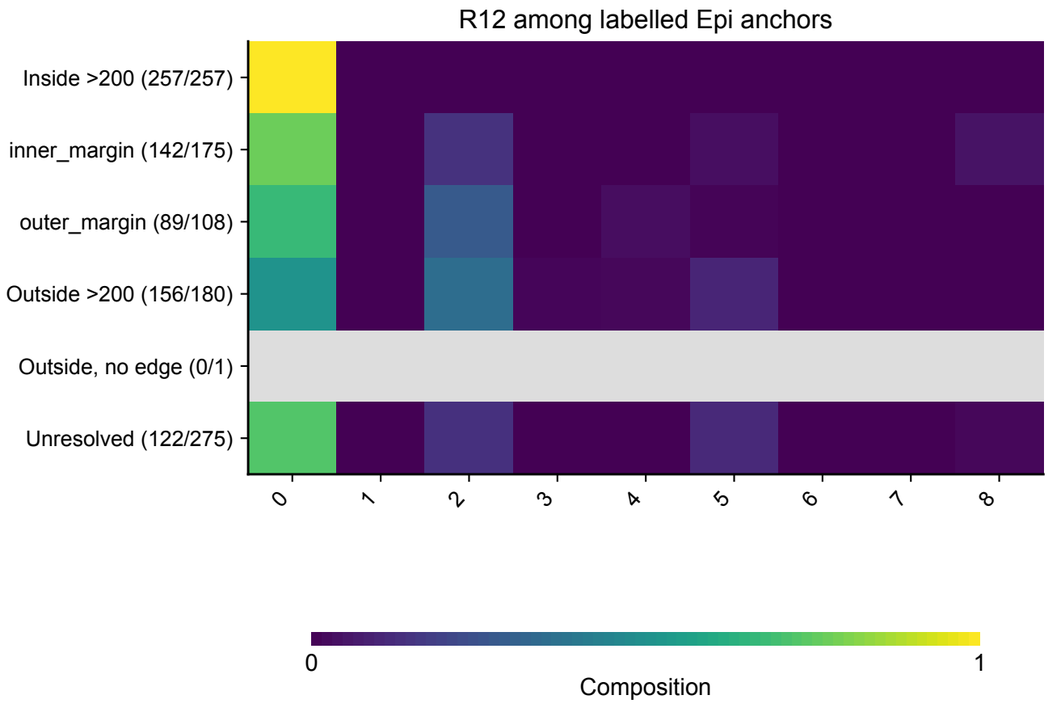
Orange = zero distance
Line = within-section linear fit

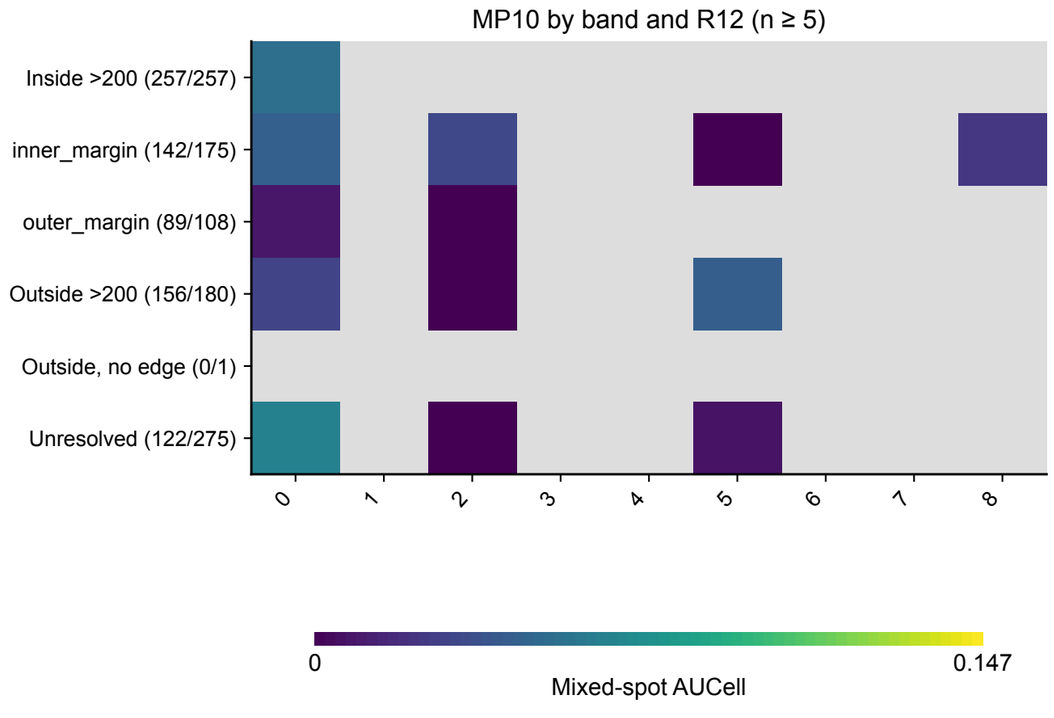
NPJ_sample1 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





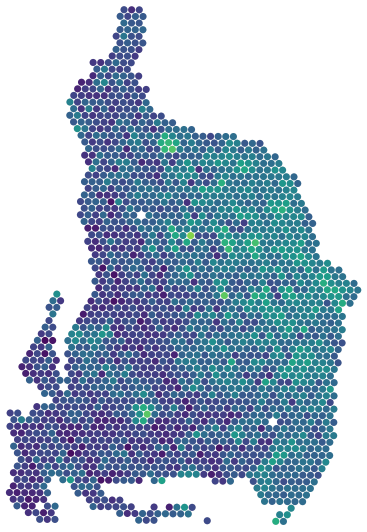




NPJ_sample1 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

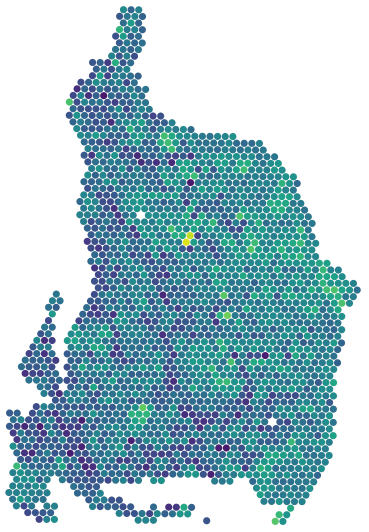
E2F Targets



0 0.152

UCell score

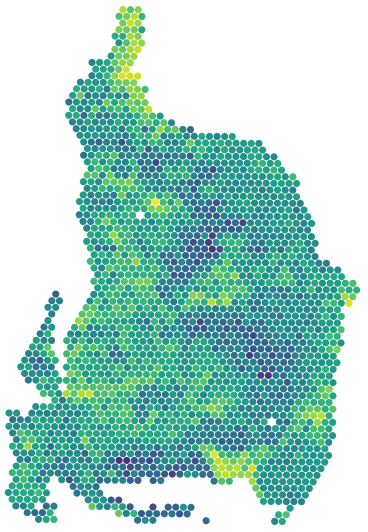
G2M Checkpoint



0 0.134

UCell score

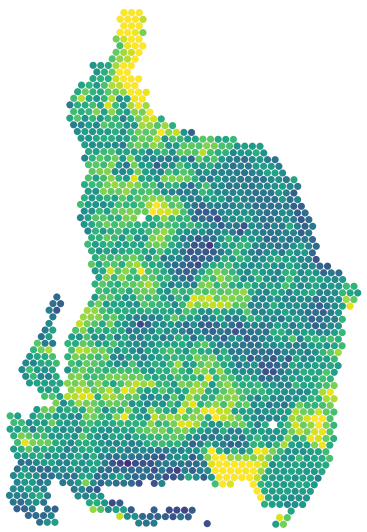
Hypoxia



0 0.211

UCell score

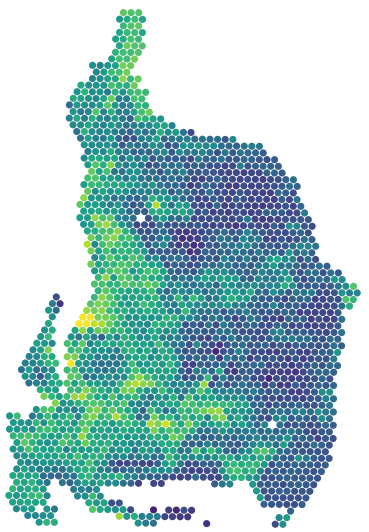
Tnfa Signaling Via Nfkb



0 0.275

UCell score

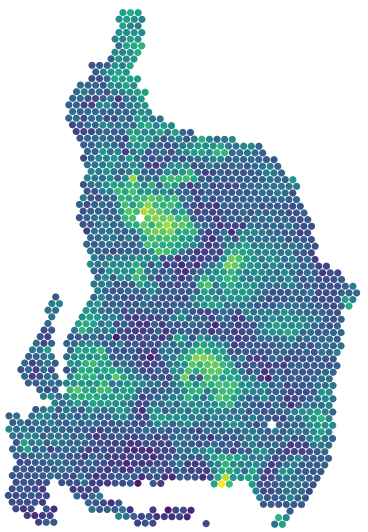
Emt



0 0.386

UCell score

Interferon Gamma Response



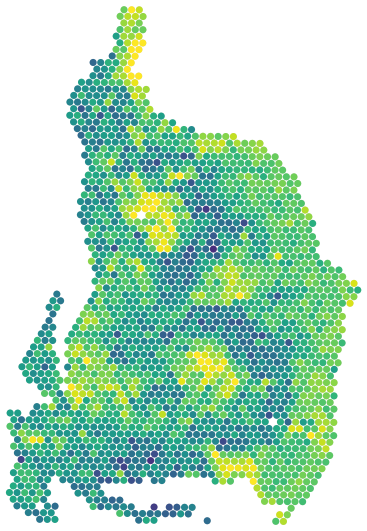
0 0.275

UCell score

NPJ_sample1 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

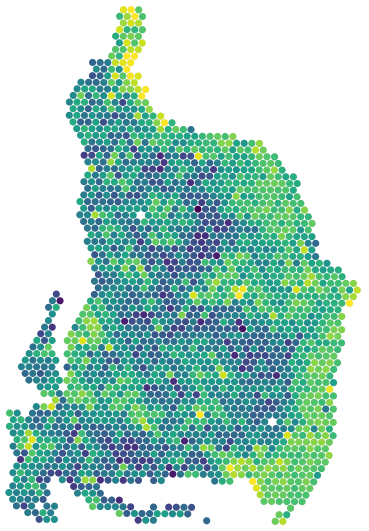
Il2 Stat5 Signaling



0 0.144

UCell score

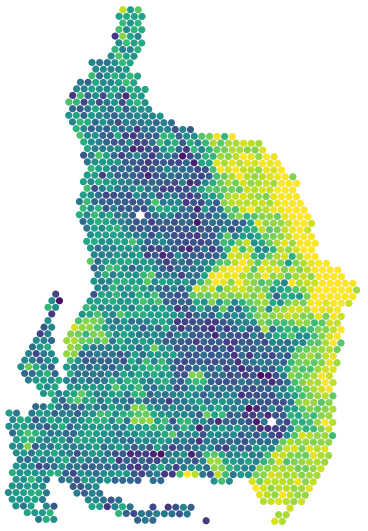
Tgf Beta Signaling



0 0.232

UCell score

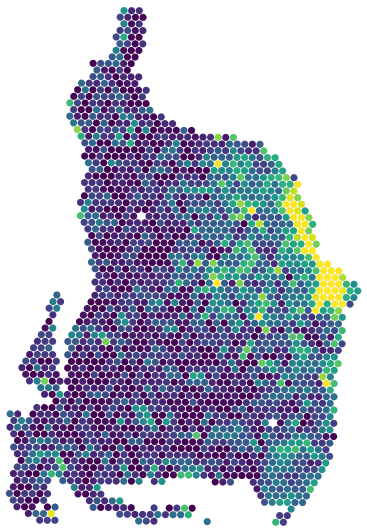
Cholesterol Homeostasis



0 0.21

UCell score

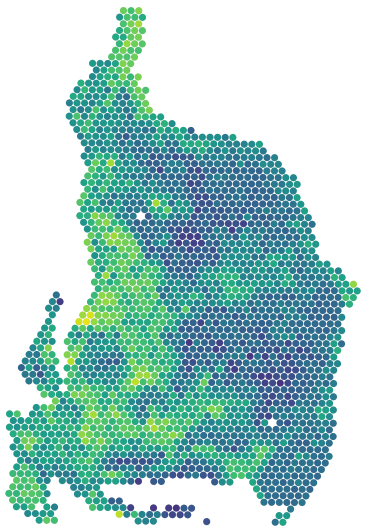
Cholesterol Biosynthesis



0 0.166

UCell score

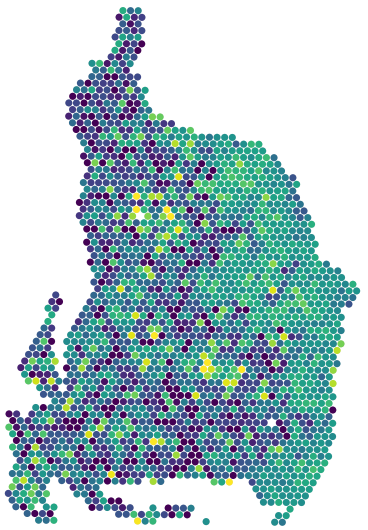
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



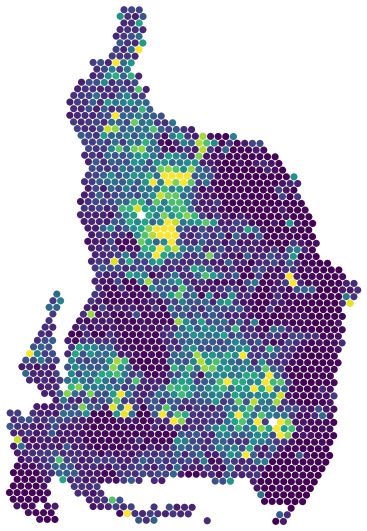
0 0.151

UCell score

NPJ_sample1 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

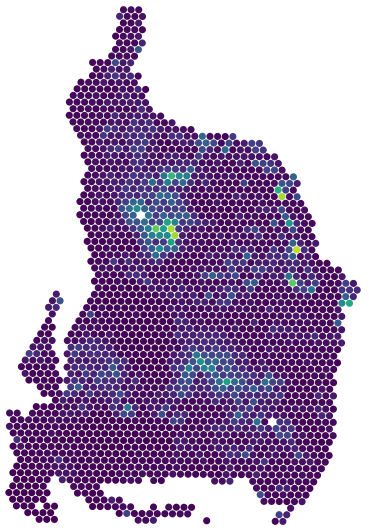
Treg composition



0 0.102

Fraction of all 30

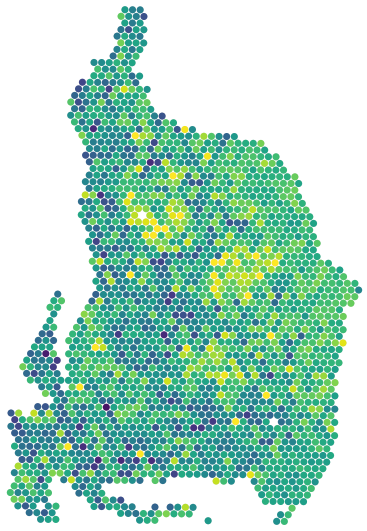
Treg estimated abundance



0 2.85

Posterior mean estimate

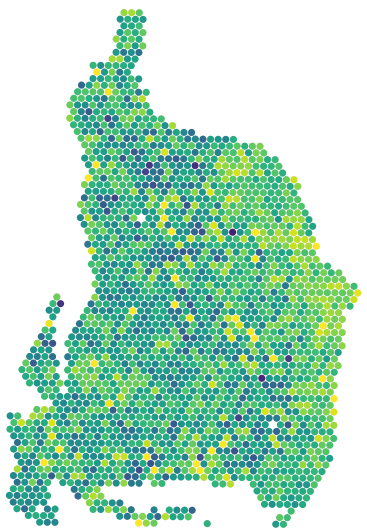
Treg Resting Vs Activated Dn



0 0.0778

UCell score

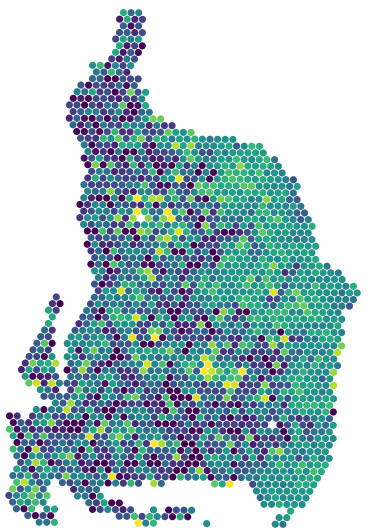
Treg Resting Vs Activated Up



0 0.0609

UCell score

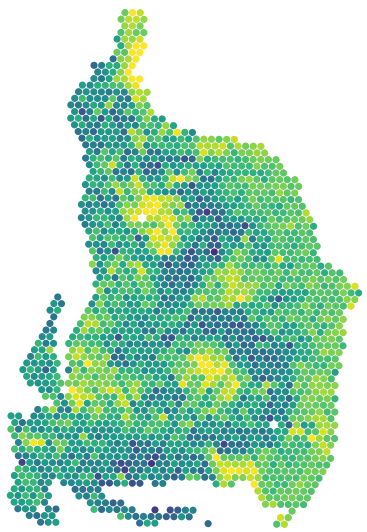
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

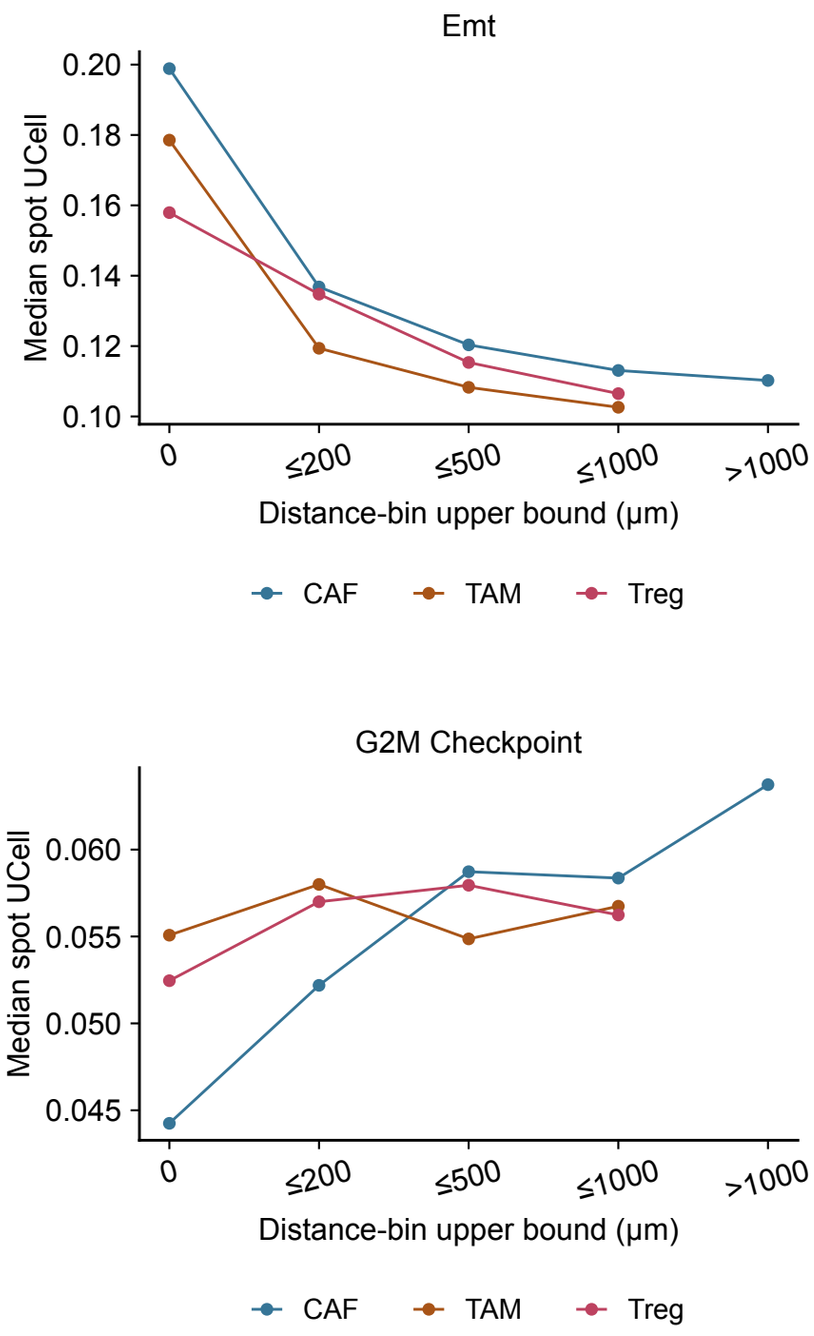


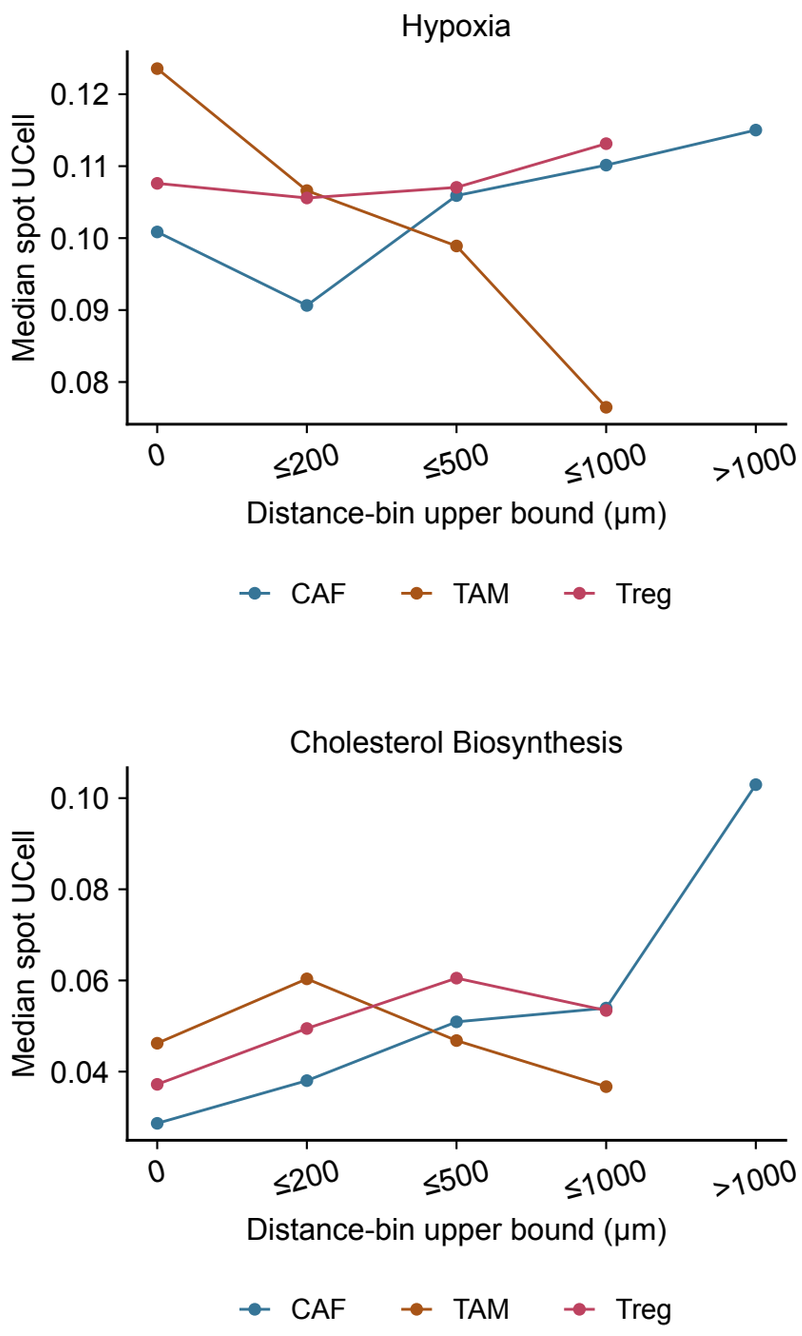
0 0.144

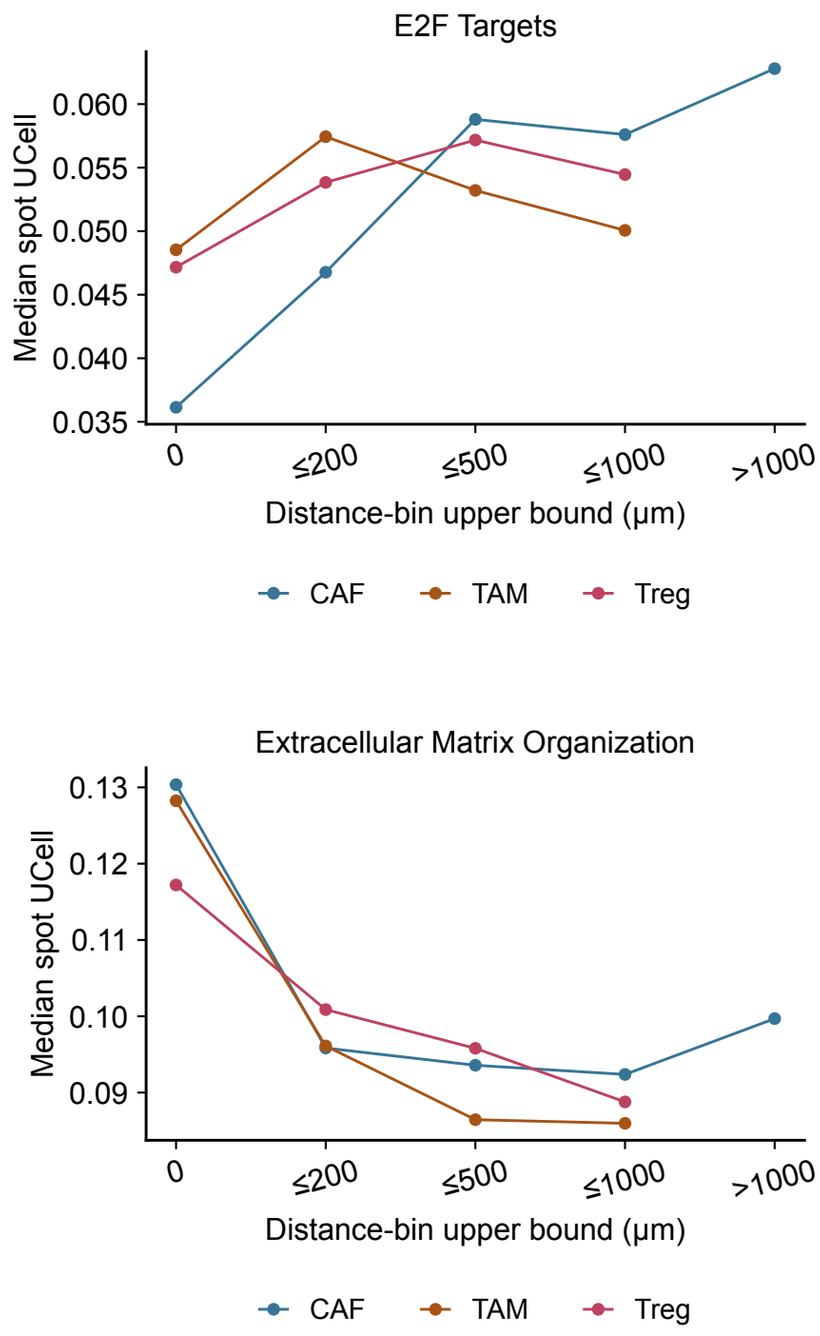
UCell score

NPJ_sample1 | Programs by target distance

Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

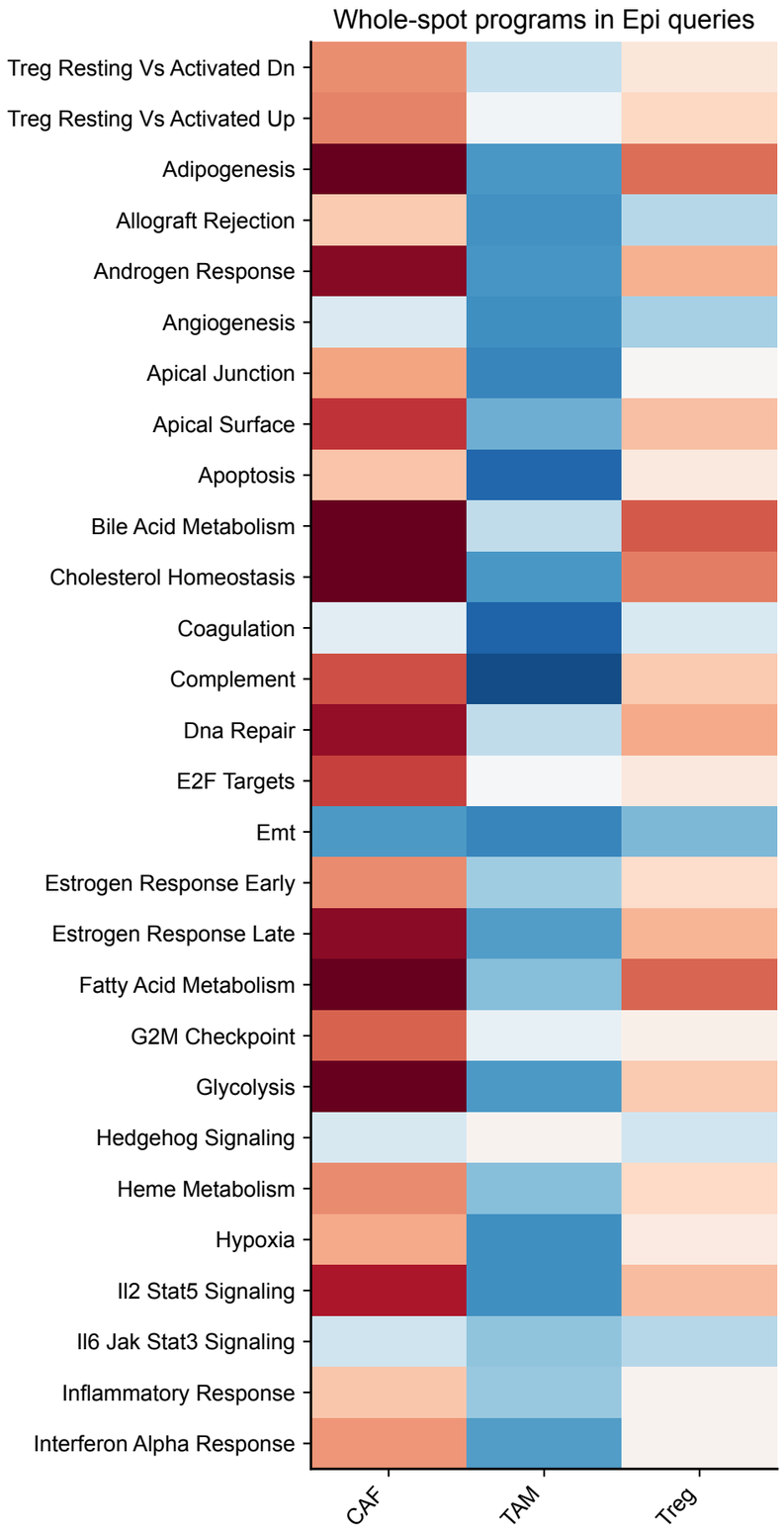


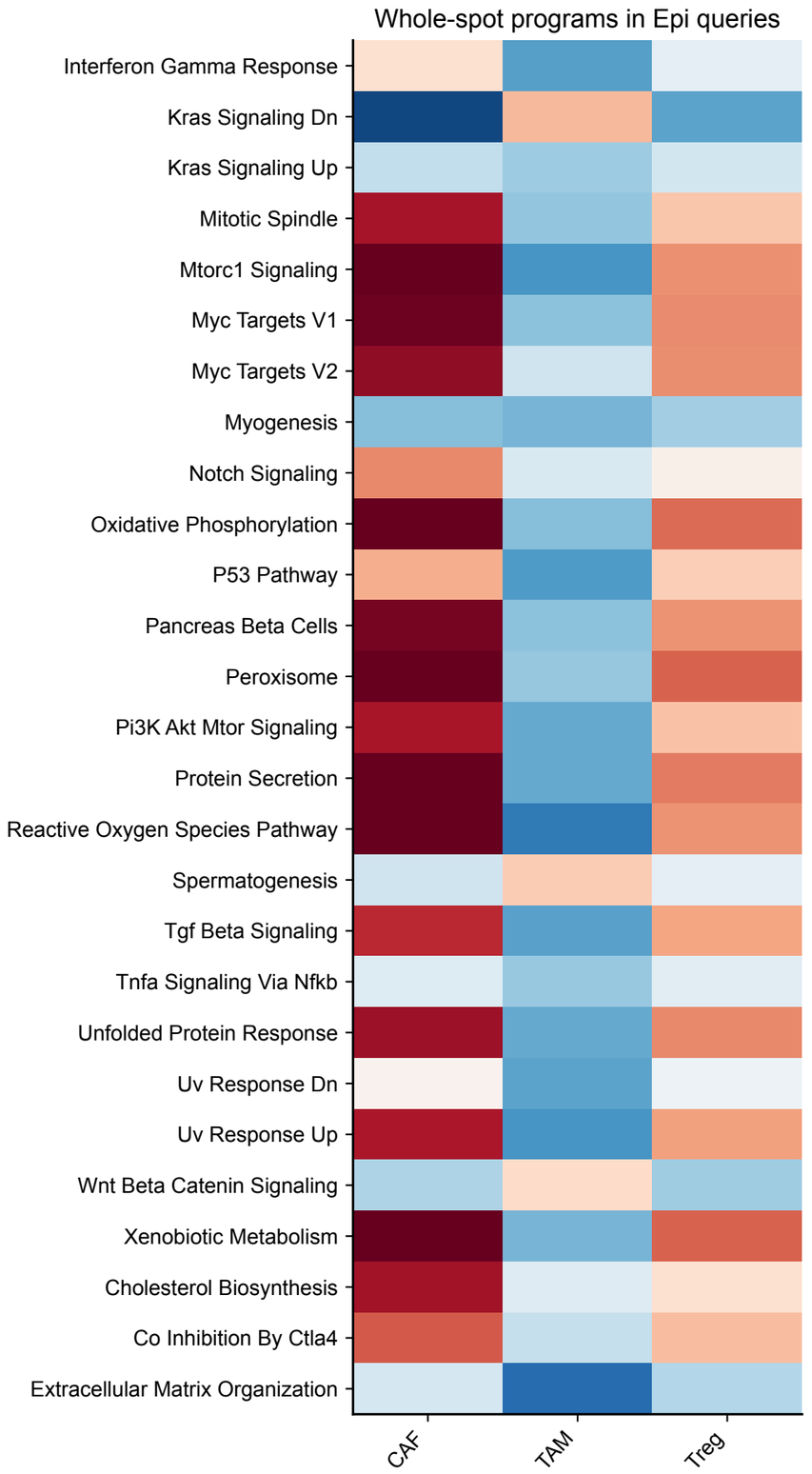




NPJ_sample1 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6





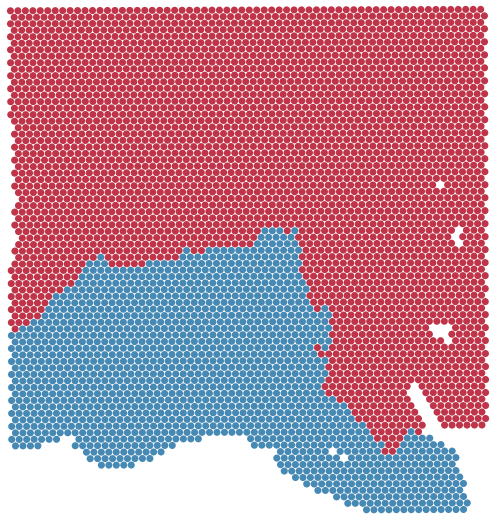
NPJ_sample2 | Tissue context

Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge

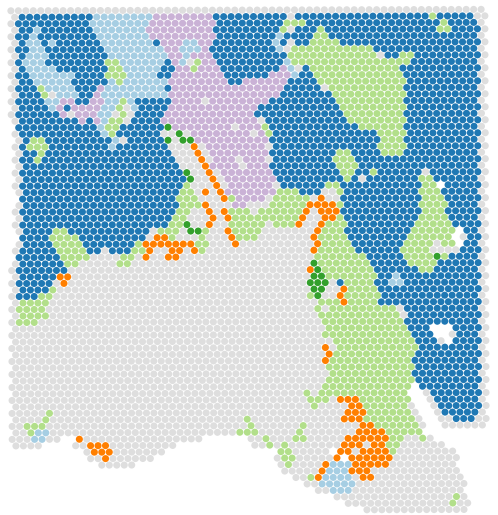
Black line: observed candidate outer edge

Final candidate location



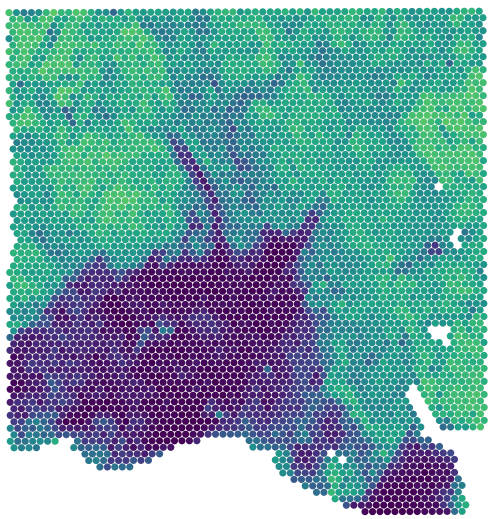
inside outside

Accepted R12 (original 0–8)



0 2 3 5 8 1

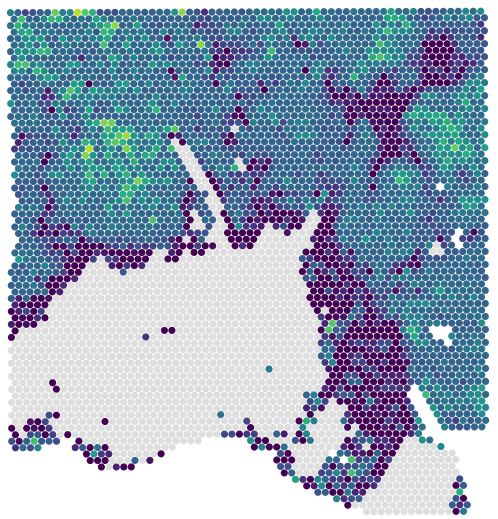
Epithelial support



0 1

Fraction of all 30

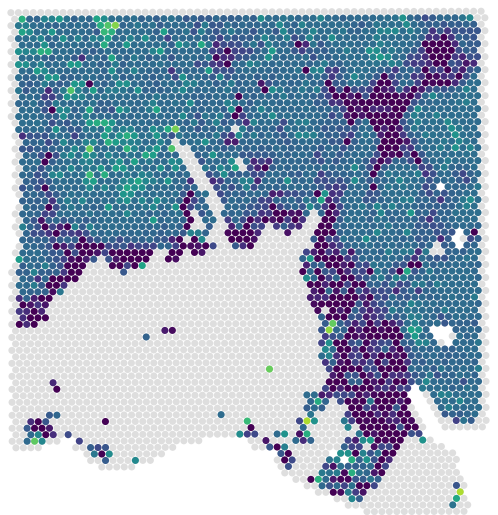
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



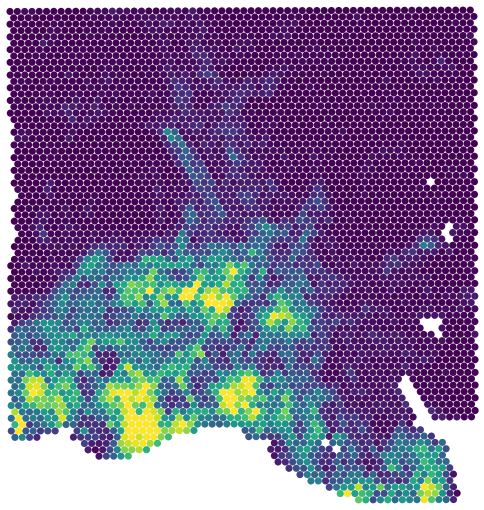
0 0.14

Matched-model AUCell

NPJ_sample2 | Candidate neighborhoods

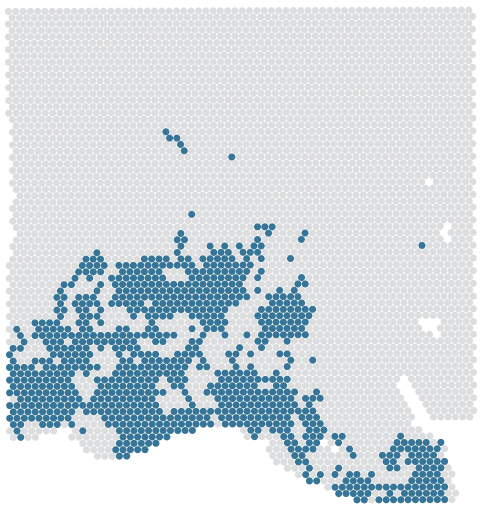
Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

CAF composition



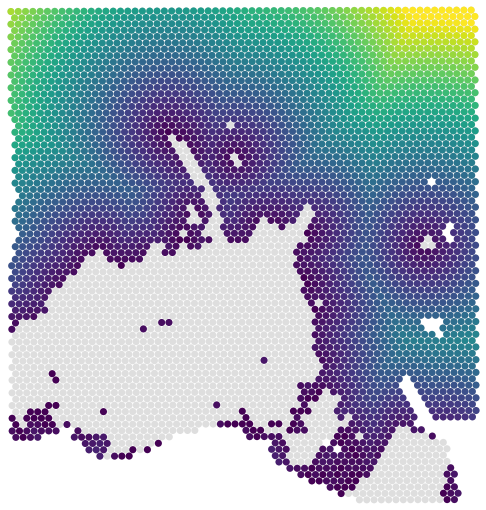
0 0.418
Fraction of all 30

CAF P80 targets



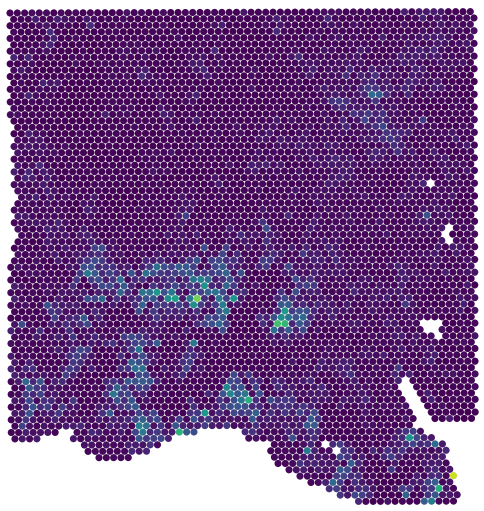
False True

CAF distance



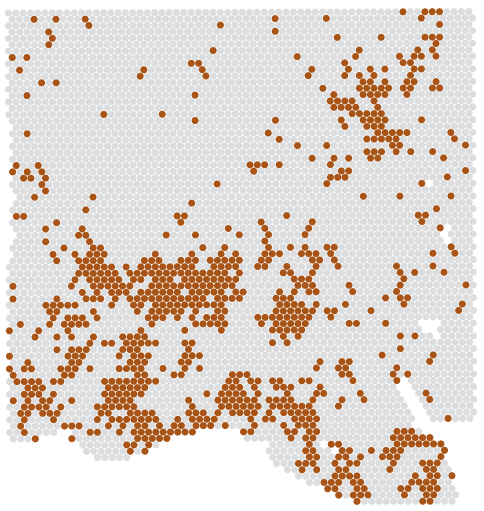
0 3.1e+03
Nearest spot center (μm)

TAM composition



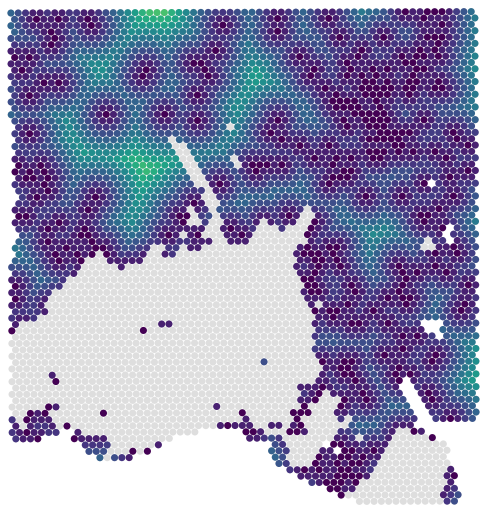
0 0.291
Fraction of all 30

TAM P80 targets



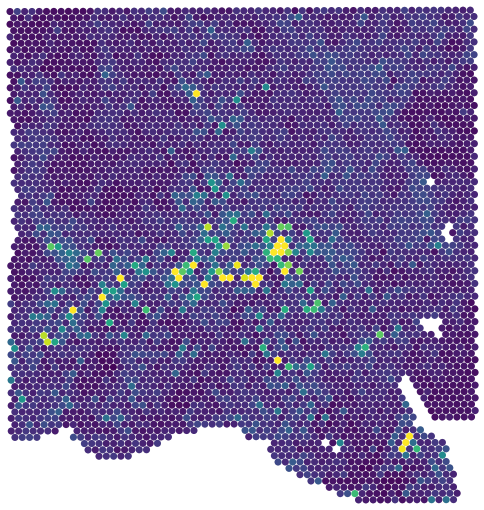
False True

TAM distance



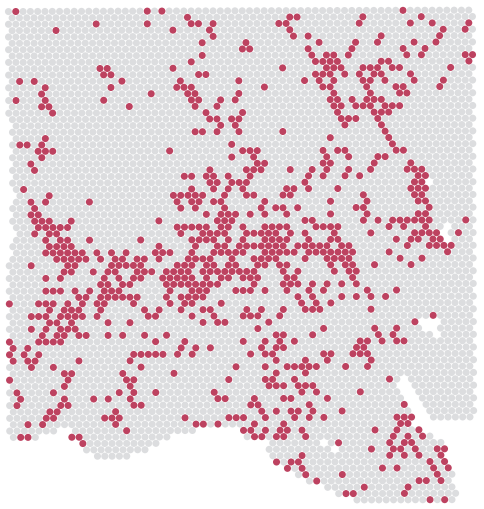
0 1.13e+03
Nearest spot center (μm)

Treg composition



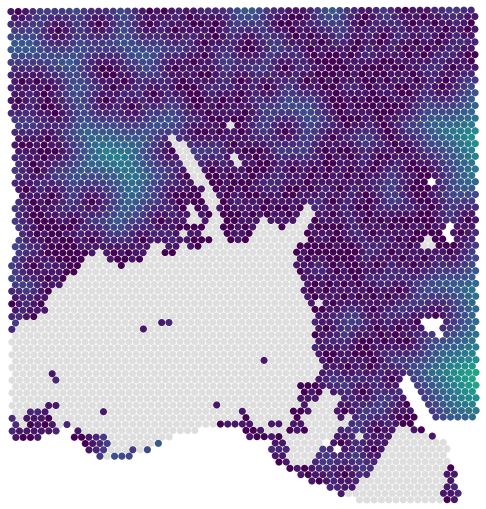
0 0.102
Fraction of all 30

Treg P80 targets



False True

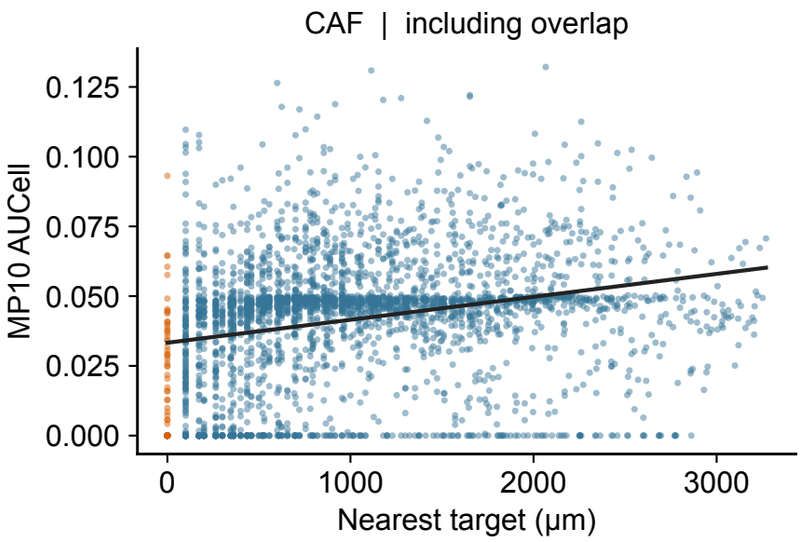
Treg distance



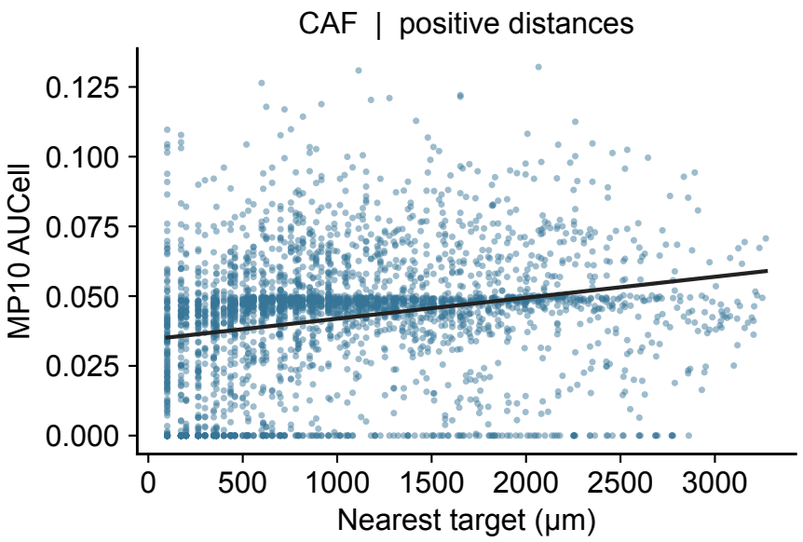
0 1.61e+03
Nearest spot center (μm)

NPJ_sample2 | MP10 versus nearest target

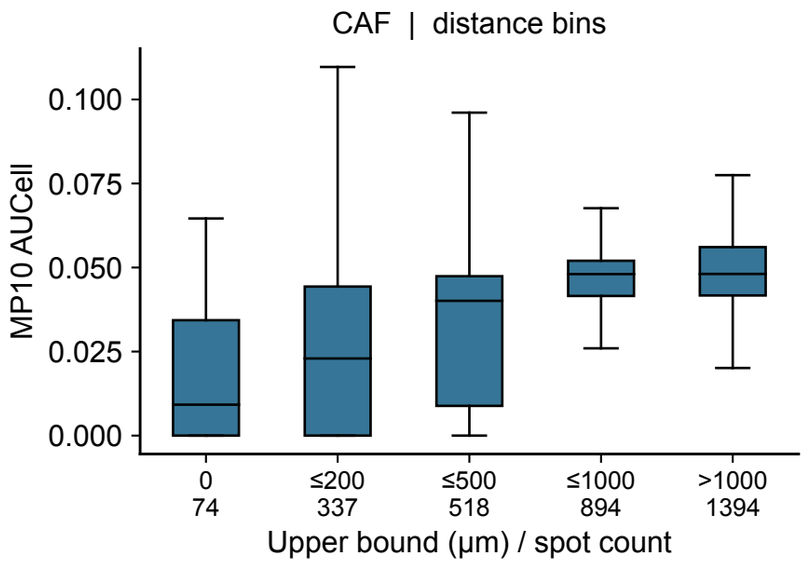
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



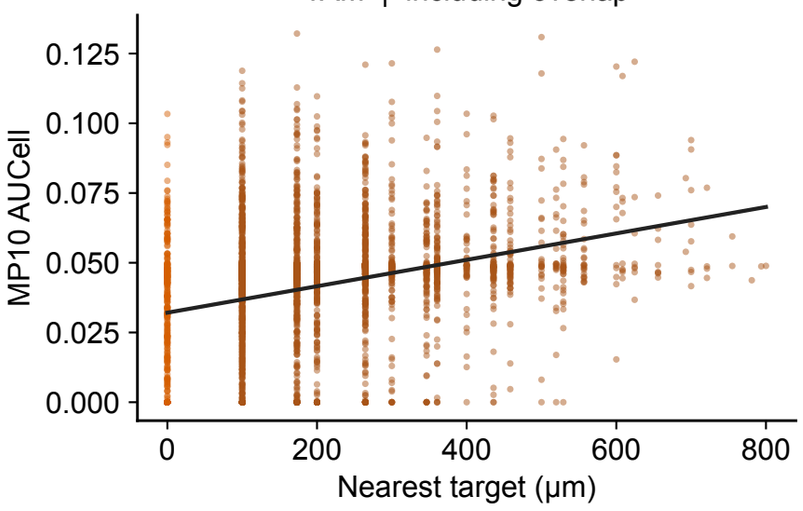
r +0.26 ρ +0.31
n 3217/3217 queries • 902 targets



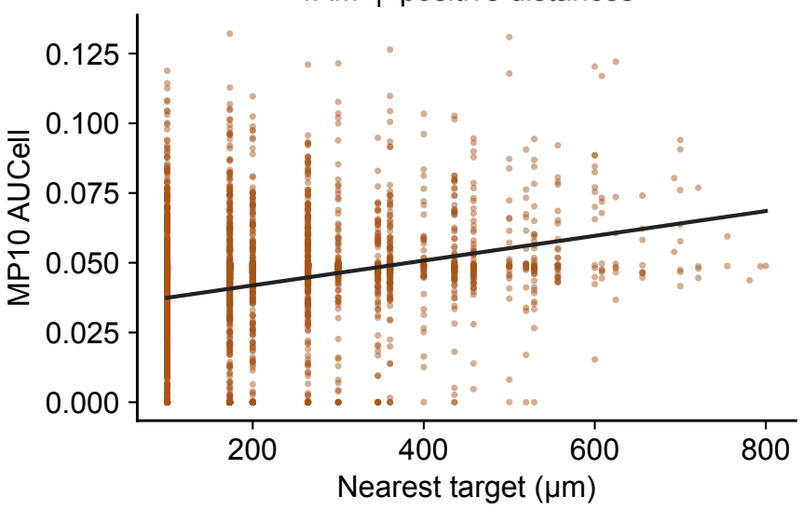
r +0.24 ρ +0.29
n 3143/3217 queries • 902 targets



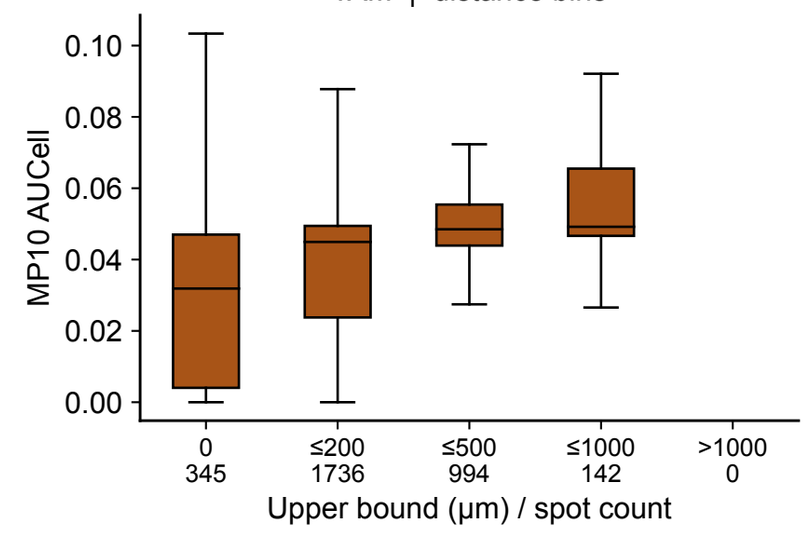
Orange = zero distance
Line = within-section linear fit



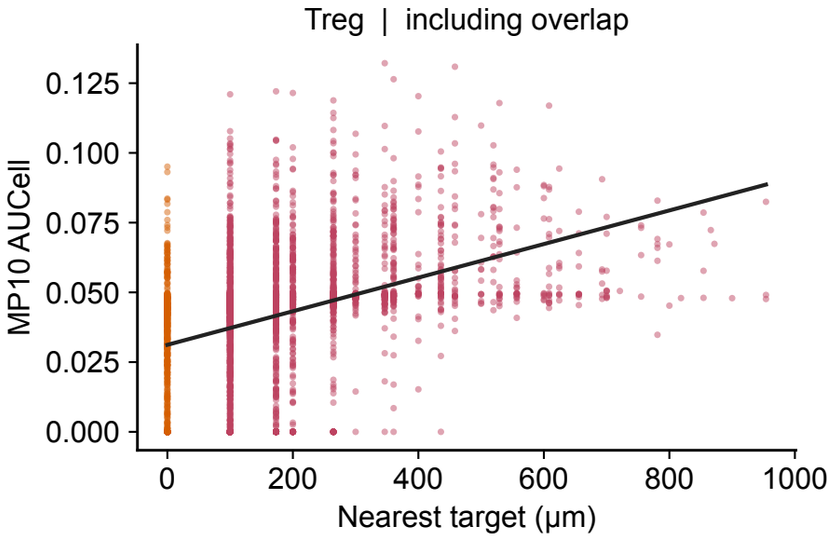
r +0.29 ρ +0.32
n 3217/3217 queries • 902 targets



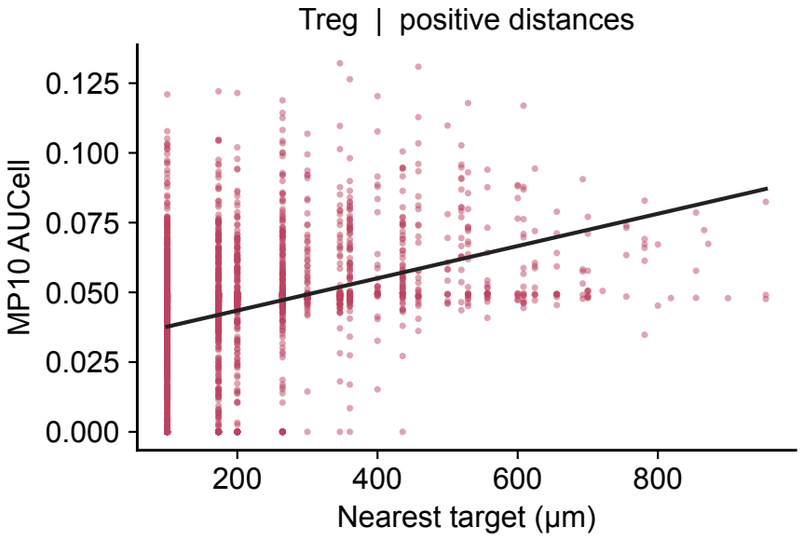
r +0.26 ρ +0.29
n 2872/3217 queries • 902 targets



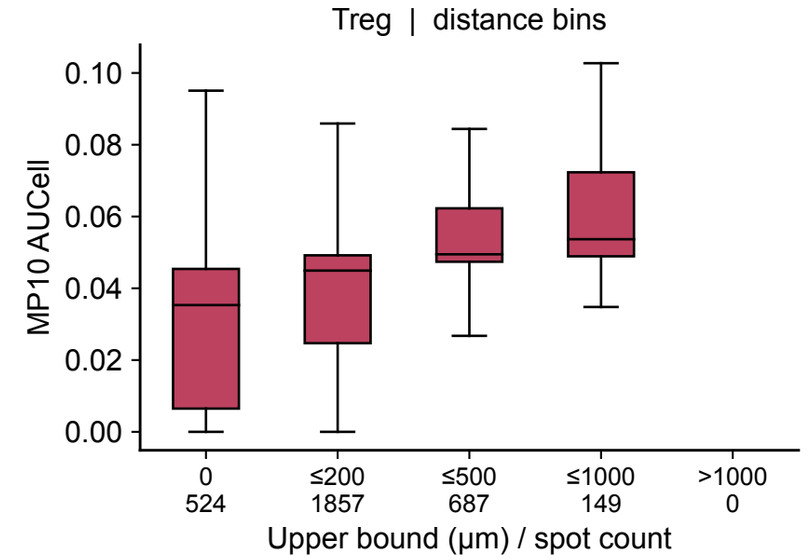
Orange = zero distance
Line = within-section linear fit



r +0.39 ρ +0.46
n 3217/3217 queries • 902 targets



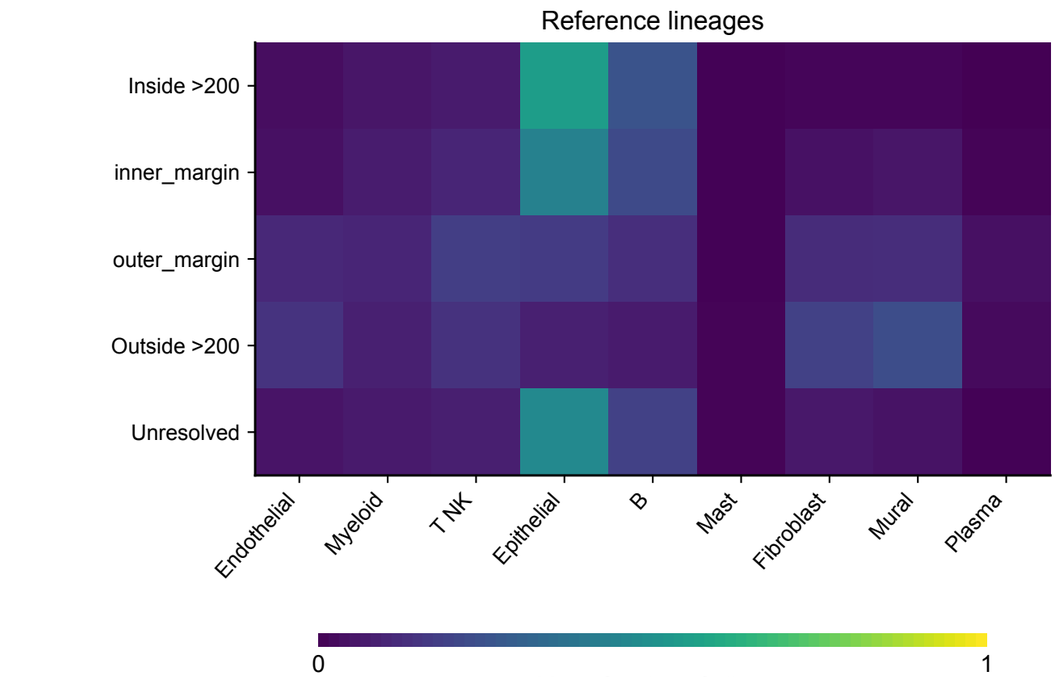
r +0.36 ρ +0.43
n 2693/3217 queries • 902 targets

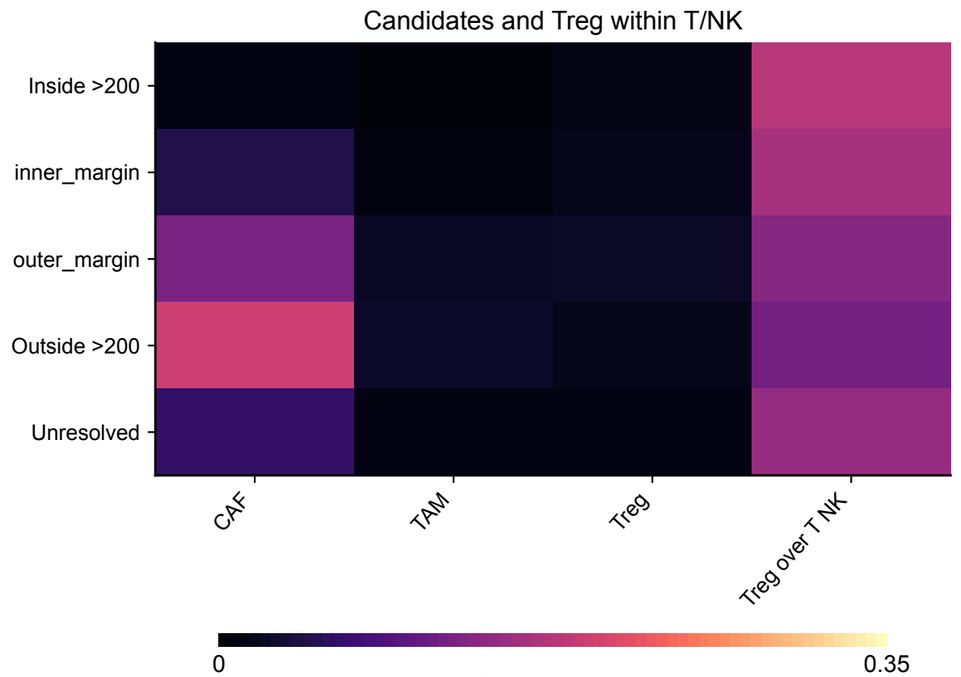


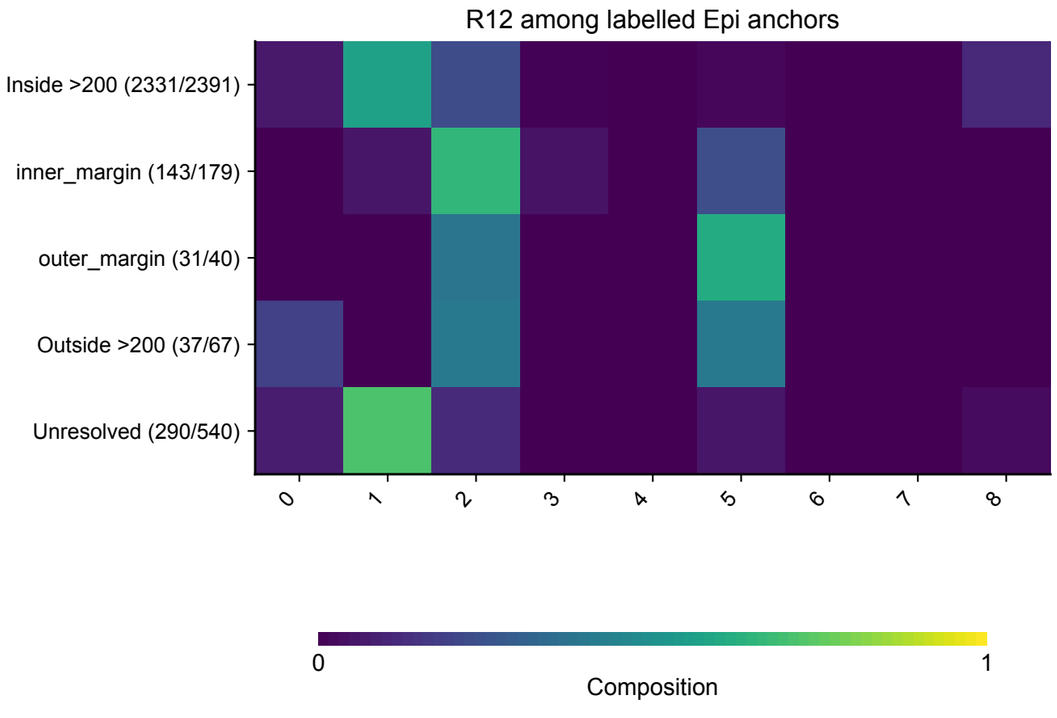
Orange = zero distance
Line = within-section linear fit

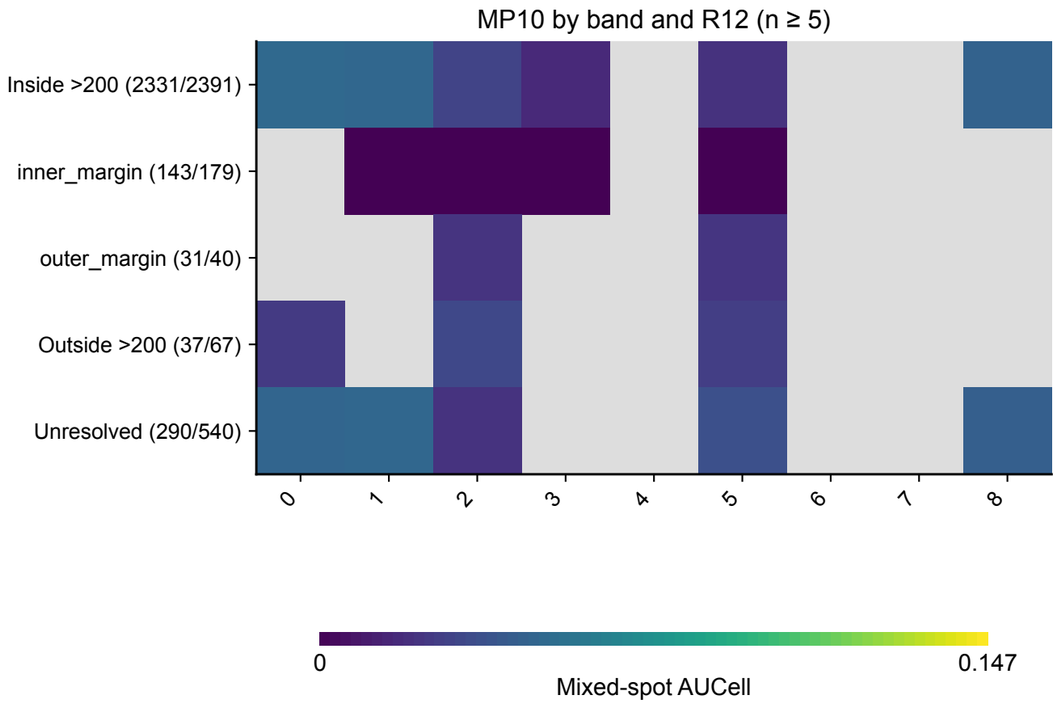
NPJ_sample2 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





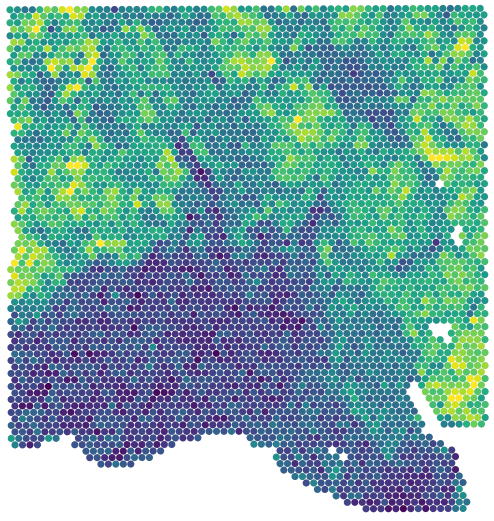




NPJ_sample2 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

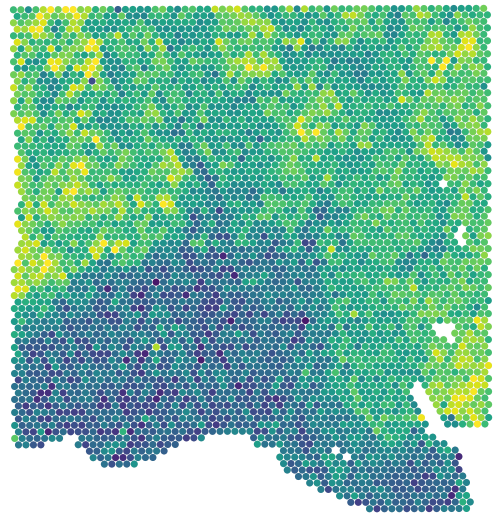
E2F Targets



0 0.152

UCell score

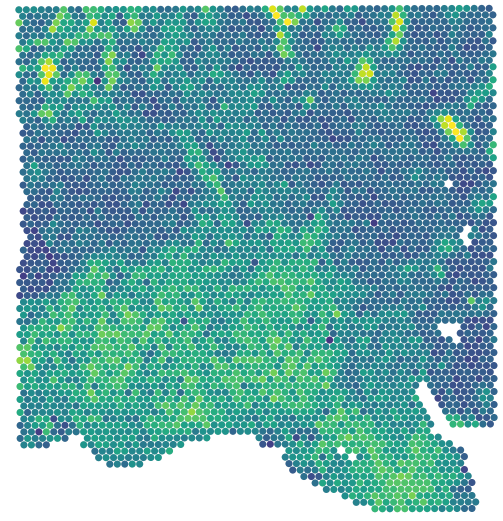
G2M Checkpoint



0 0.134

UCell score

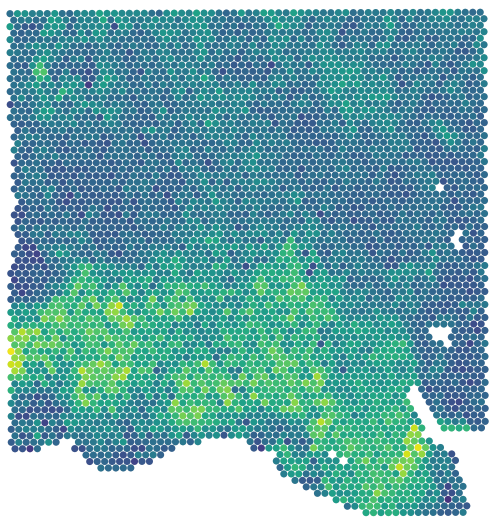
Hypoxia



0 0.211

UCell score

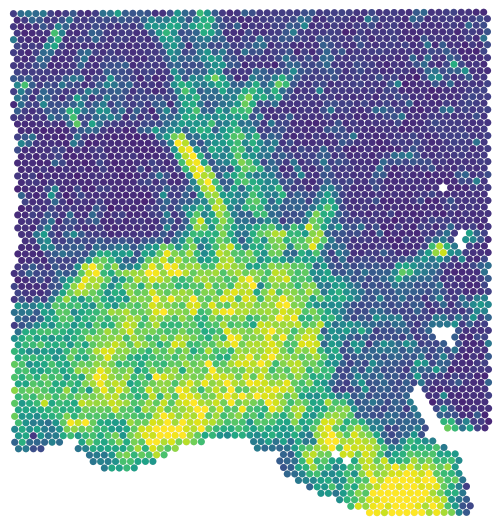
Tnfa Signaling Via Nfkb



0 0.275

UCell score

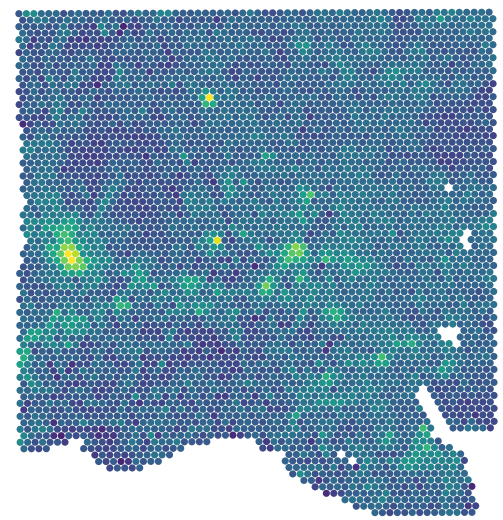
Emt



0 0.386

UCell score

Interferon Gamma Response



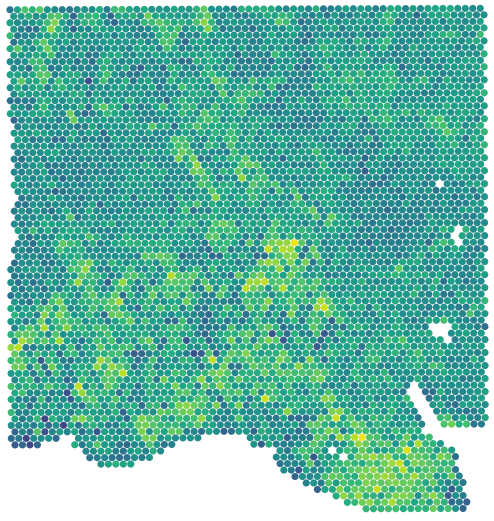
0 0.275

UCell score

NPJ_sample2 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

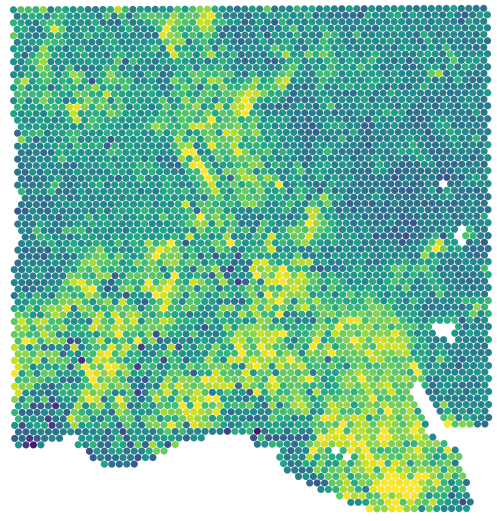
Il2 Stat5 Signaling



0 0.144

UCell score

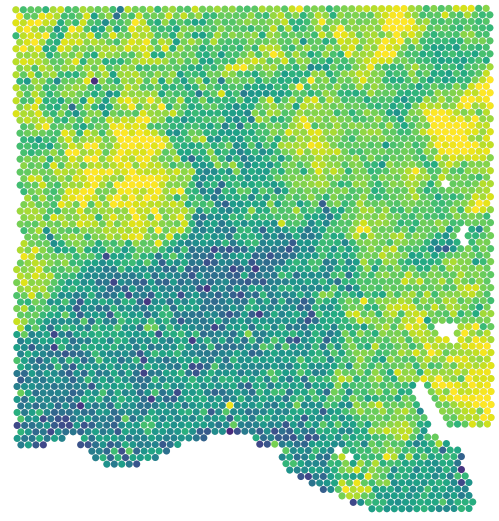
Tgf Beta Signaling



0 0.232

UCell score

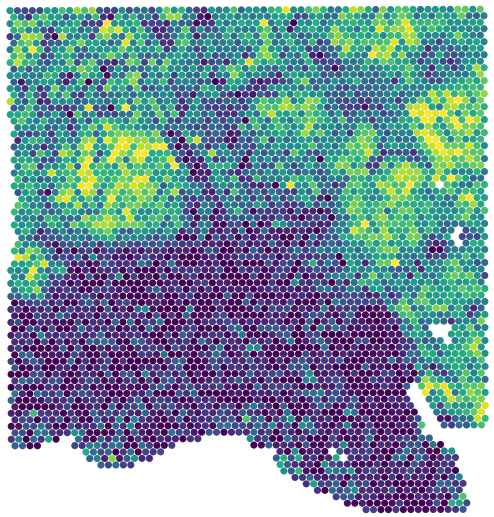
Cholesterol Homeostasis



0 0.21

UCell score

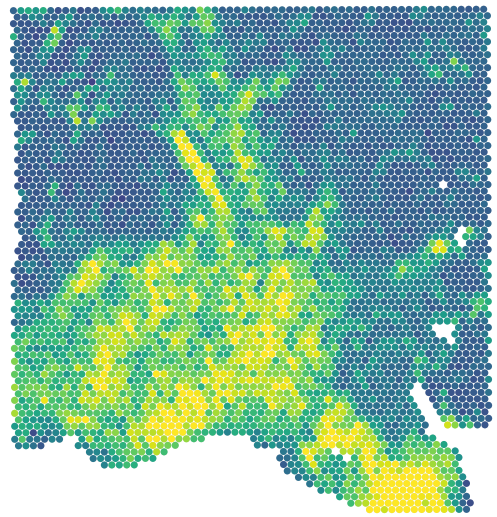
Cholesterol Biosynthesis



0 0.166

UCell score

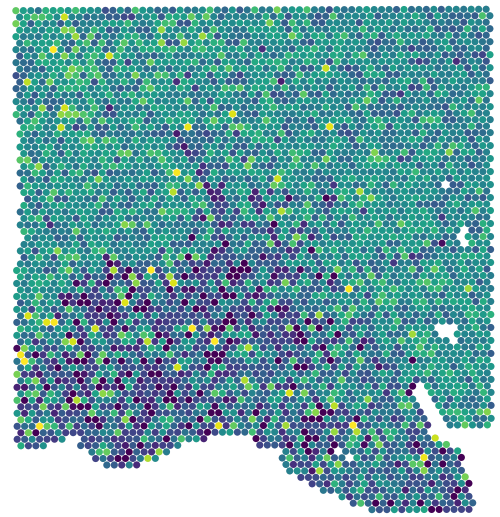
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



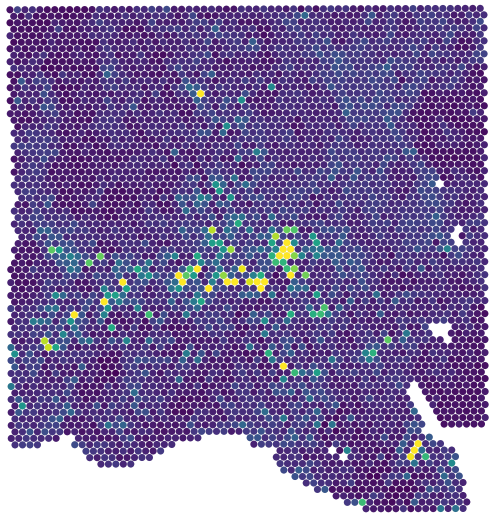
0 0.151

UCell score

NPJ_sample2 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

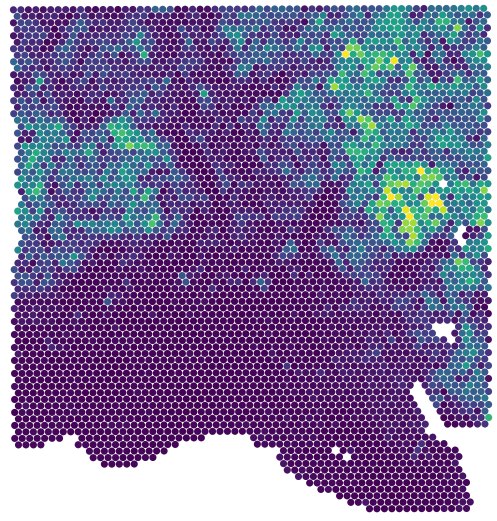
Treg composition



0 0.102

Fraction of all 30

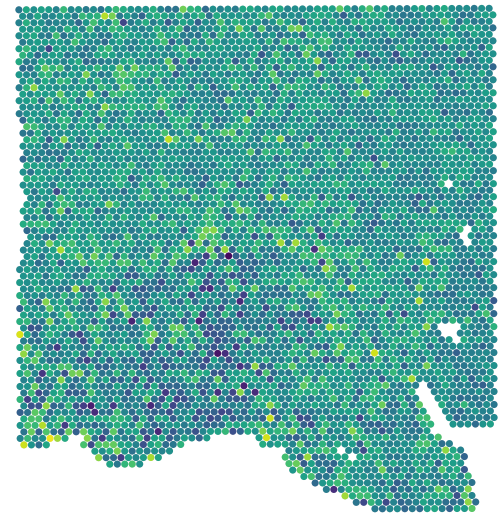
Treg estimated abundance



0 2.85

Posterior mean estimate

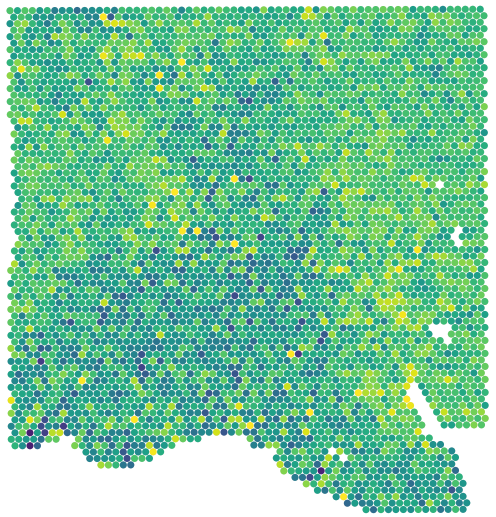
Treg Resting Vs Activated Dn



0 0.0778

UCell score

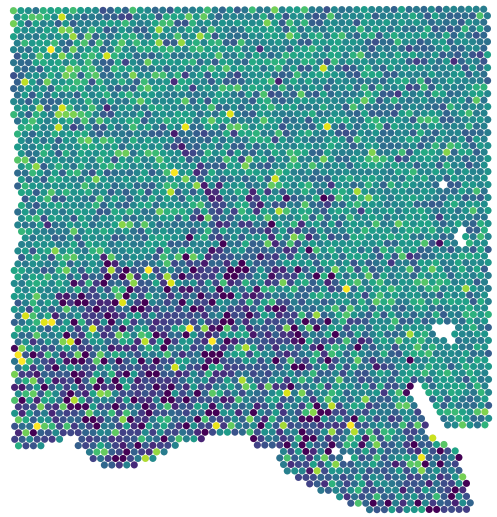
Treg Resting Vs Activated Up



0 0.0609

UCell score

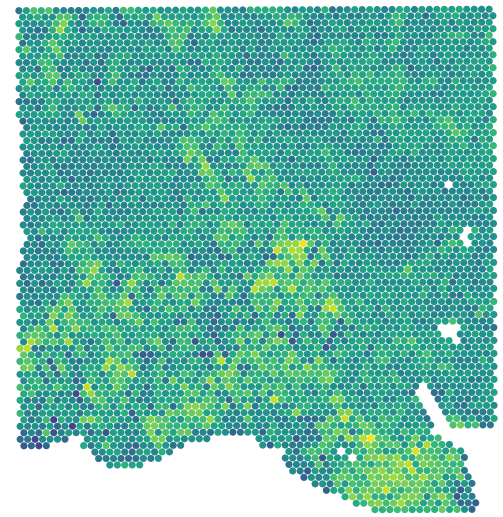
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

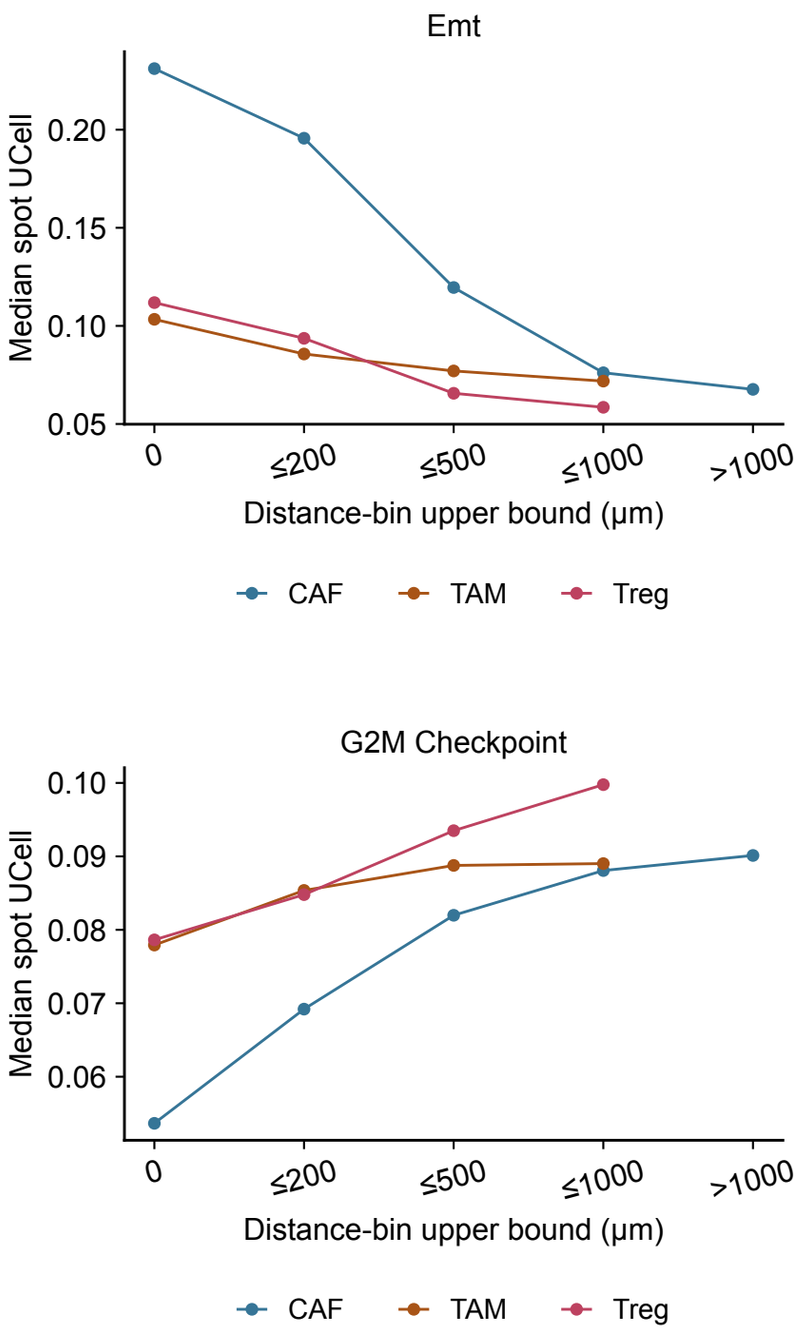


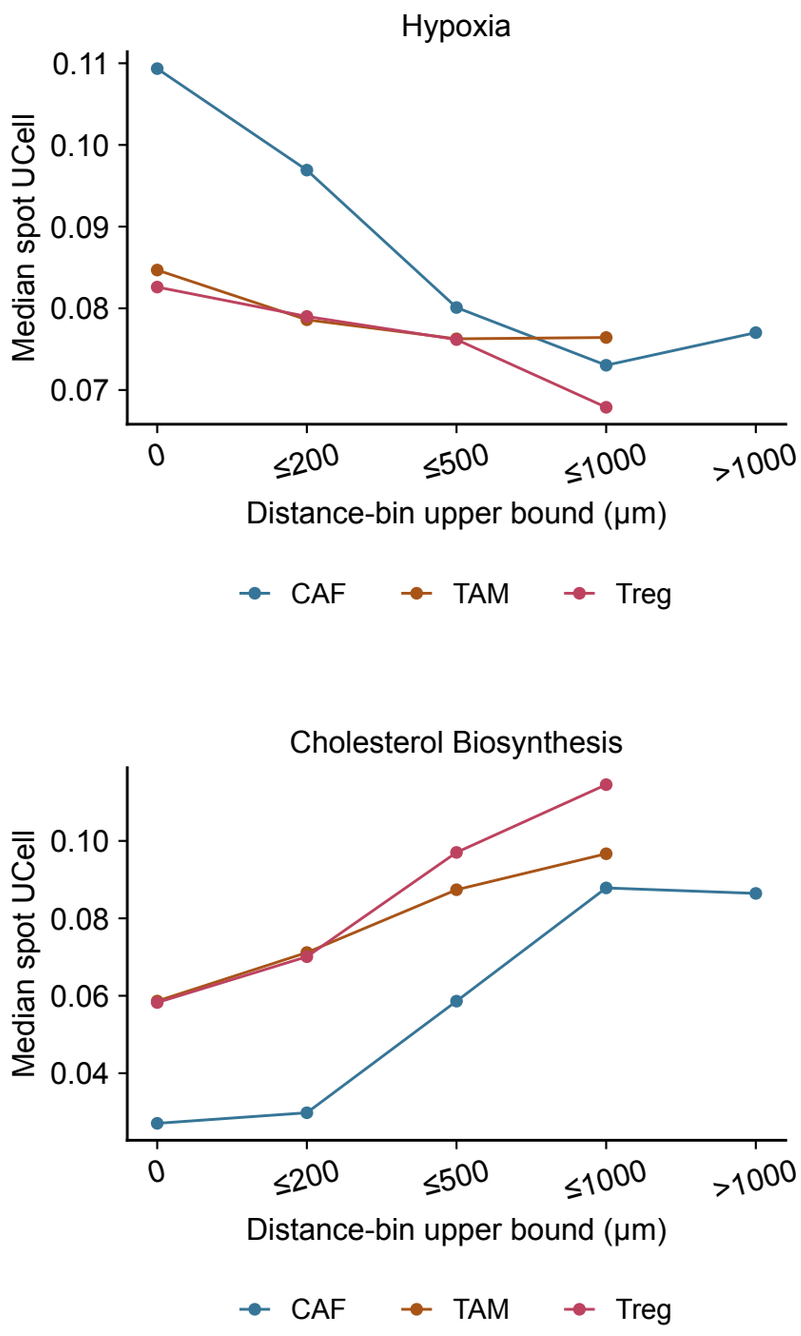
0 0.144

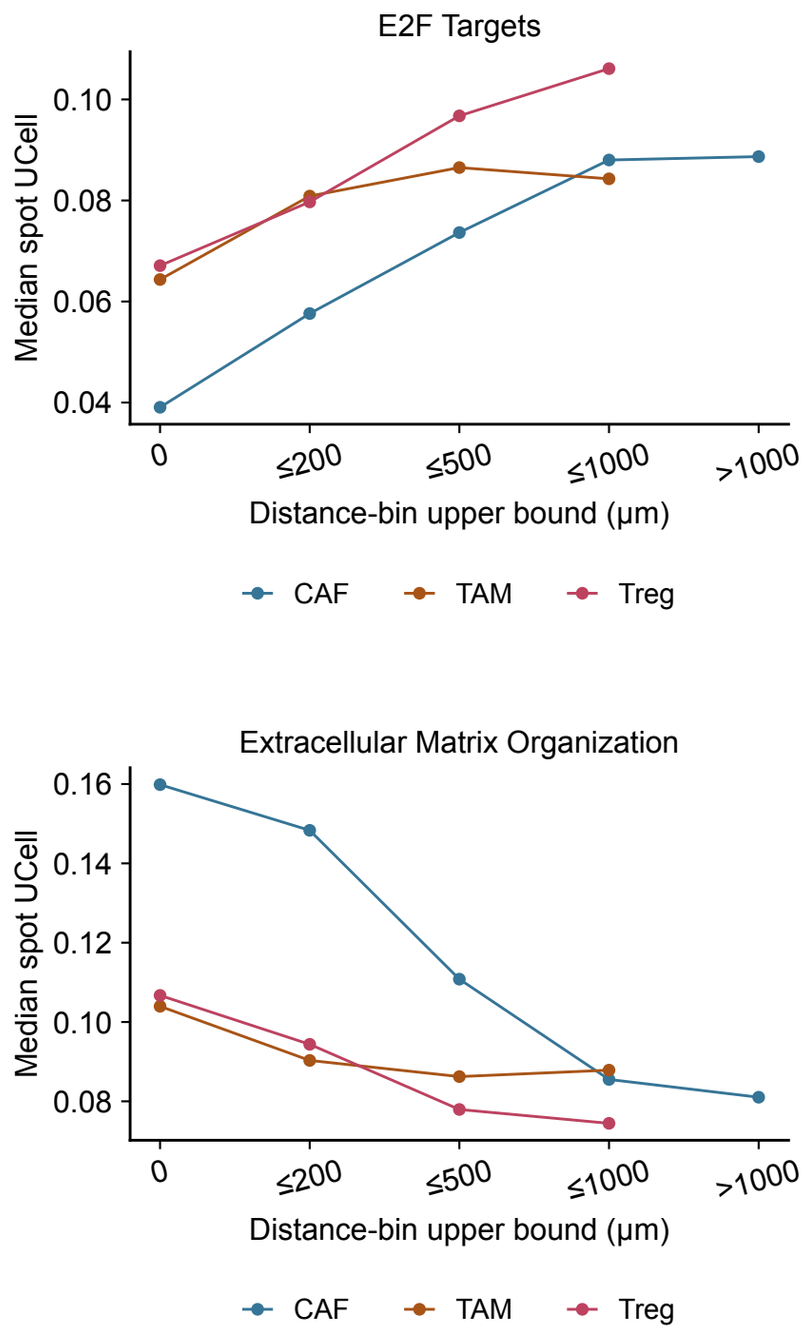
UCell score

NPJ_sample2 | Programs by target distance

Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

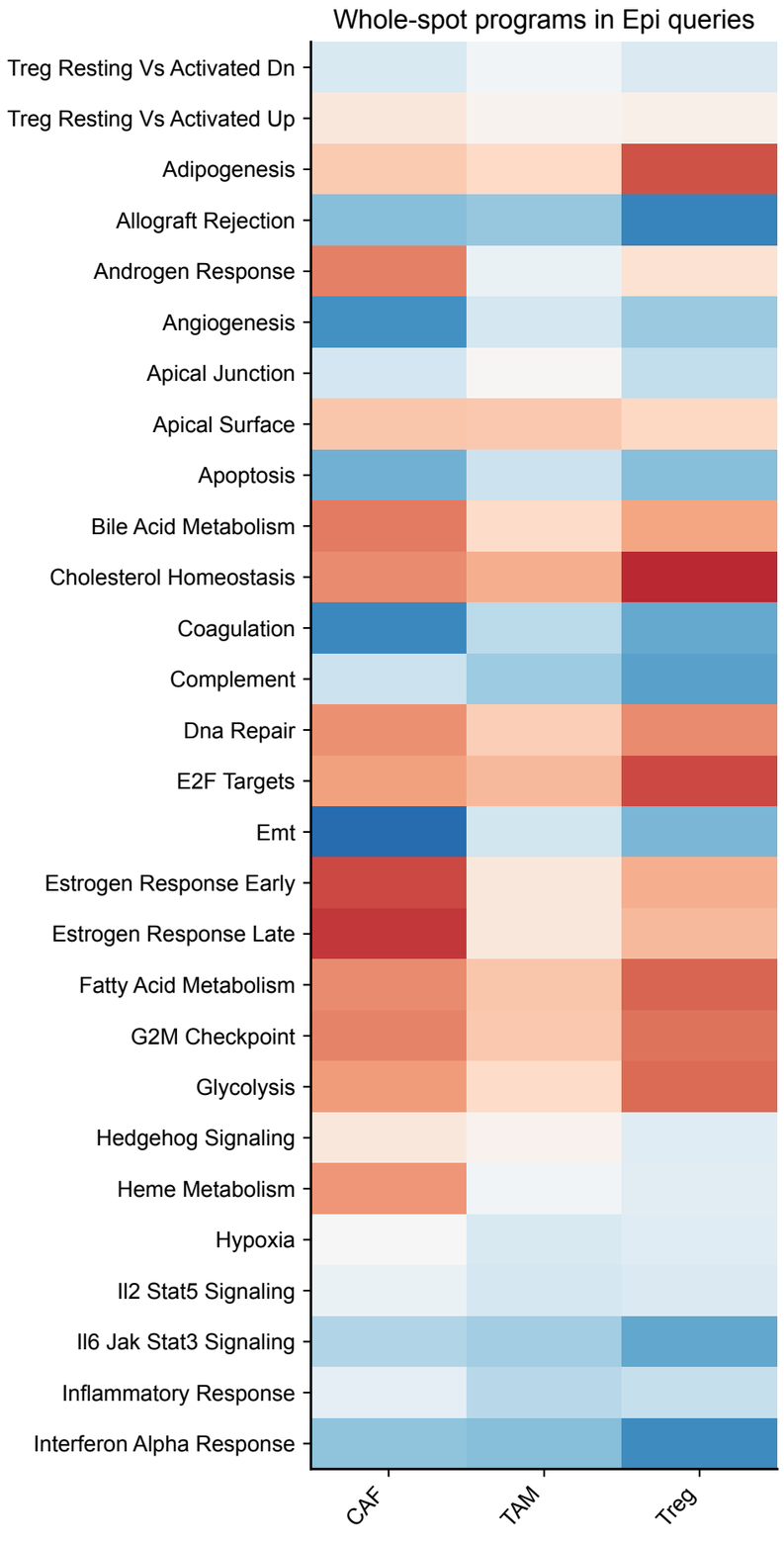


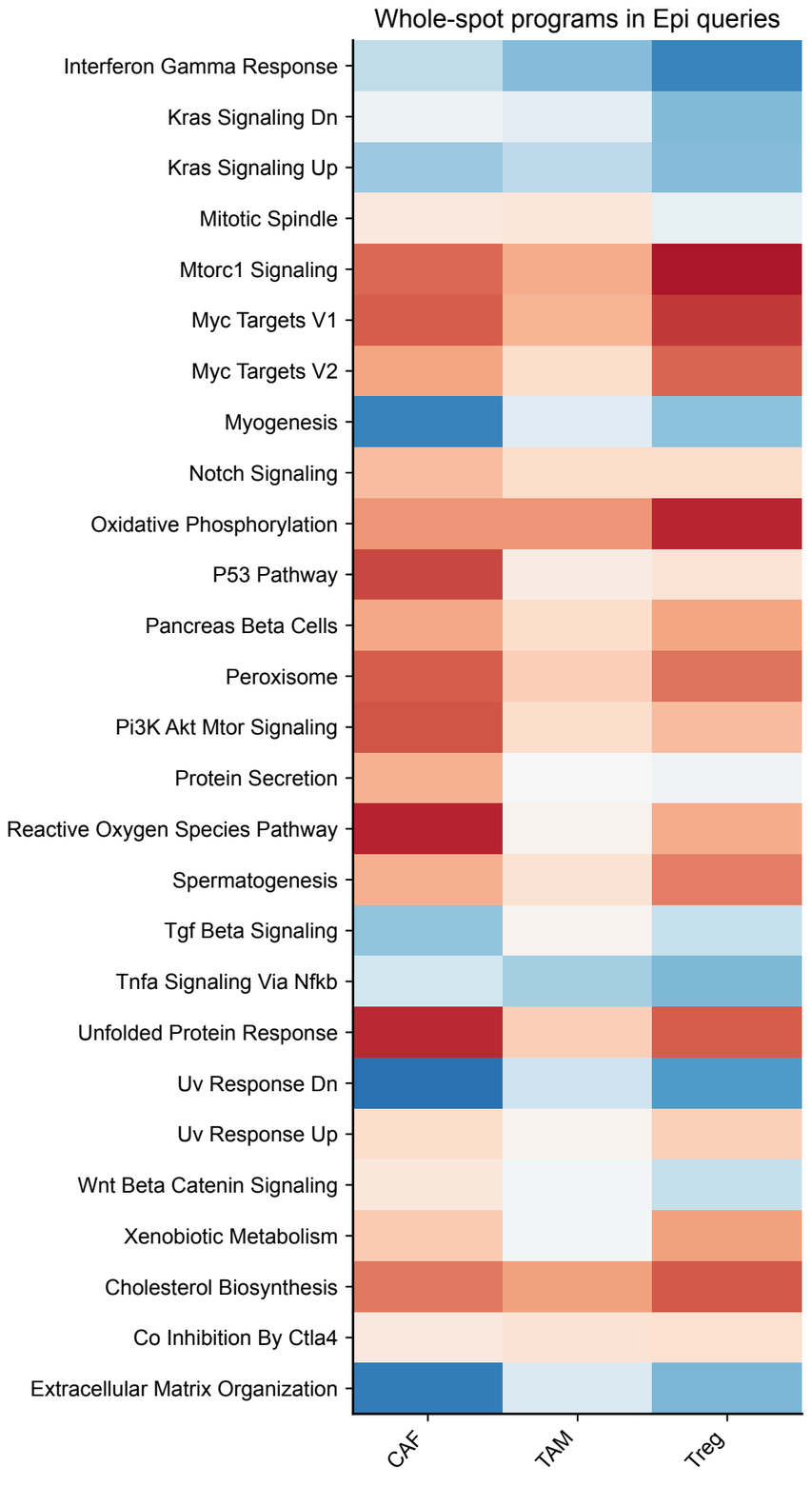




NPJ_sample2 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

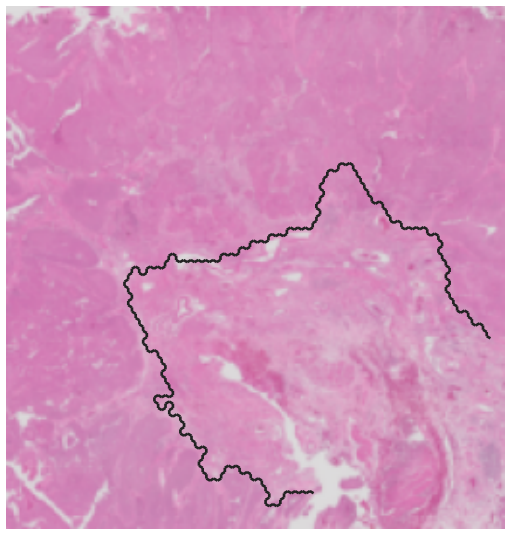




NPJ_sample3 | Tissue context

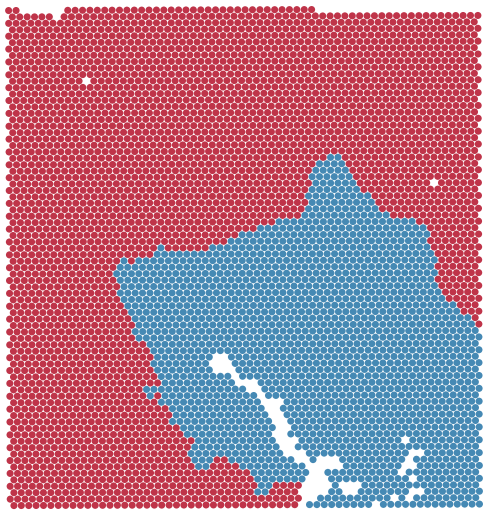
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



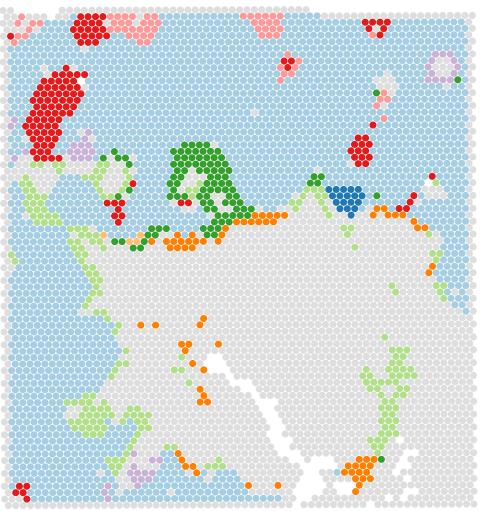
Black line: observed candidate outer edge

Final candidate location



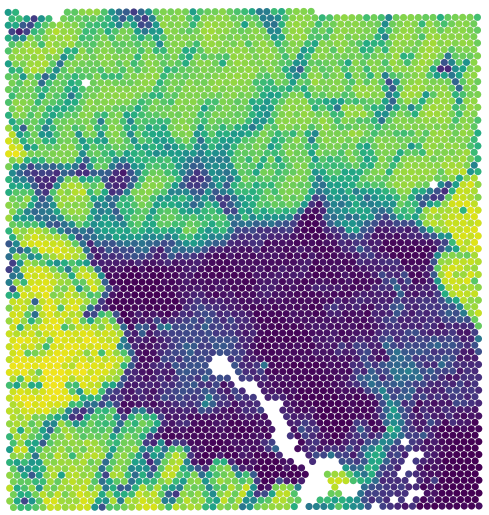
inside outside

Accepted R12 (original 0–8)



0 1 2 3 4 5 6 7 8

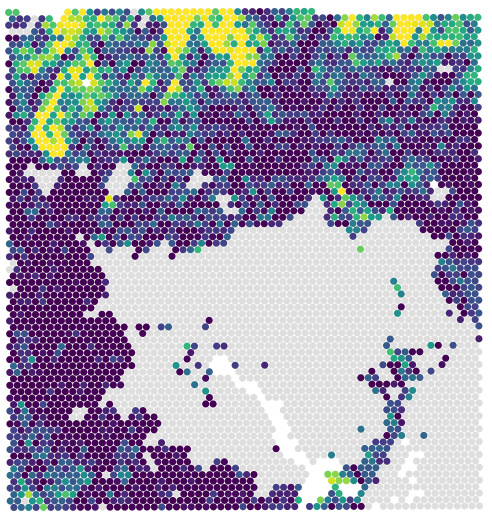
Epithelial support



0 1

Fraction of all 30

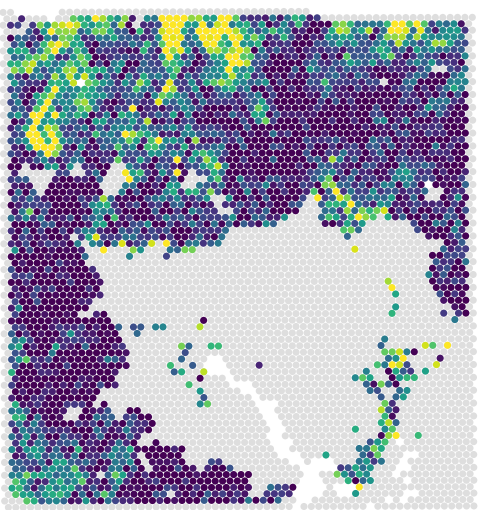
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



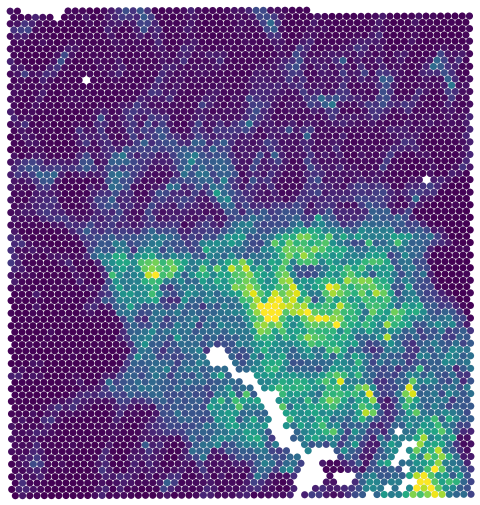
0 0.14

Matched-model AUCell

NPJ_sample3 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

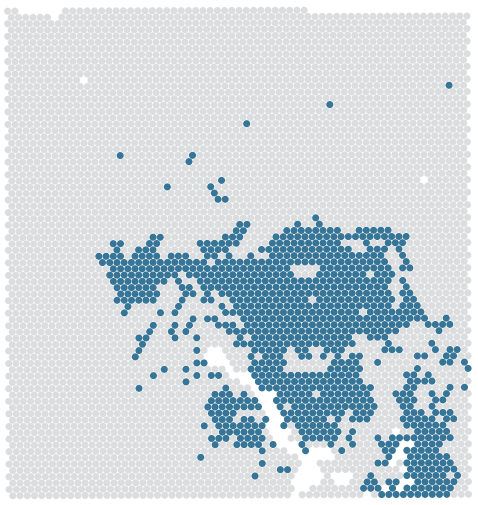
CAF composition



0 0.418

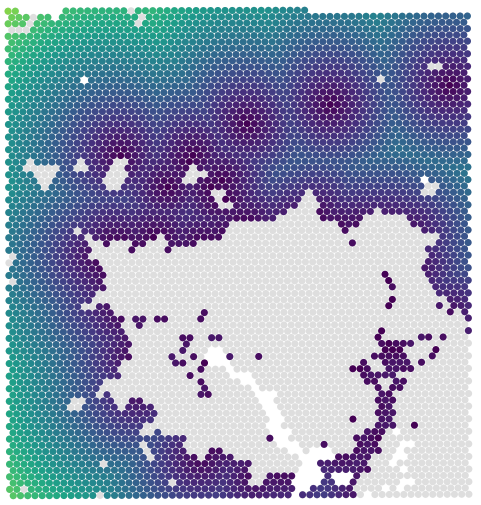
Fraction of all 30

CAF P80 targets



False True

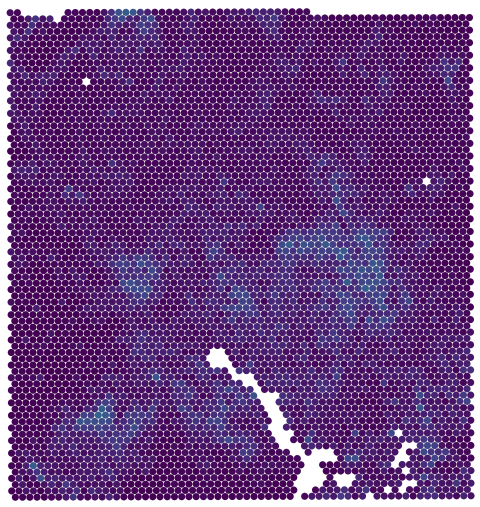
CAF distance



0 3.1e+03

Nearest spot center (μm)

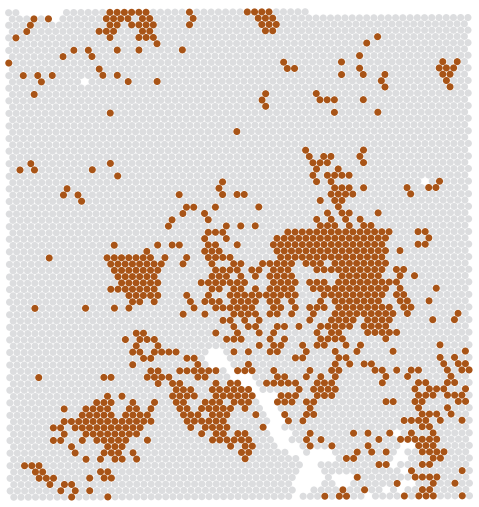
TAM composition



0 0.291

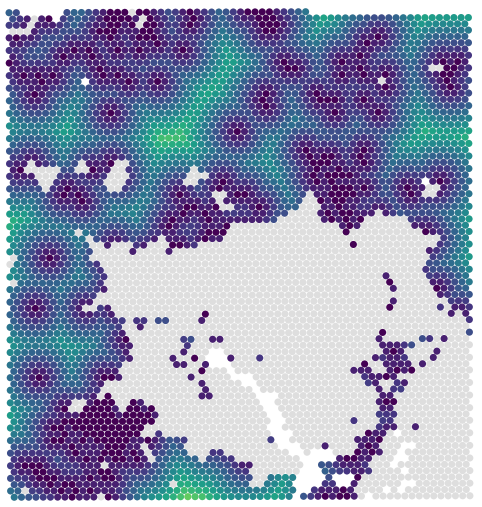
Fraction of all 30

TAM P80 targets



False True

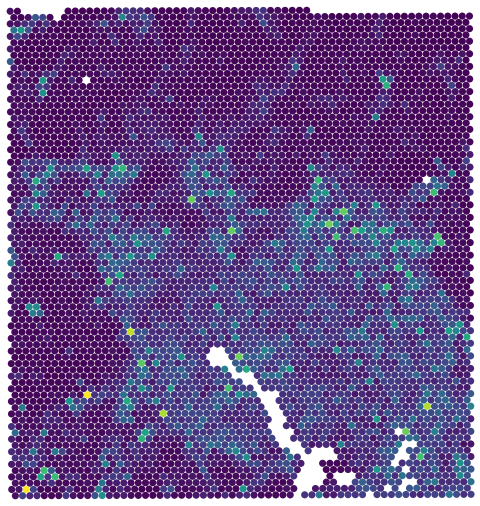
TAM distance



0 1.13e+03

Nearest spot center (μm)

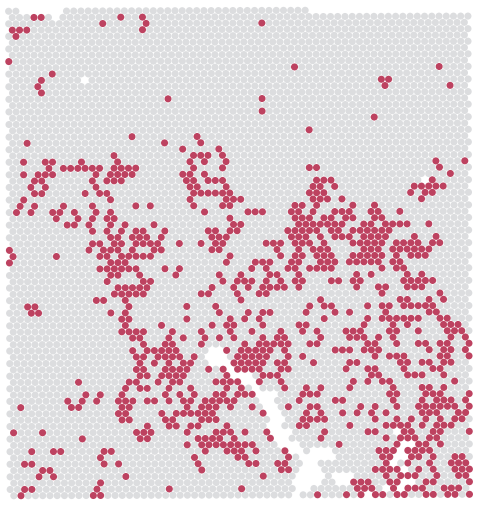
Treg composition



0 0.102

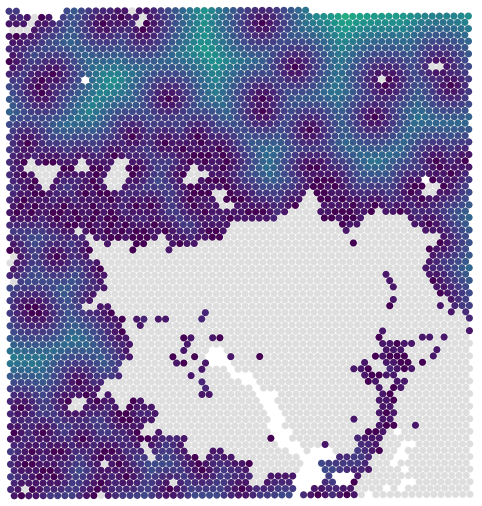
Fraction of all 30

Treg P80 targets



False True

Treg distance

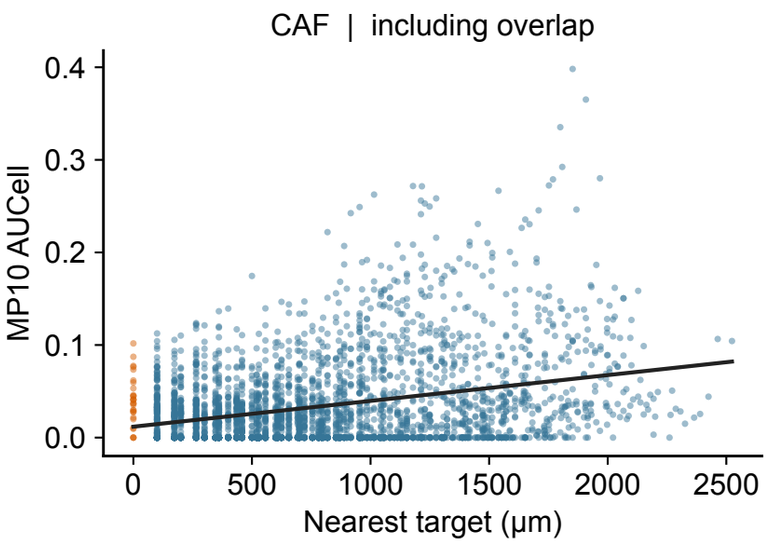


0 1.61e+03

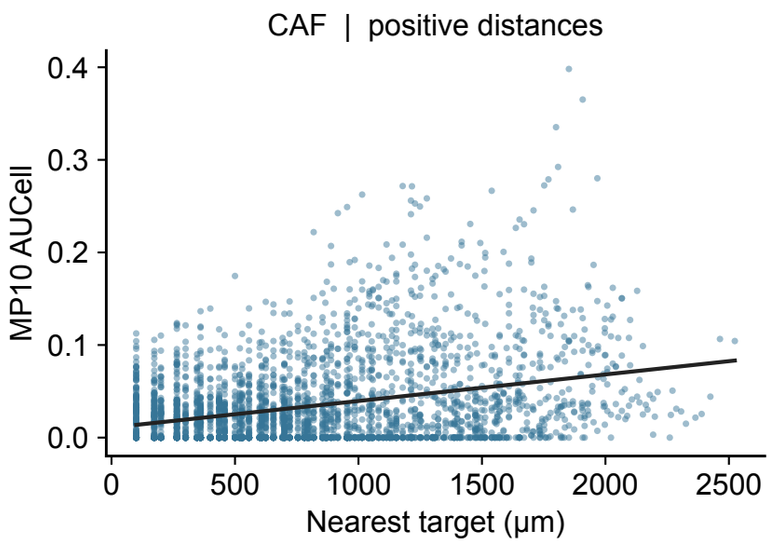
Nearest spot center (μm)

NPJ_sample3 | MP10 versus nearest target

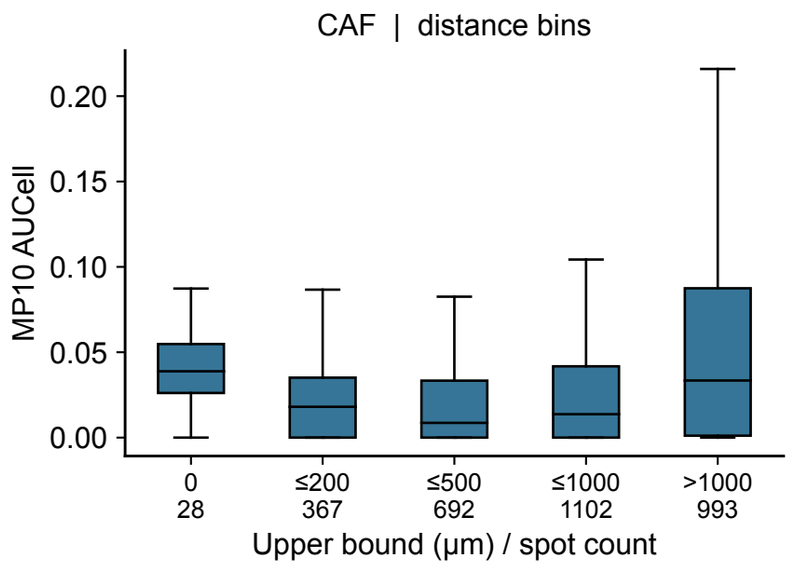
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



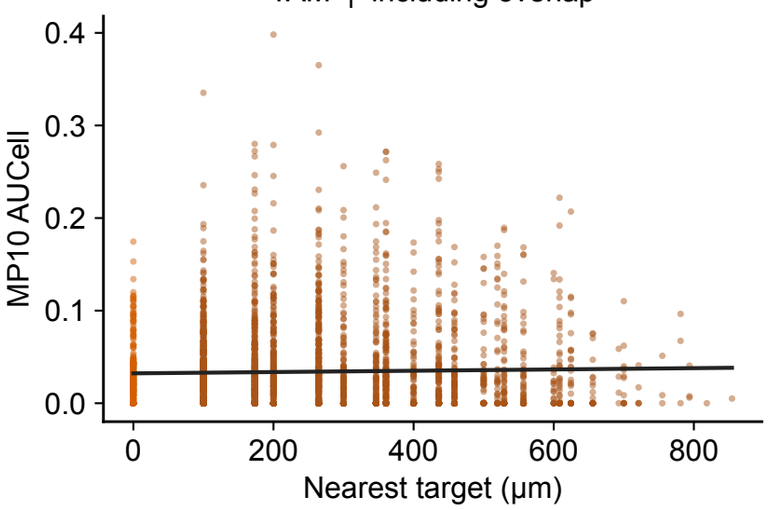
r +0.31 ρ +0.23
n 3182/3182 queries • 977 targets



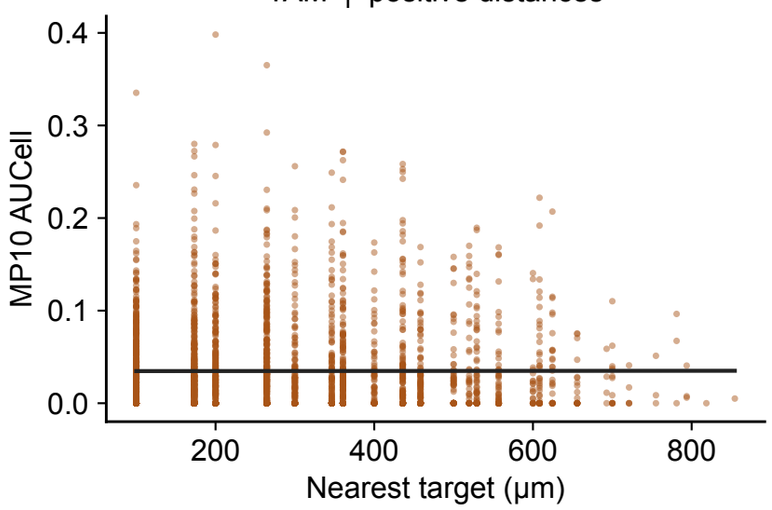
r +0.31 ρ +0.24
n 3154/3182 queries • 977 targets



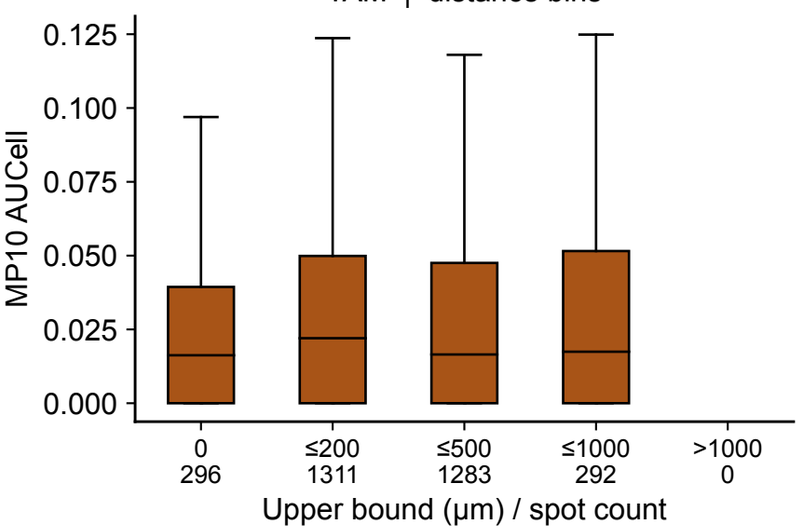
Orange = zero distance
Line = within-section linear fit



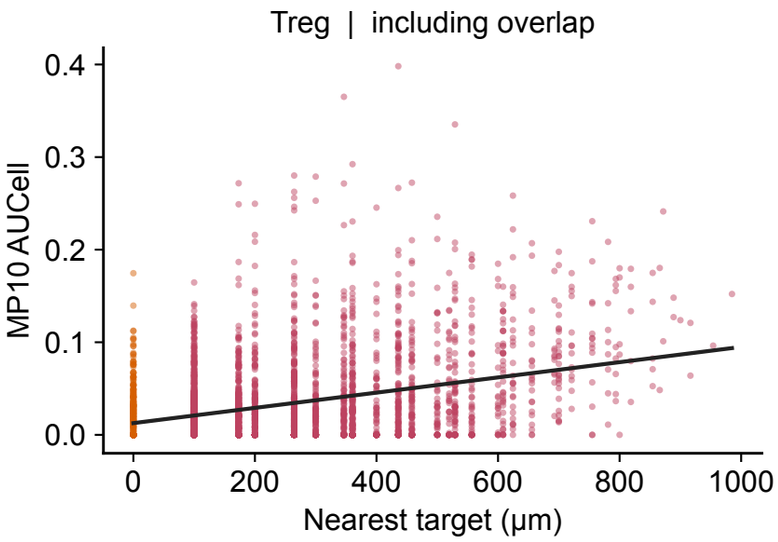
r +0.03 ρ -0.01
n 3182/3182 queries • 977 targets



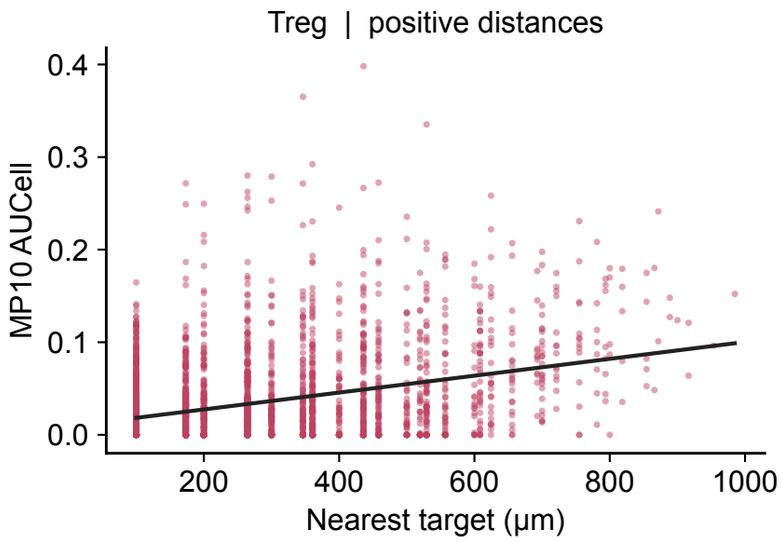
r +0.00 ρ -0.03
n 2886/3182 queries • 977 targets



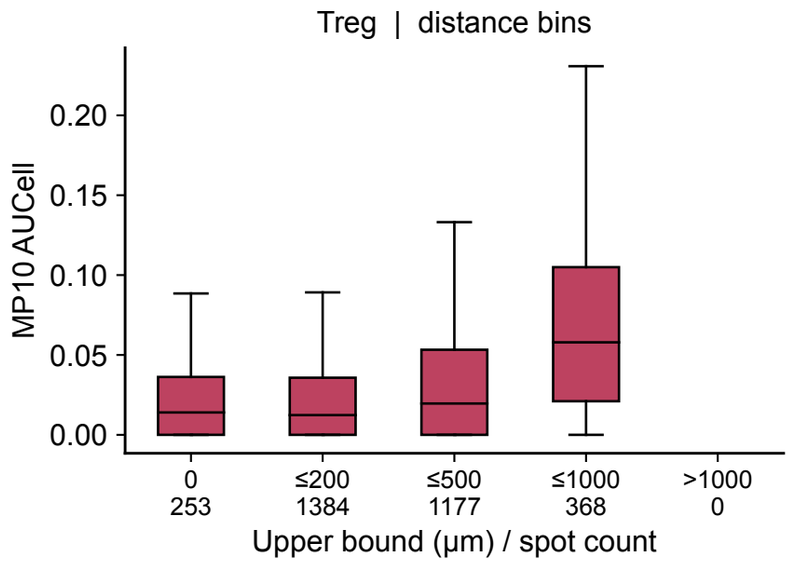
Orange = zero distance
Line = within-section linear fit



r +0.33 ρ +0.23
n 3182/3182 queries • 977 targets



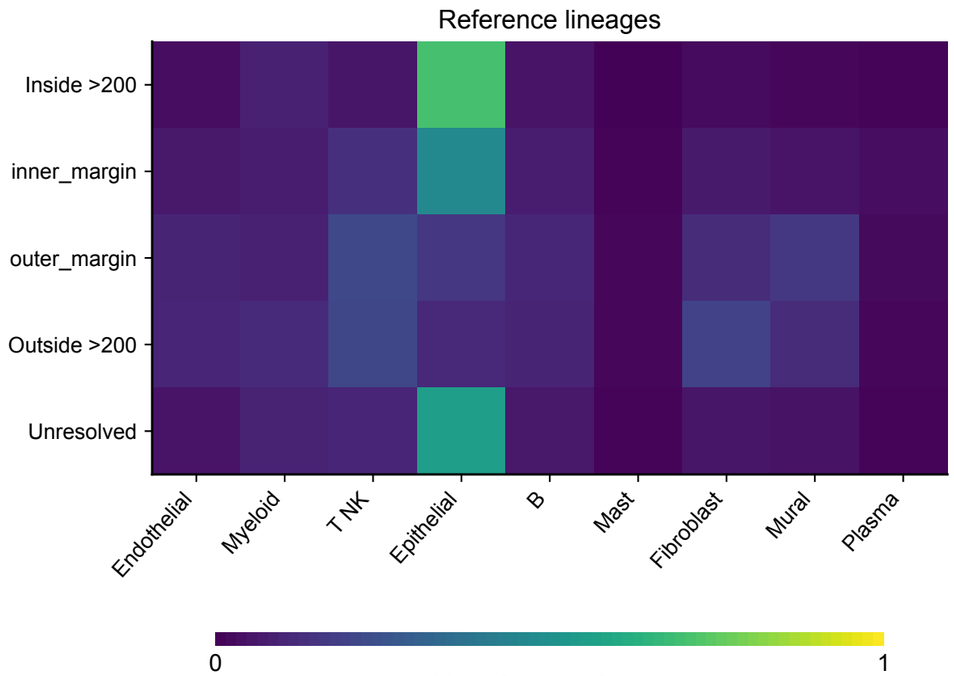
r +0.34 ρ +0.25
n 2929/3182 queries • 977 targets

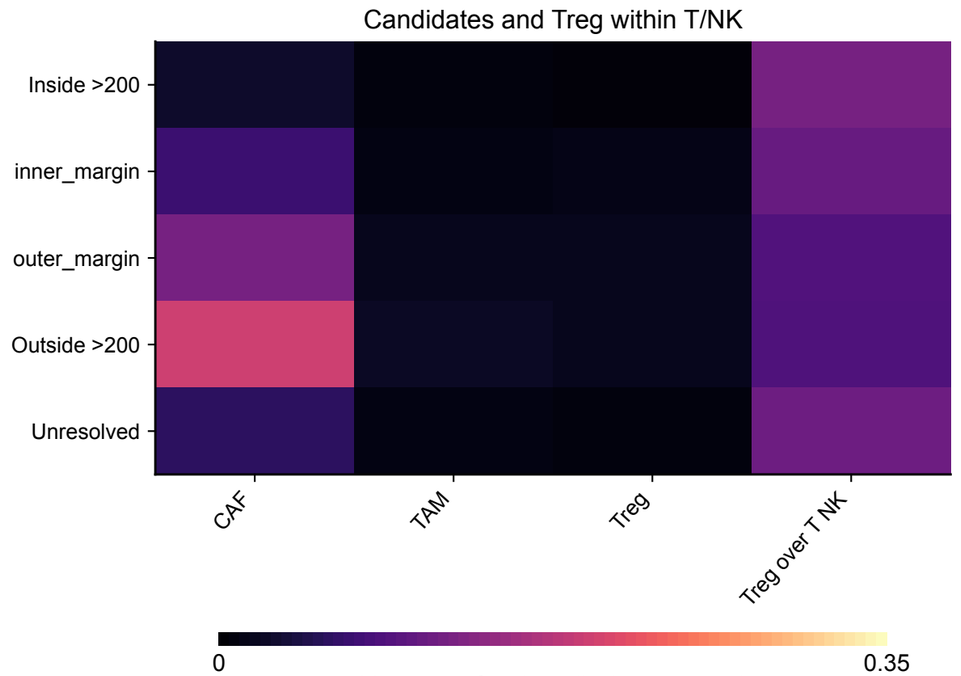


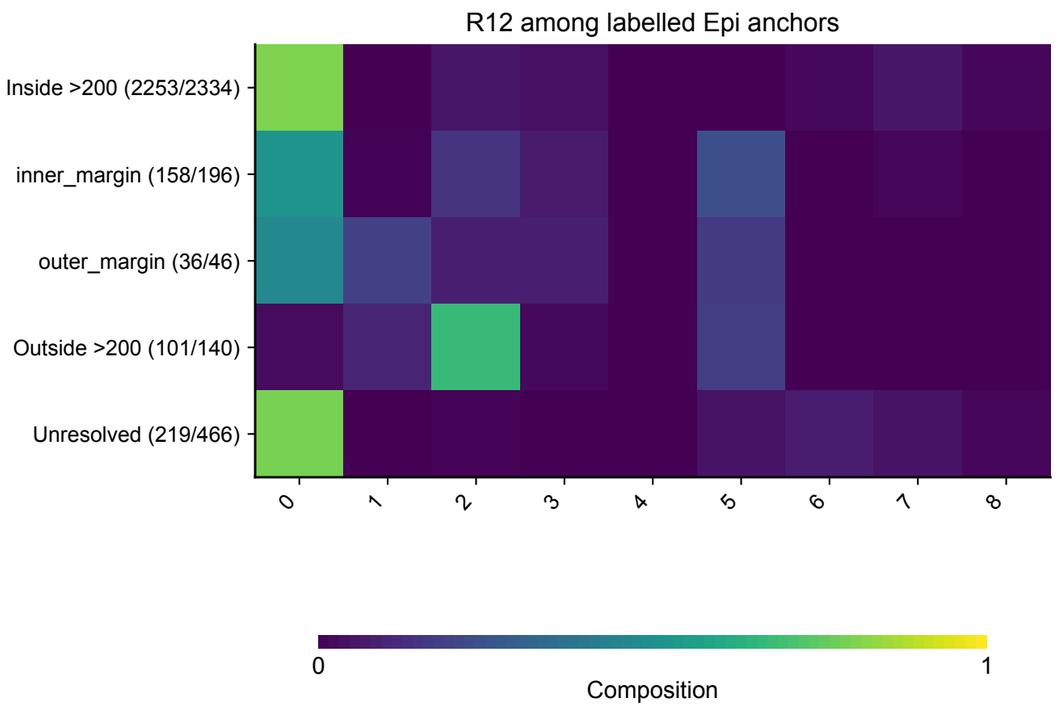
Orange = zero distance
Line = within-section linear fit

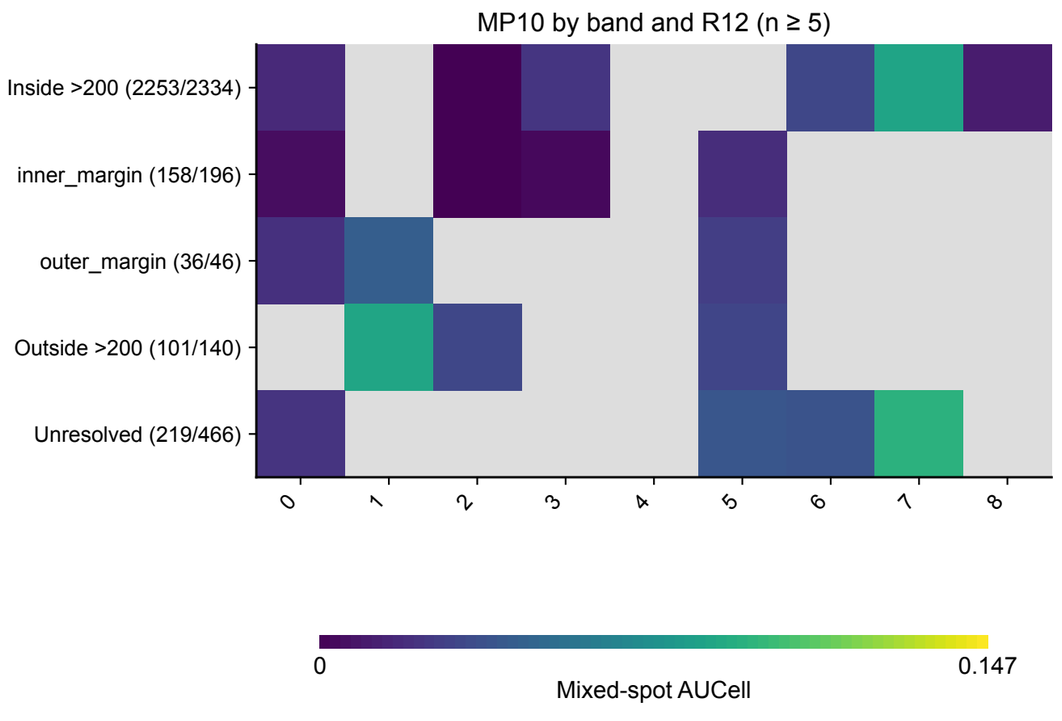
NPJ_sample3 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





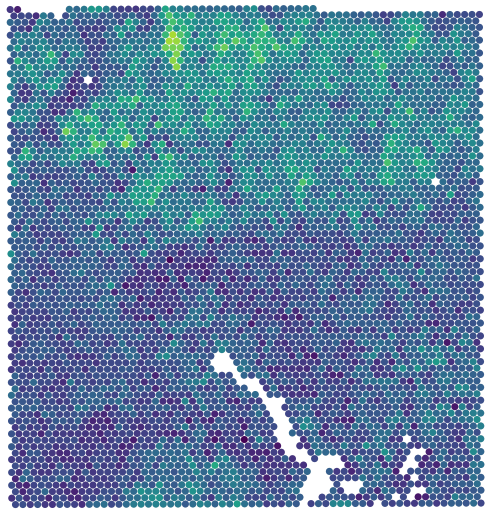




NPJ_sample3 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

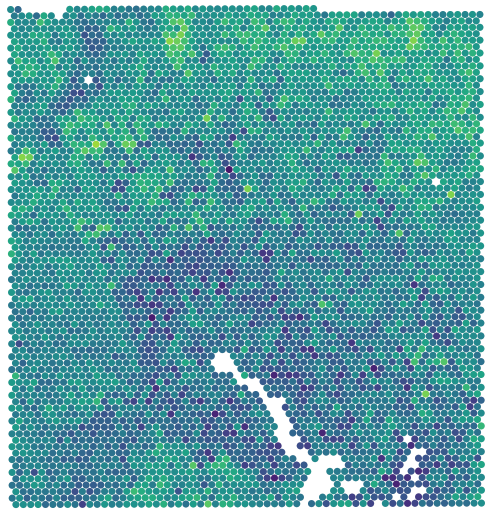
E2F Targets



0 0.152

UCell score

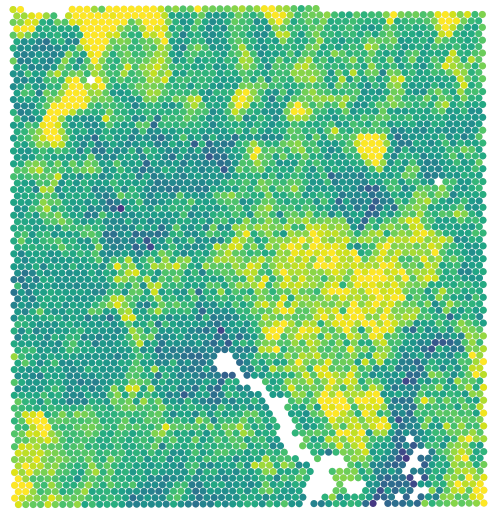
G2M Checkpoint



0 0.134

UCell score

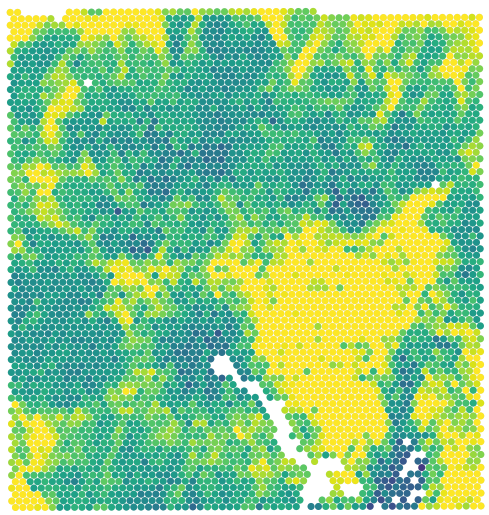
Hypoxia



0 0.211

UCell score

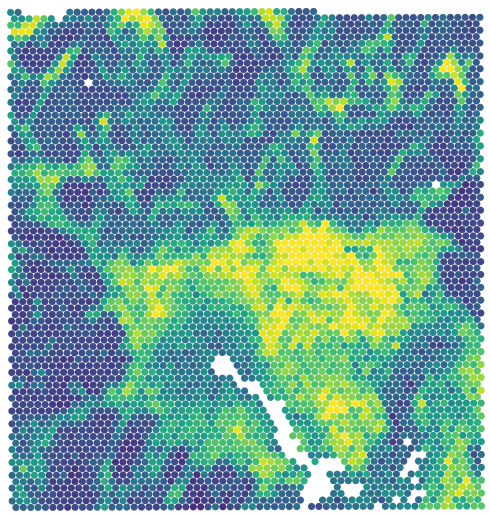
Tnfa Signaling Via Nfkb



0 0.275

UCell score

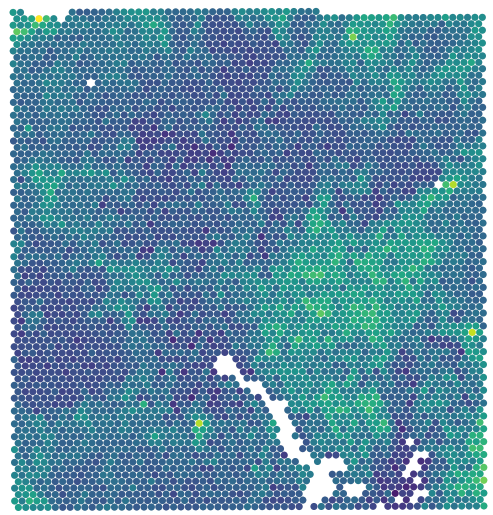
Emt



0 0.386

UCell score

Interferon Gamma Response



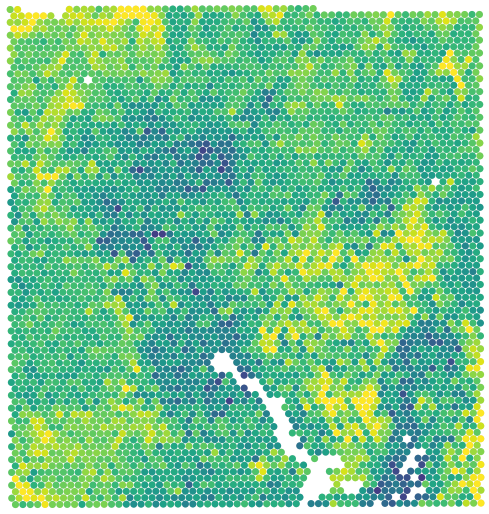
0 0.275

UCell score

NPJ_sample3 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

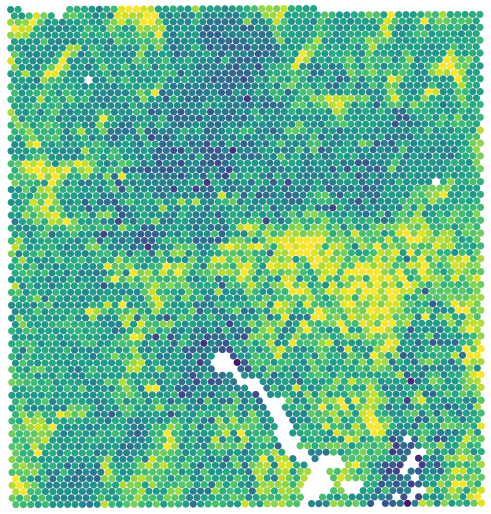
Il2 Stat5 Signaling



0 0.144

UCell score

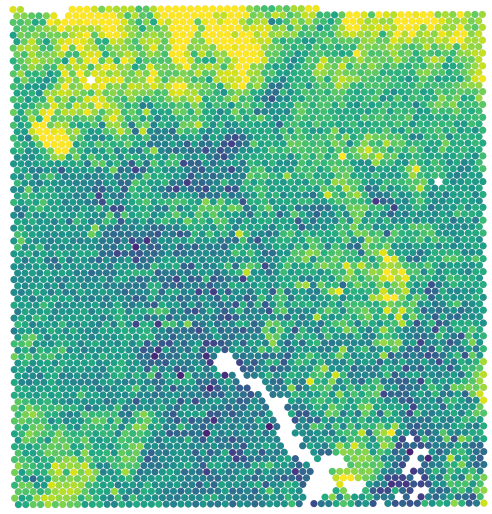
Tgf Beta Signaling



0 0.232

UCell score

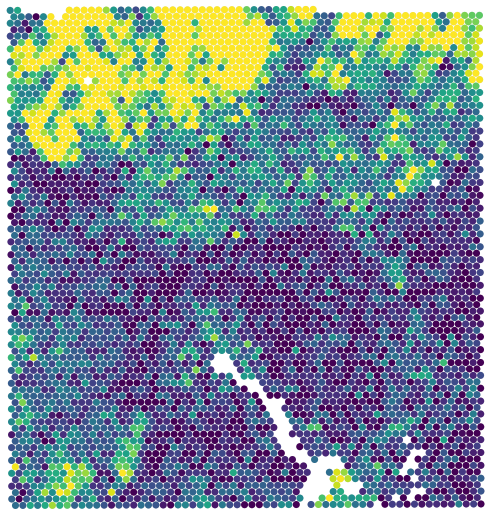
Cholesterol Homeostasis



0 0.21

UCell score

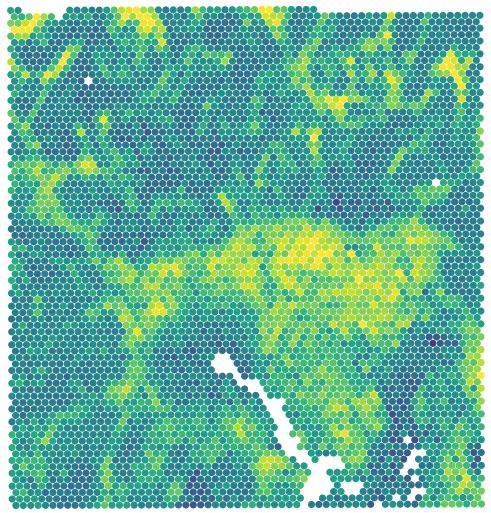
Cholesterol Biosynthesis



0 0.166

UCell score

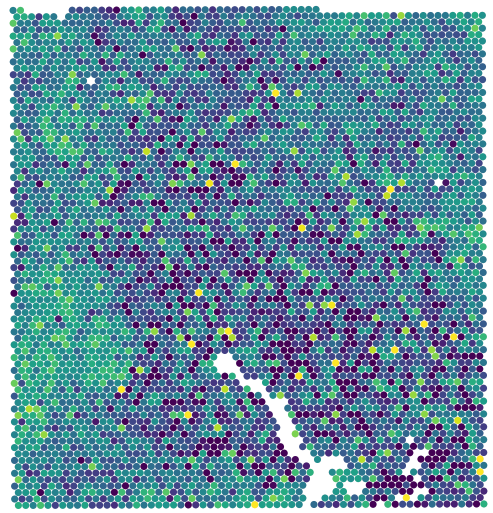
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



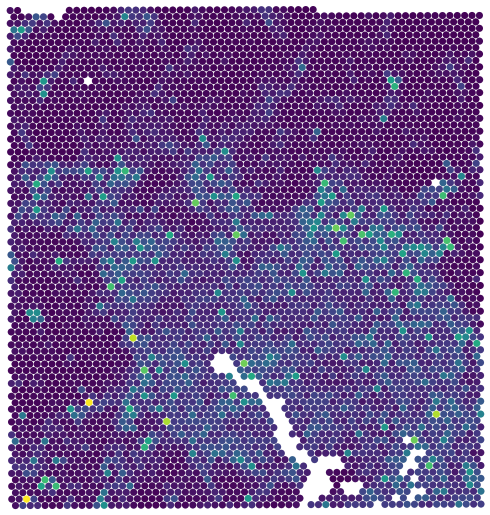
0 0.151

UCell score

NPJ_sample3 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

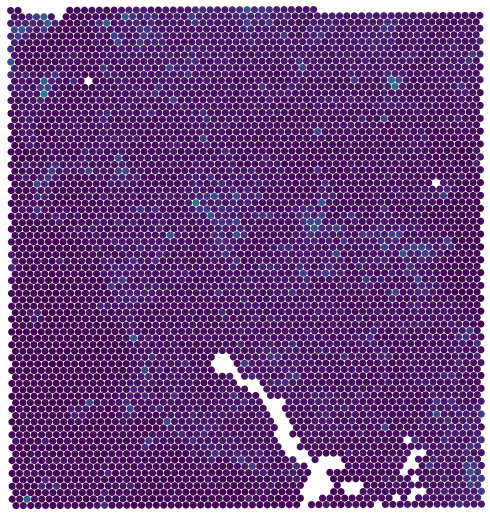
Treg composition



0 0.102

Fraction of all 30

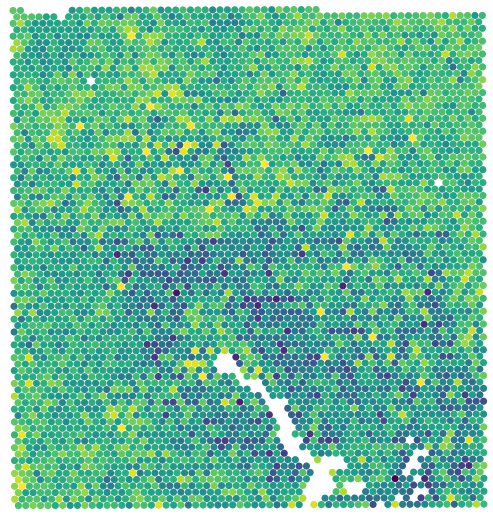
Treg estimated abundance



0 2.85

Posterior mean estimate

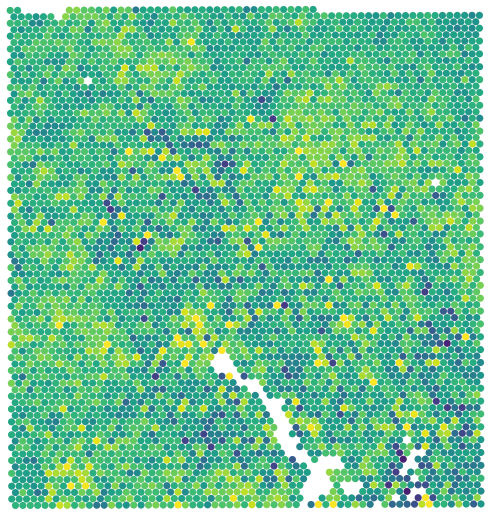
Treg Resting Vs Activated Dn



0 0.0778

UCell score

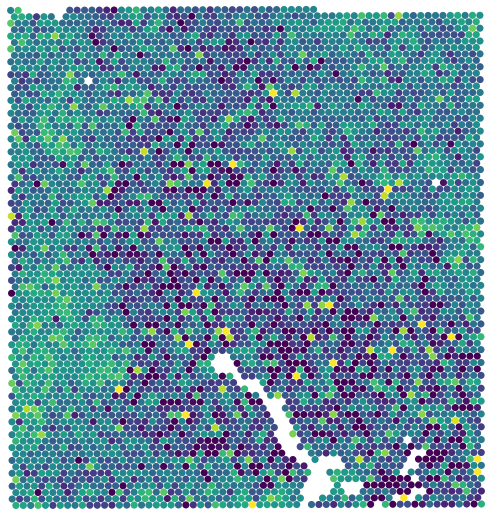
Treg Resting Vs Activated Up



0 0.0609

UCell score

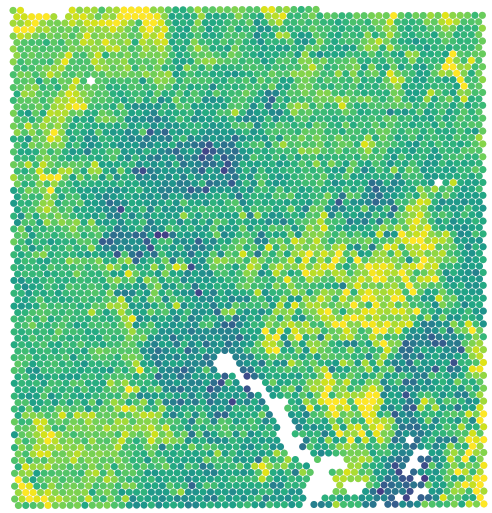
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

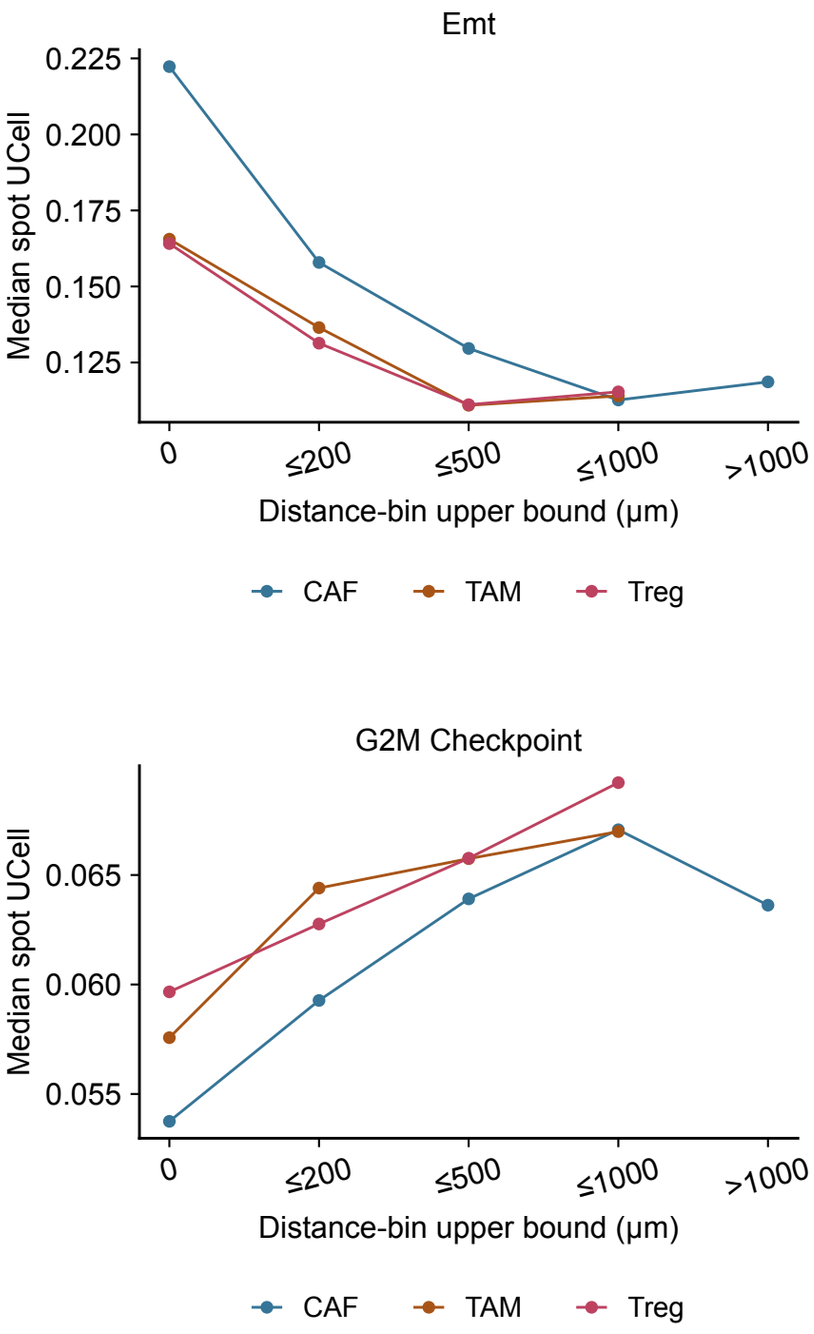


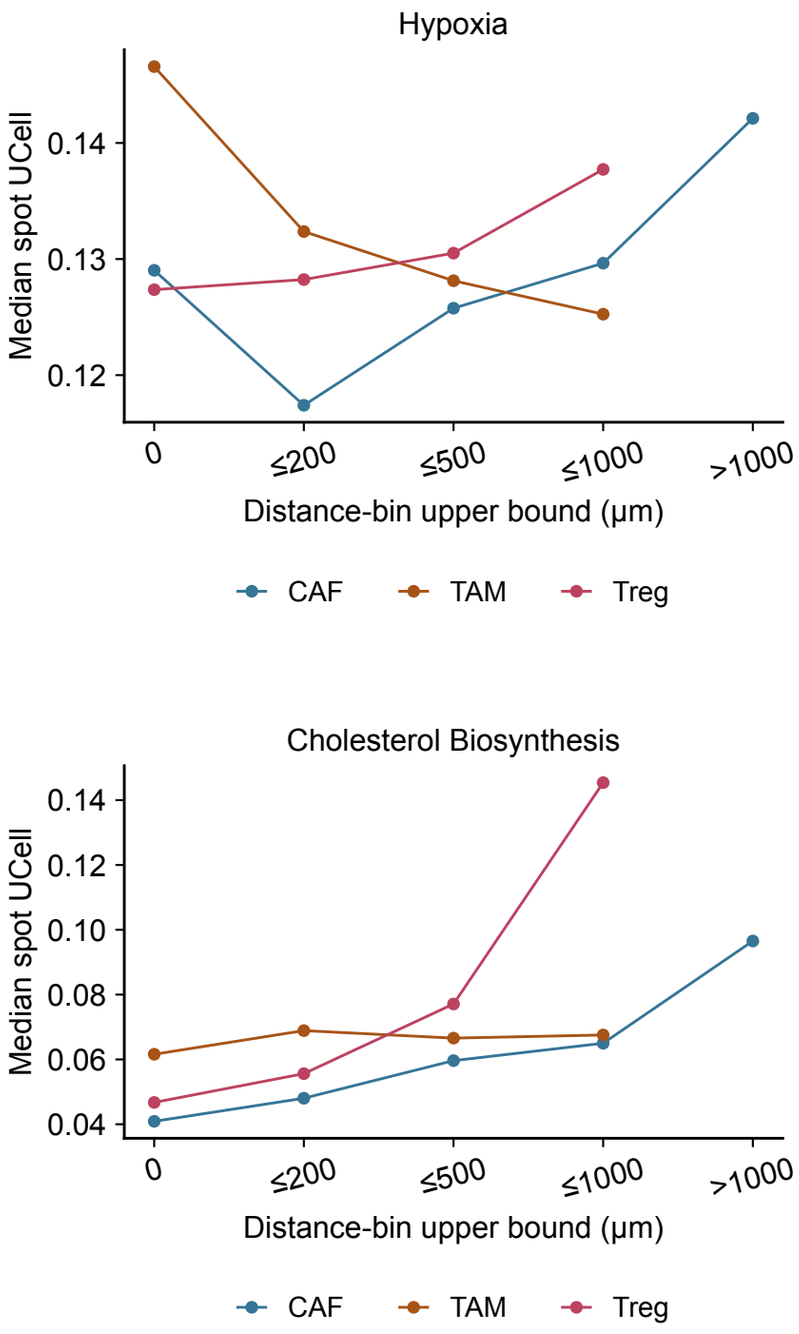
0 0.144

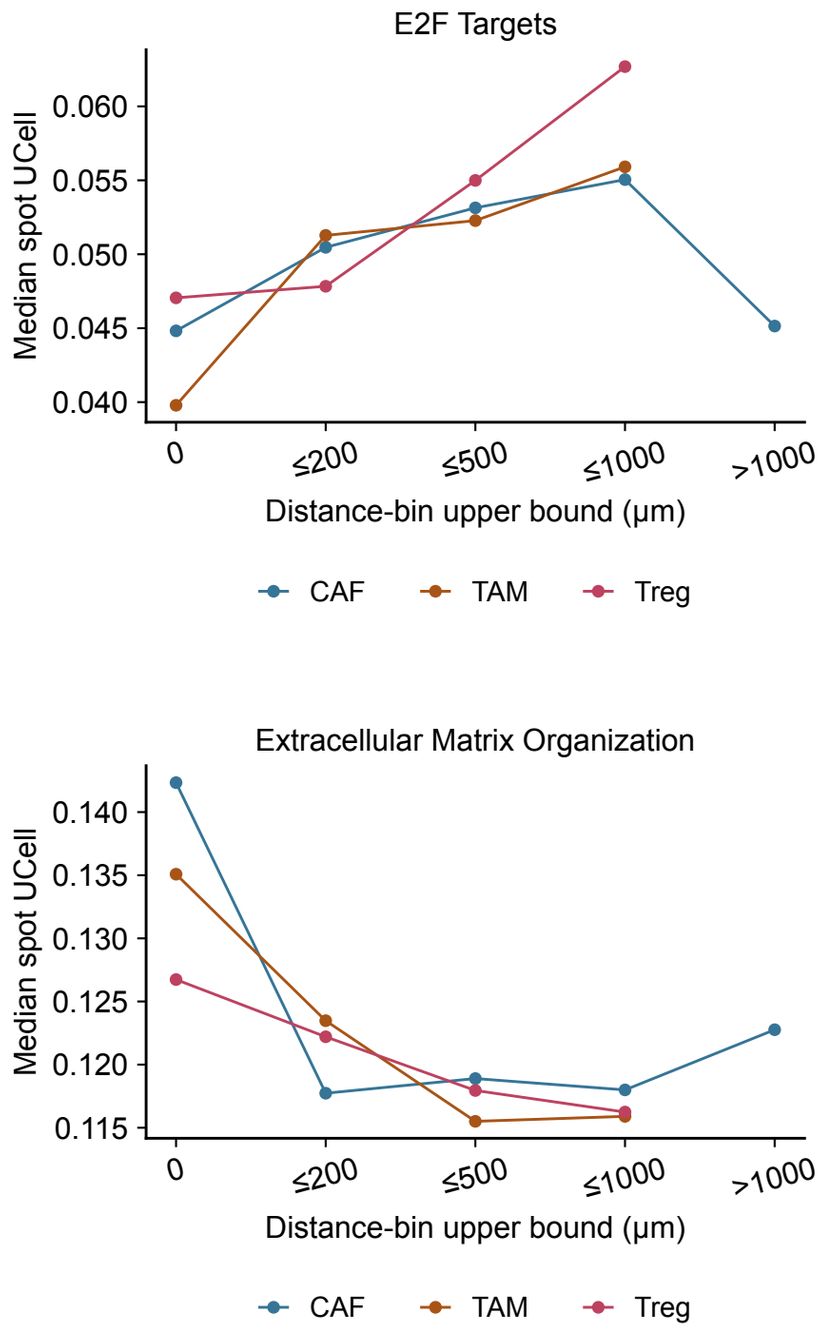
UCell score

NPJ_sample3 | Programs by target distance

Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

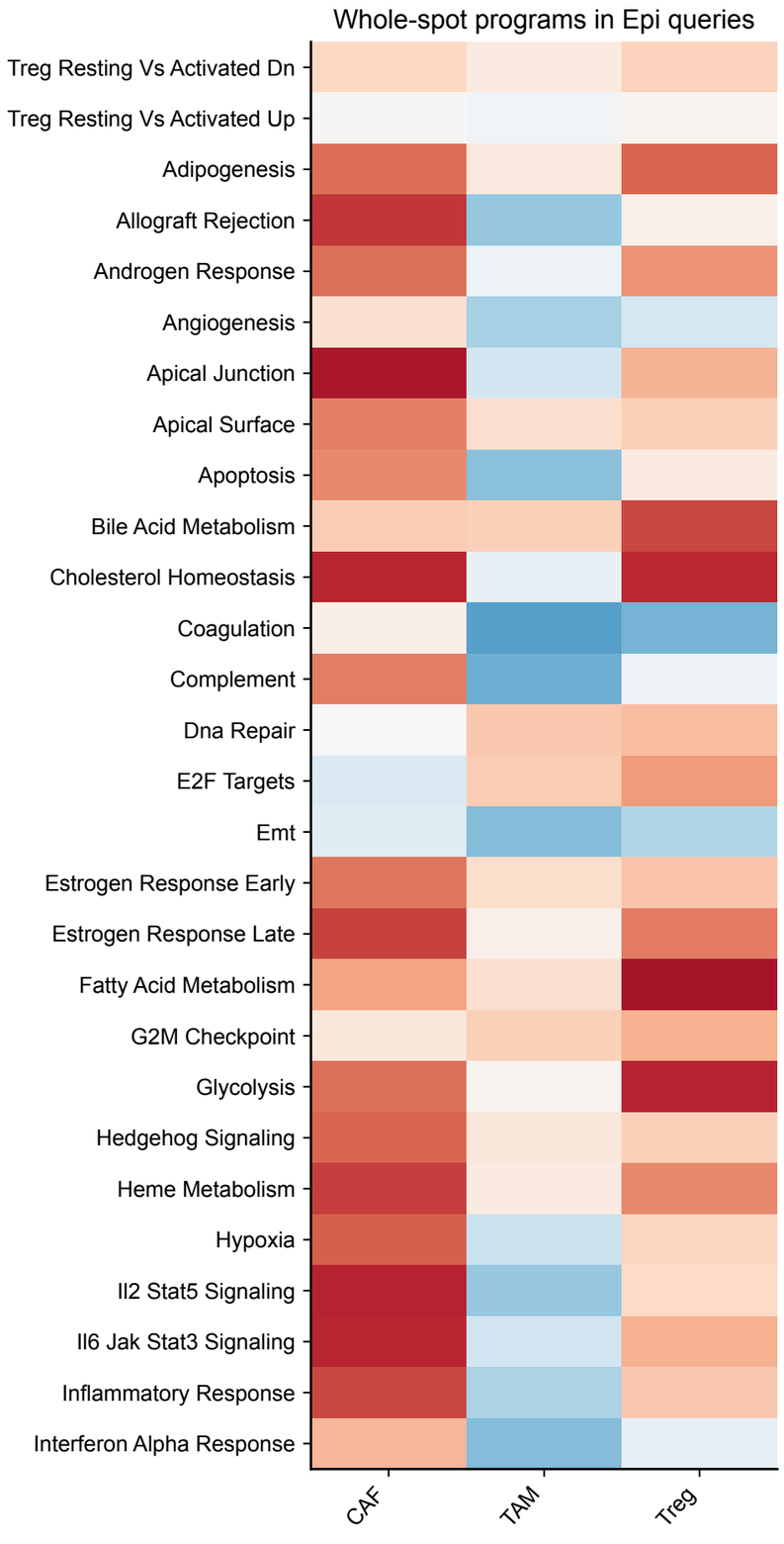


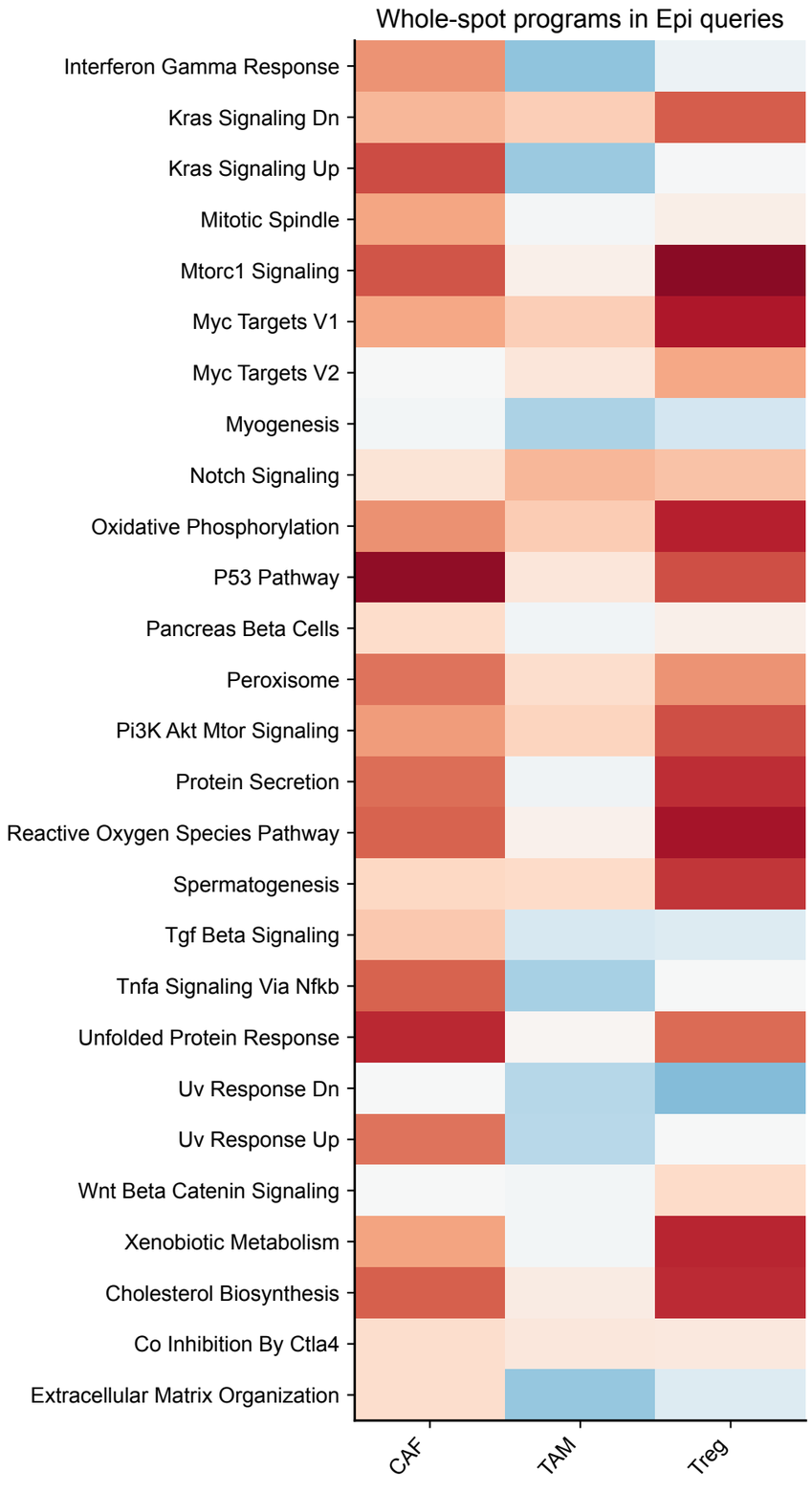




NPJ_sample3 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

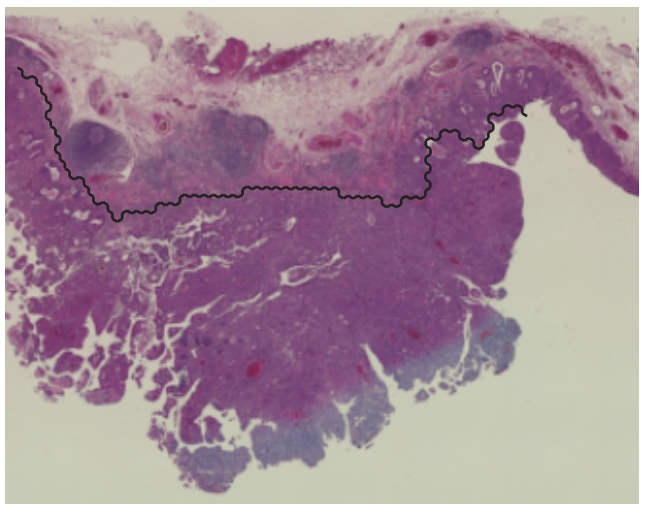




NPJ_sample4 | Tissue context

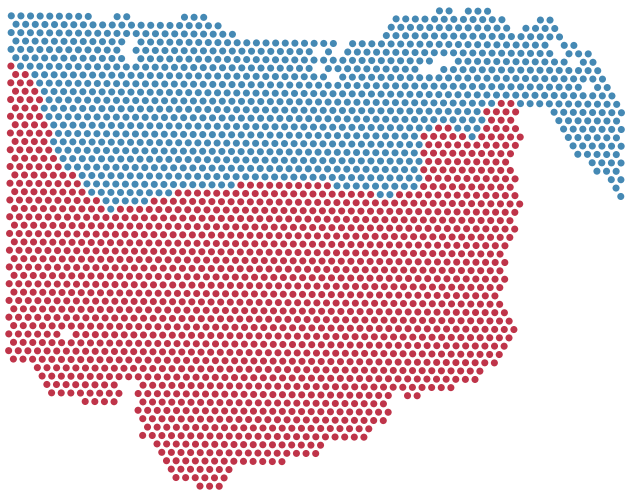
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



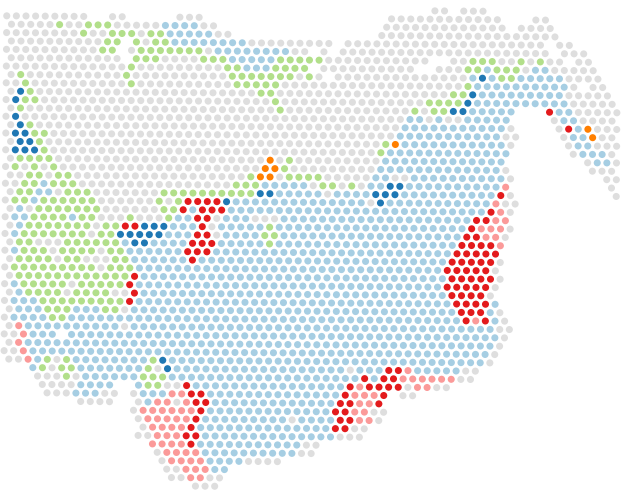
Black line: observed candidate outer edge

Final candidate location



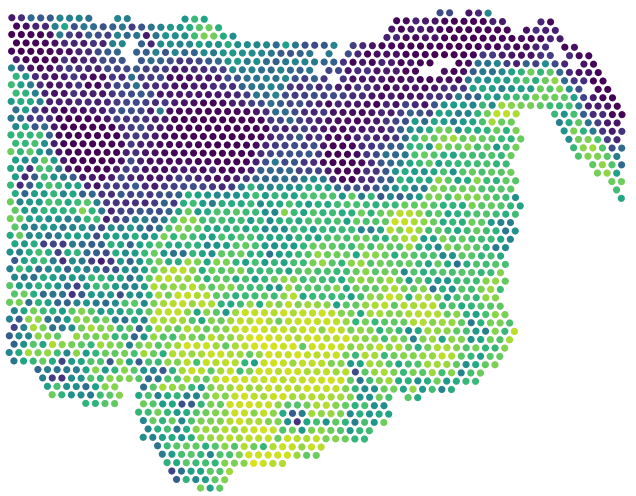
inside outside

Accepted R12 (original 0–8)



0 1 2 5 6 7 8

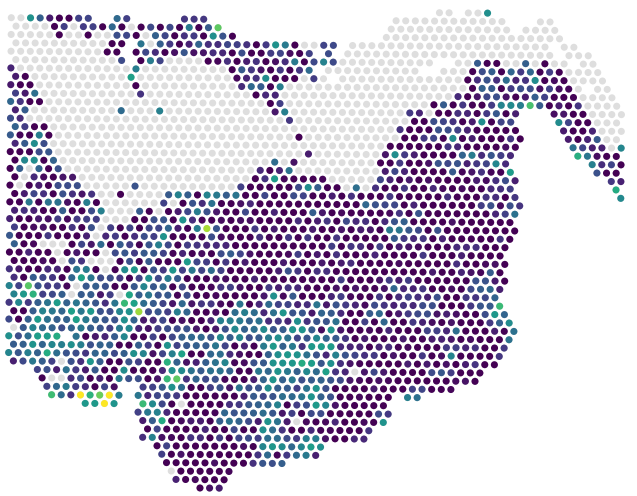
Epithelial support



0 1

Fraction of all 30

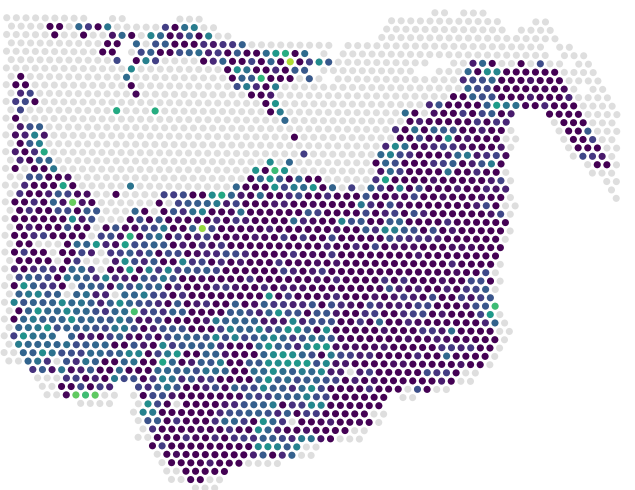
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



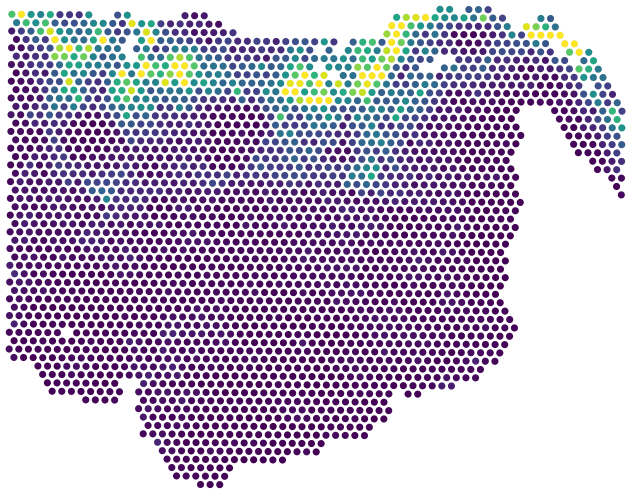
0 0.14

Matched-model AUCell

NPJ_sample4 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

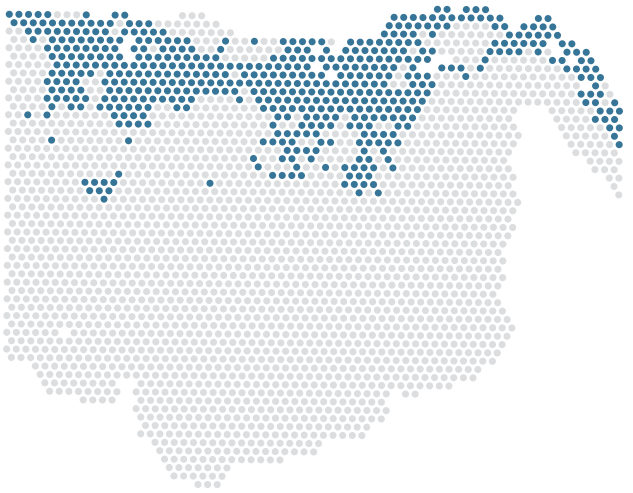
CAF composition



0 0.418

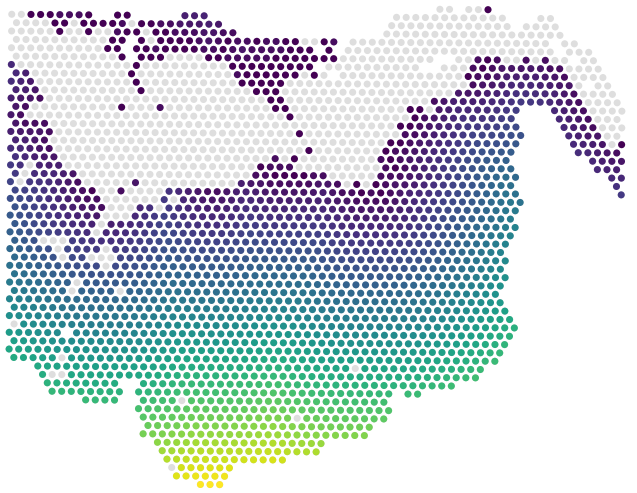
Fraction of all 30

CAF P80 targets



False True

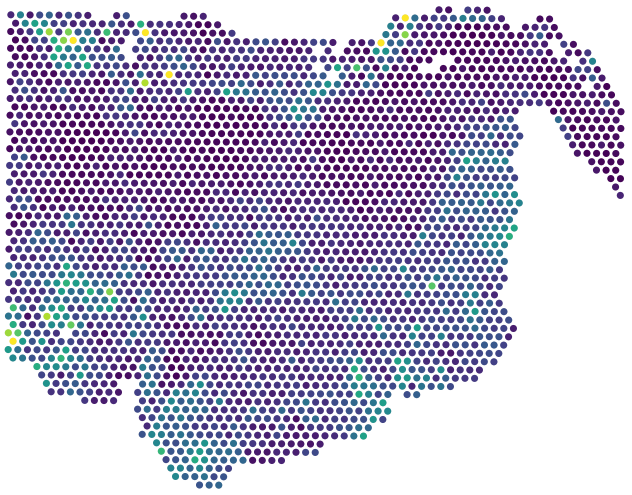
CAF distance



0 3.1e+03

Nearest spot center (µm)

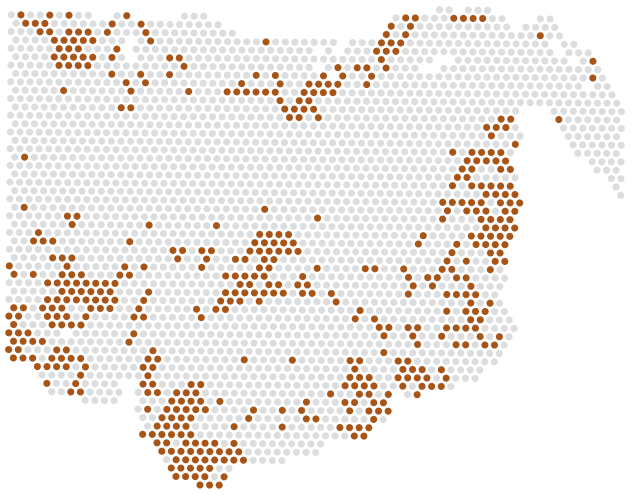
TAM composition



0 0.291

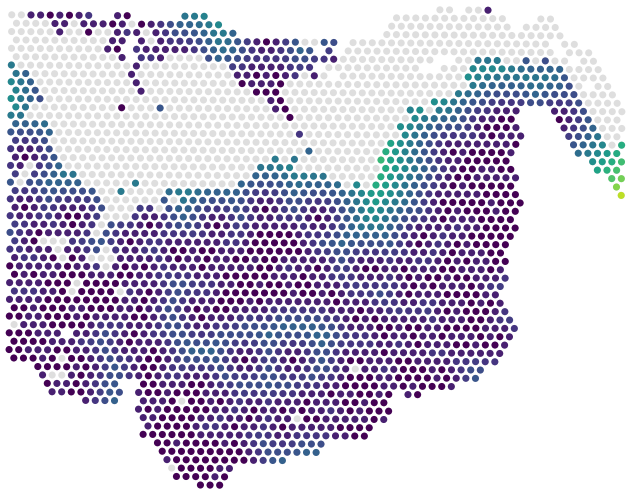
Fraction of all 30

TAM P80 targets



False True

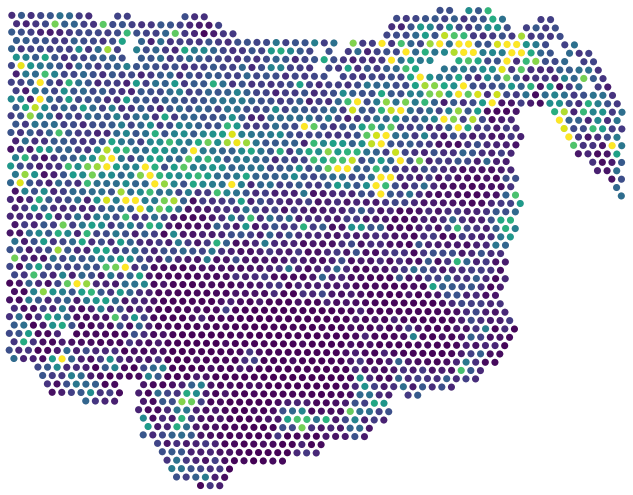
TAM distance



0 1.13e+03

Nearest spot center (µm)

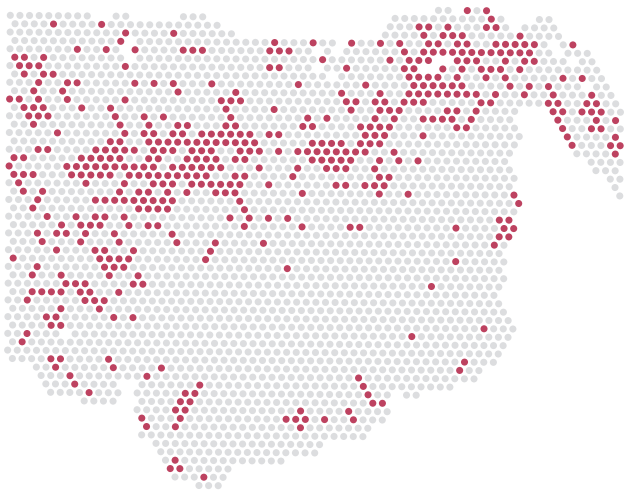
Treg composition



0 0.102

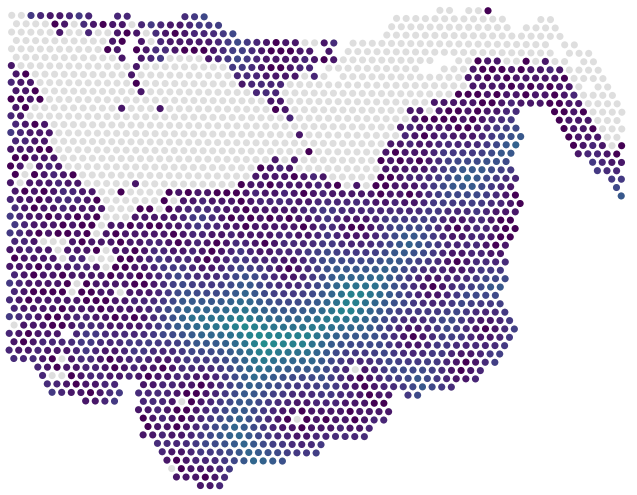
Fraction of all 30

Treg P80 targets



False True

Treg distance

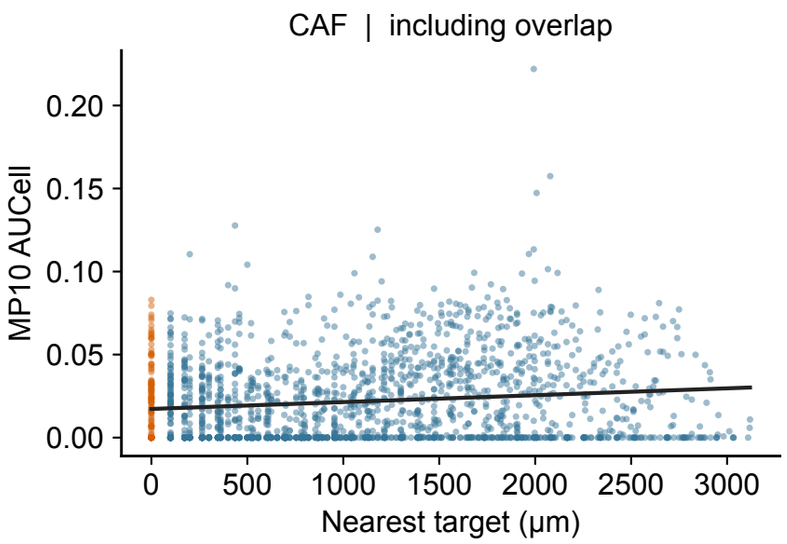


0 1.61e+03

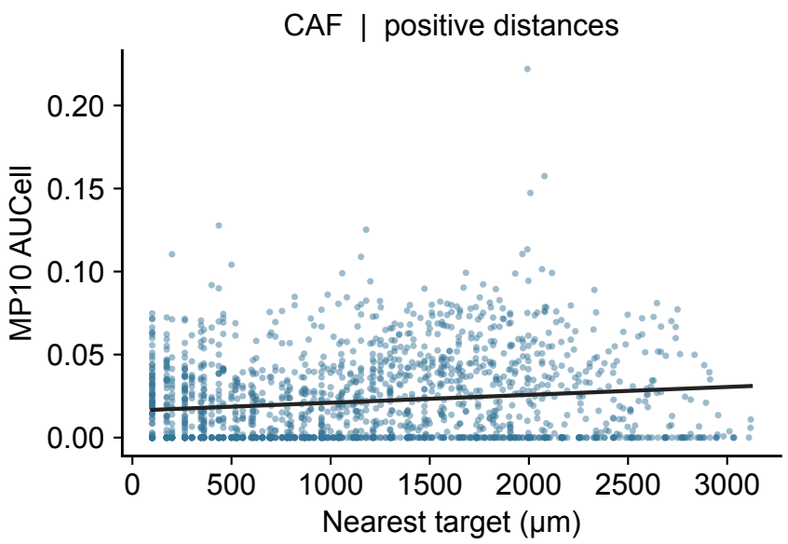
Nearest spot center (µm)

NPJ_sample4 | MP10 versus nearest target

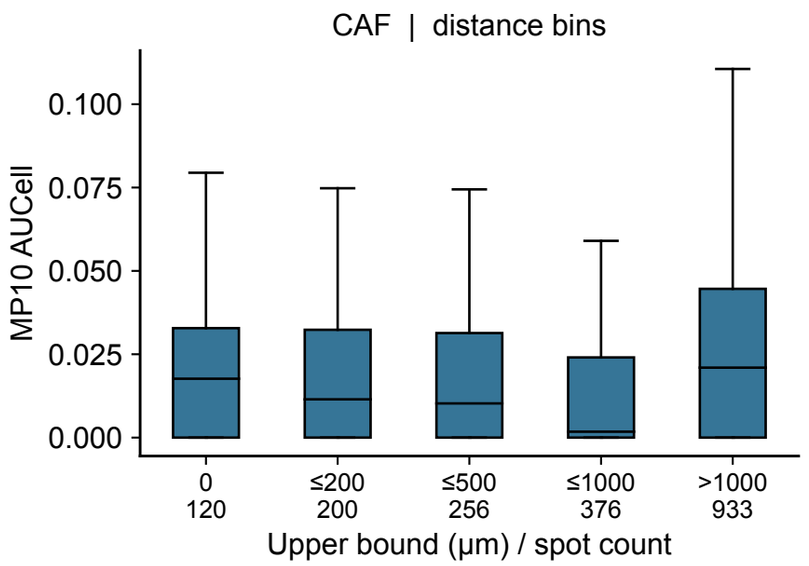
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



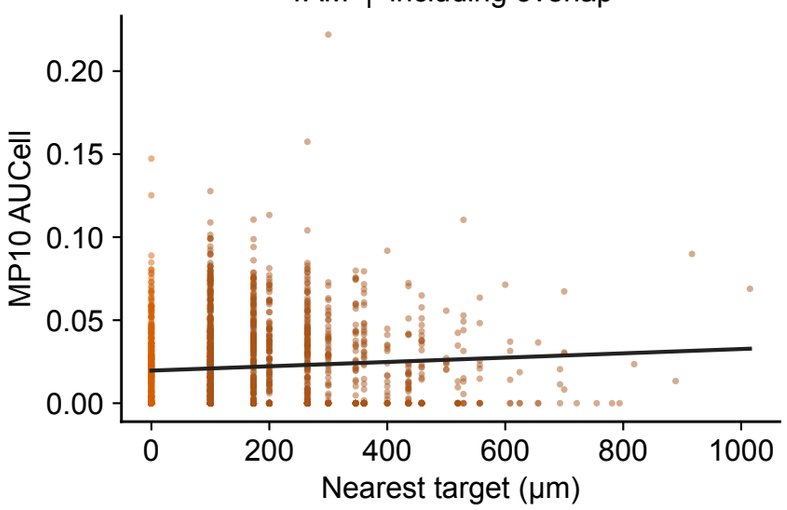
r +0.13 ρ +0.12
n 1885/1885 queries • 528 targets



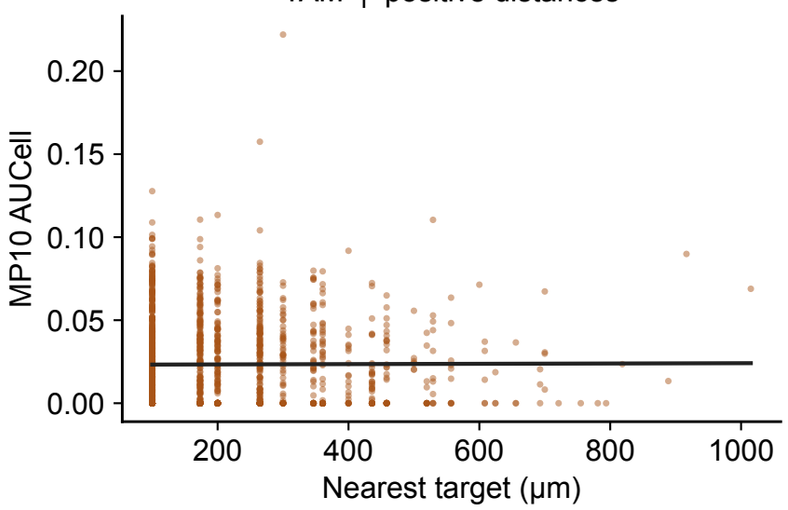
r +0.14 ρ +0.14
n 1765/1885 queries • 528 targets



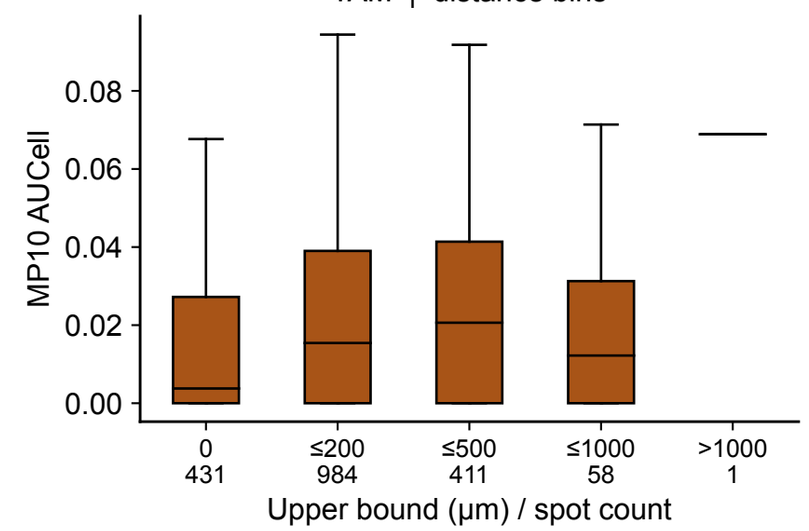
Orange = zero distance
Line = within-section linear fit



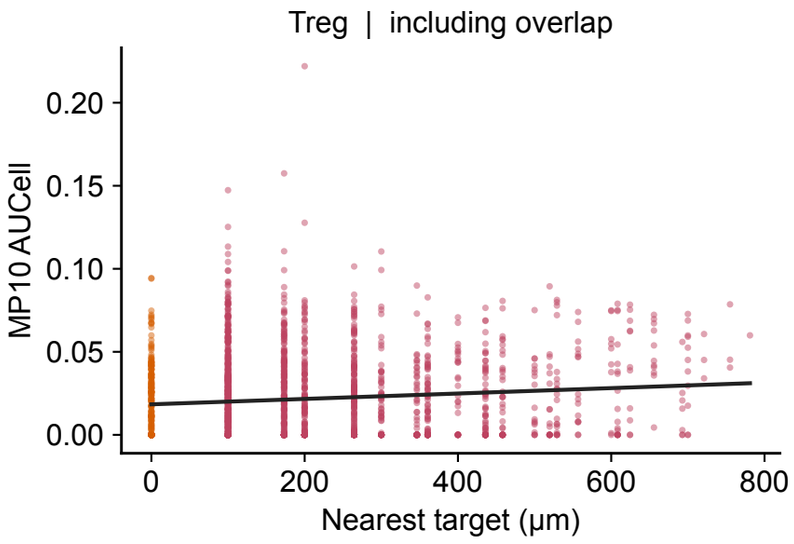
r +0.08 ρ +0.12
n 1885/1885 queries • 528 targets



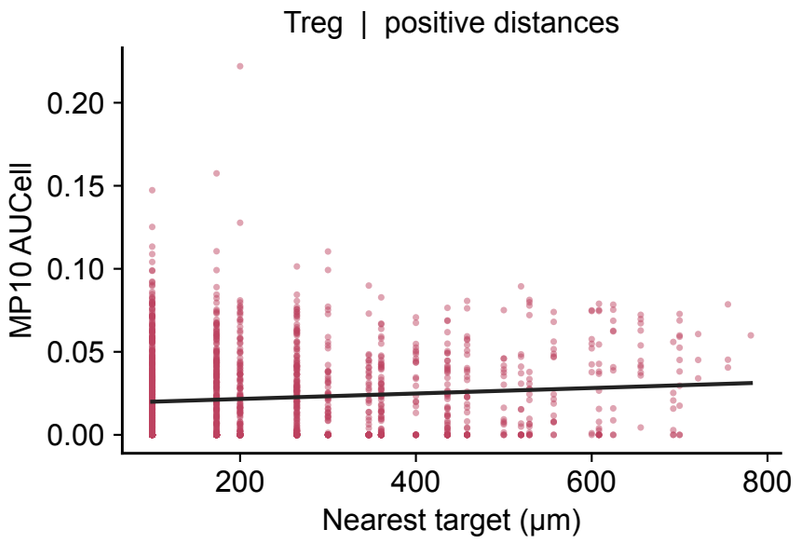
r +0.00 ρ +0.04
n 1454/1885 queries • 528 targets



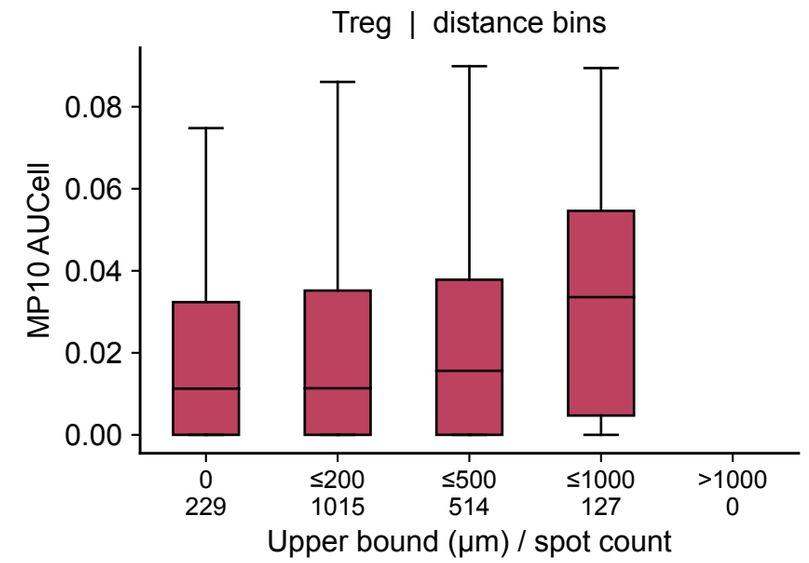
Orange = zero distance
Line = within-section linear fit



r +0.10 ρ +0.09
n 1885/1885 queries • 528 targets



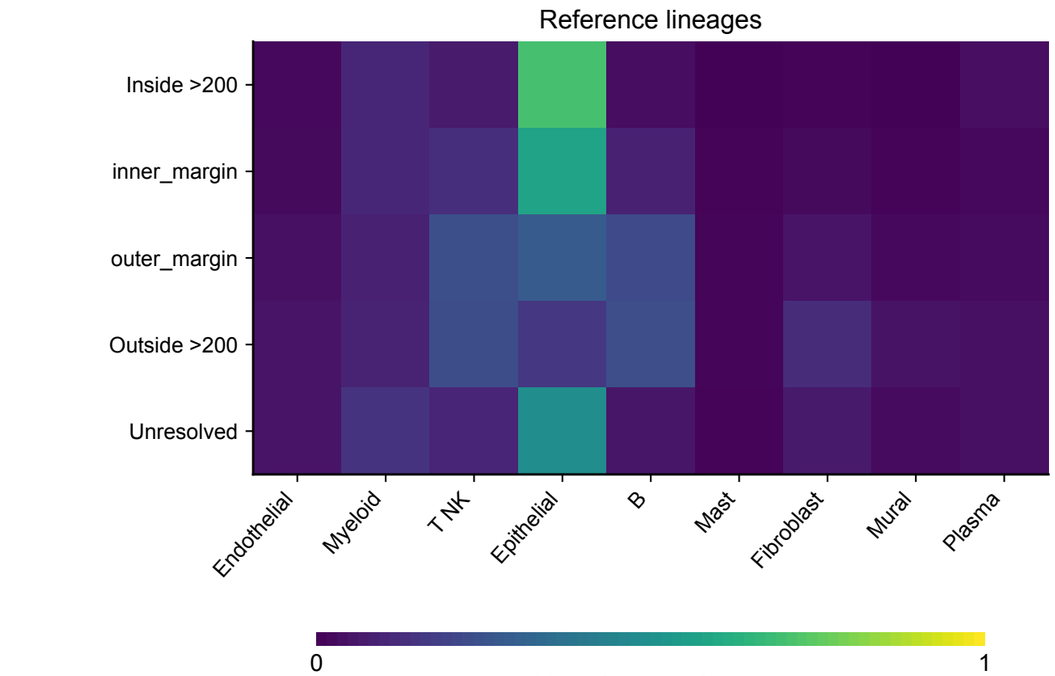
r +0.10 ρ +0.09
n 1656/1885 queries • 528 targets

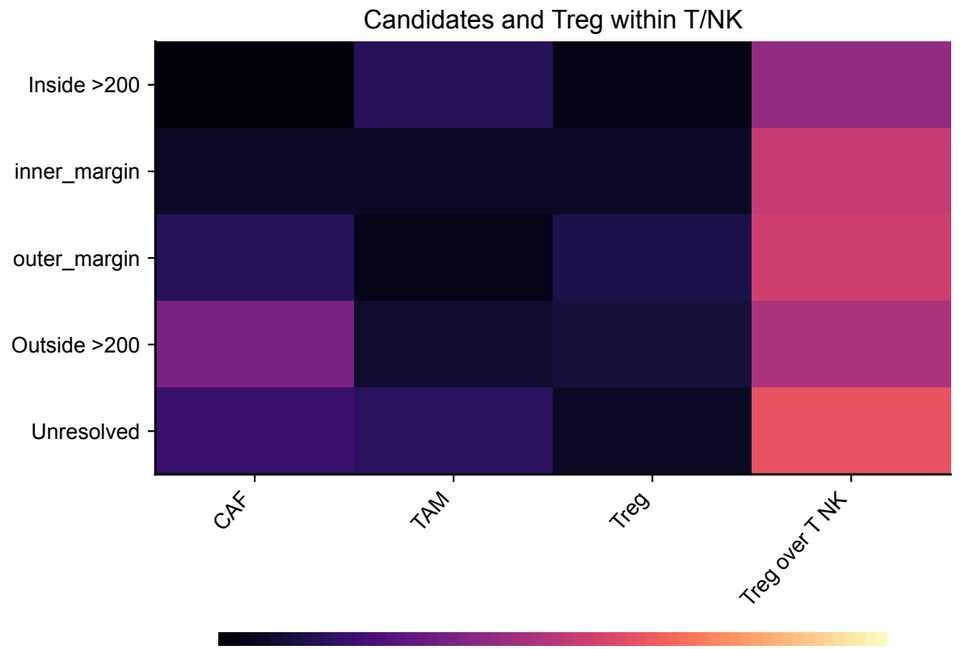


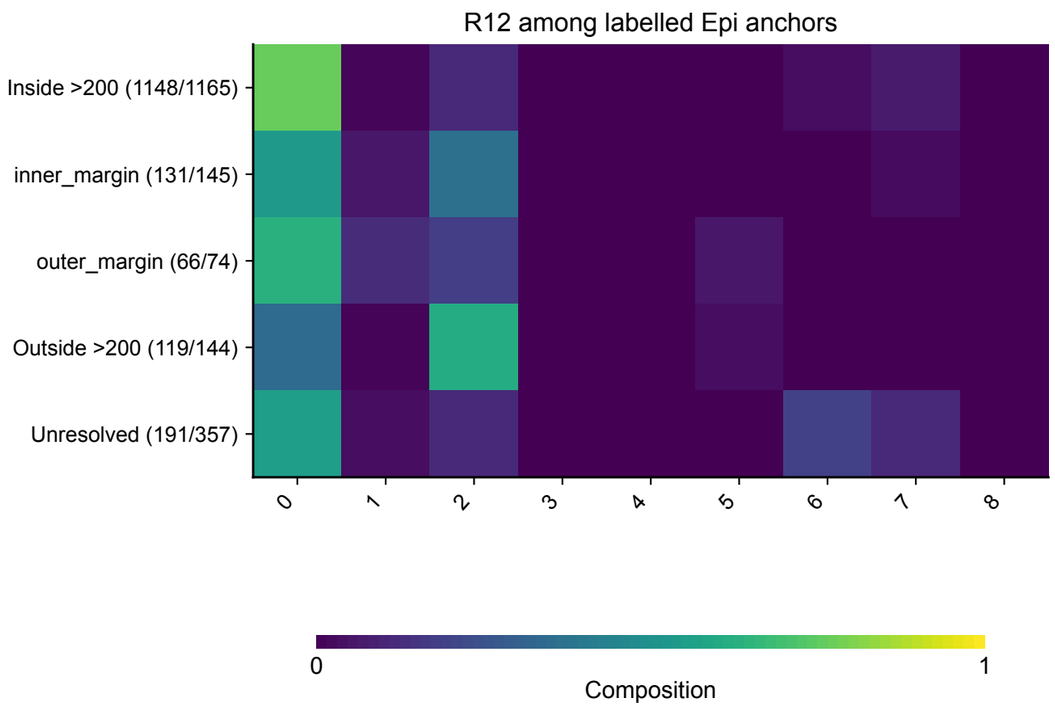
Orange = zero distance
Line = within-section linear fit

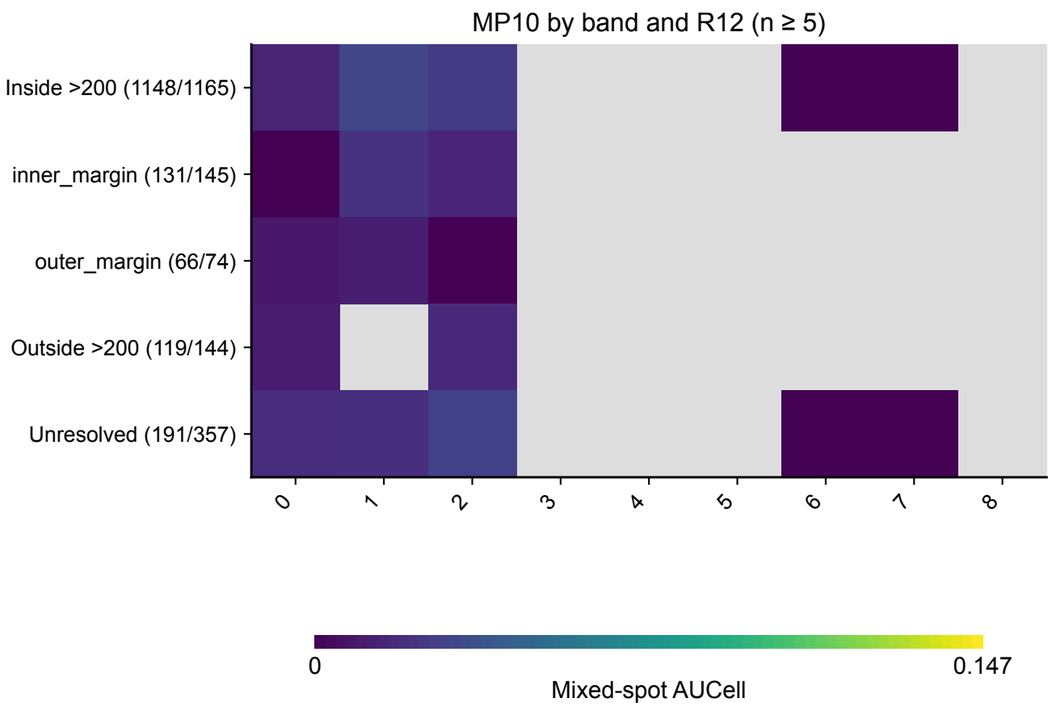
NPJ_sample4 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





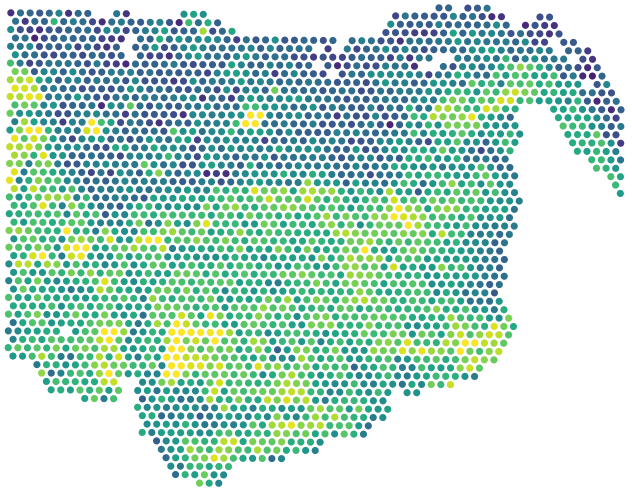




NPJ_sample4 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

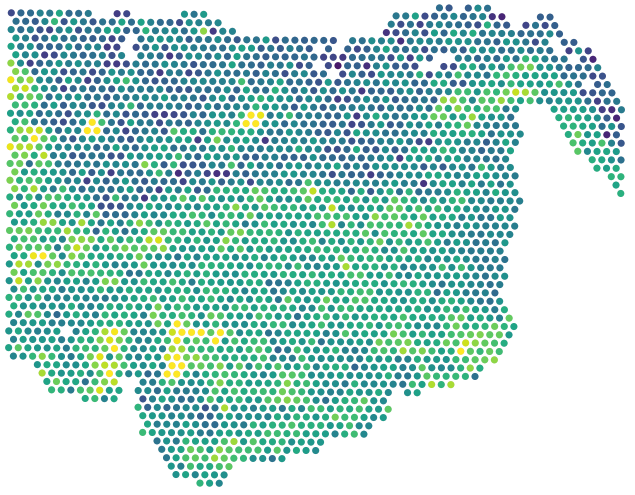
E2F Targets



0 0.152

UCell score

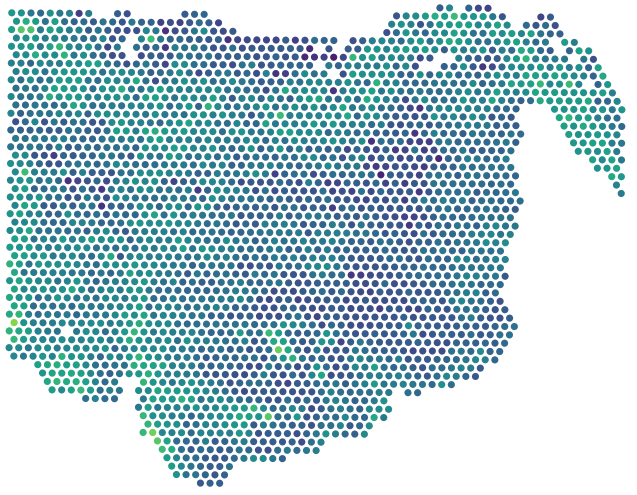
G2M Checkpoint



0 0.134

UCell score

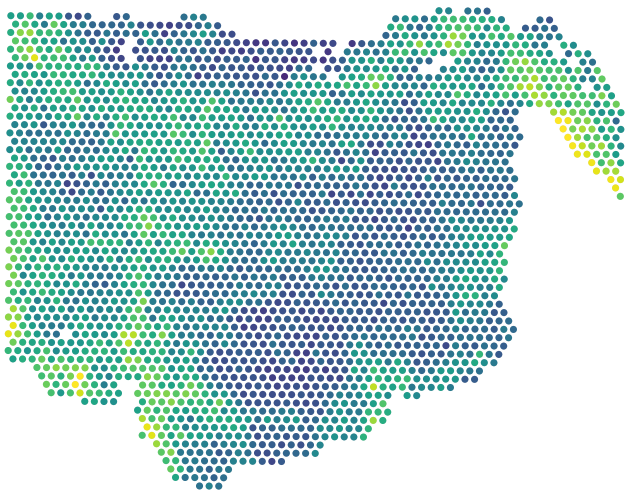
Hypoxia



0 0.211

UCell score

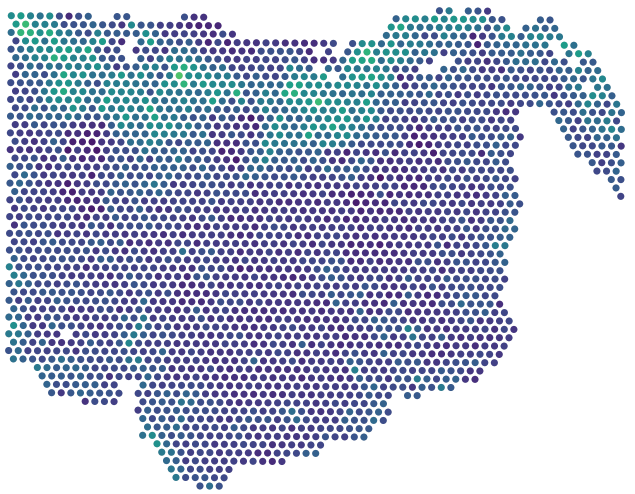
Tnfa Signaling Via Nfkb



0 0.275

UCell score

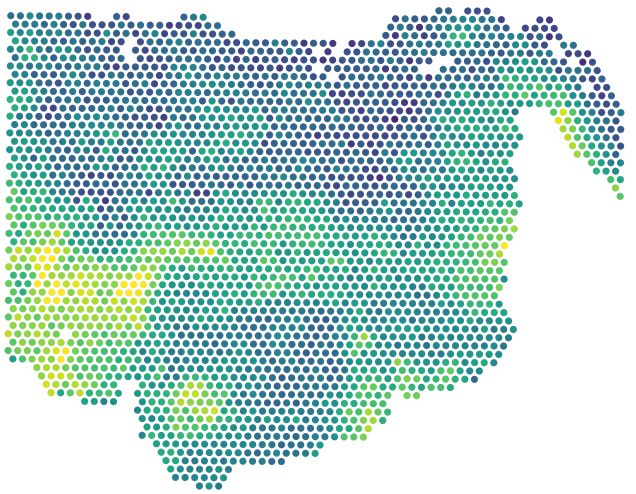
Emt



0 0.386

UCell score

Interferon Gamma Response



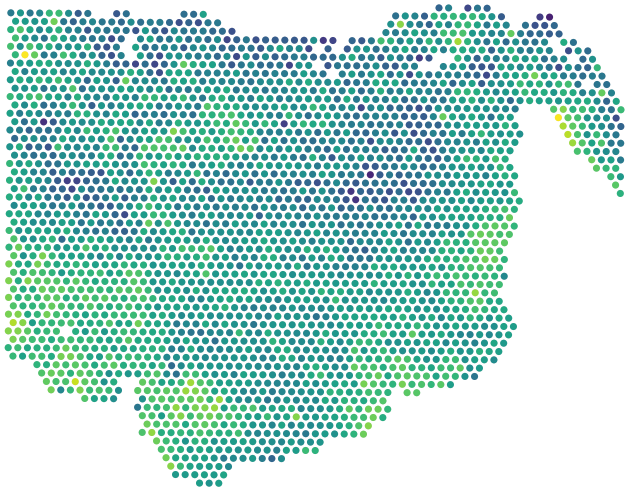
0 0.275

UCell score

NPJ_sample4 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

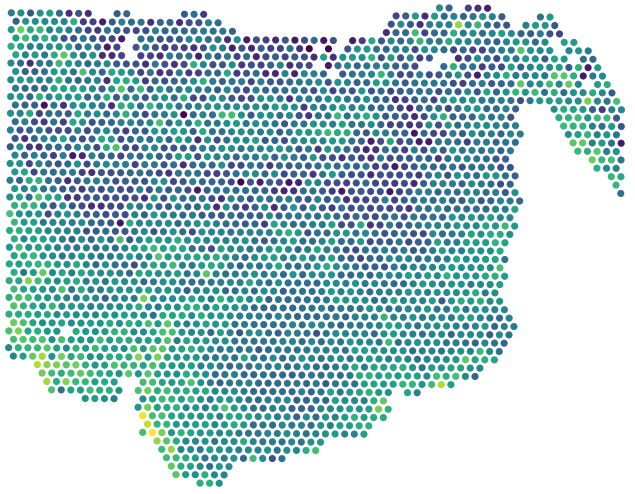
Il2 Stat5 Signaling



0 0.144

UCell score

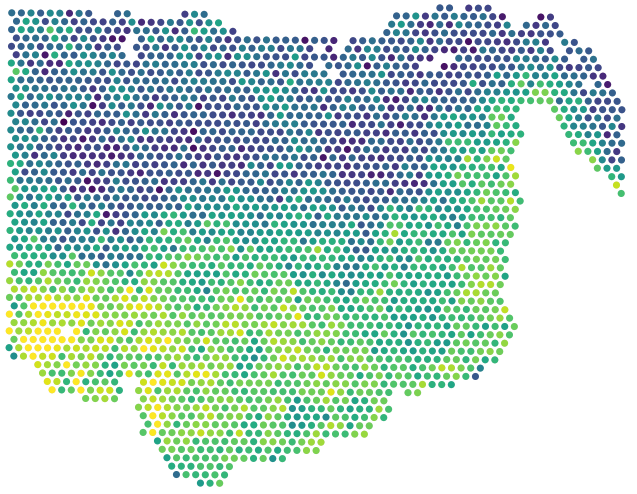
Tgf Beta Signaling



0 0.232

UCell score

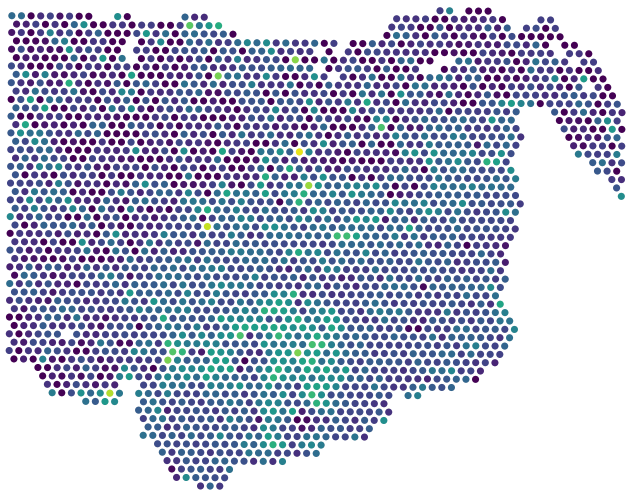
Cholesterol Homeostasis



0 0.21

UCell score

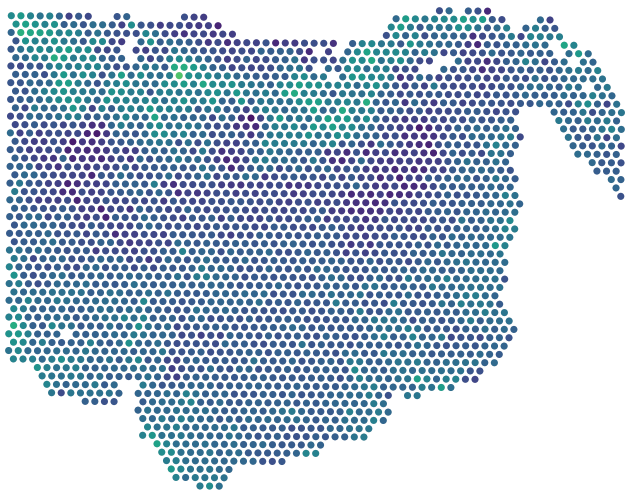
Cholesterol Biosynthesis



0 0.166

UCell score

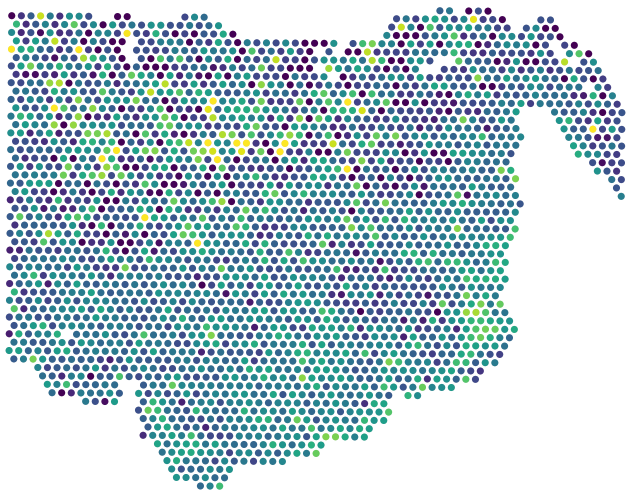
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



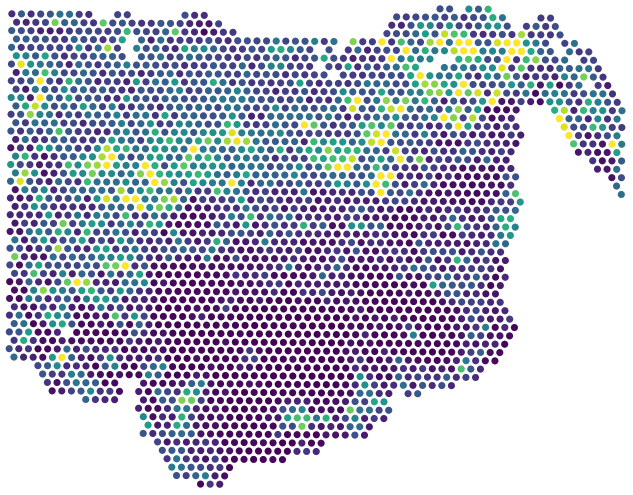
0 0.151

UCell score

NPJ_sample4 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

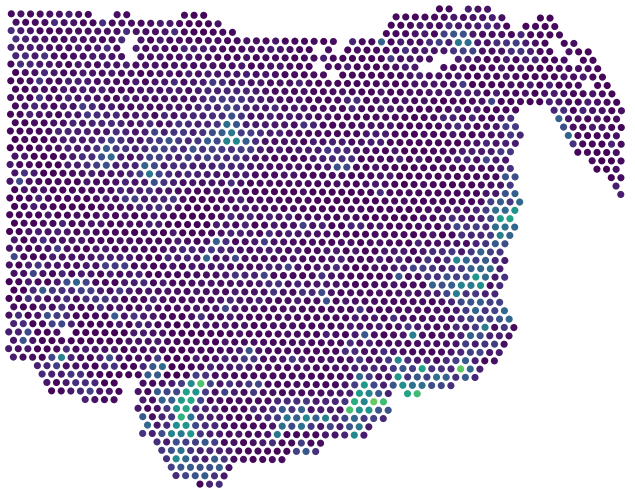
Treg composition



0 0.102

Fraction of all 30

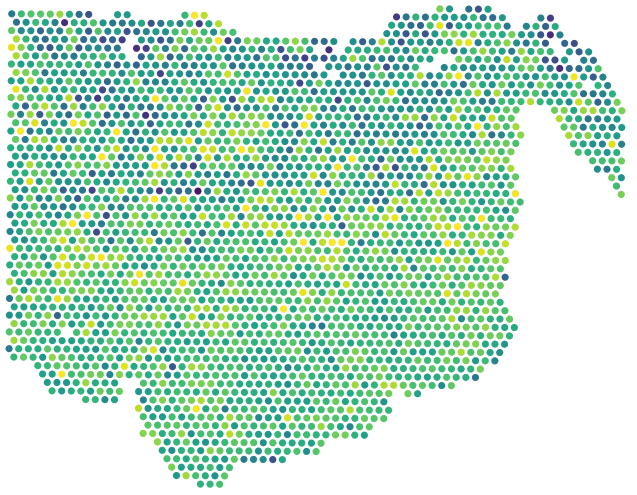
Treg estimated abundance



0 2.85

Posterior mean estimate

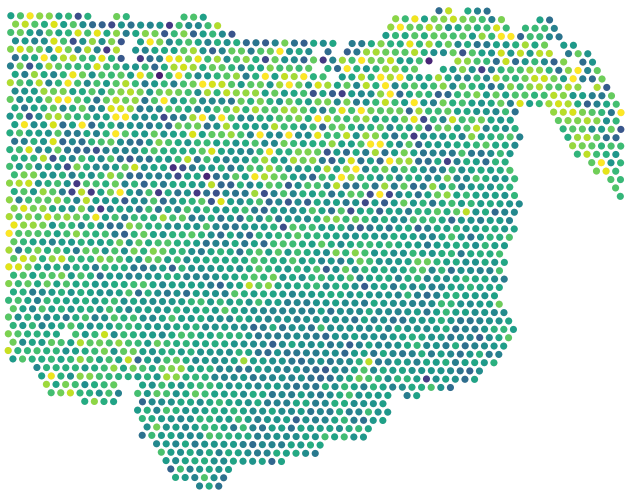
Treg Resting Vs Activated Dn



0 0.0778

UCell score

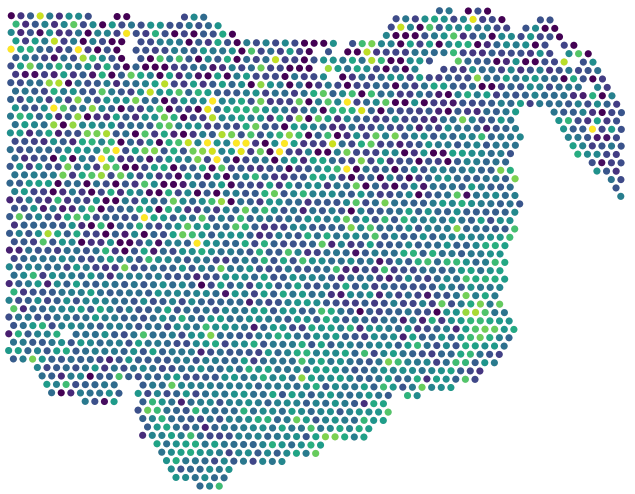
Treg Resting Vs Activated Up



0 0.0609

UCell score

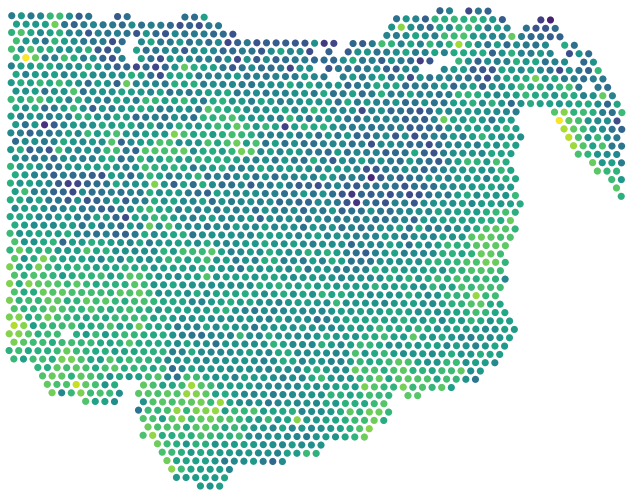
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

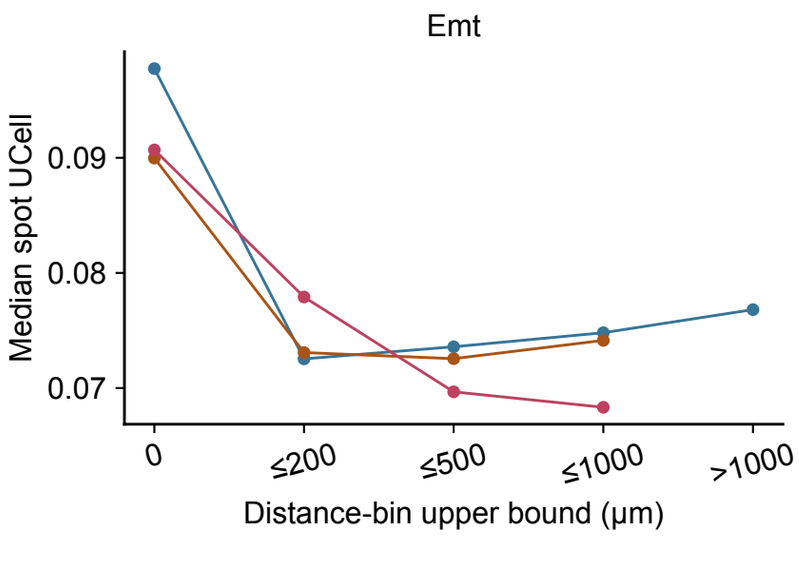


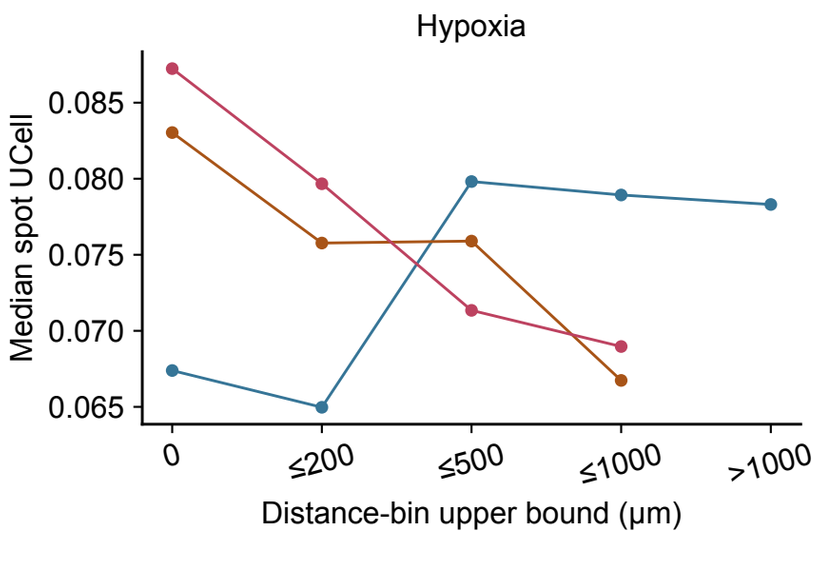
0 0.144

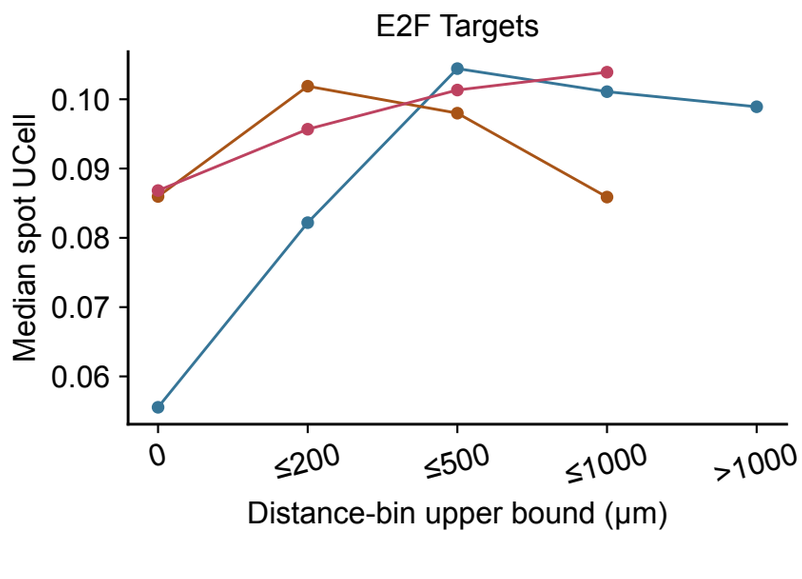
UCell score

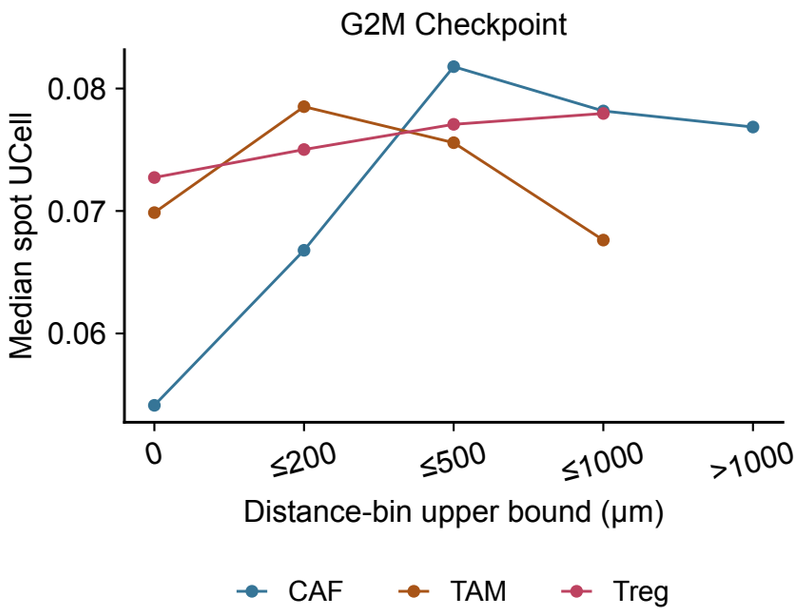
NPJ_sample4 | Programs by target distance

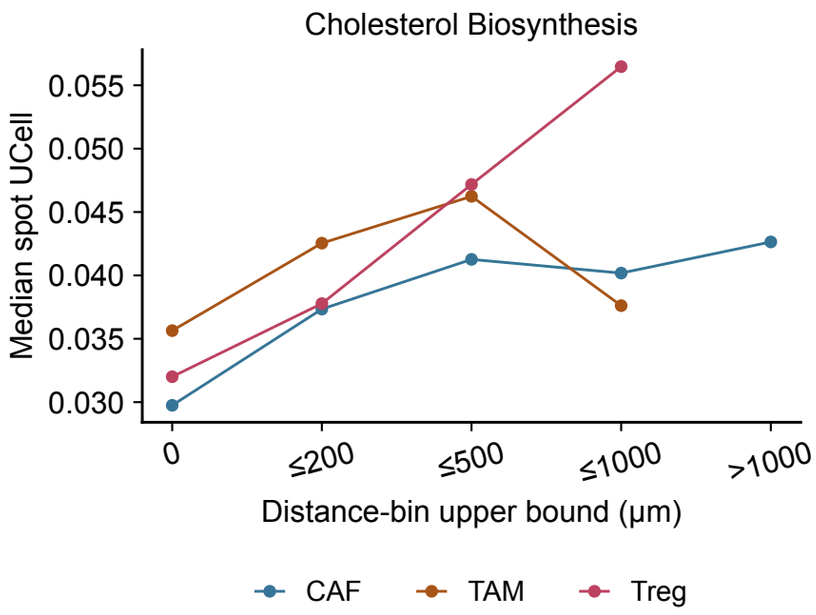
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

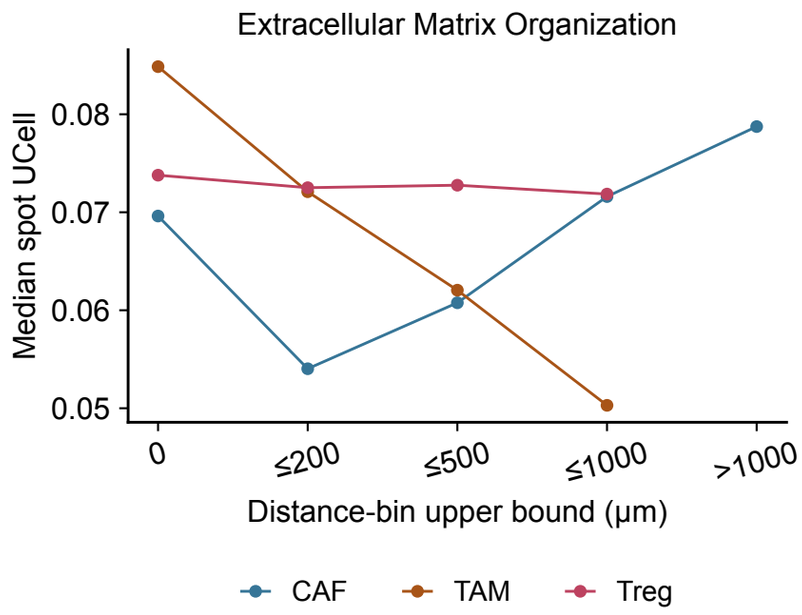






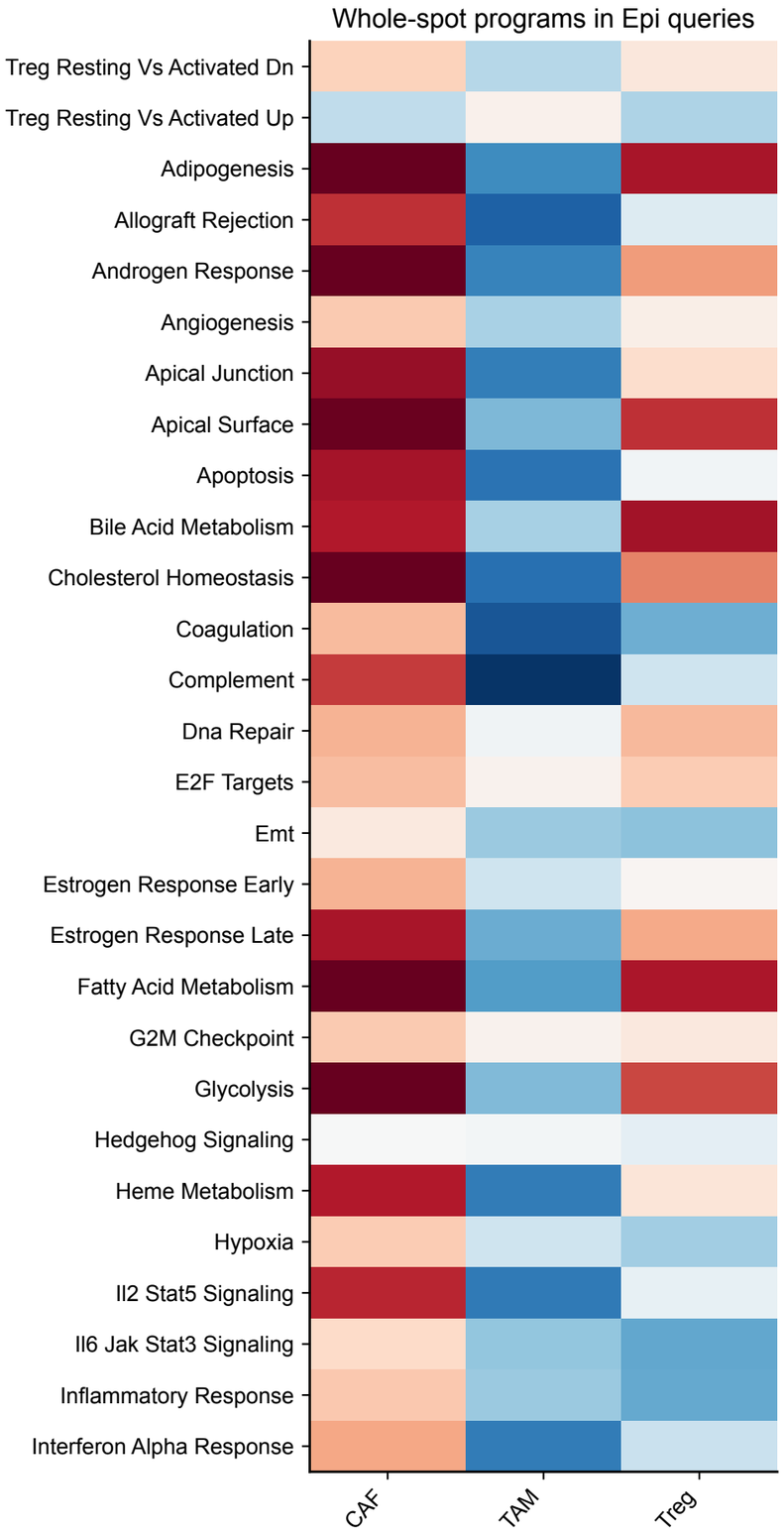


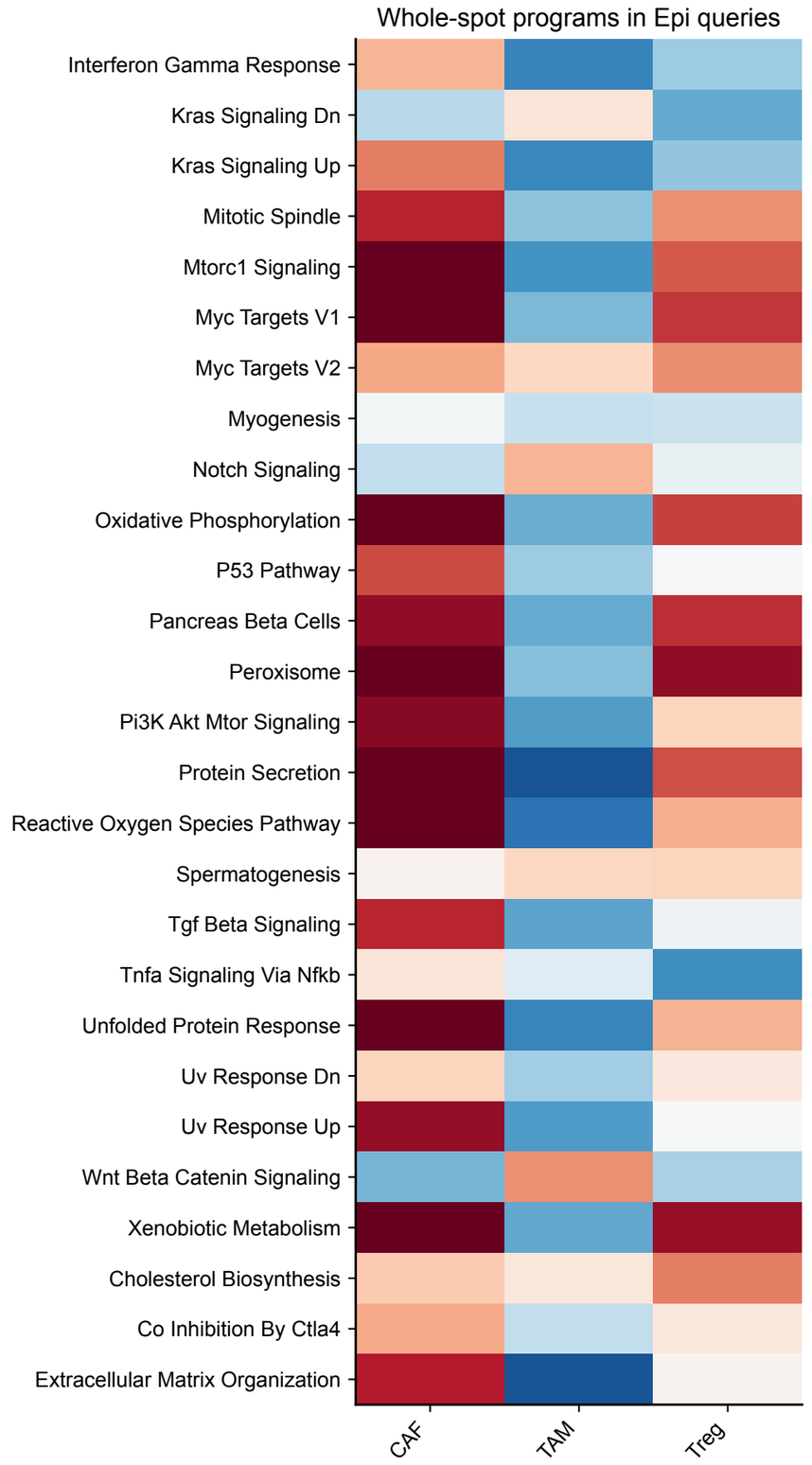




NPJ_sample4 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

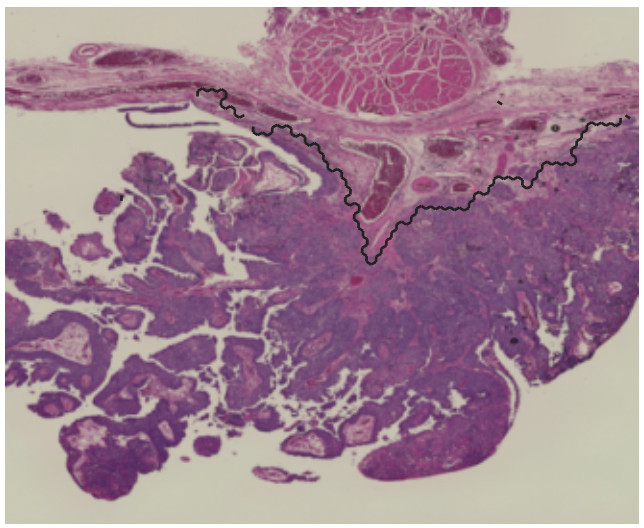




NPJ_sample5 | Tissue context

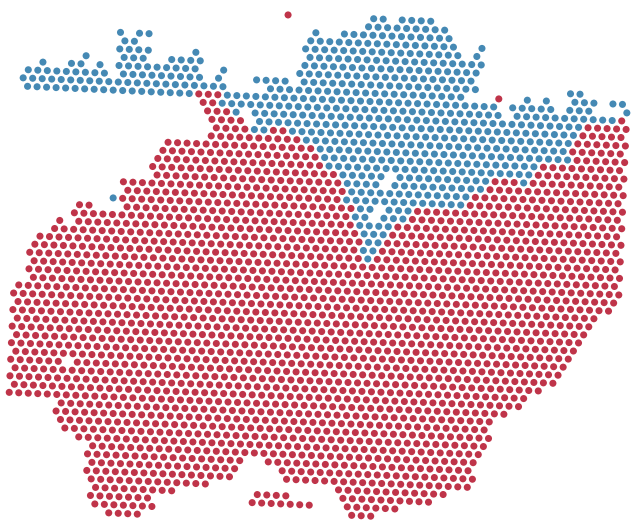
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



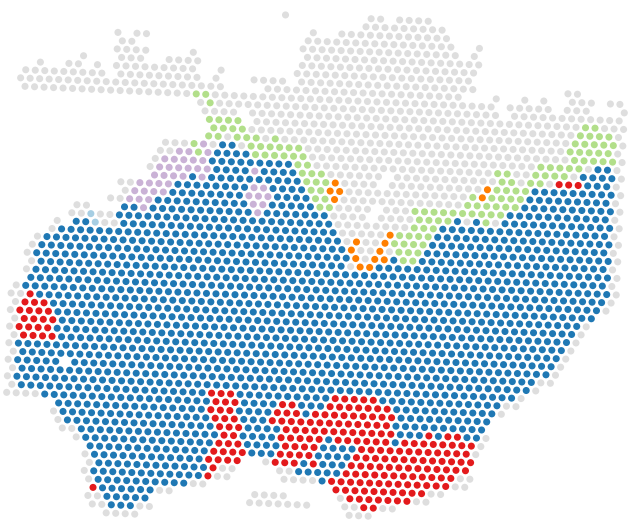
Black line: observed candidate outer edge

Final candidate location



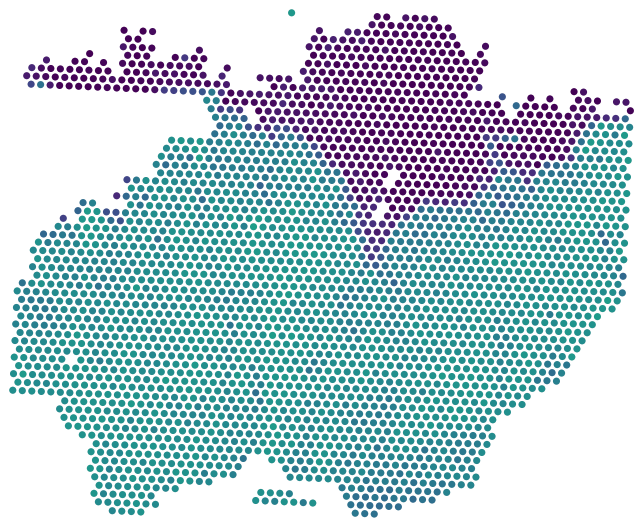
inside outside

Accepted R12 (original 0–8)



0 2 5 7 8
1

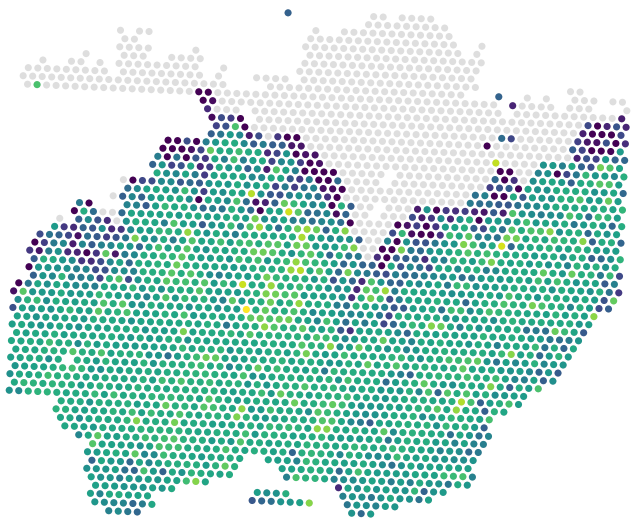
Epithelial support



0 1

Fraction of all 30

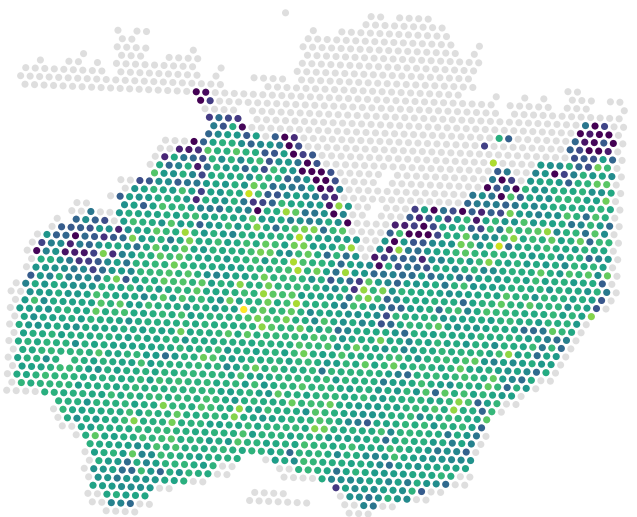
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



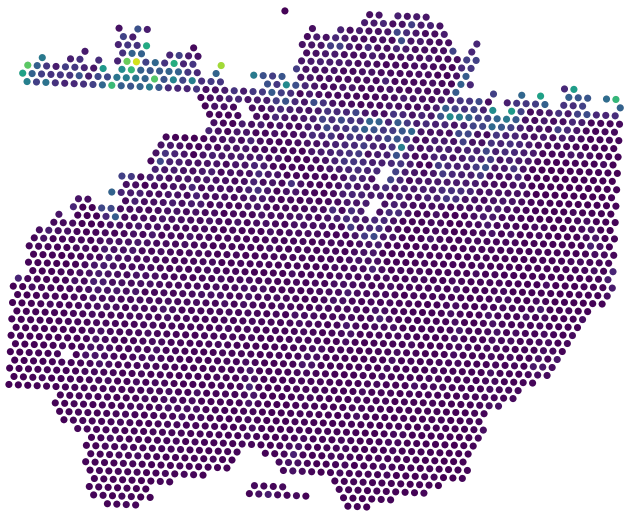
0 0.14

Matched-model AUCell

NPJ_sample5 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

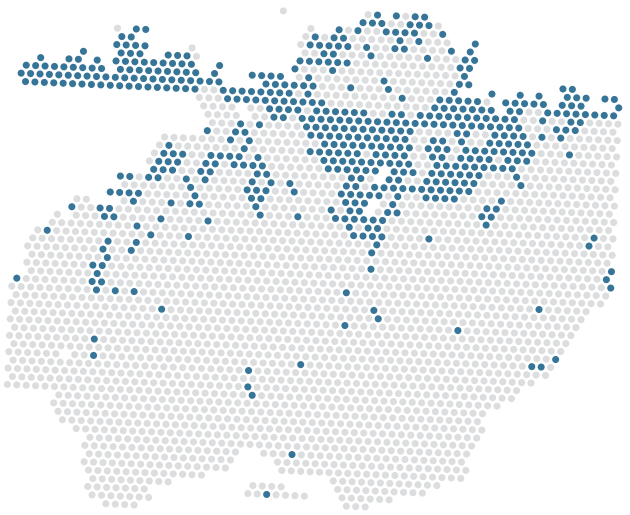
CAF composition



0 0.418

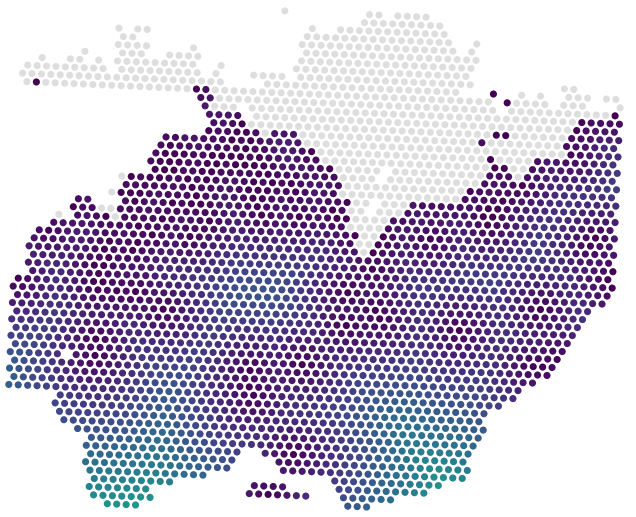
Fraction of all 30

CAF P80 targets



False True

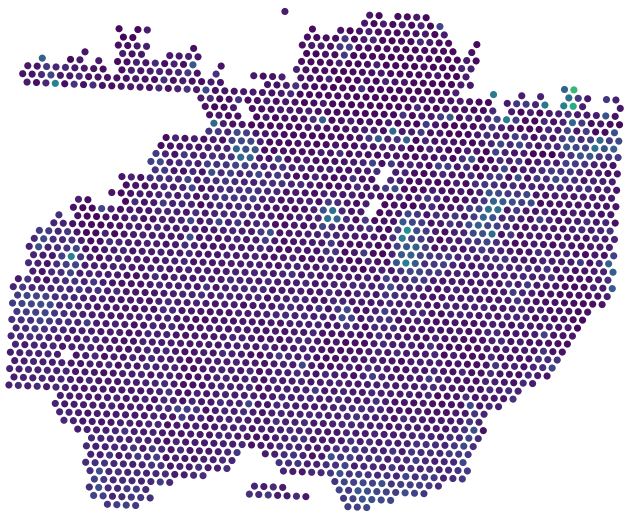
CAF distance



0 3.1e+03

Nearest spot center (μm)

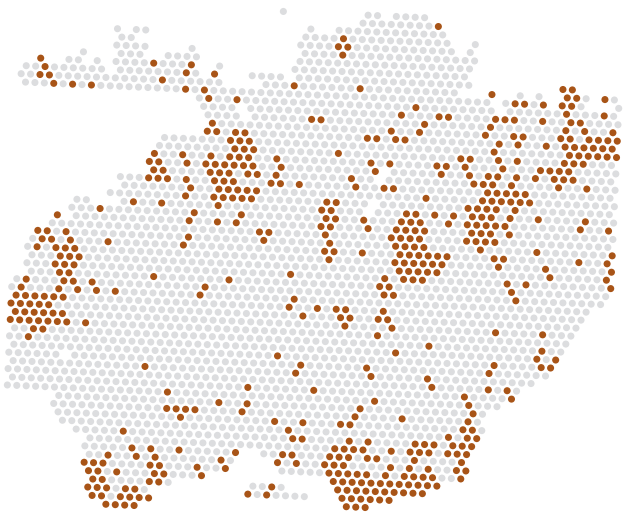
TAM composition



0 0.291

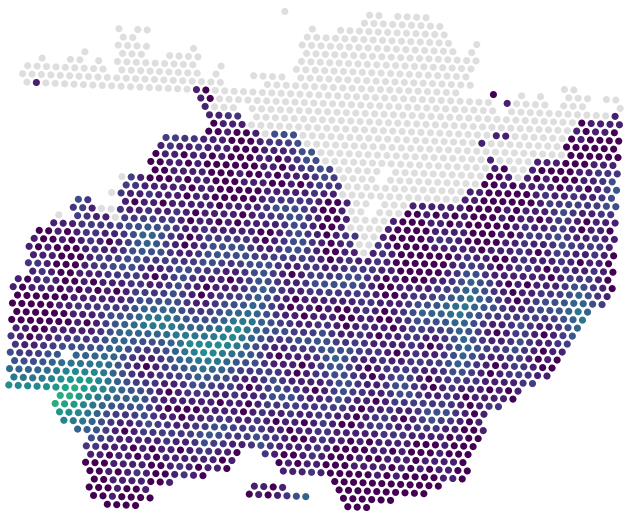
Fraction of all 30

TAM P80 targets



False True

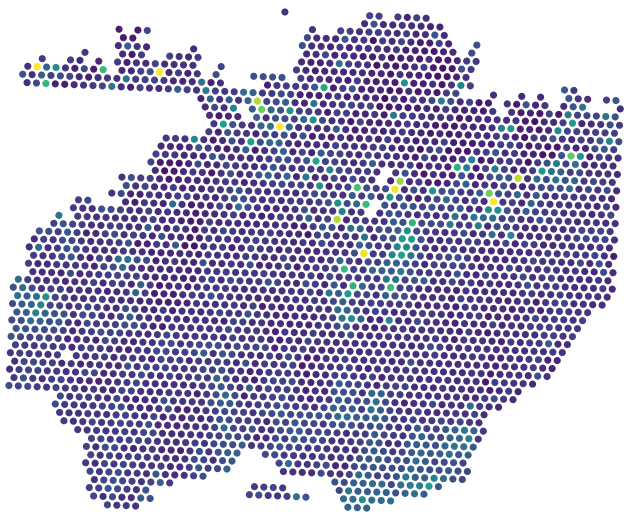
TAM distance



0 1.13e+03

Nearest spot center (μm)

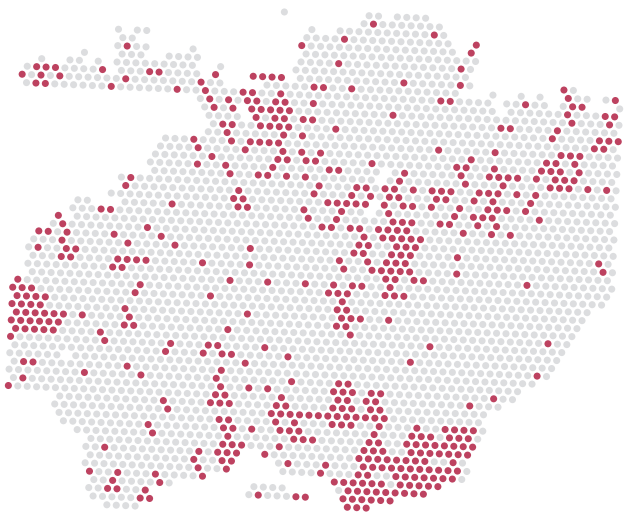
Treg composition



0 0.102

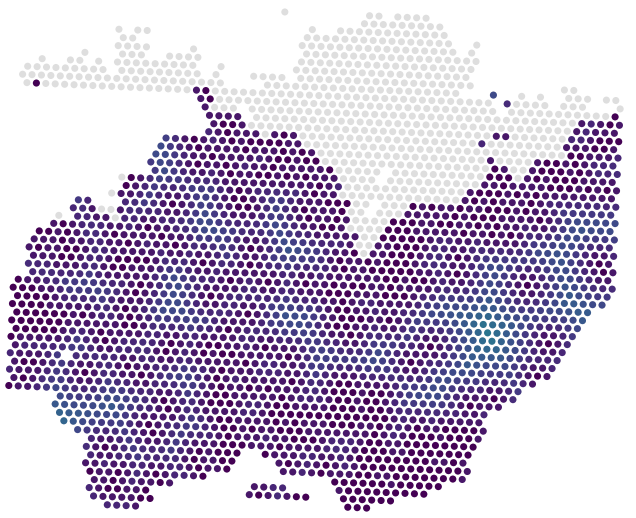
Fraction of all 30

Treg P80 targets



False True

Treg distance

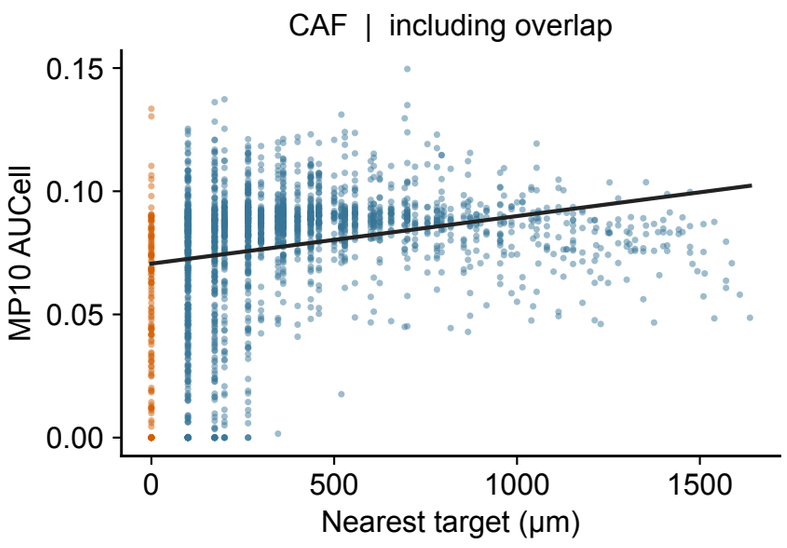


0 1.61e+03

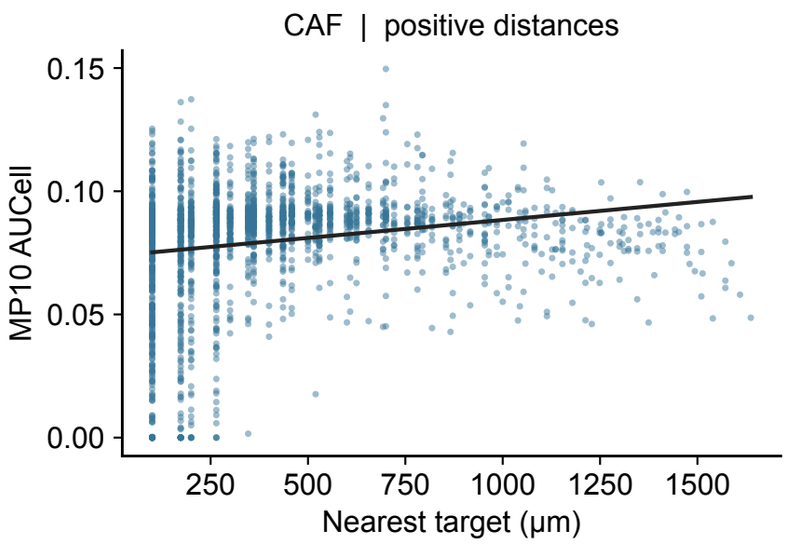
Nearest spot center (μm)

NPJ_sample5 | MP10 versus nearest target

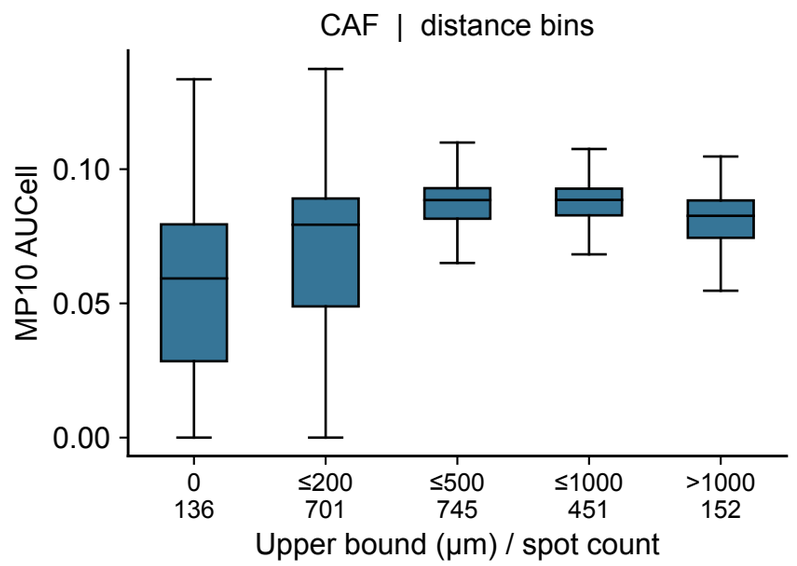
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



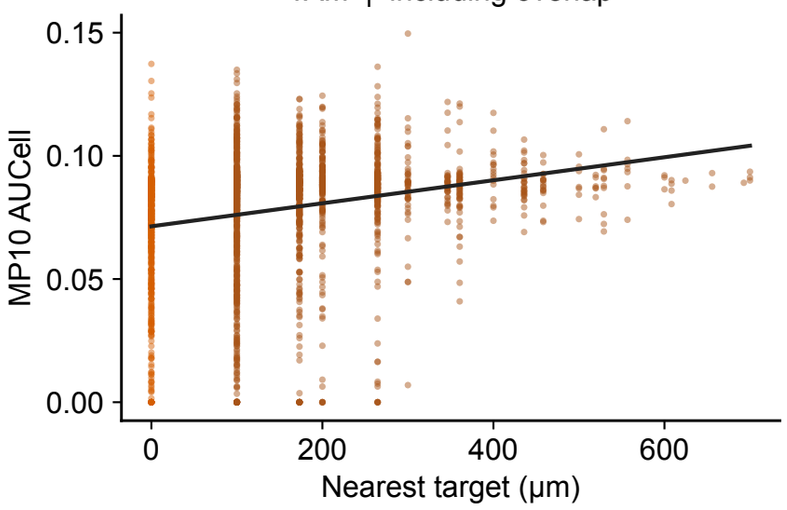
r +0.25 ρ +0.30
n 2185/2186 queries • 563 targets



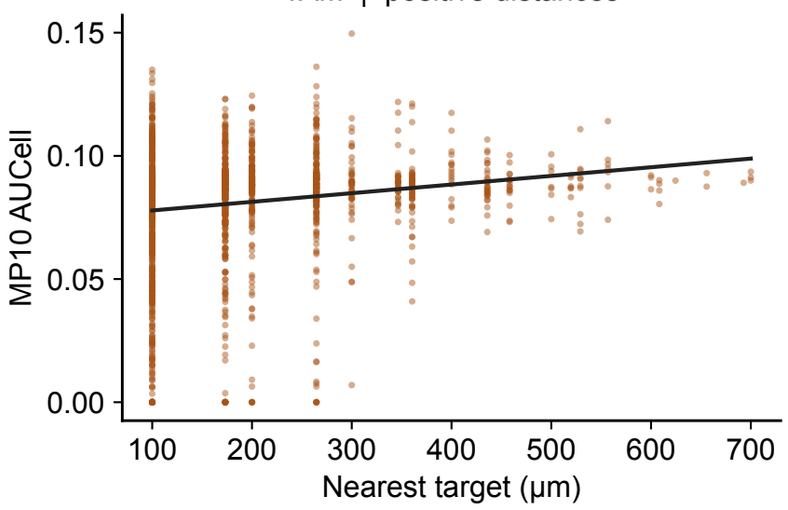
r +0.20 ρ +0.24
n 2049/2186 queries • 563 targets



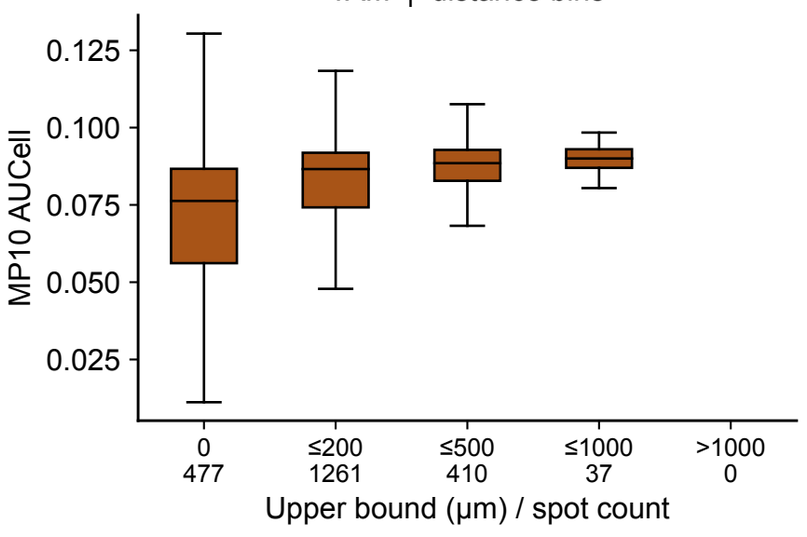
Orange = zero distance
Line = within-section linear fit



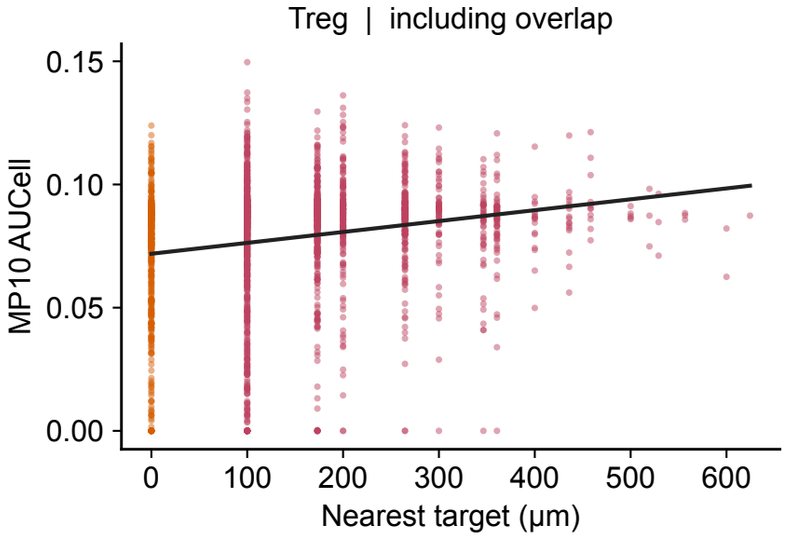
r +0.23 ρ +0.28
n 2185/2186 queries • 563 targets



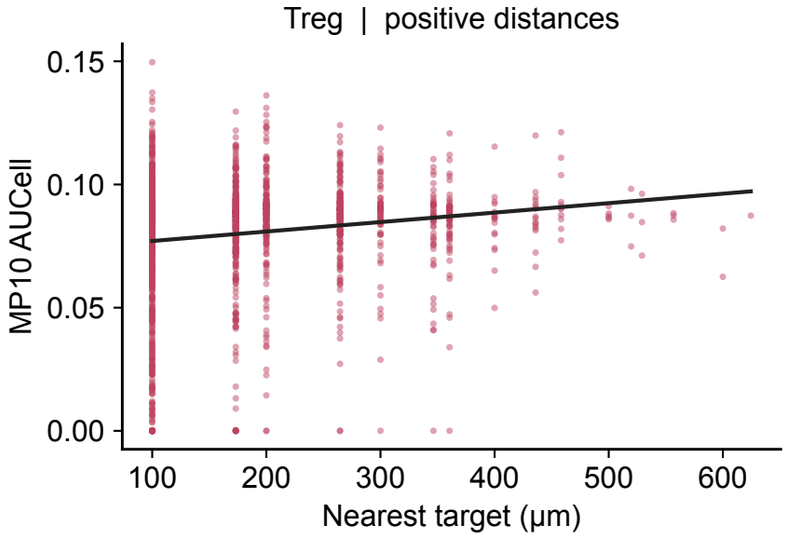
r +0.16 ρ +0.16
n 1708/2186 queries • 563 targets



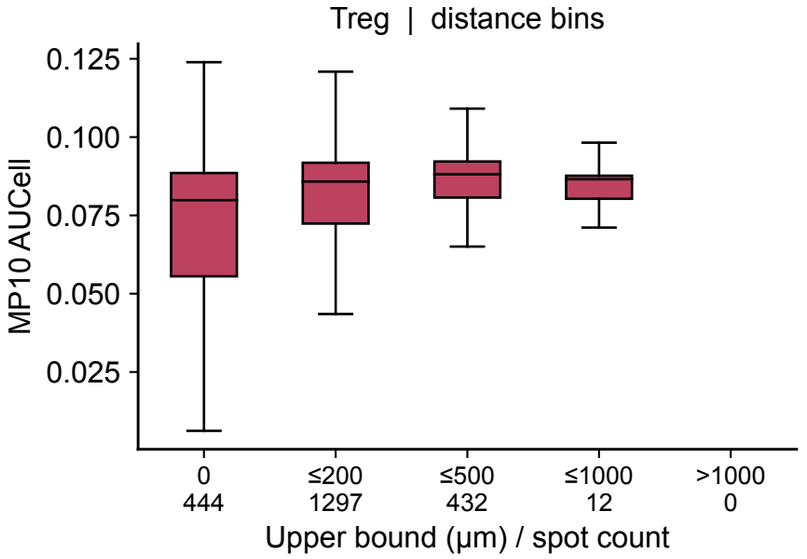
Orange = zero distance
Line = within-section linear fit



r +0.20 ρ +0.22
n 2185/2186 queries • 563 targets



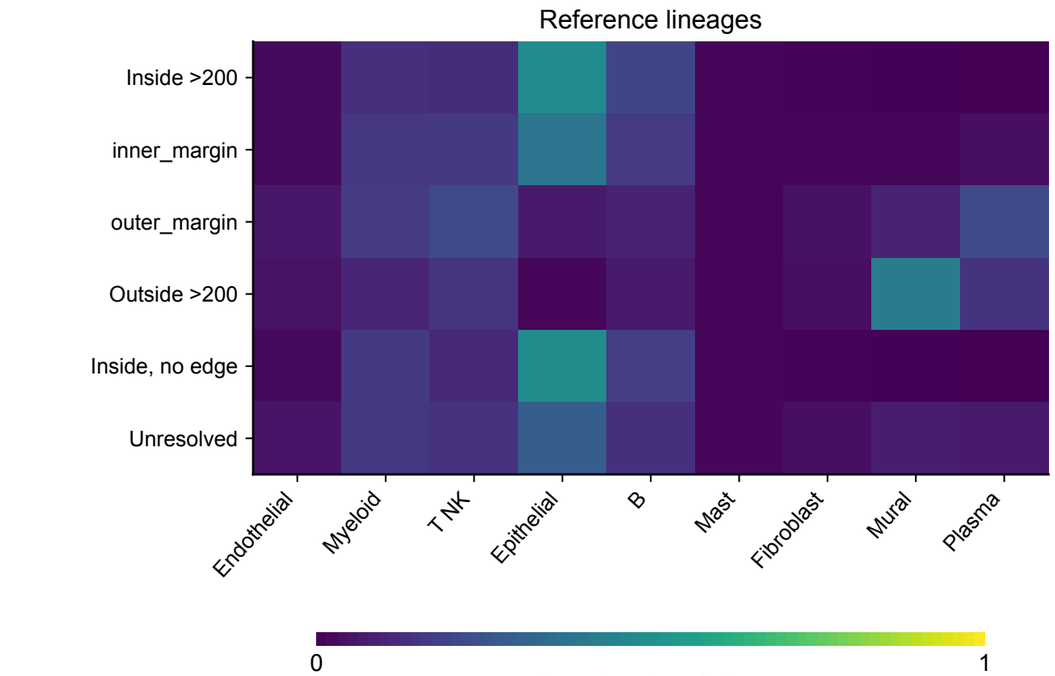
r +0.15 ρ +0.16
n 1741/2186 queries • 563 targets

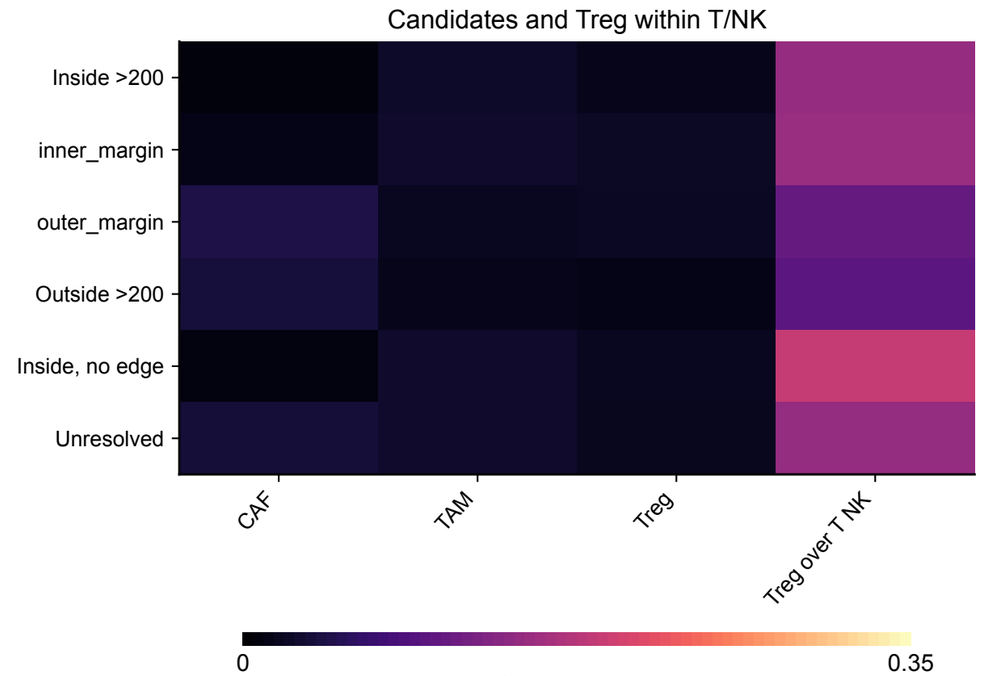


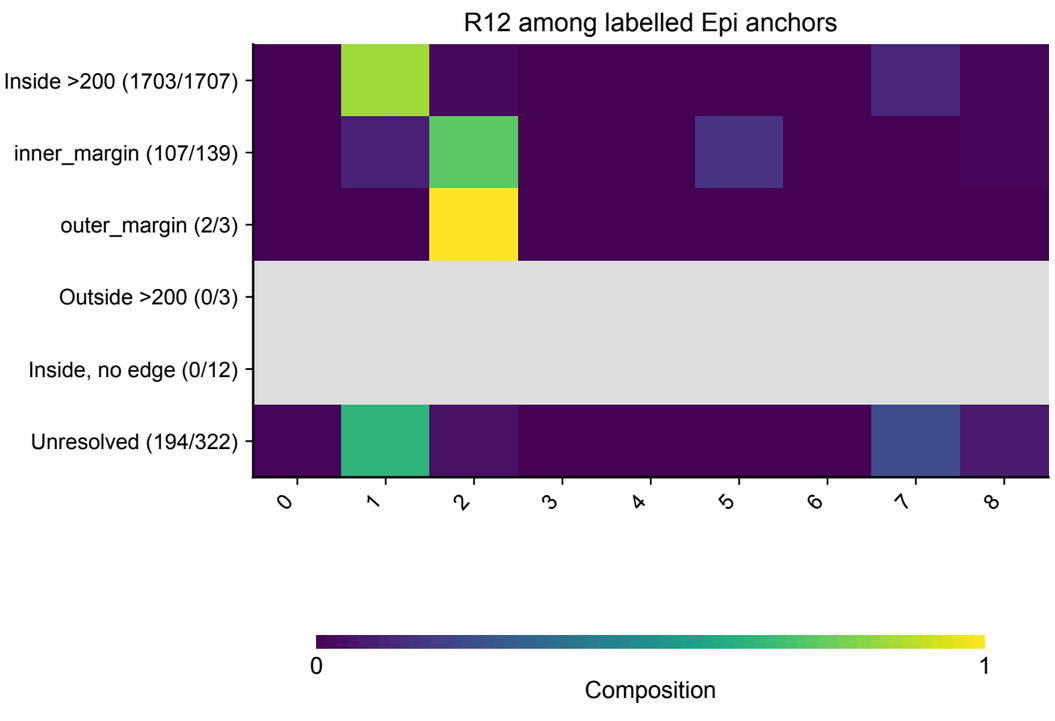
Orange = zero distance
Line = within-section linear fit

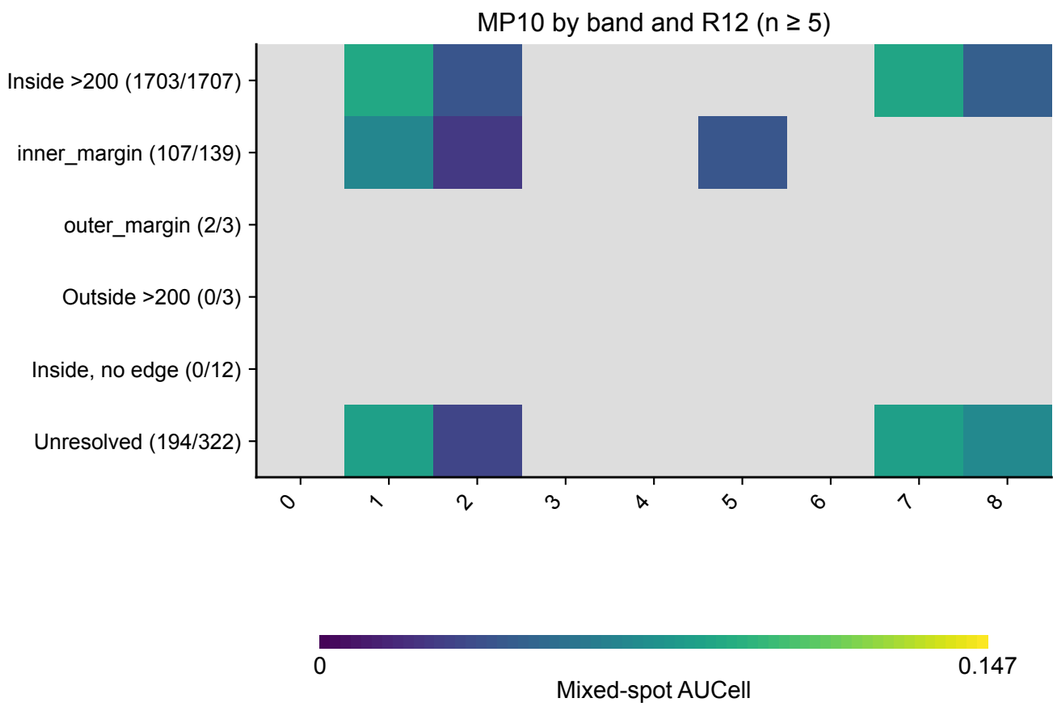
NPJ_sample5 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





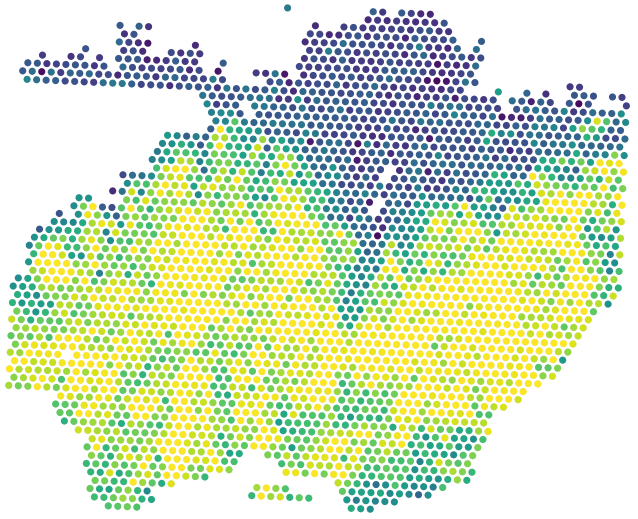




NPJ_sample5 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

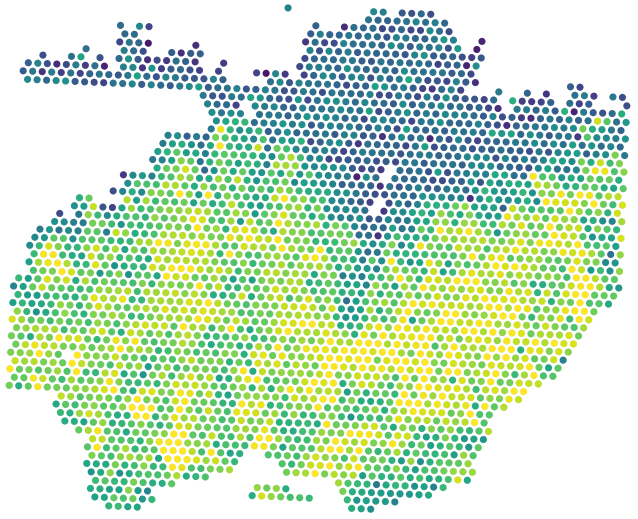
E2F Targets



0 0.152

UCell score

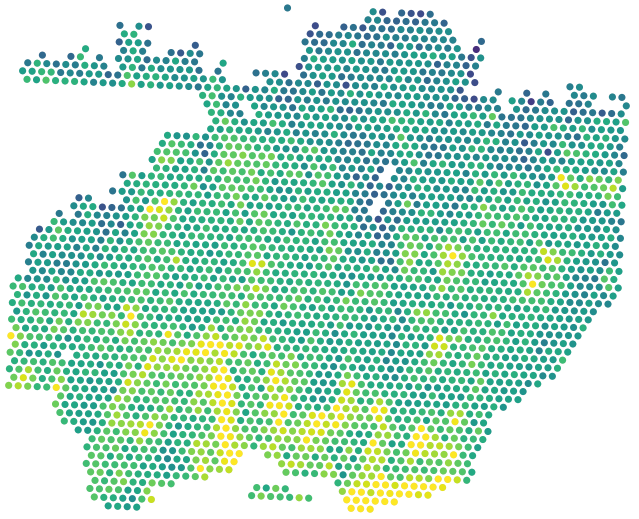
G2M Checkpoint



0 0.134

UCell score

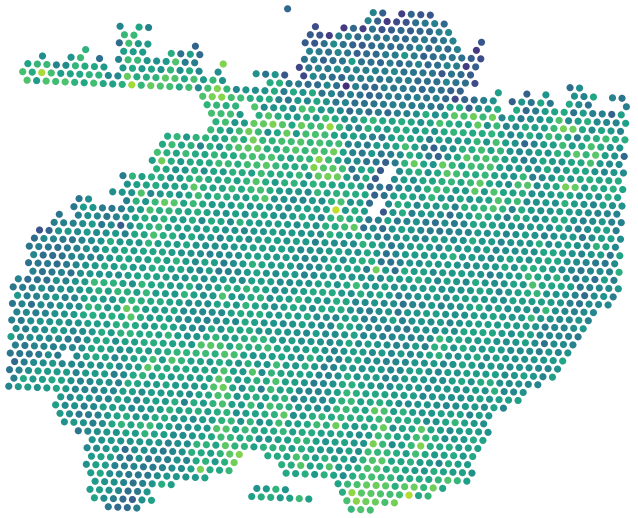
Hypoxia



0 0.211

UCell score

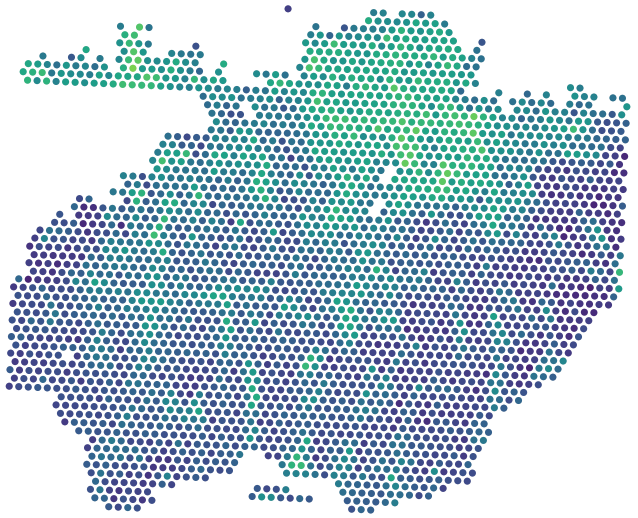
Tnfa Signaling Via Nfkb



0 0.275

UCell score

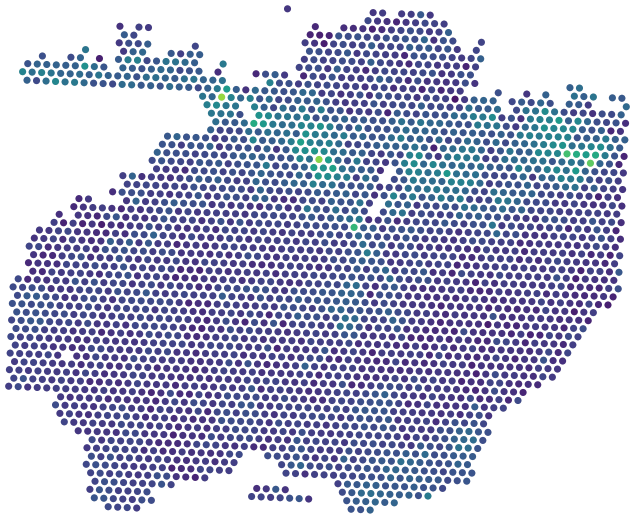
Emt



0 0.386

UCell score

Interferon Gamma Response



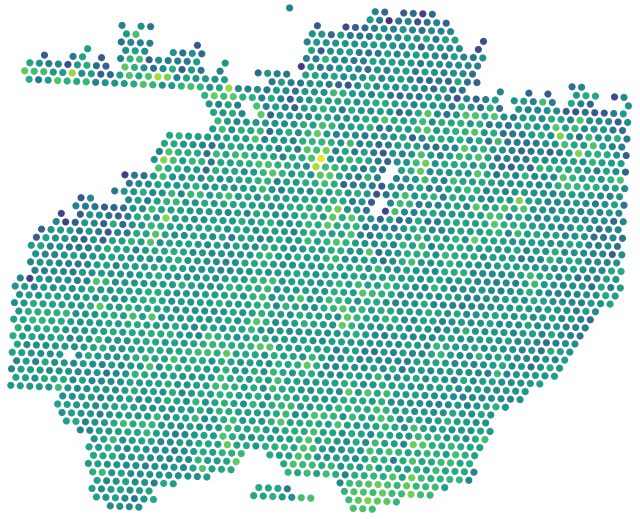
0 0.275

UCell score

NPJ_sample5 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

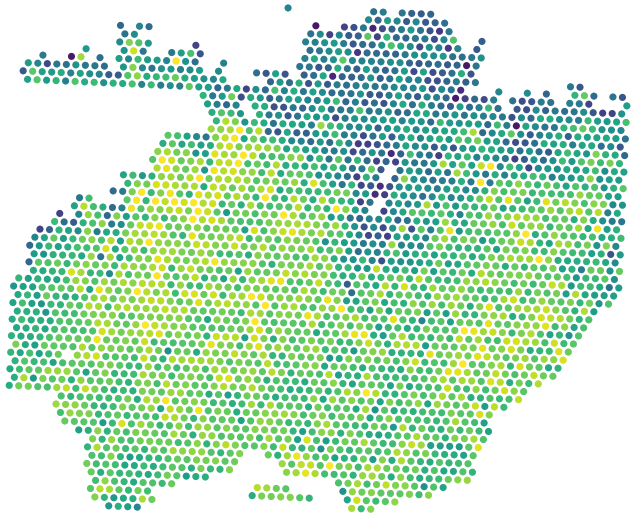
Il2 Stat5 Signaling



0 0.144

UCell score

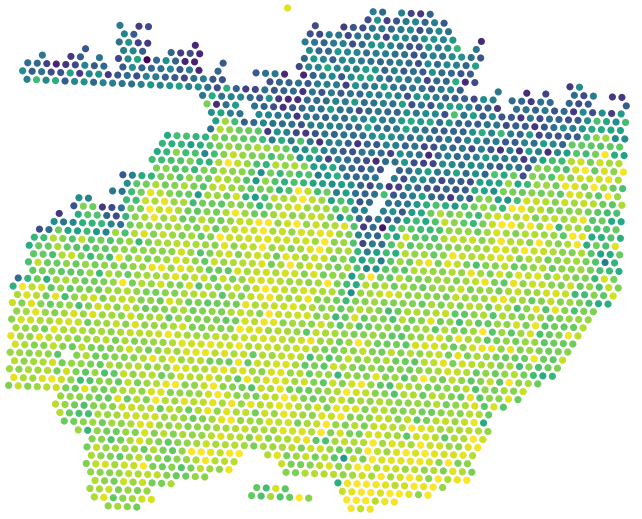
Tgf Beta Signaling



0 0.232

UCell score

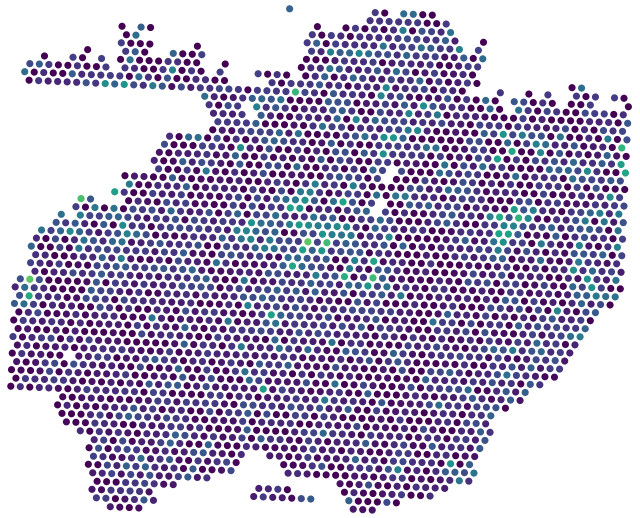
Cholesterol Homeostasis



0 0.21

UCell score

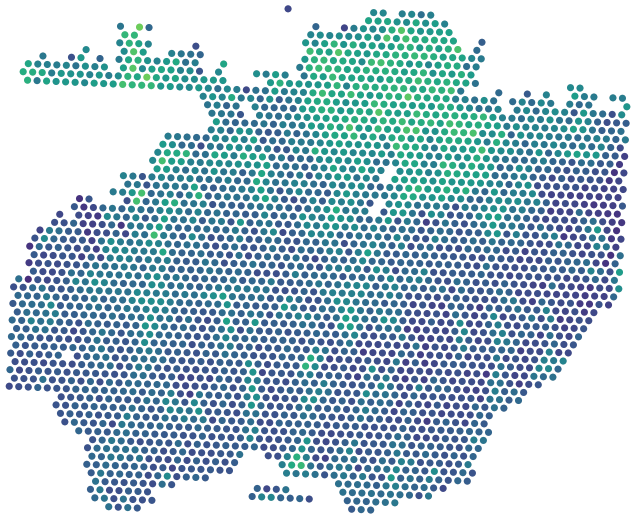
Cholesterol Biosynthesis



0 0.166

UCell score

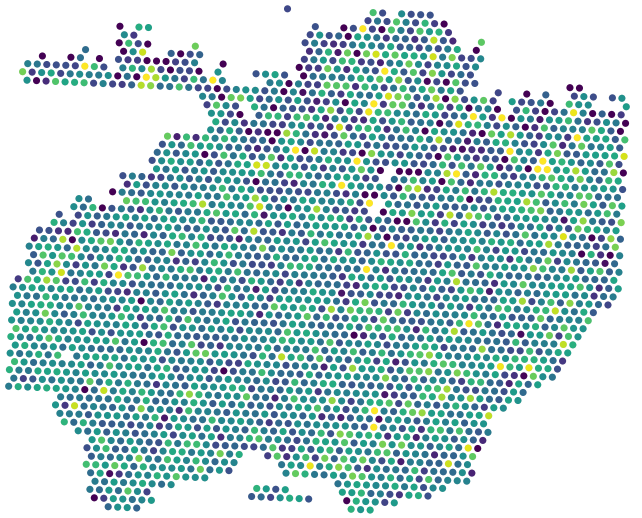
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



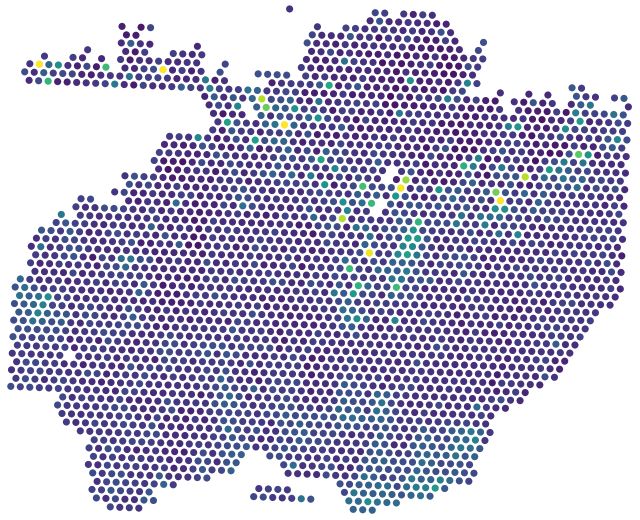
0 0.151

UCell score

NPJ_sample5 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

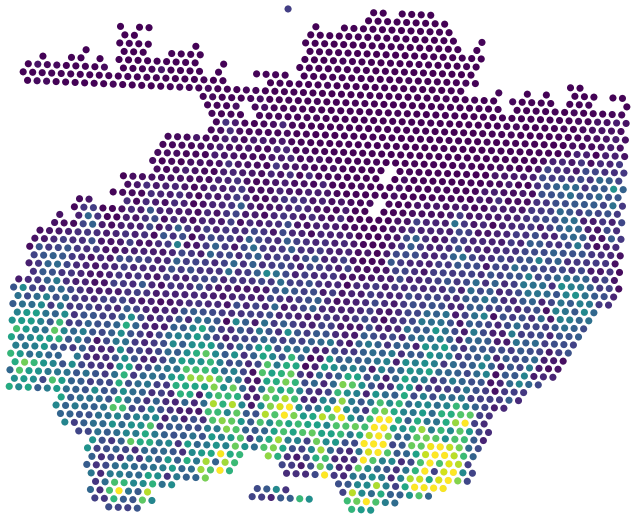
Treg composition



0 0.102

Fraction of all 30

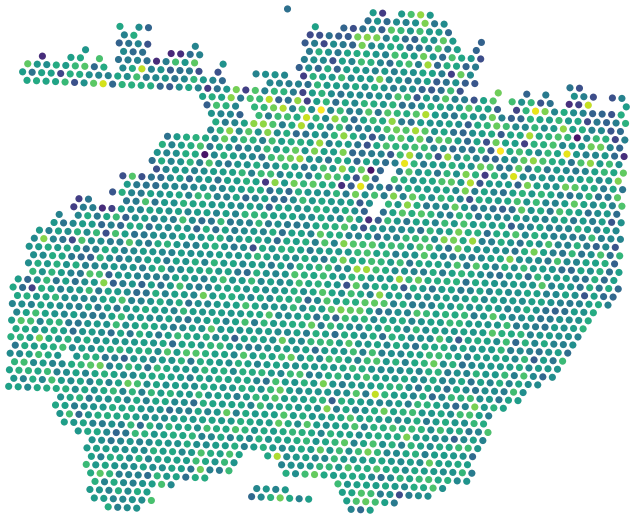
Treg estimated abundance



0 2.85

Posterior mean estimate

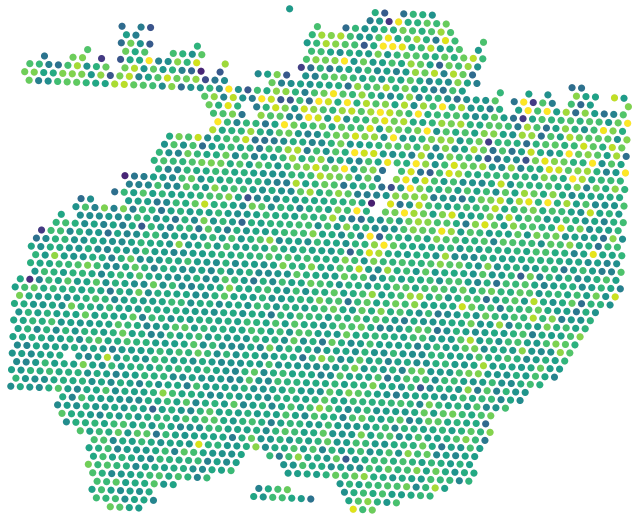
Treg Resting Vs Activated Dn



0 0.0778

UCell score

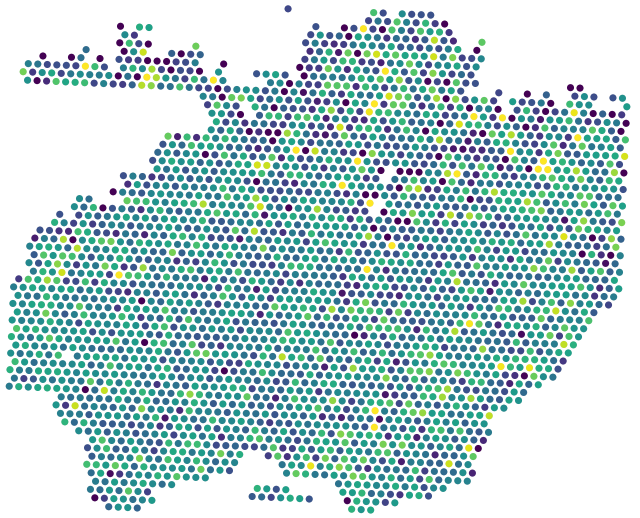
Treg Resting Vs Activated Up



0 0.0609

UCell score

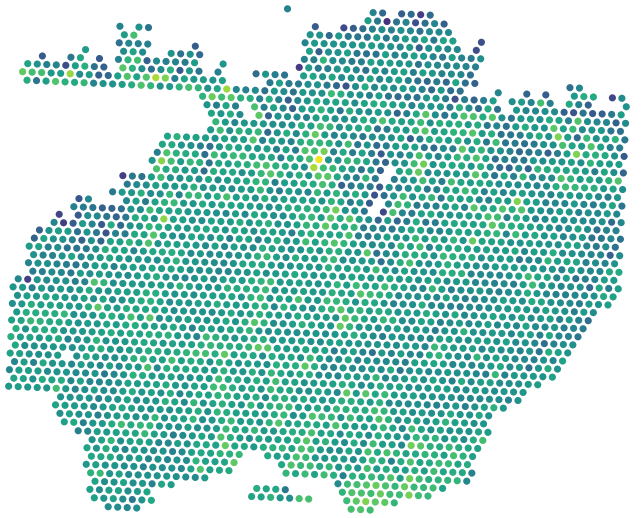
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

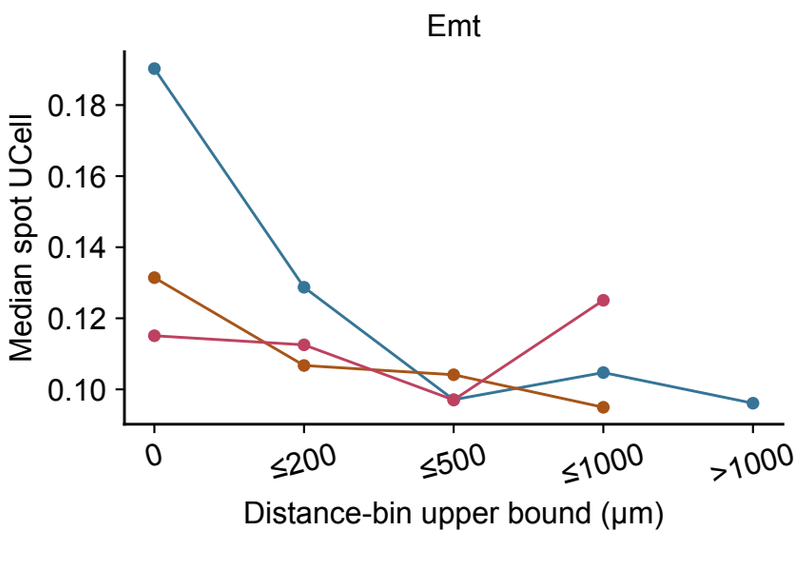


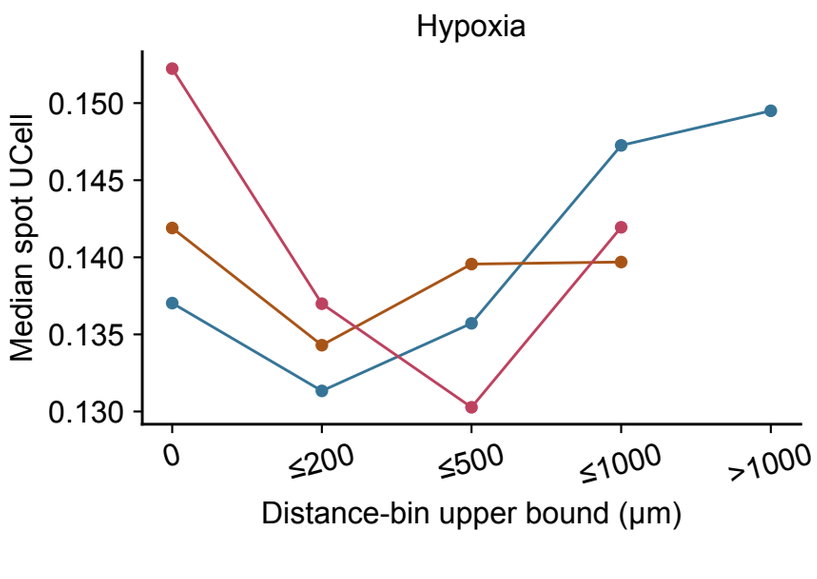
0 0.144

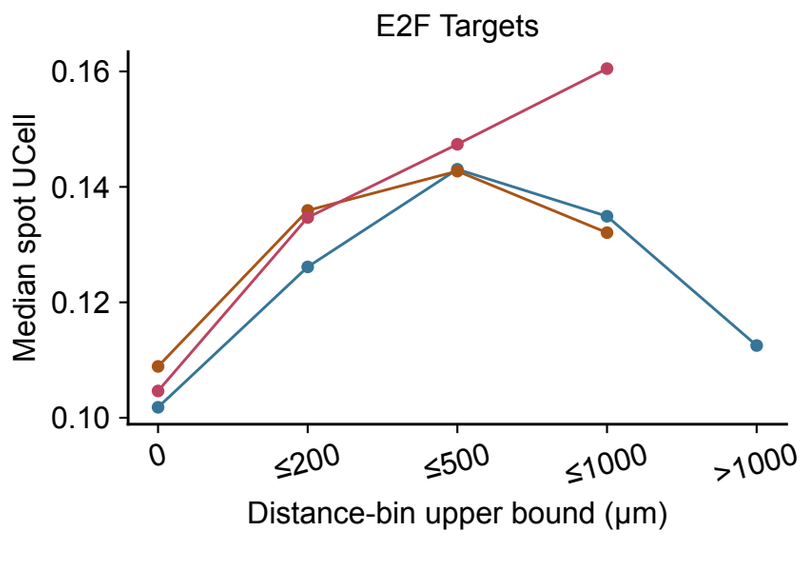
UCell score

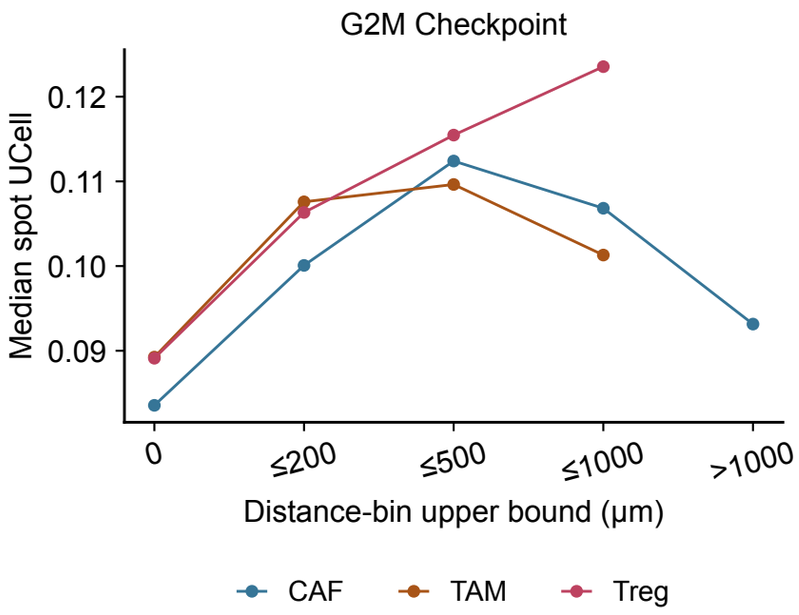
NPJ_sample5 | Programs by target distance

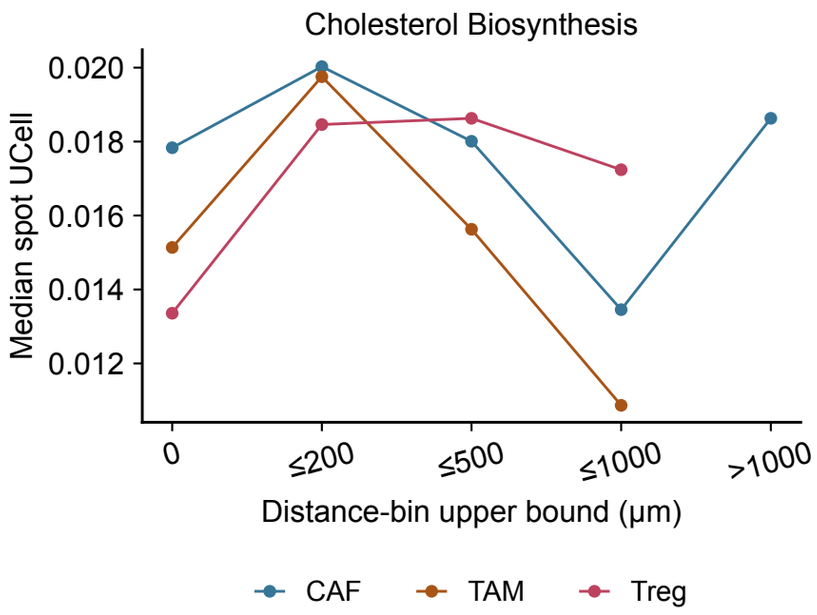
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

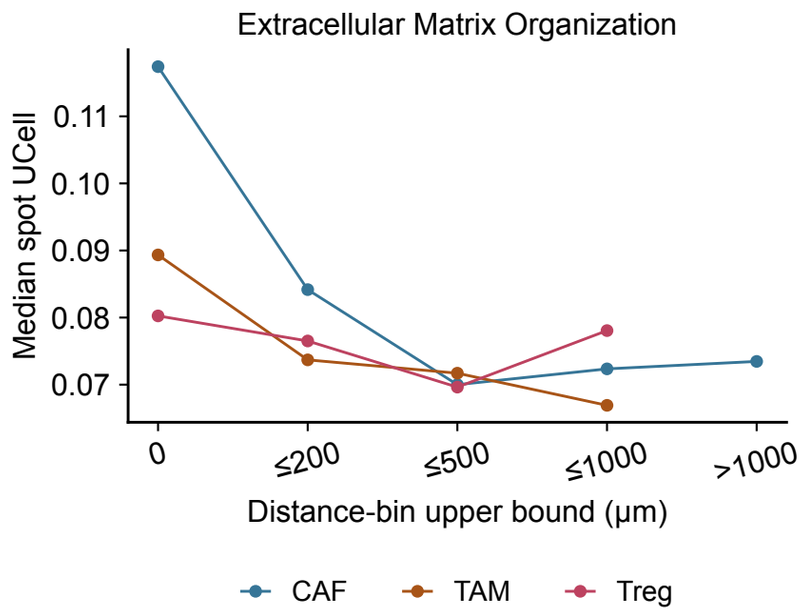






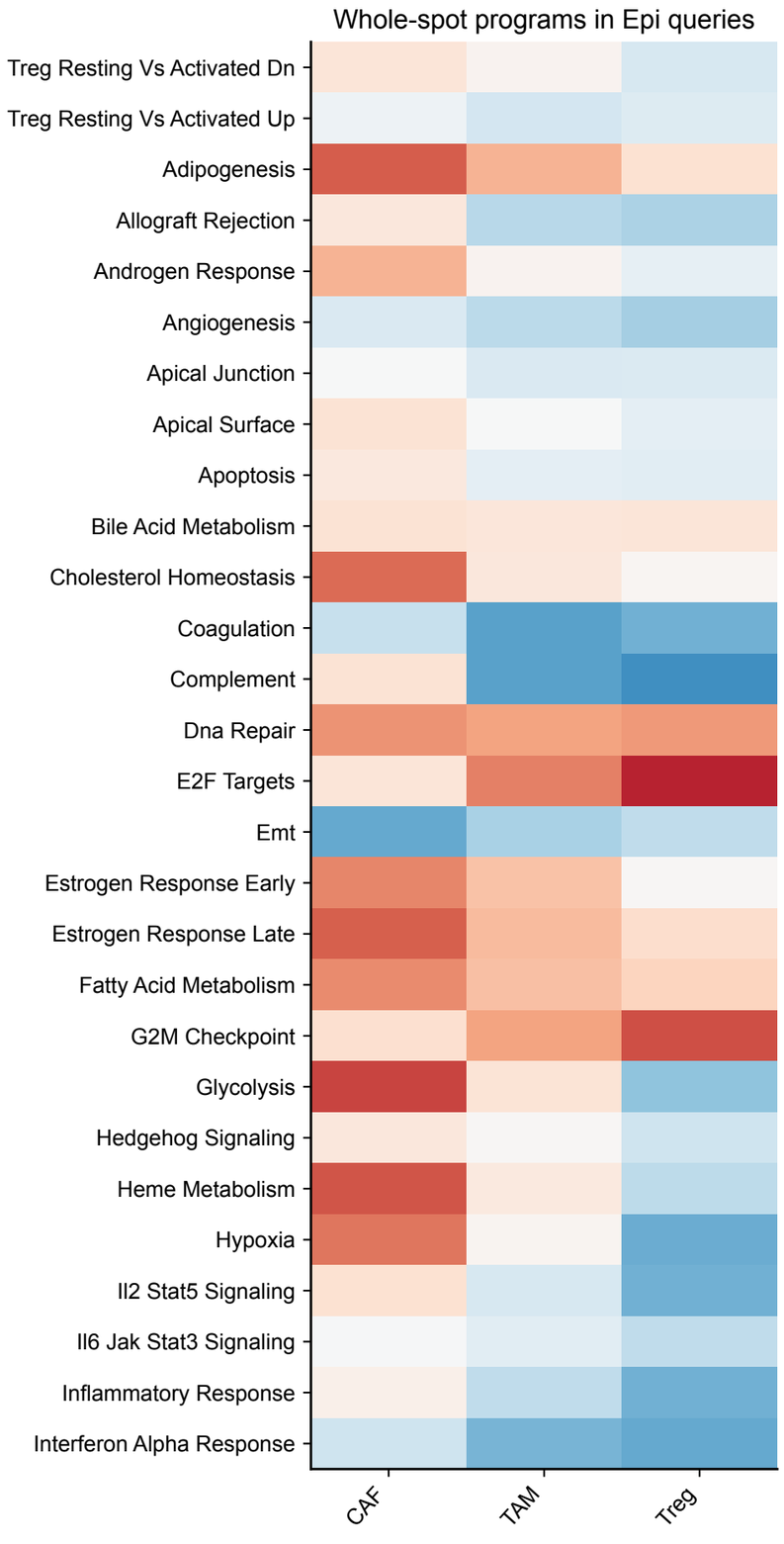


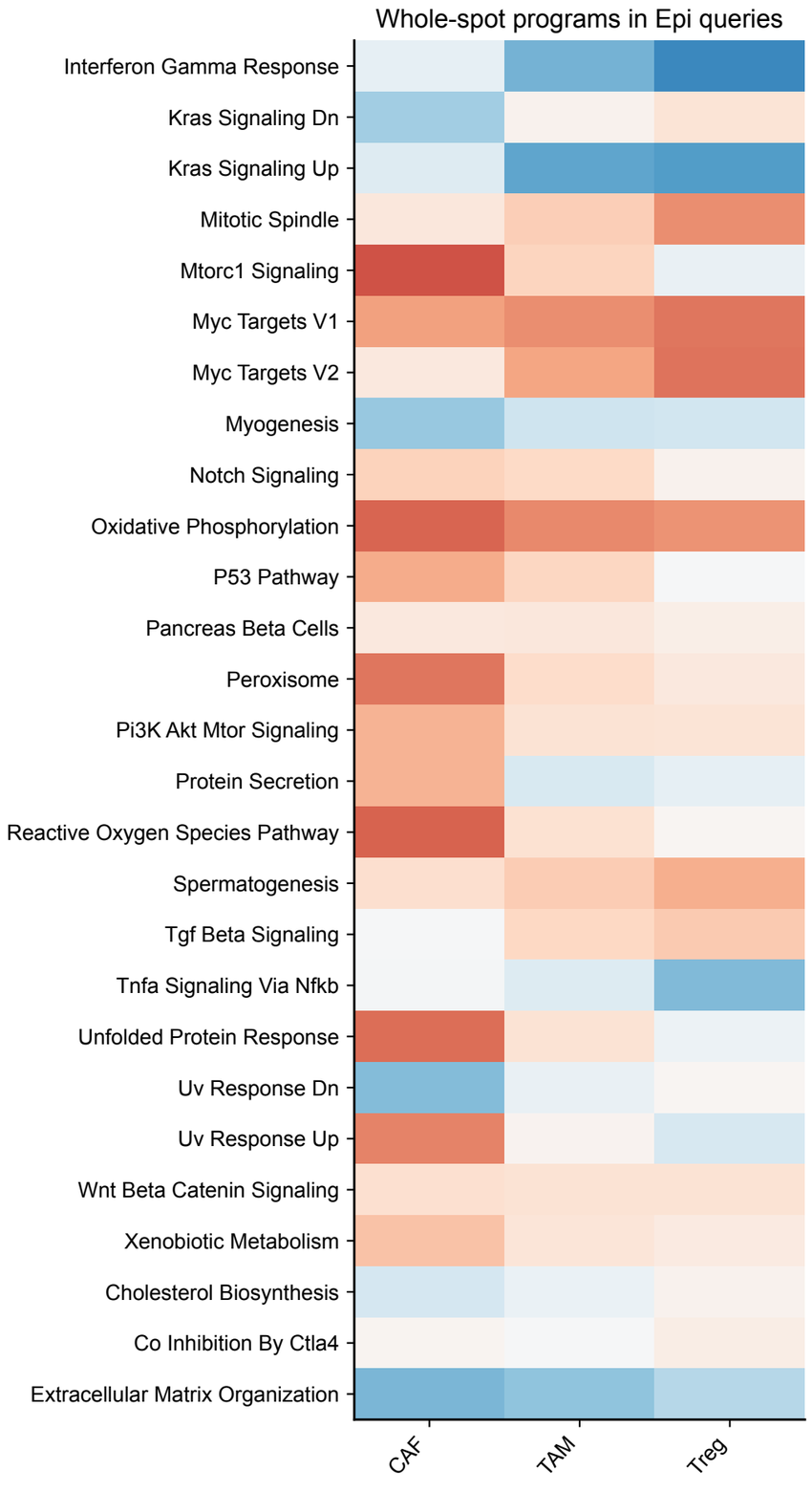




NPJ_sample5 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

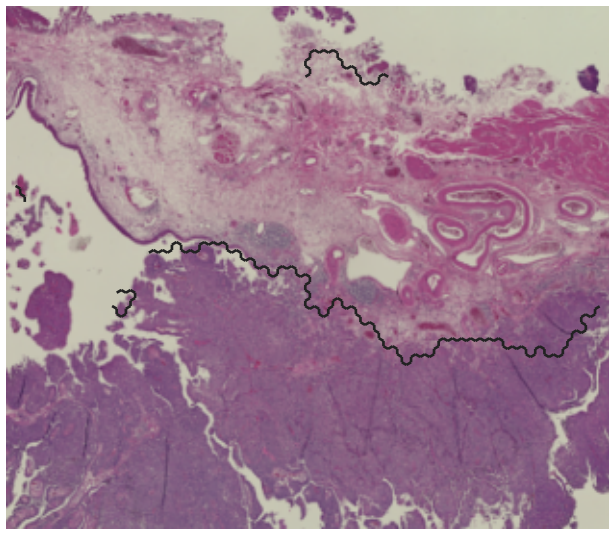




NPJ_sample6 | Tissue context

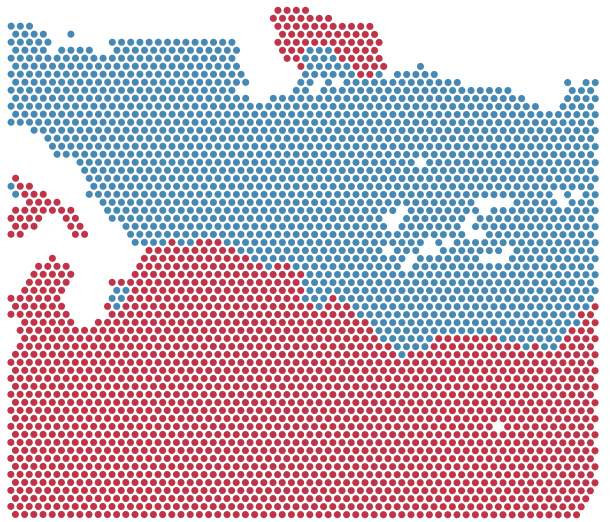
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



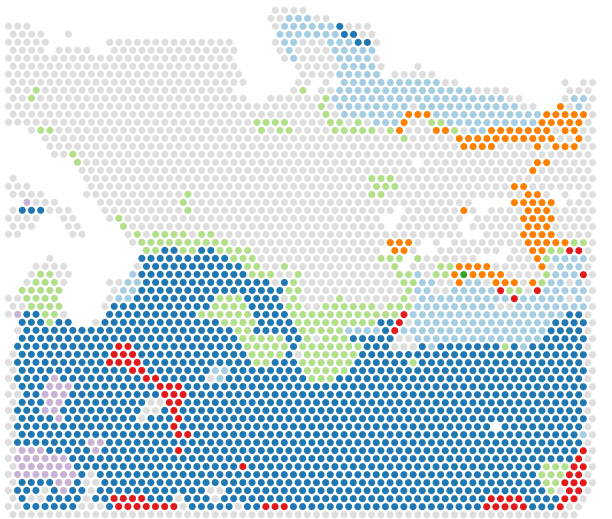
Black line: observed candidate outer edge

Final candidate location



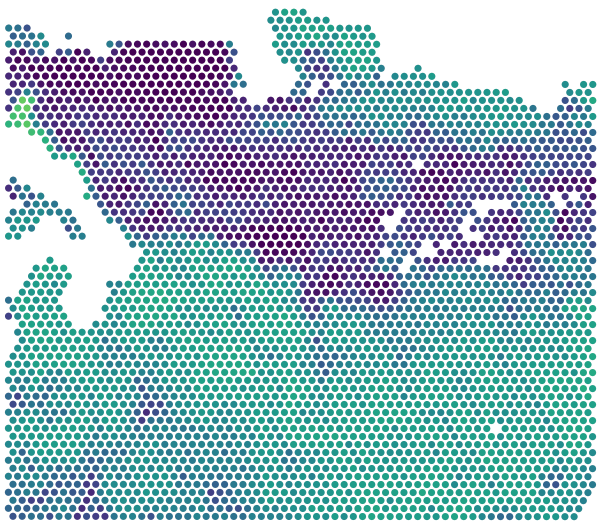
inside outside

Accepted R12 (original 0–8)



0 1 2 5 7 8
1 3

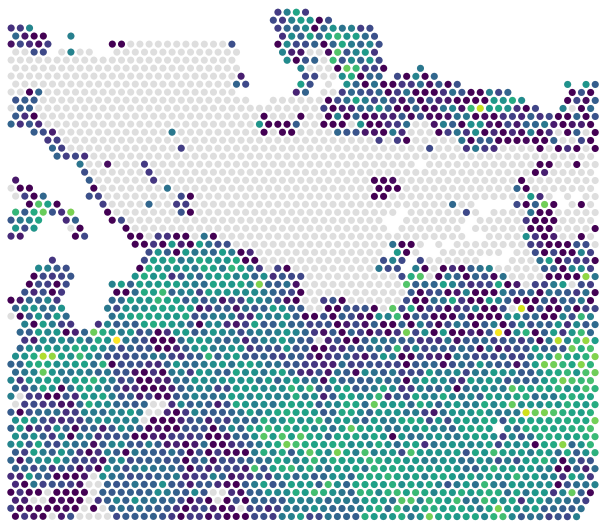
Epithelial support



0 1

Fraction of all 30

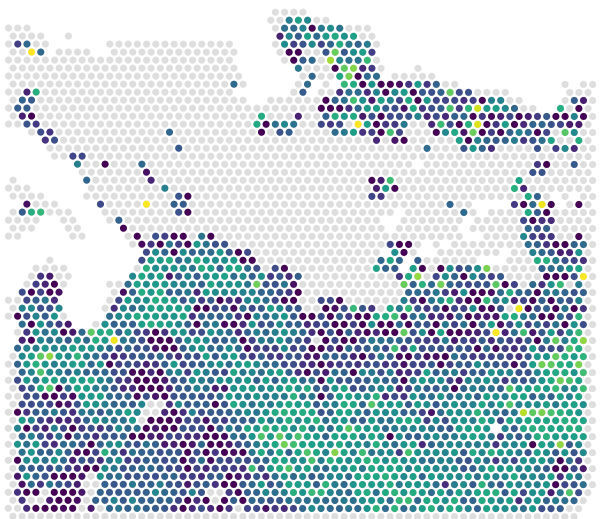
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)

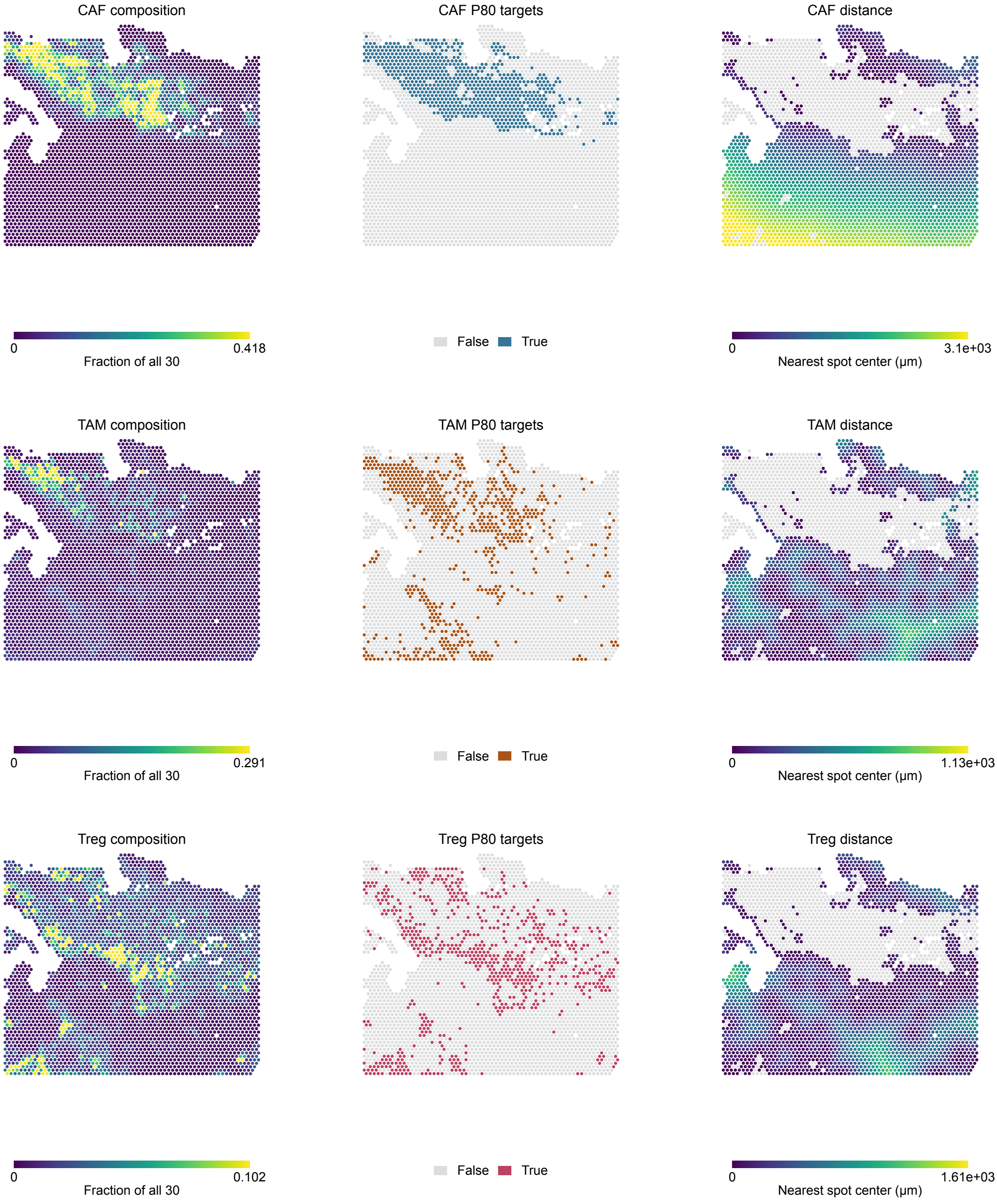


0 0.14

Matched-model AUCell

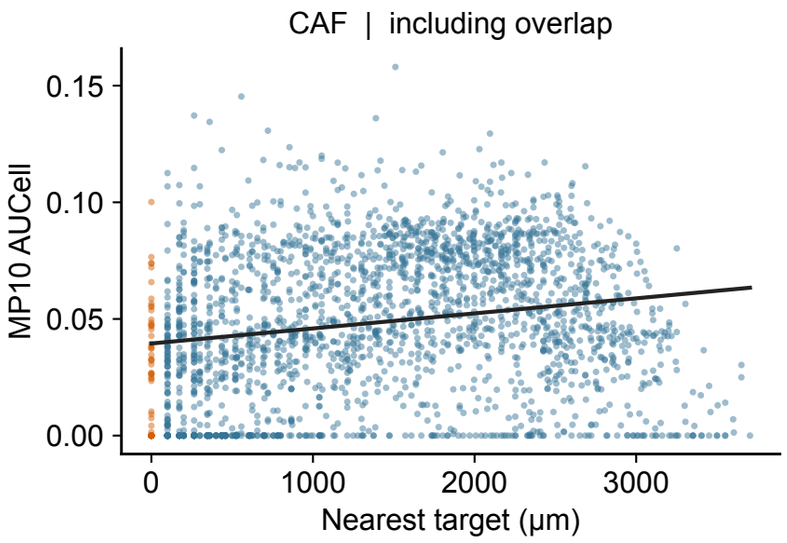
NPJ_sample6 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

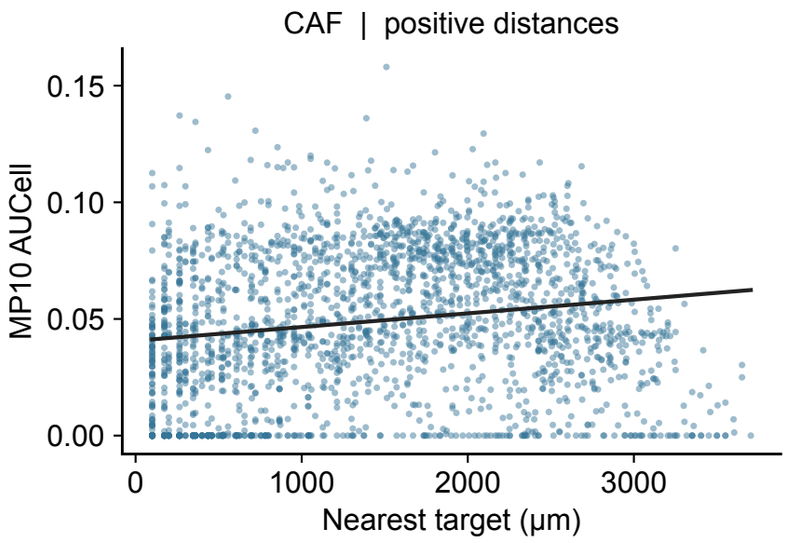


NPJ_sample6 | MP10 versus nearest target

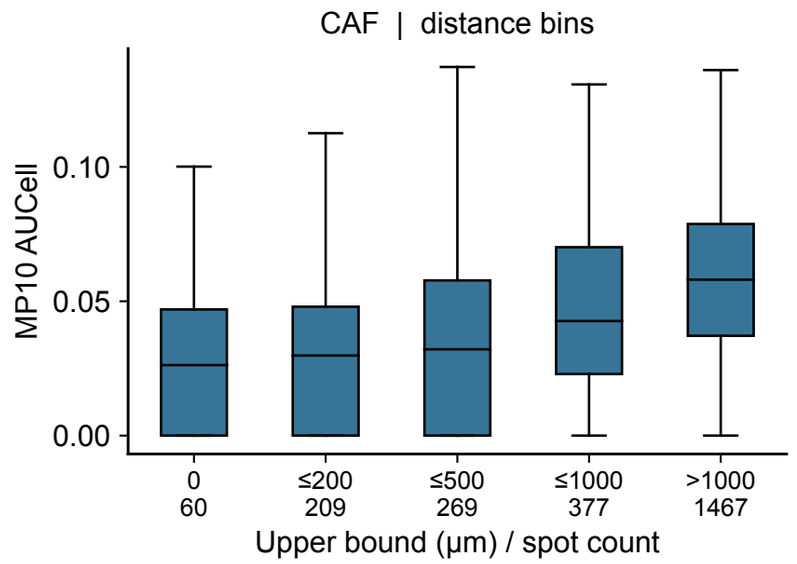
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



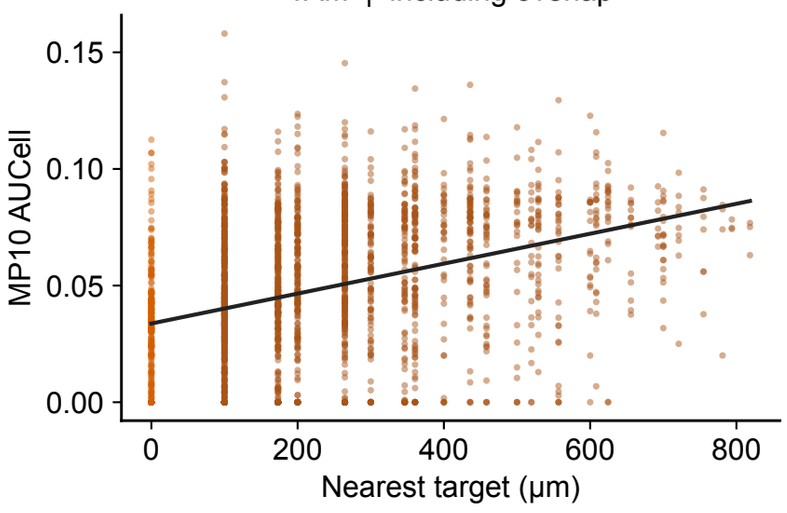
r +0.19 ρ +0.22
n 2382/2414 queries • 700 targets



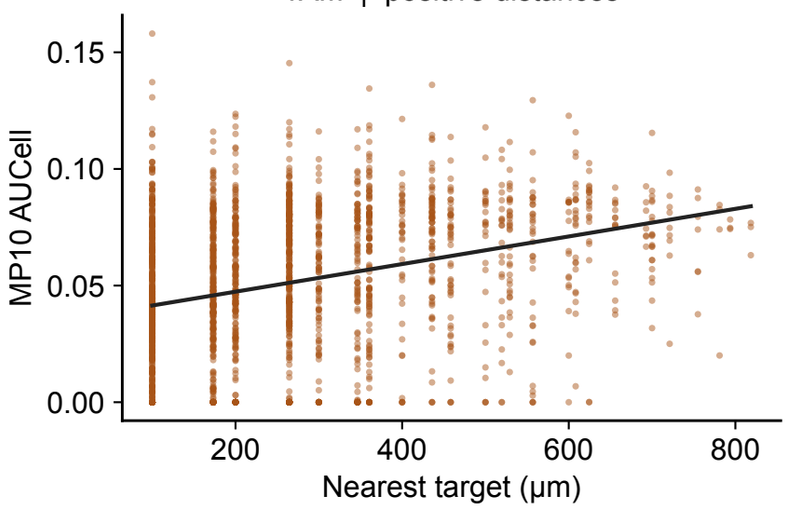
r +0.17 ρ +0.20
n 2322/2414 queries • 700 targets



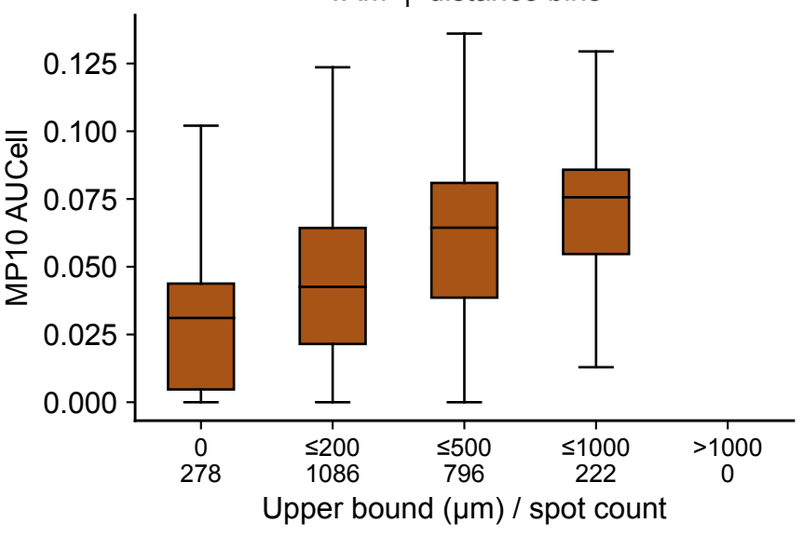
Orange = zero distance
Line = within-section linear fit



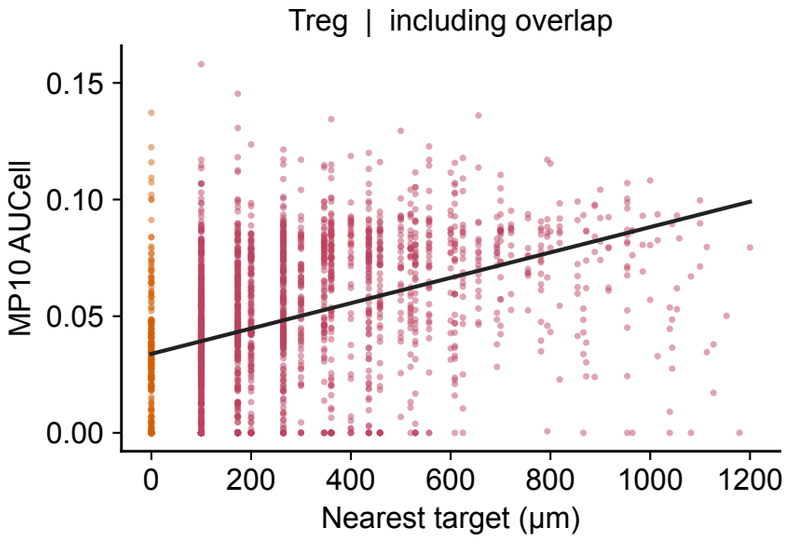
r +0.36 ρ +0.39
n 2382/2414 queries • 700 targets



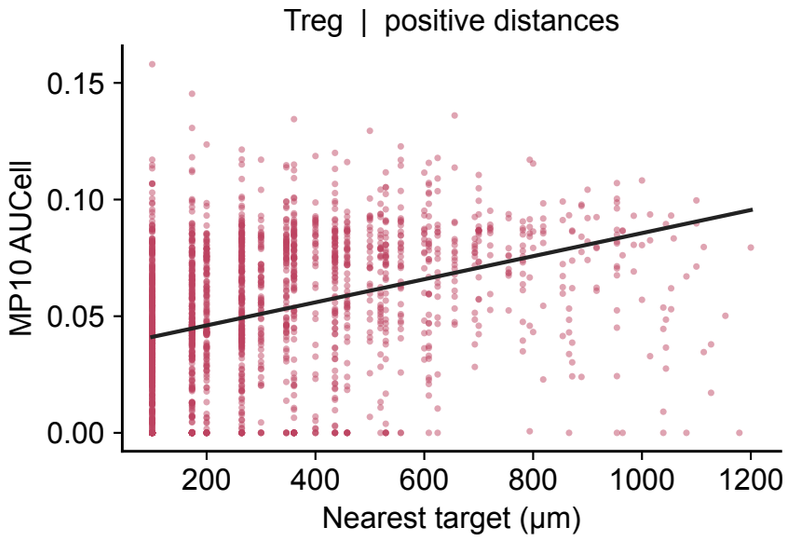
r +0.31 ρ +0.34
n 2104/2414 queries • 700 targets



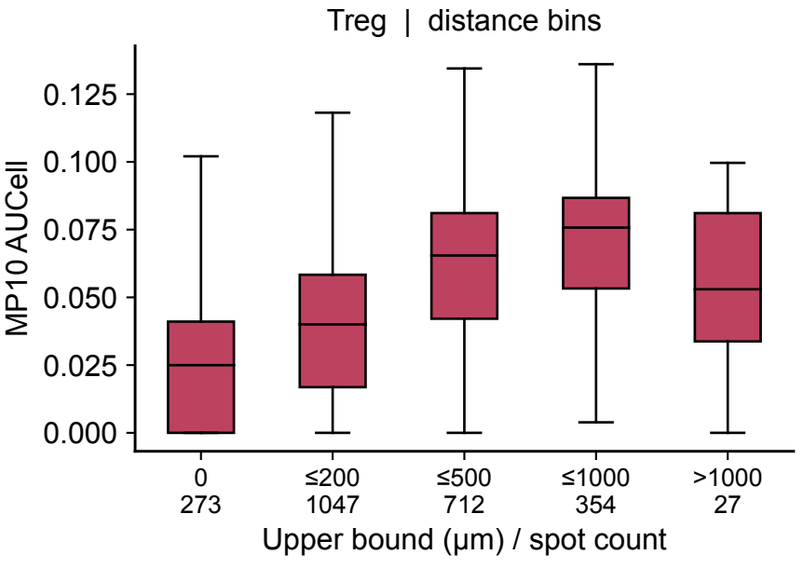
Orange = zero distance
Line = within-section linear fit



r +0.41 ρ +0.48
n 2413/2414 queries • 700 targets



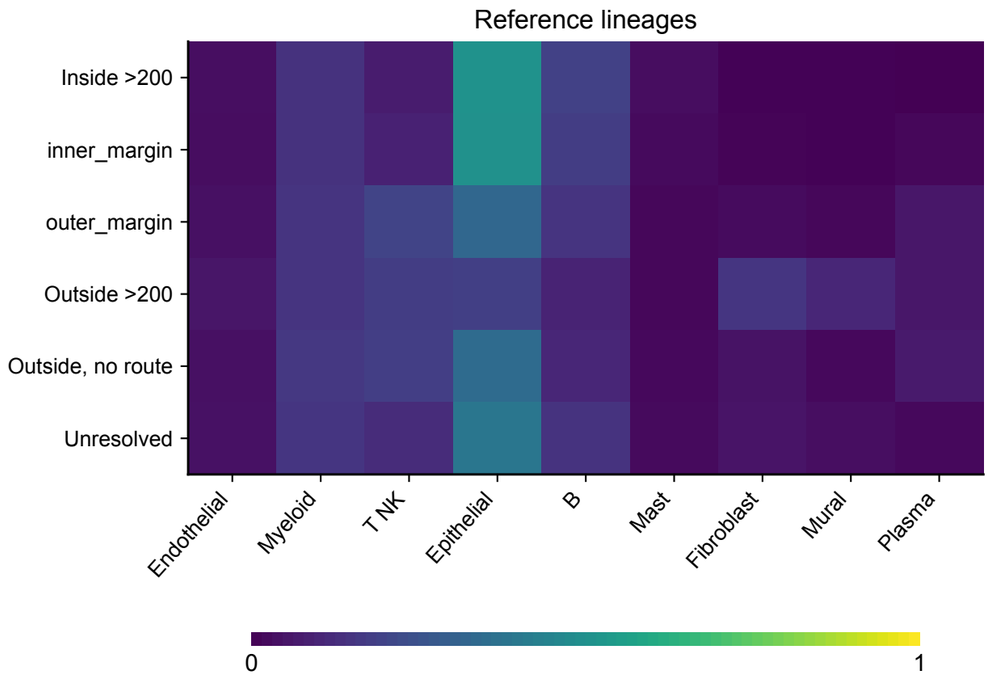
r +0.37 ρ +0.44
n 2140/2414 queries • 700 targets

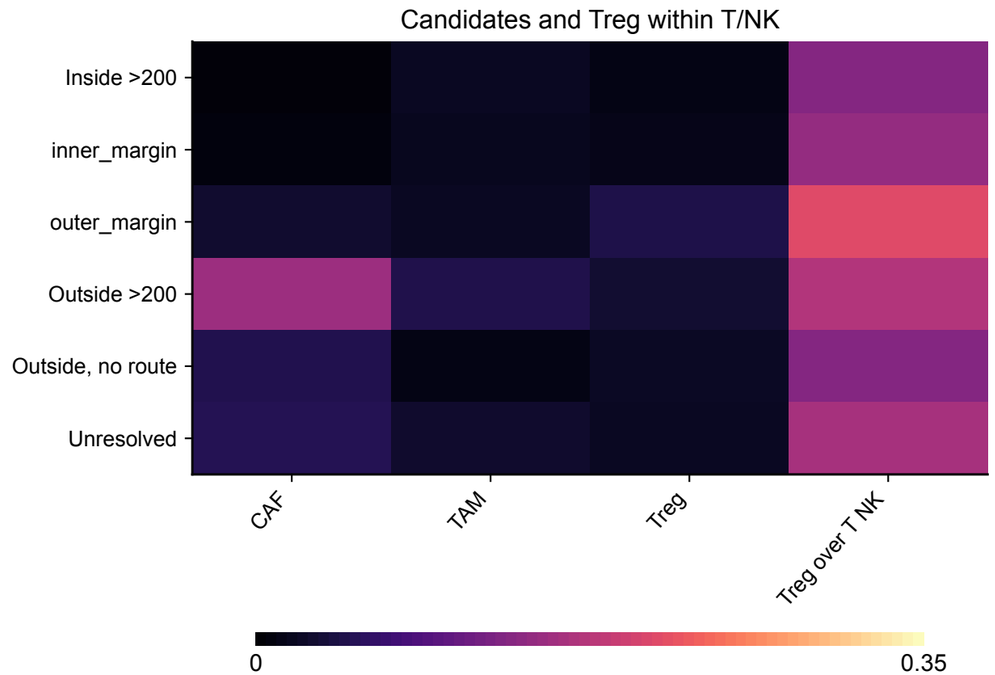


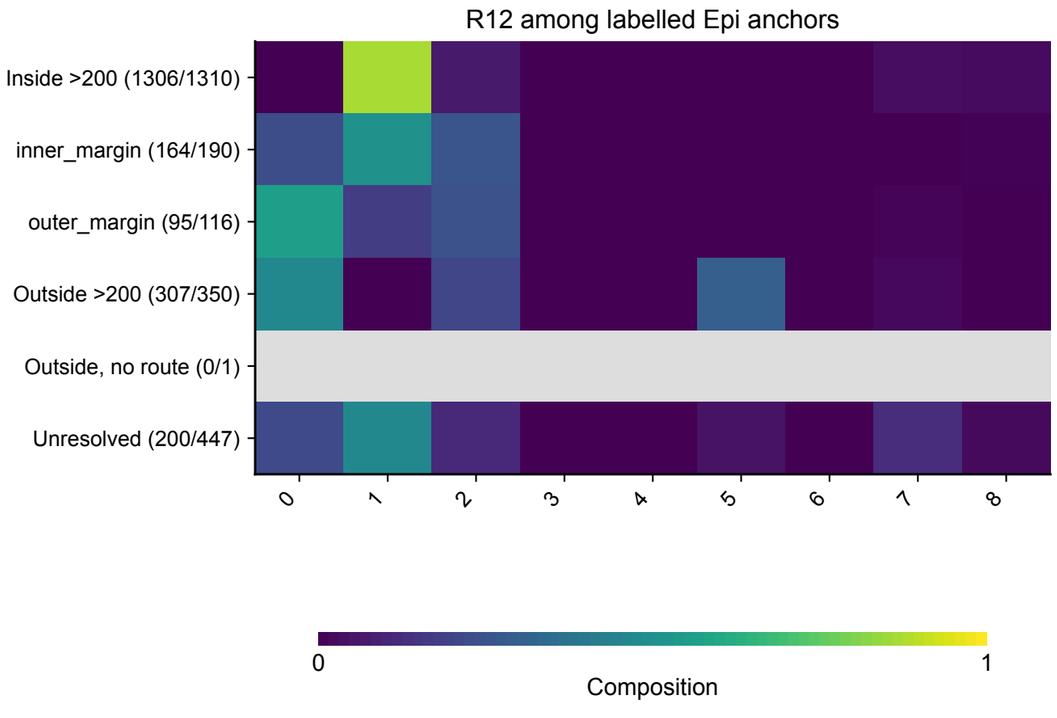
Orange = zero distance
Line = within-section linear fit

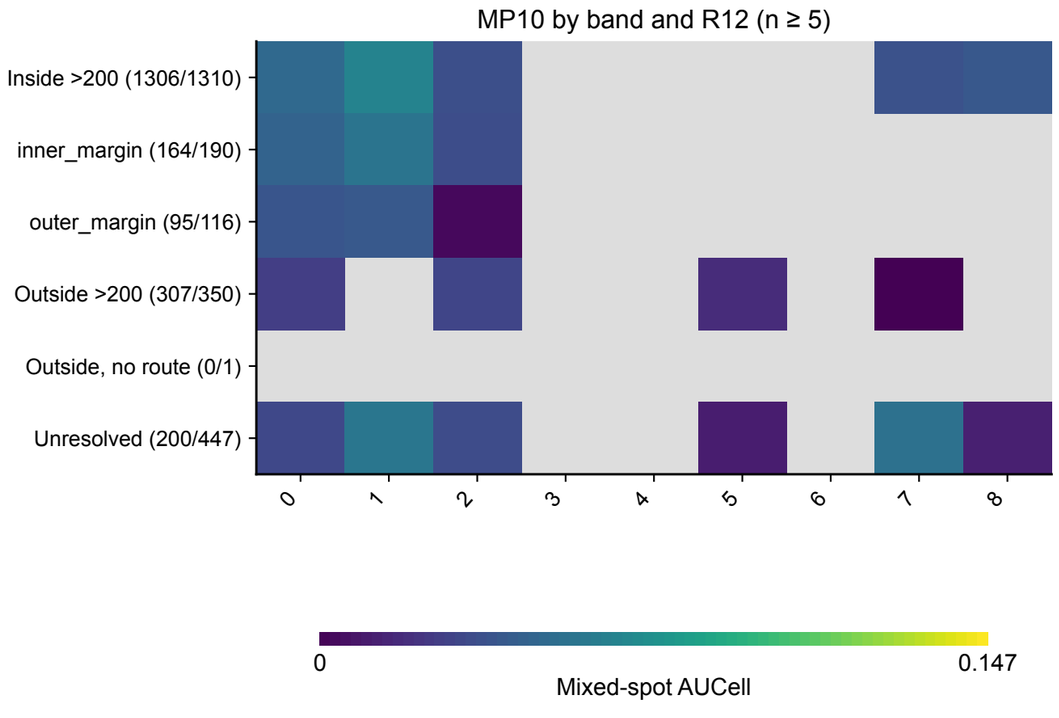
NPJ_sample6 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





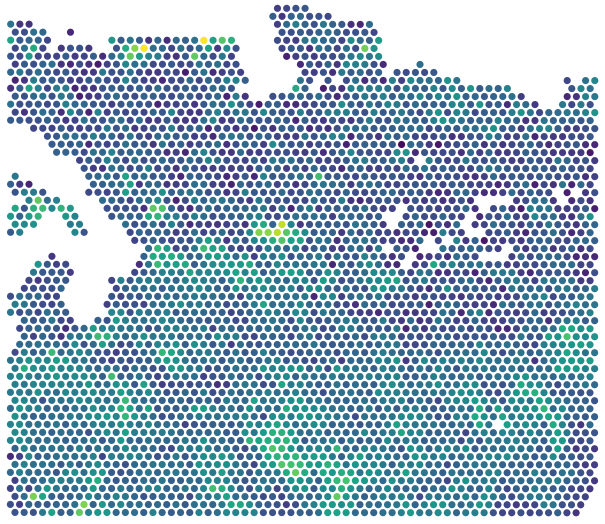




NPJ_sample6 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

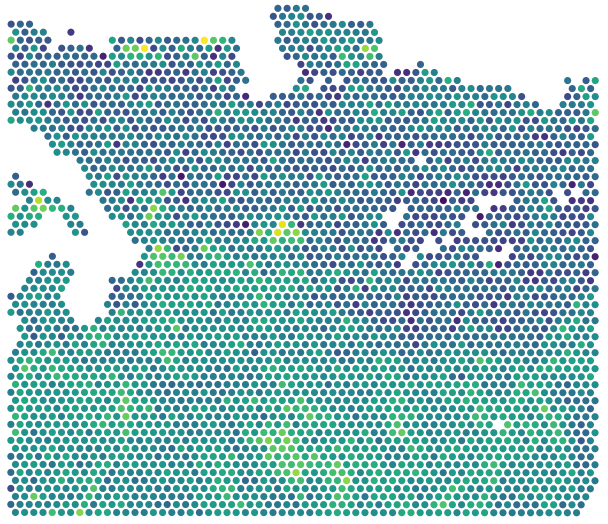
E2F Targets



0 0.152

UCell score

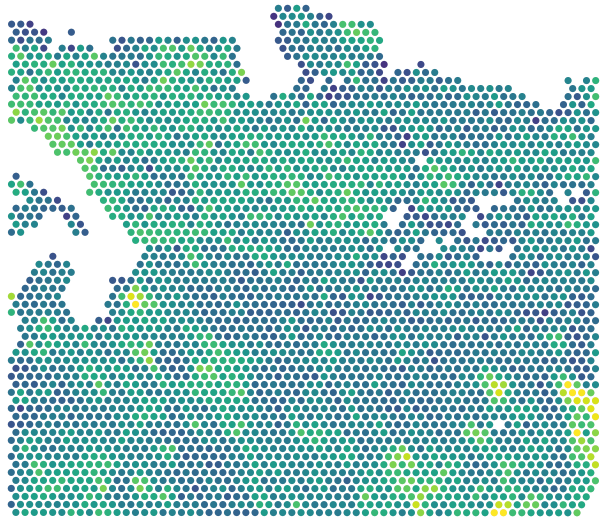
G2M Checkpoint



0 0.134

UCell score

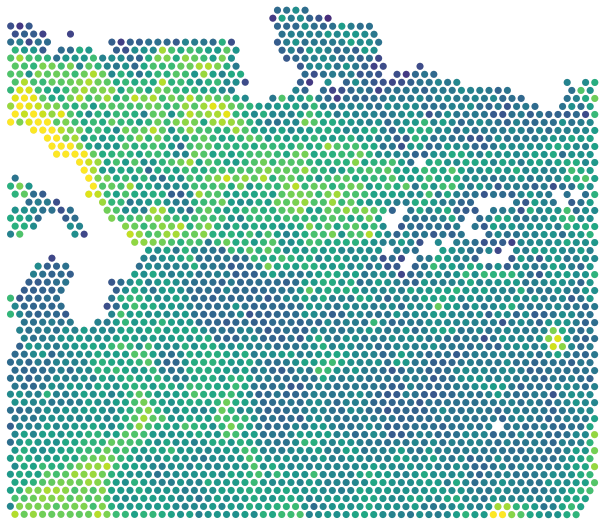
Hypoxia



0 0.211

UCell score

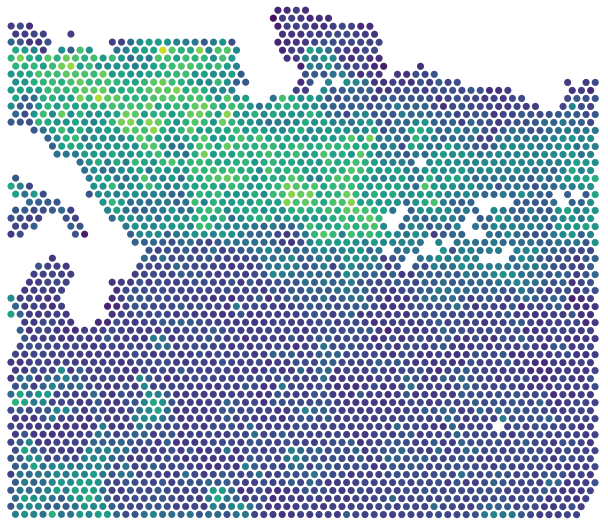
Tnfa Signaling Via Nfkb



0 0.275

UCell score

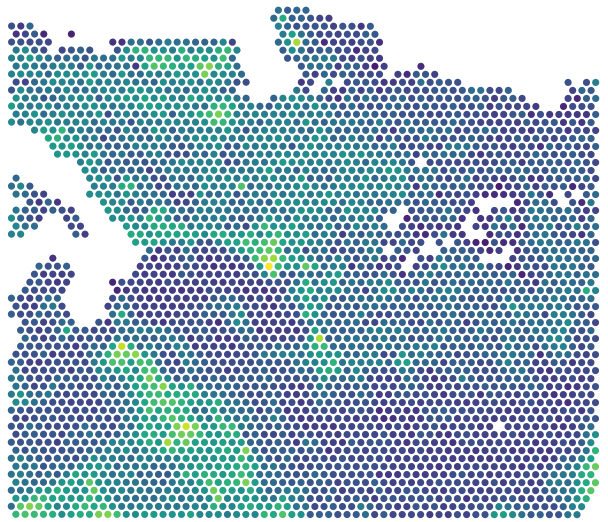
Emt



0 0.386

UCell score

Interferon Gamma Response



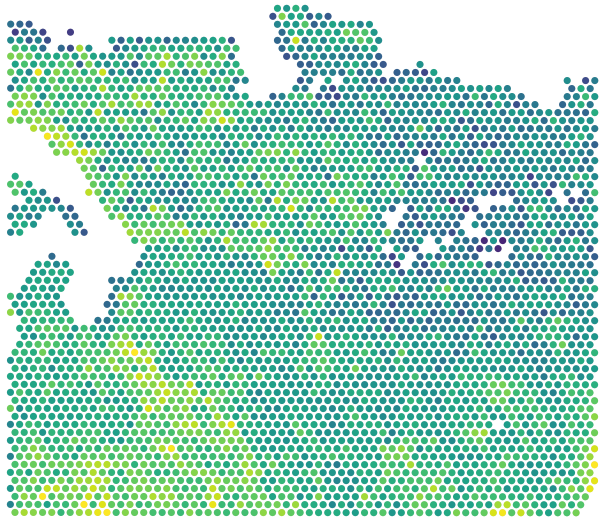
0 0.275

UCell score

NPJ_sample6 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

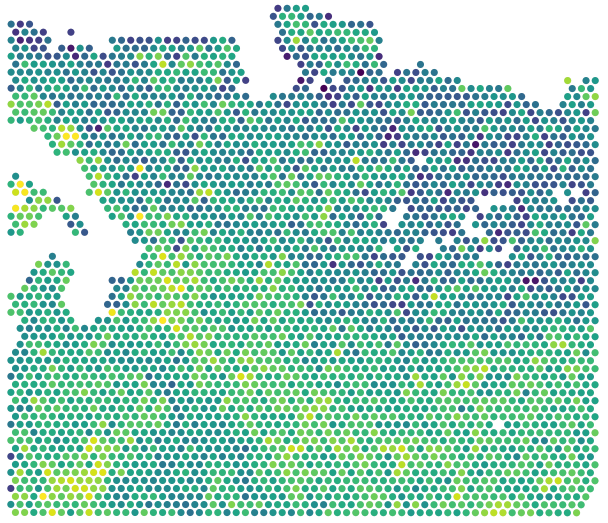
Il2 Stat5 Signaling



0 0.144

UCell score

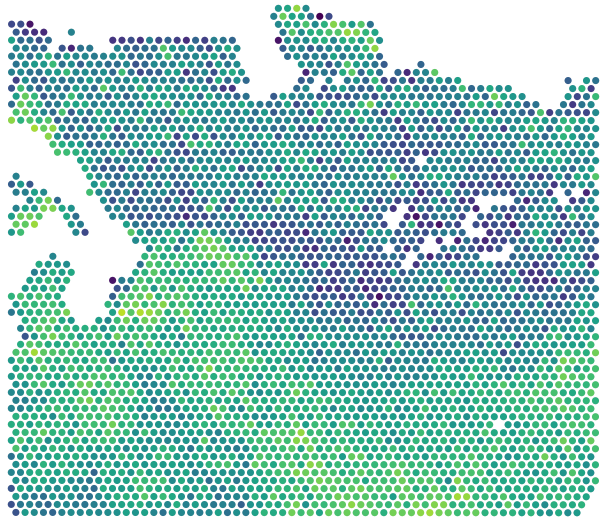
Tgf Beta Signaling



0 0.232

UCell score

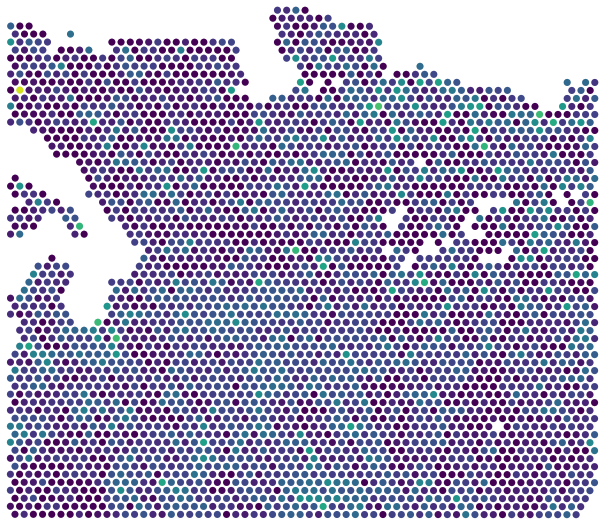
Cholesterol Homeostasis



0 0.21

UCell score

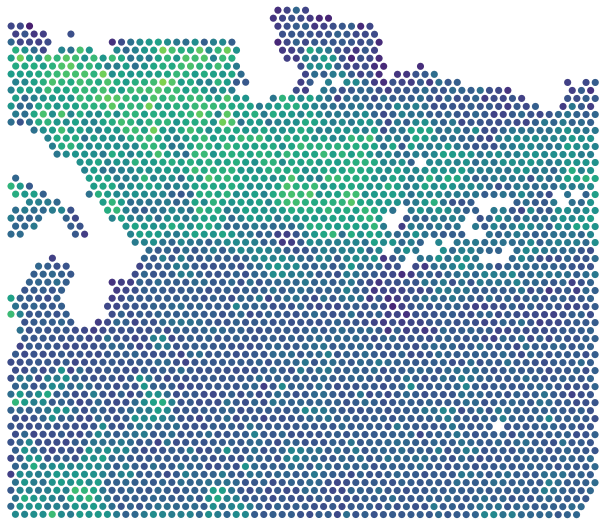
Cholesterol Biosynthesis



0 0.166

UCell score

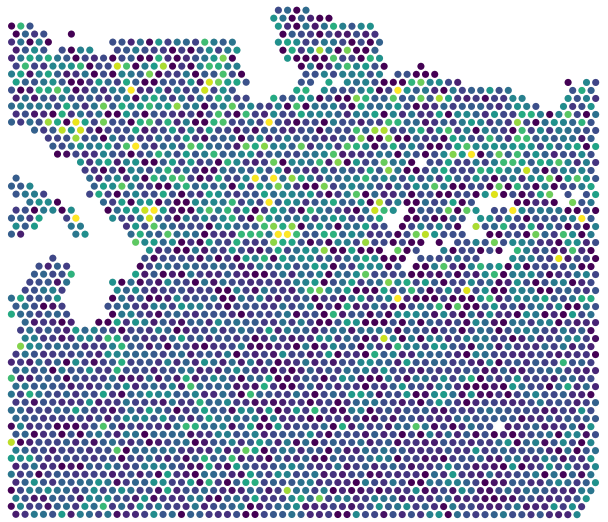
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



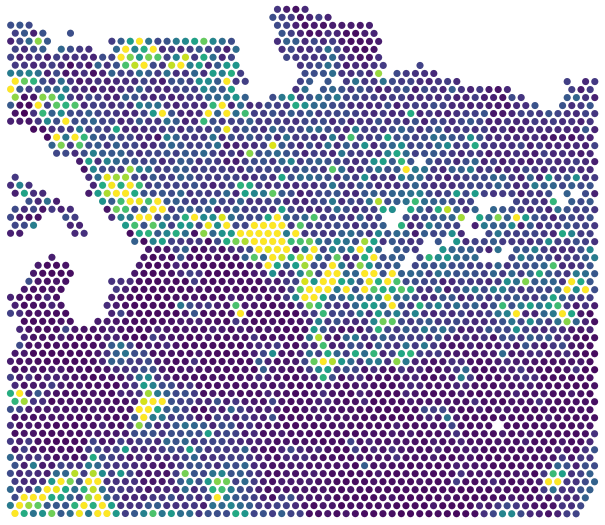
0 0.151

UCell score

NPJ_sample6 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

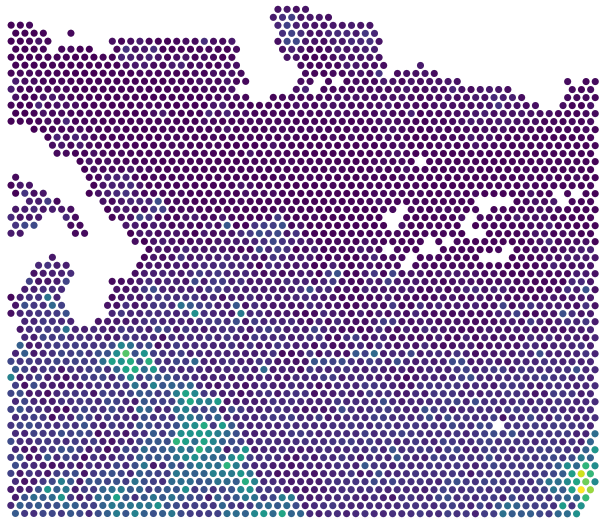
Treg composition



0 0.102

Fraction of all 30

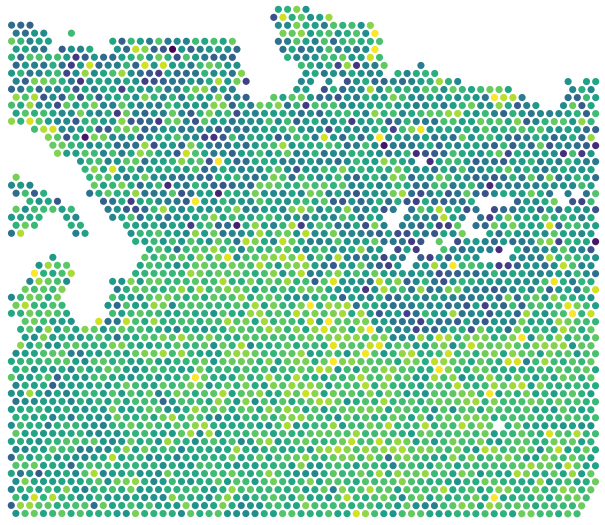
Treg estimated abundance



0 2.85

Posterior mean estimate

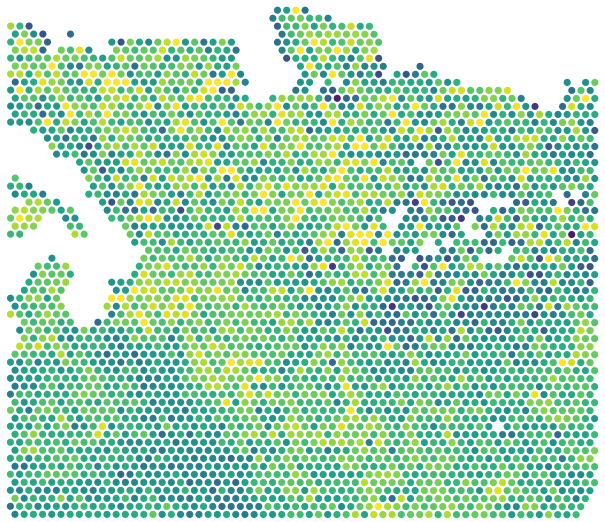
Treg Resting Vs Activated Dn



0 0.0778

UCell score

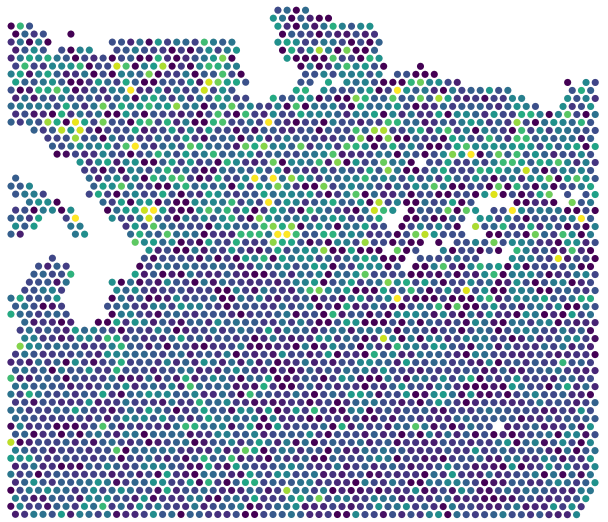
Treg Resting Vs Activated Up



0 0.0609

UCell score

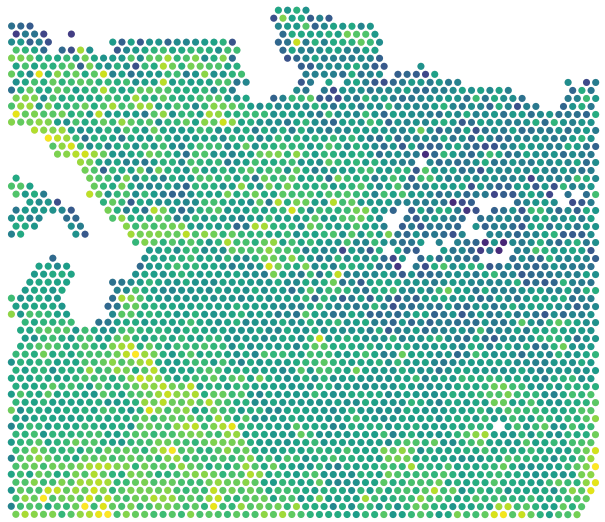
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

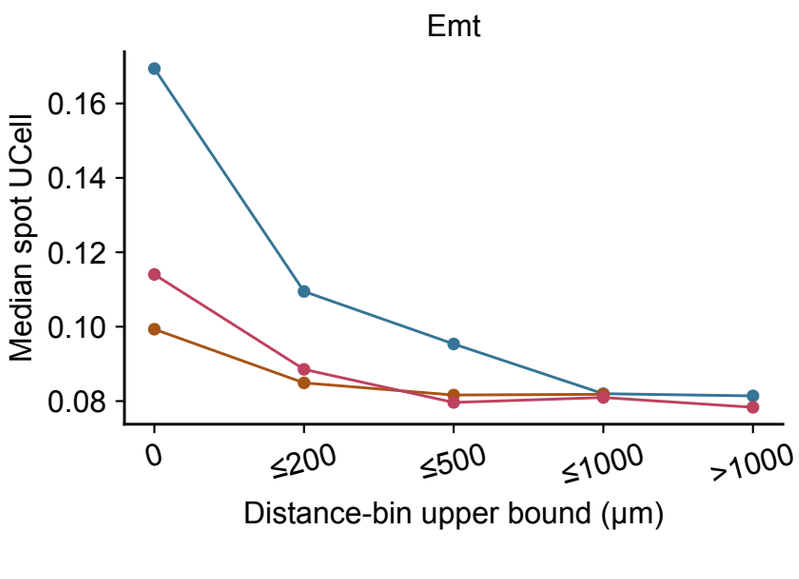


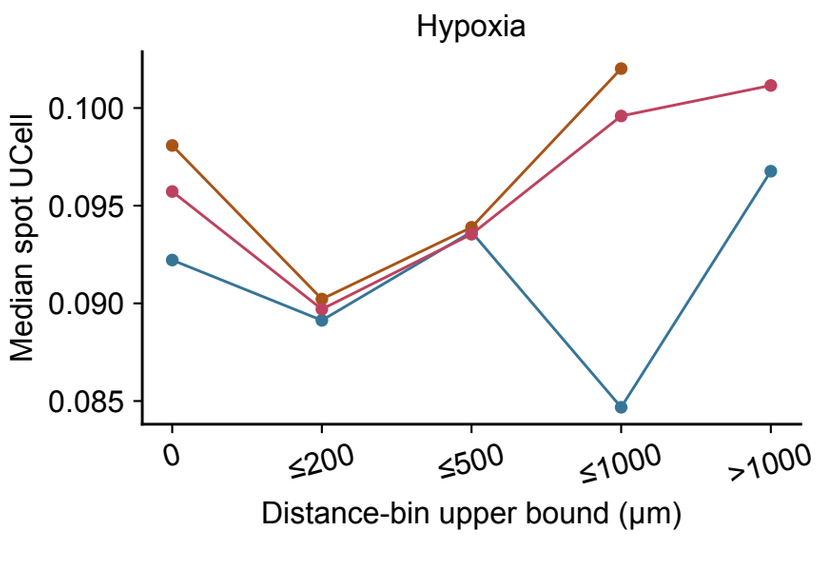
0 0.144

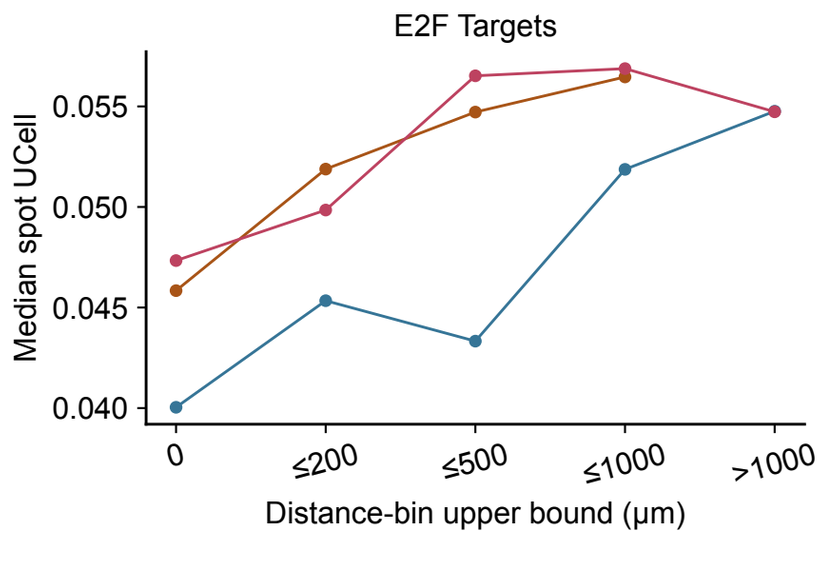
UCell score

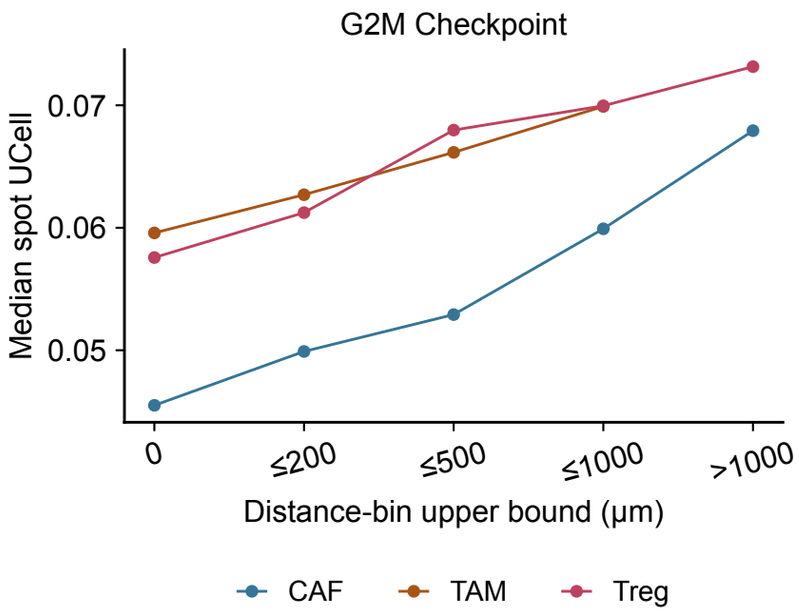
NPJ_sample6 | Programs by target distance

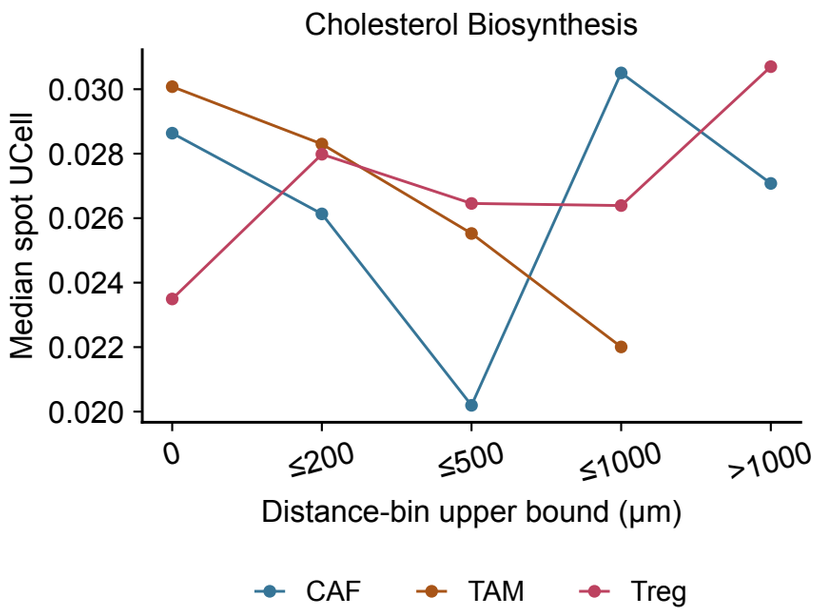
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

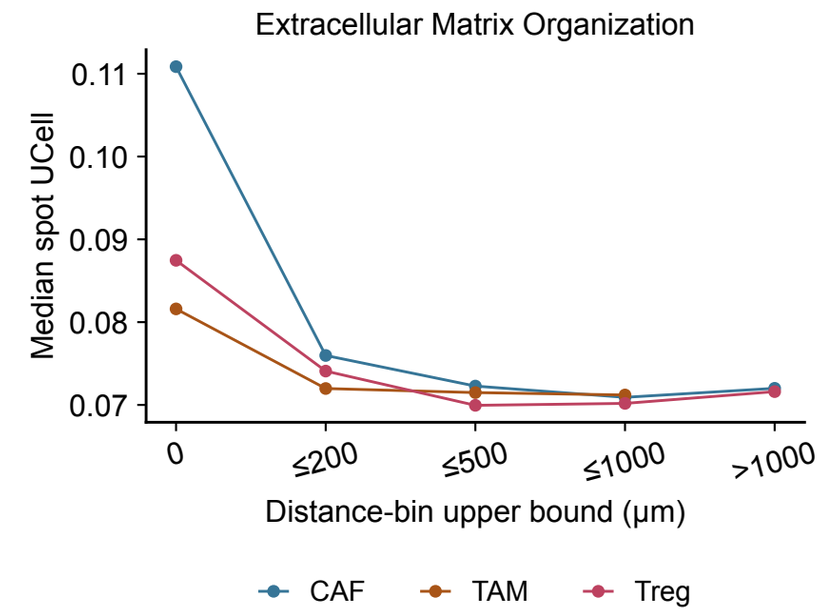






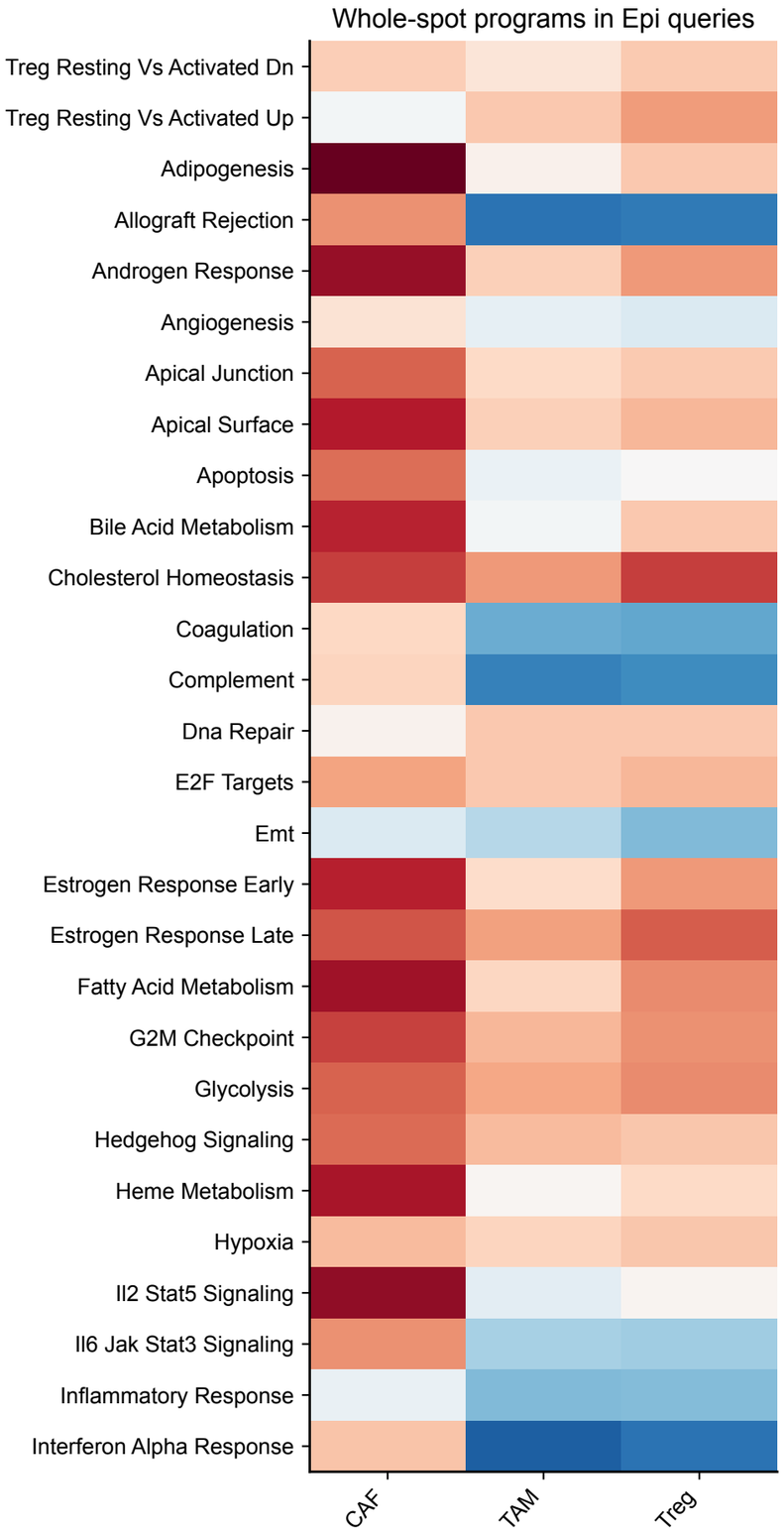


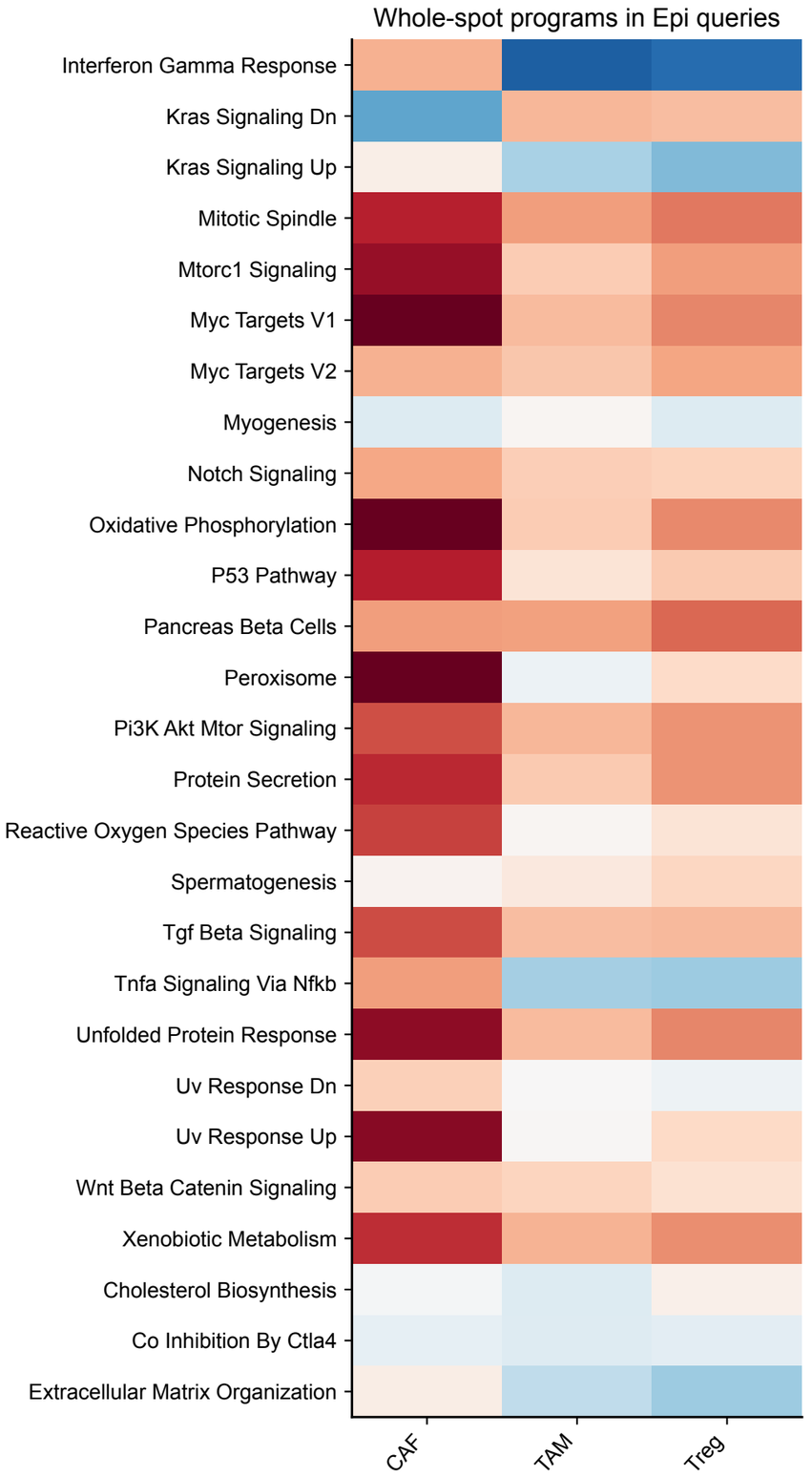




NPJ_sample6 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

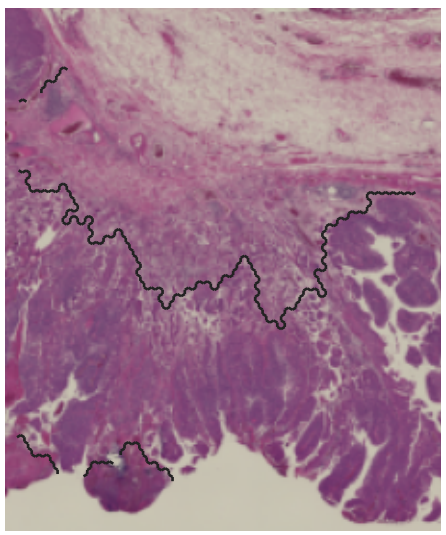




NPJ_sample7 | Tissue context

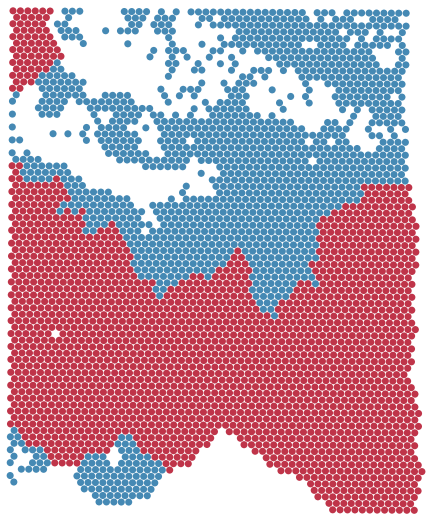
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



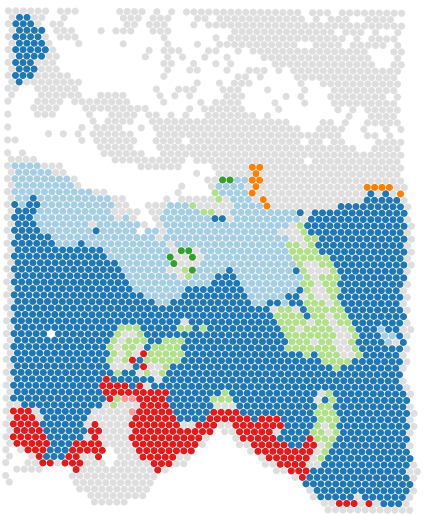
Black line: observed candidate outer edge

Final candidate location



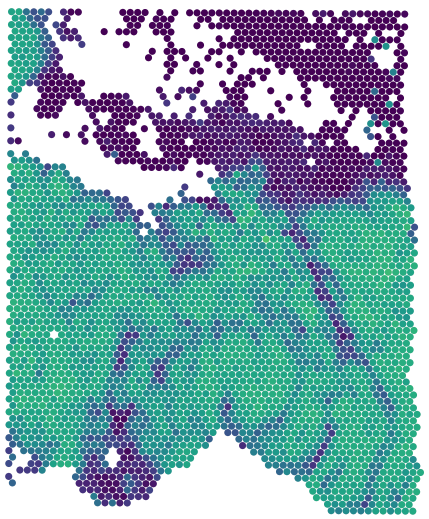
inside outside

Accepted R12 (original 0–8)



0 1 2 3 5 6 7

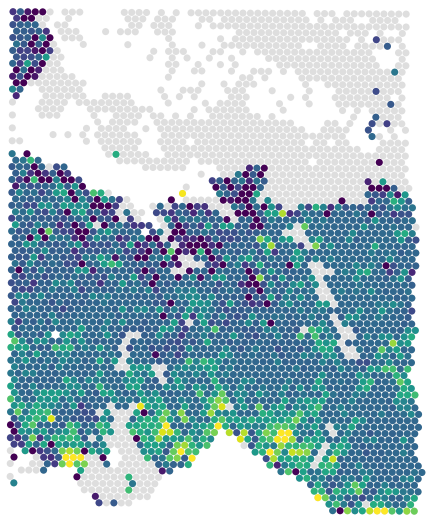
Epithelial support



0 1

Fraction of all 30

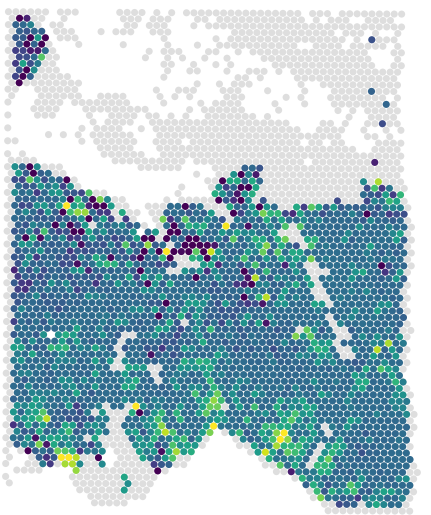
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



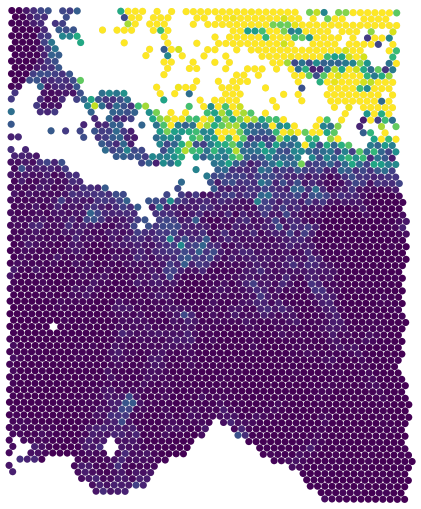
0 0.14

Matched-model AUCell

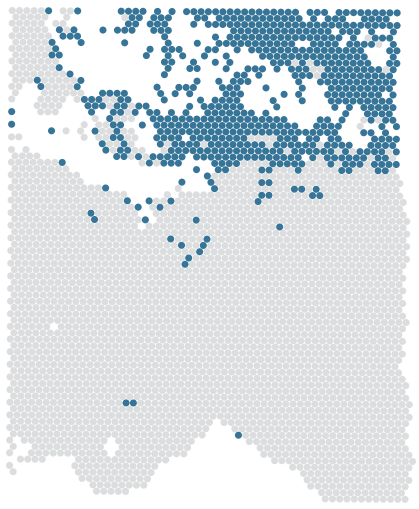
NPJ_sample7 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

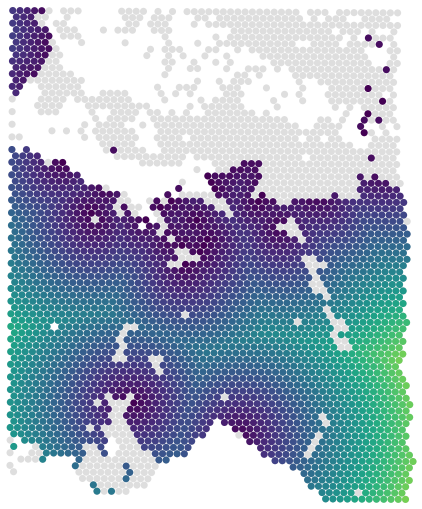
CAF composition



CAF P80 targets



CAF distance

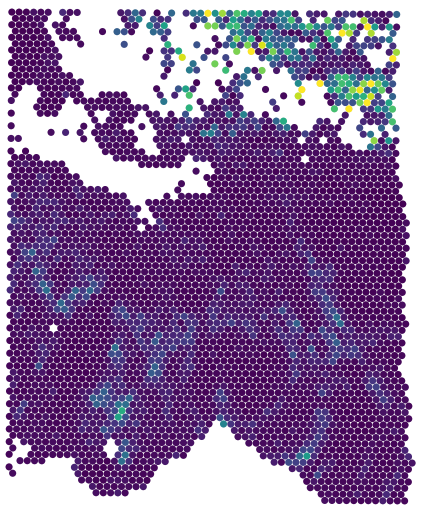


0 0.418
Fraction of all 30

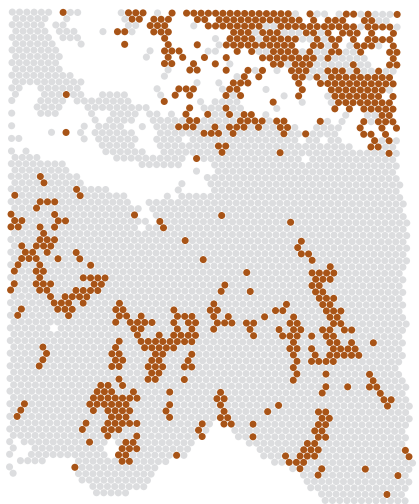
False True

0 3.1e+03
Nearest spot center (μm)

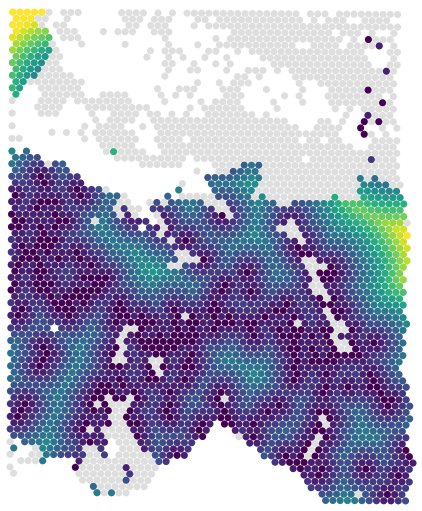
TAM composition



TAM P80 targets



TAM distance

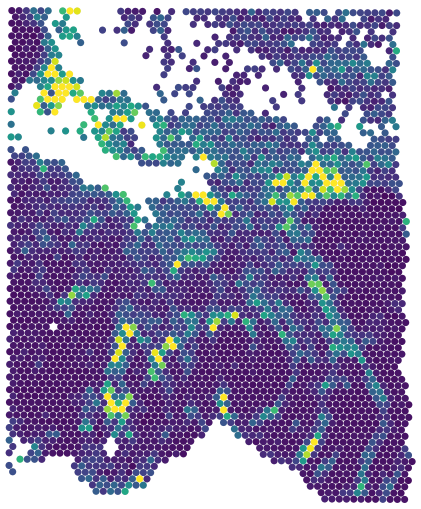


0 0.291
Fraction of all 30

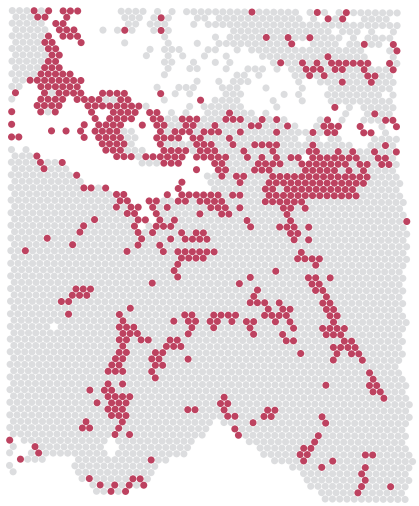
False True

0 1.13e+03
Nearest spot center (μm)

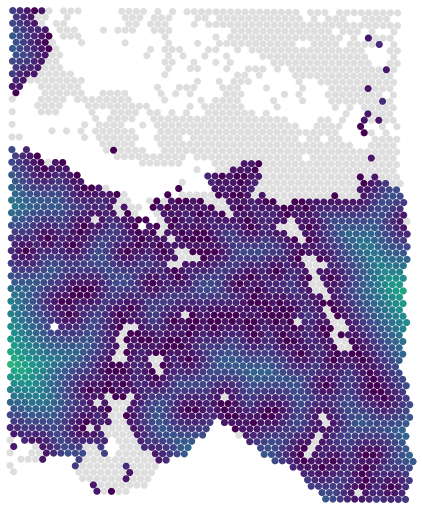
Treg composition



Treg P80 targets



Treg distance



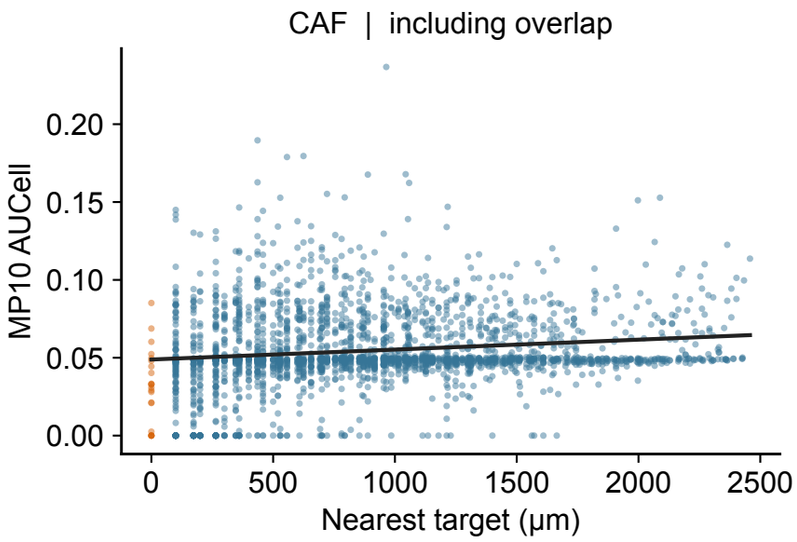
0 0.102
Fraction of all 30

False True

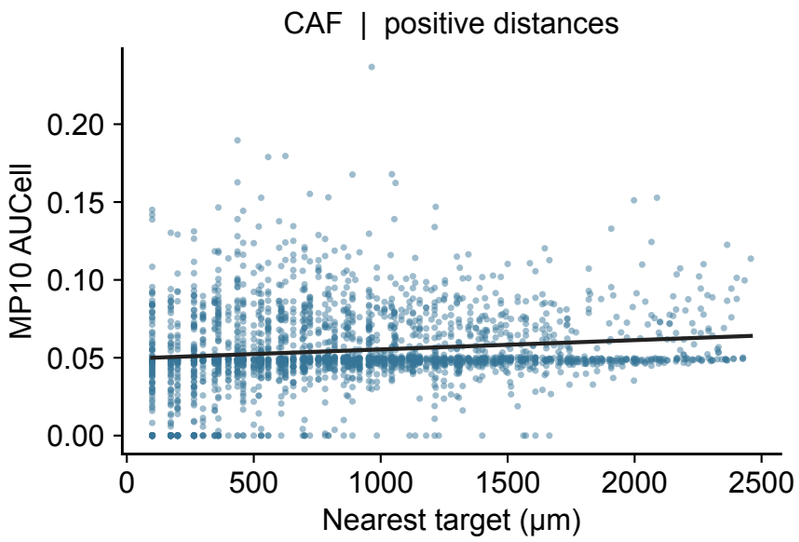
0 1.61e+03
Nearest spot center (μm)

NPJ_sample7 | MP10 versus nearest target

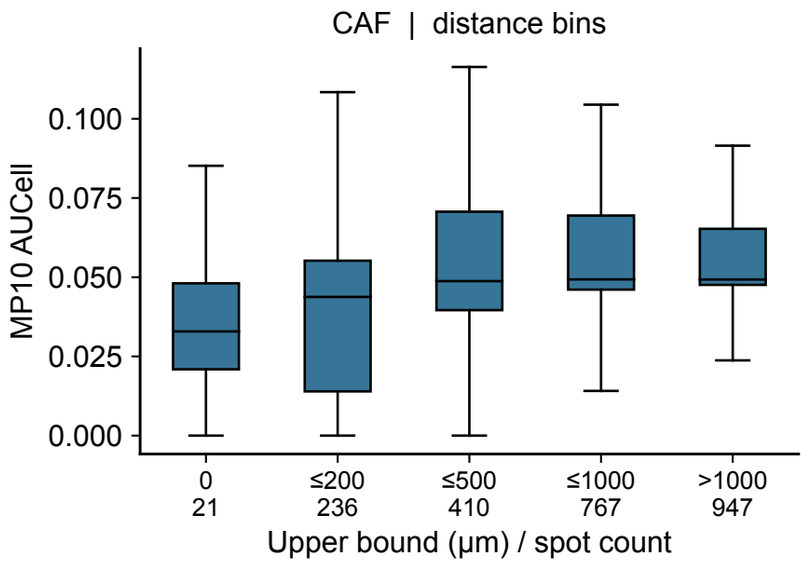
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



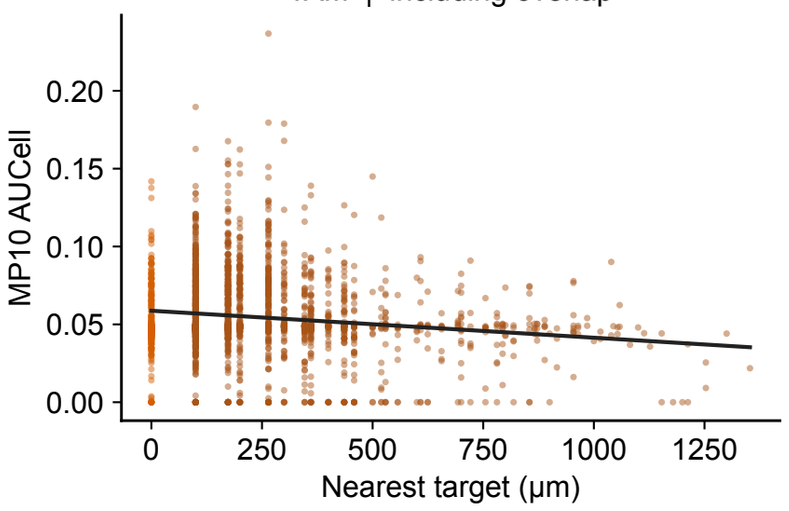
r +0.14 ρ +0.17
n 2381/2382 queries • 678 targets



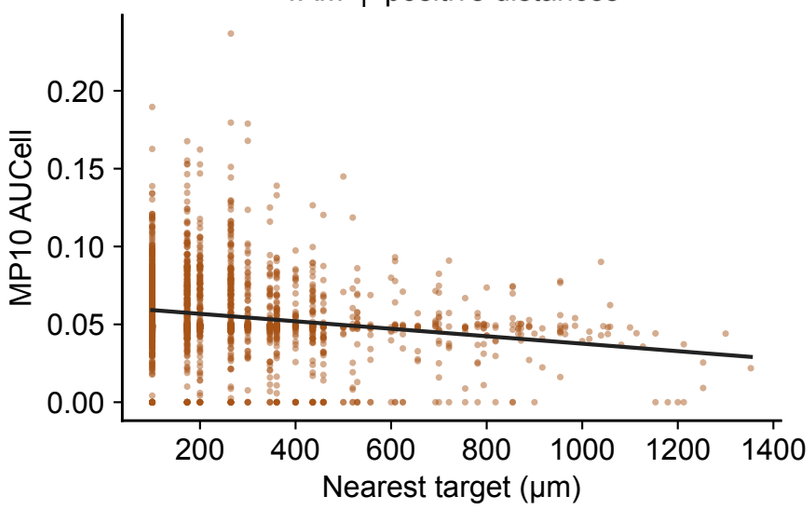
r +0.13 ρ +0.16
n 2360/2382 queries • 678 targets



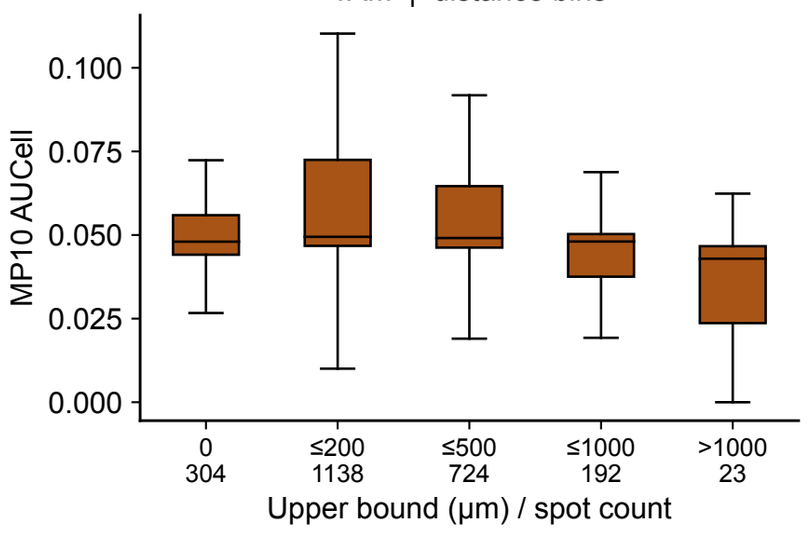
Orange = zero distance
Line = within-section linear fit



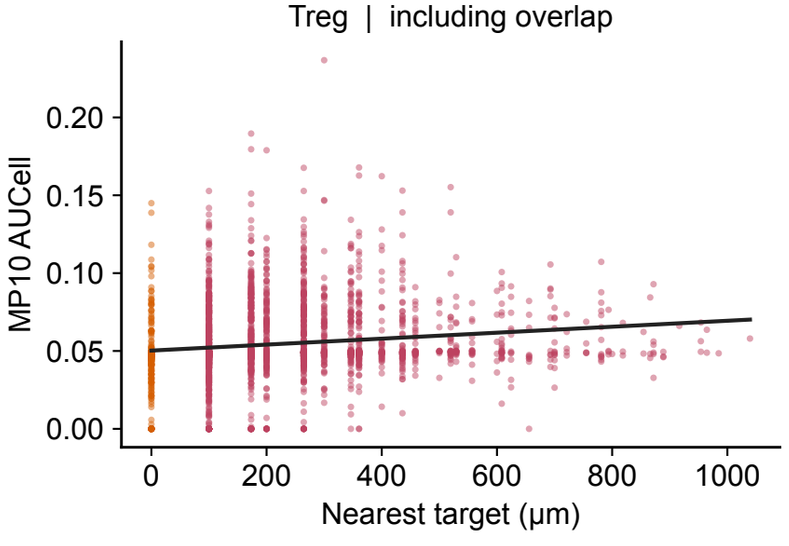
r -0.14 ρ -0.04
n 2381/2382 queries • 678 targets



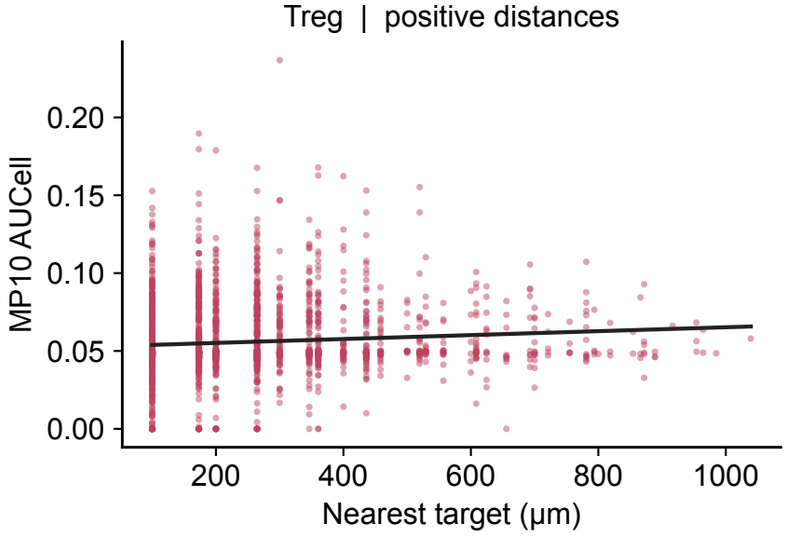
r -0.19 ρ -0.12
n 2077/2382 queries • 678 targets



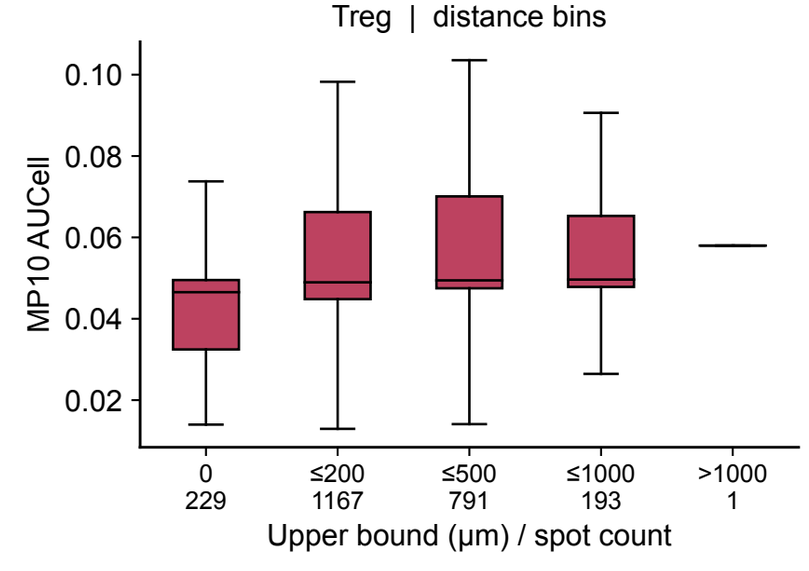
Orange = zero distance
Line = within-section linear fit



r +0.13 ρ +0.19
n 2381/2382 queries • 678 targets



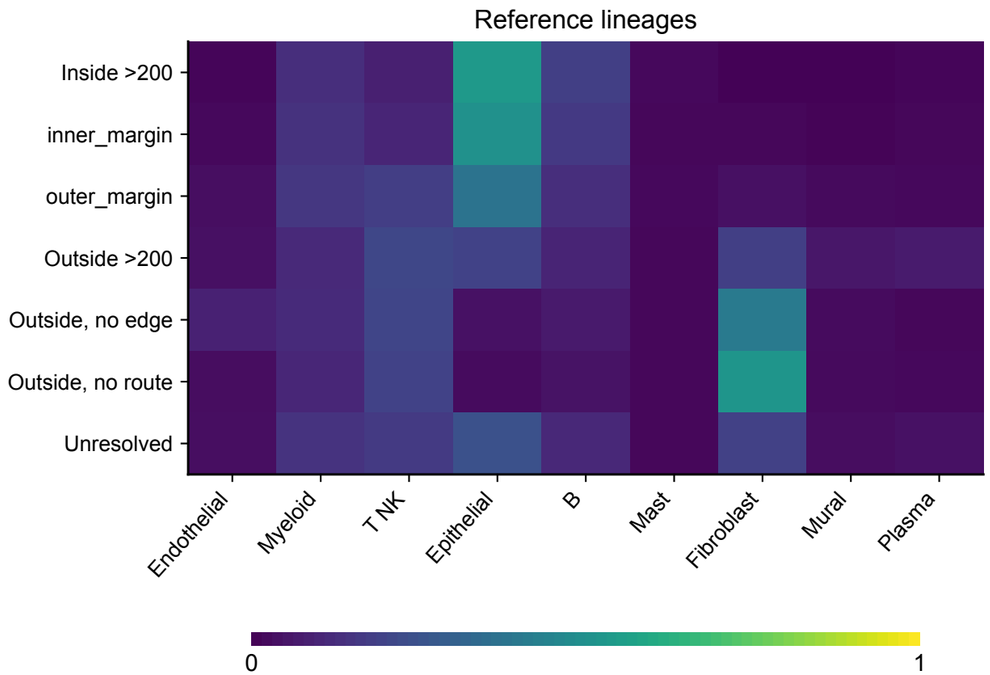
r +0.08 ρ +0.13
n 2152/2382 queries • 678 targets

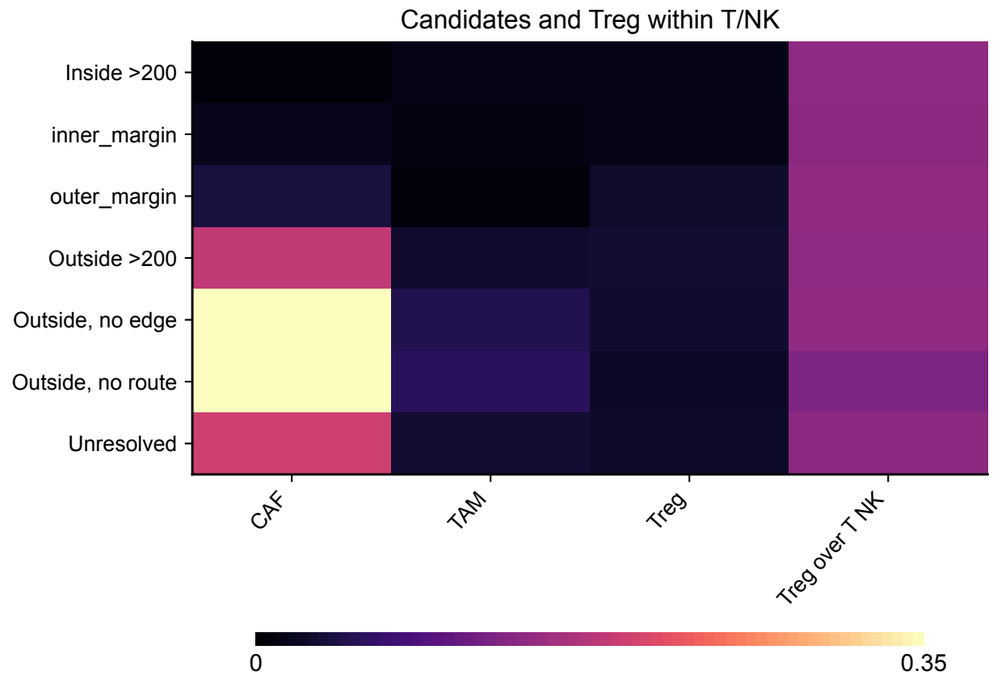


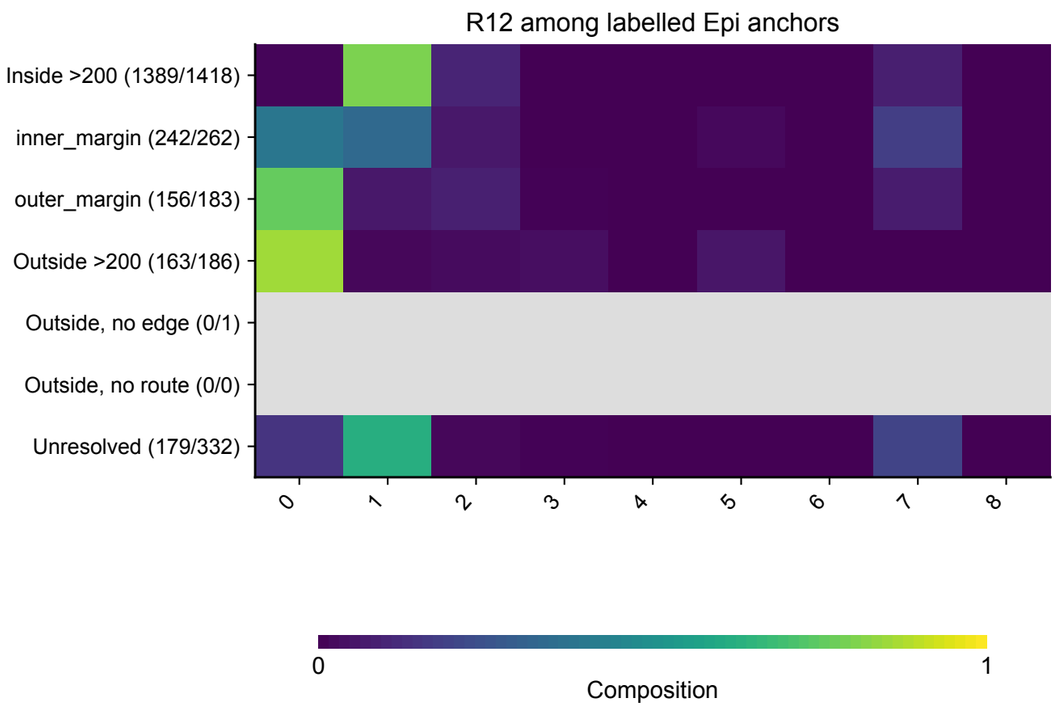
Orange = zero distance
Line = within-section linear fit

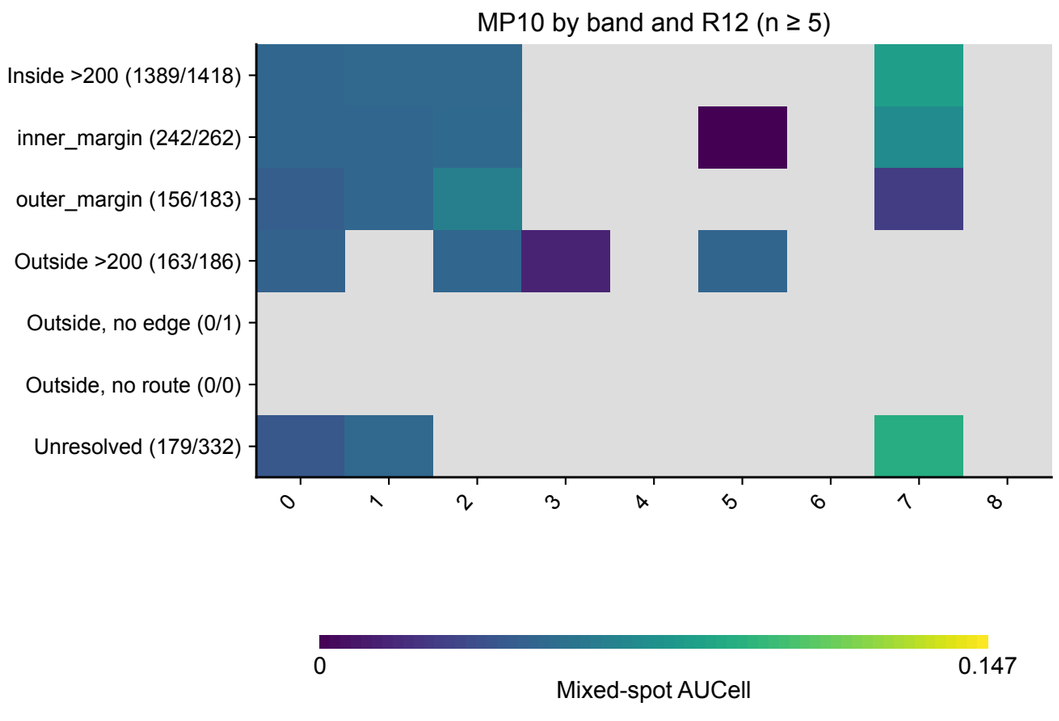
NPJ_sample7 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





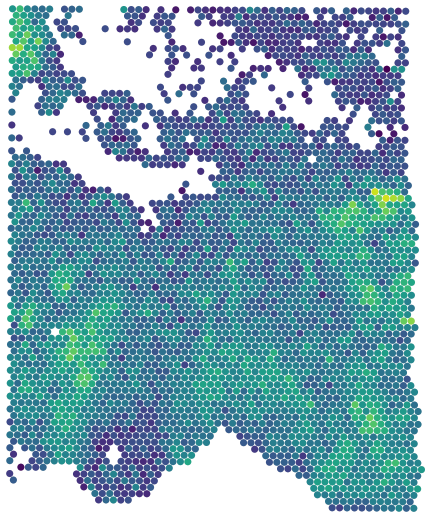




NPJ_sample7 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

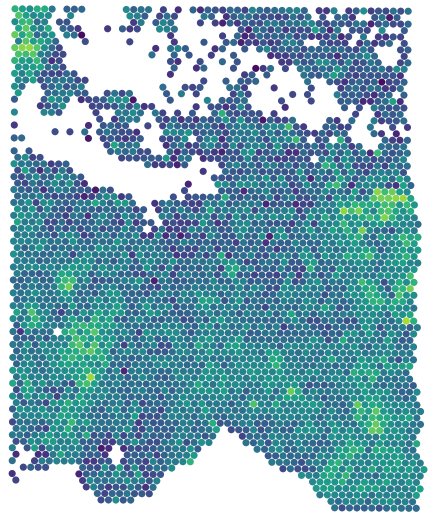
E2F Targets



0 0.152

UCell score

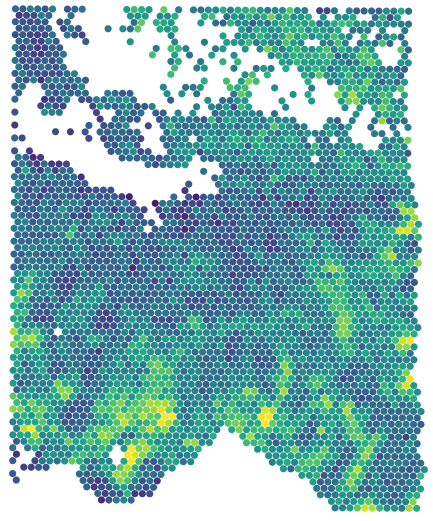
G2M Checkpoint



0 0.134

UCell score

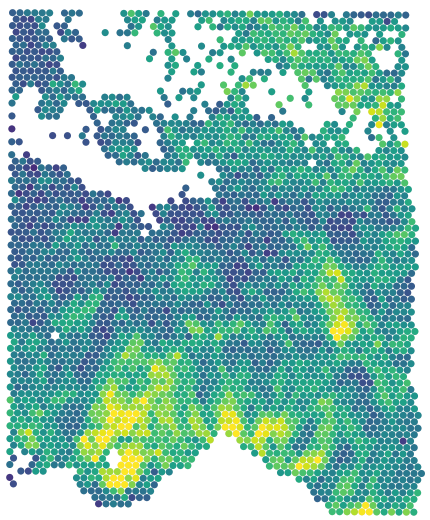
Hypoxia



0 0.211

UCell score

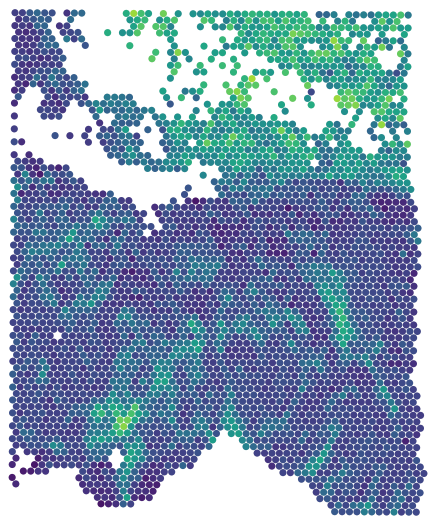
Tnfa Signaling Via Nfkb



0 0.275

UCell score

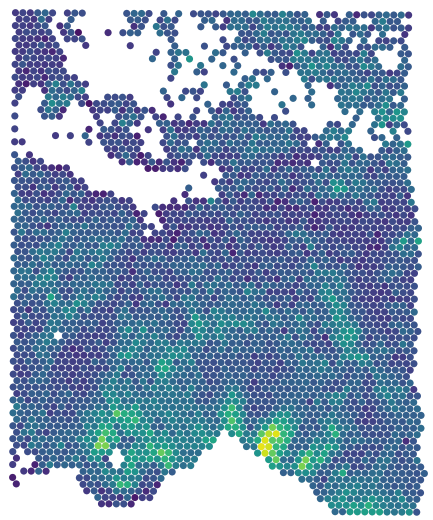
Emt



0 0.386

UCell score

Interferon Gamma Response



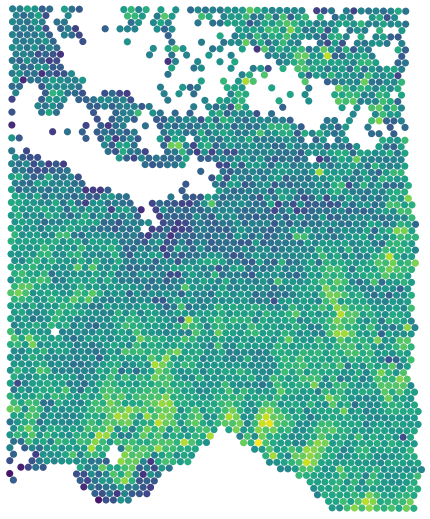
0 0.275

UCell score

NPJ_sample7 | Immune, lipid and matrix programs

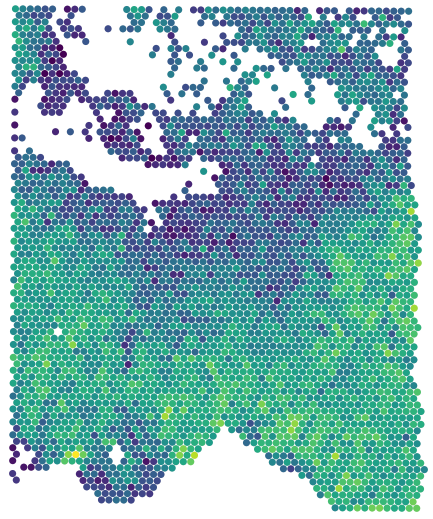
Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

Il2 Stat5 Signaling

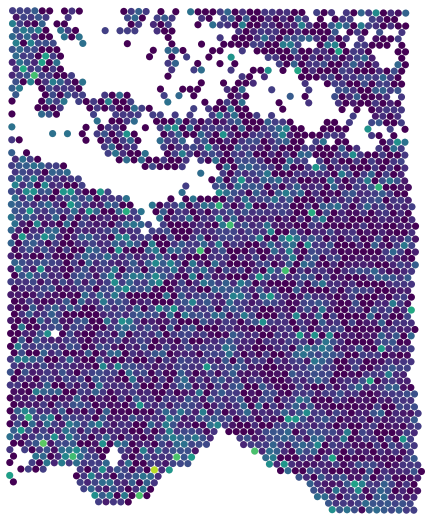


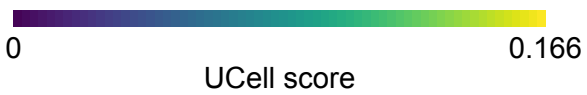
Tgf Beta Signaling

Cholesterol Homeostasis

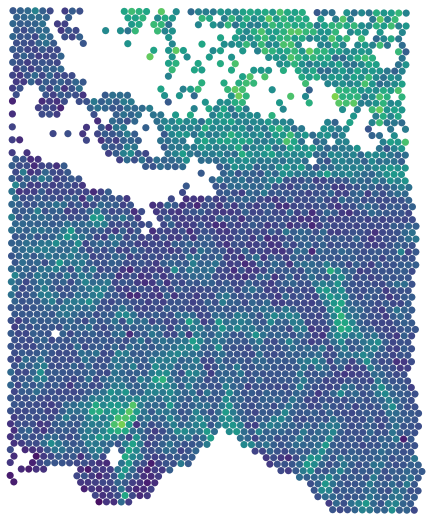


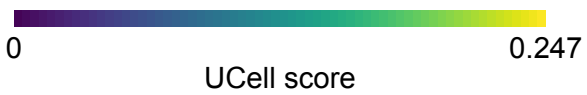
Cholesterol Biosynthesis



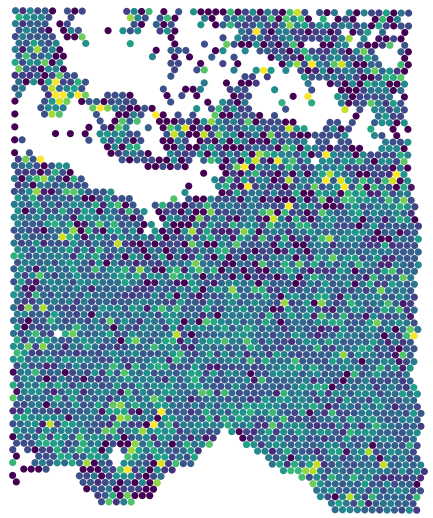


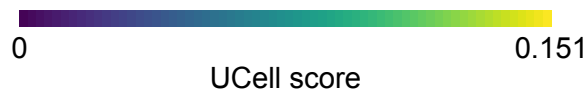
Extracellular Matrix Organization





Co Inhibition By Ctla4

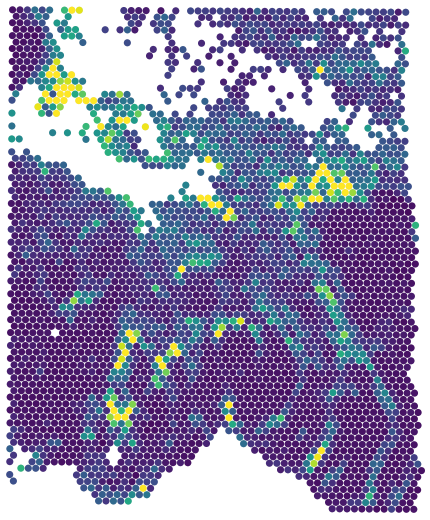




NPJ_sample7 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

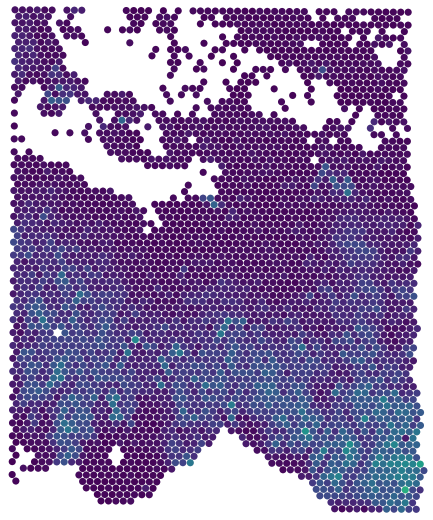
Treg composition



0 0.102

Fraction of all 30

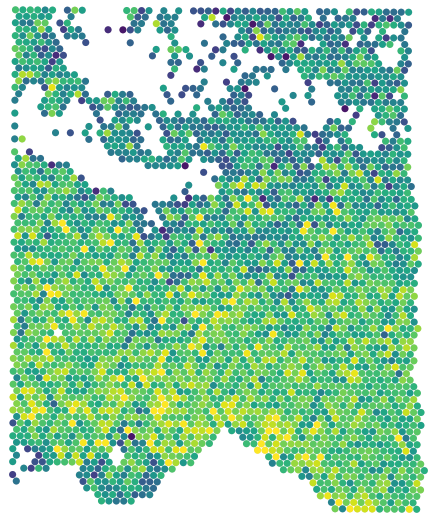
Treg estimated abundance



0 2.85

Posterior mean estimate

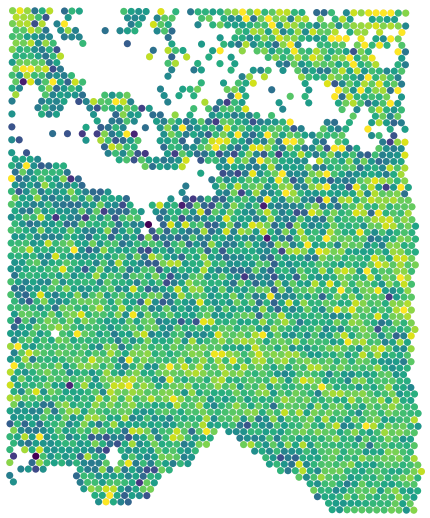
Treg Resting Vs Activated Dn



0 0.0778

UCell score

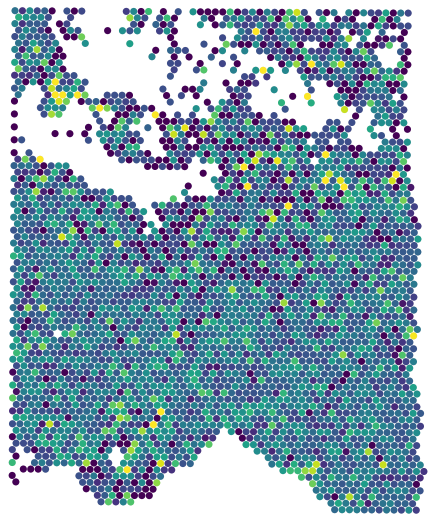
Treg Resting Vs Activated Up



0 0.0609

UCell score

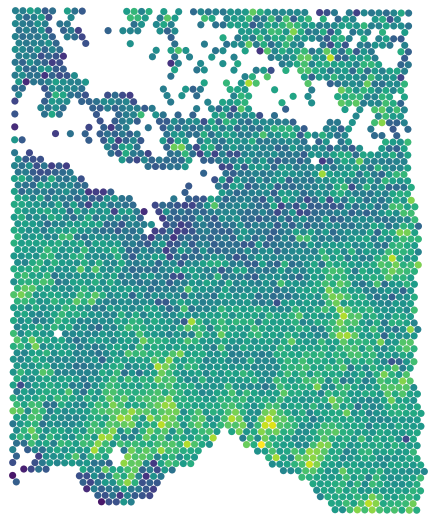
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

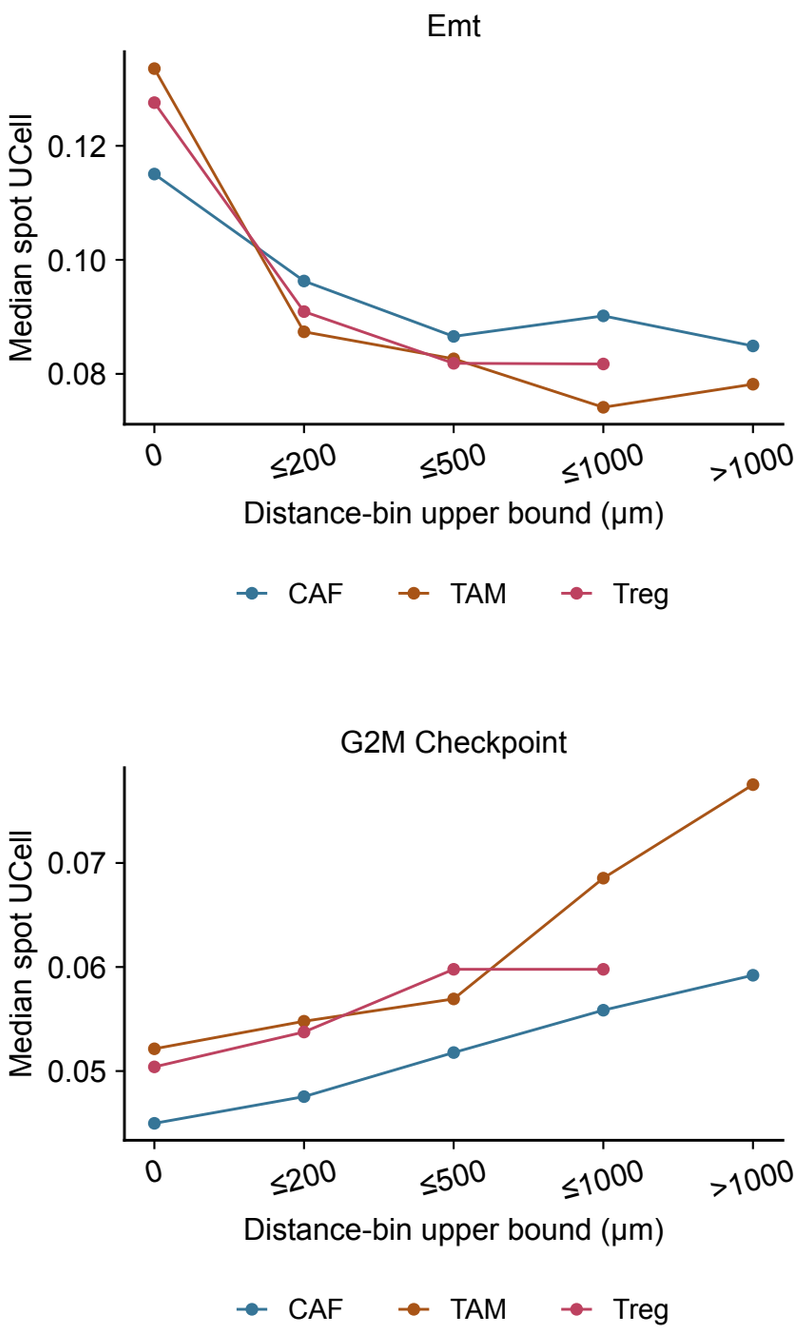


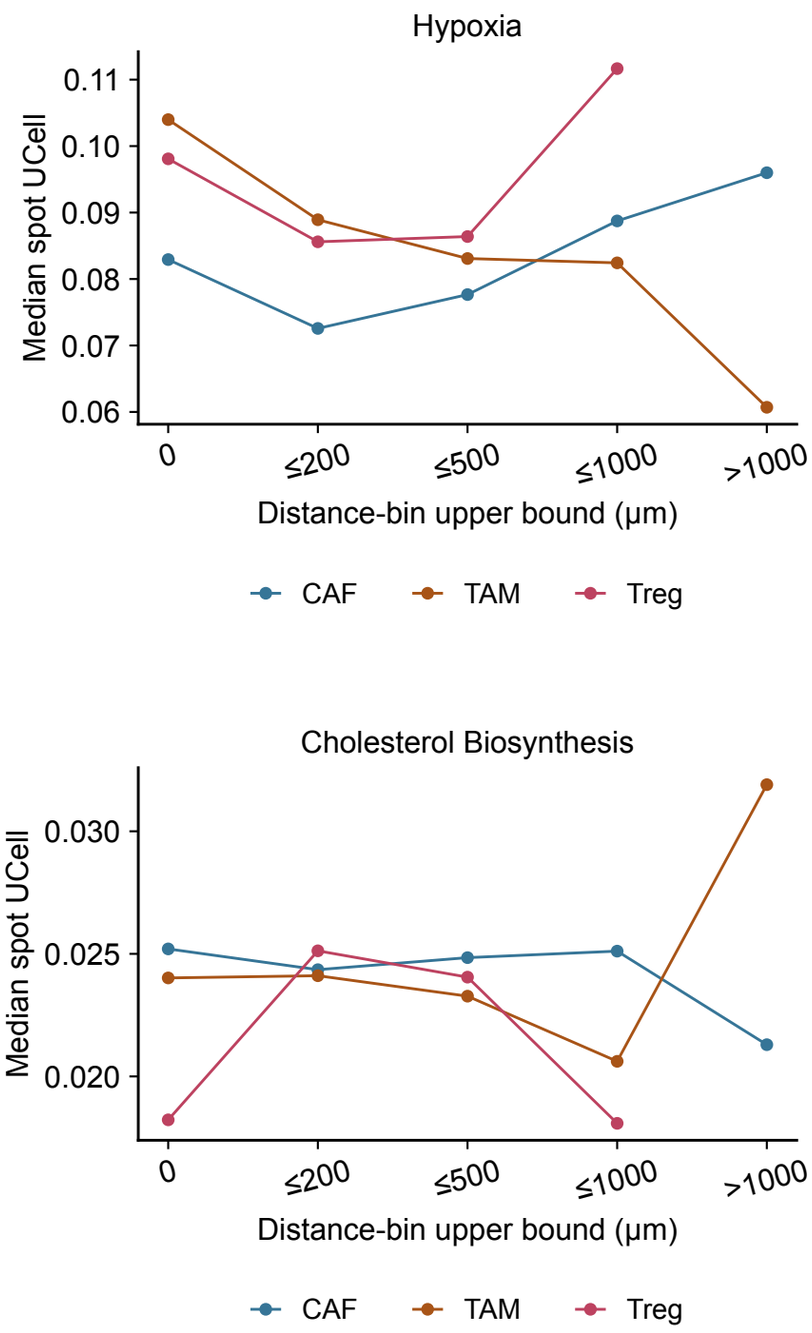
0 0.144

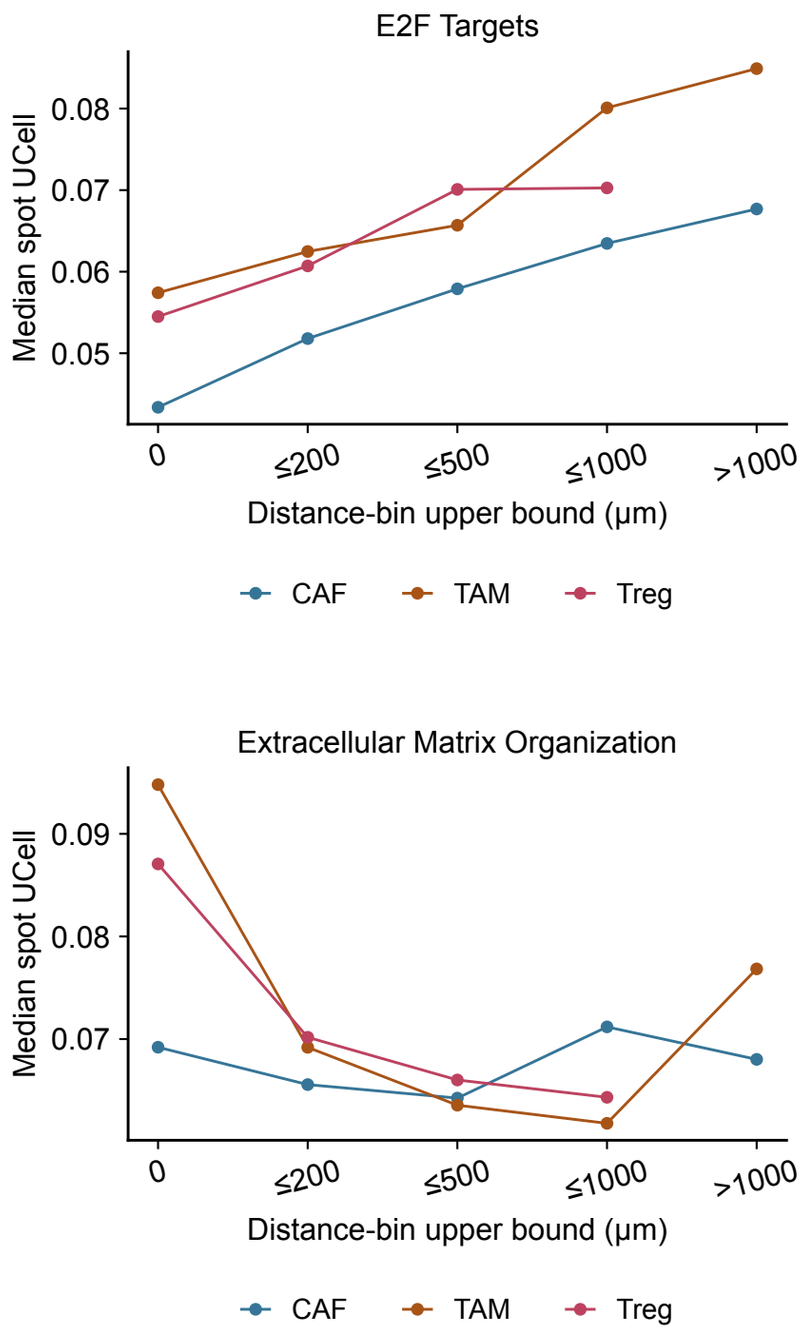
UCell score

NPJ_sample7 | Programs by target distance

Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots







NPJ_sample7 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6